



Application for Certification

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Part 1

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Tier2 Bin5 Standard**

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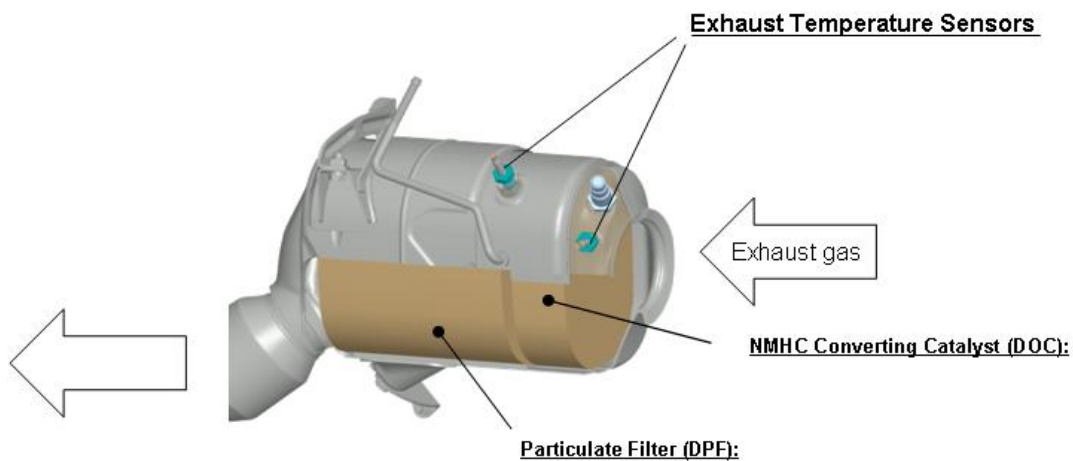
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1 NMHC Catalyst Monitoring

NMHC Catalyst	(f)(1.2.2) Catalyst	(f)(1.2.3)(A) After treatment assistance DPF regen	(f)(1.2.3)(B) After treatment assistance SCR feedgas	(f)(1.2.3)(C) After treatment assistance HC Clean up	(f)(1.2.3)(D) After treatment assistance Ammonia clean up
	P0420	P0420	P0420	n/a for MY11	n/a for MY11



1.1 Aftertreatment assistance for DPF regeneration

BMW's system uses a close coupled DPF system. This positioning guarantees high temperatures during DPF regenerations.

A deteriorated oxygen catalyst has still sufficient exothermal reaction to guarantee a proper DPF regeneration (see measurement below)

Such a deteriorated oxygen catalyst is detected by NMHC conversion efficiency monitoring during coldstart (P0420)

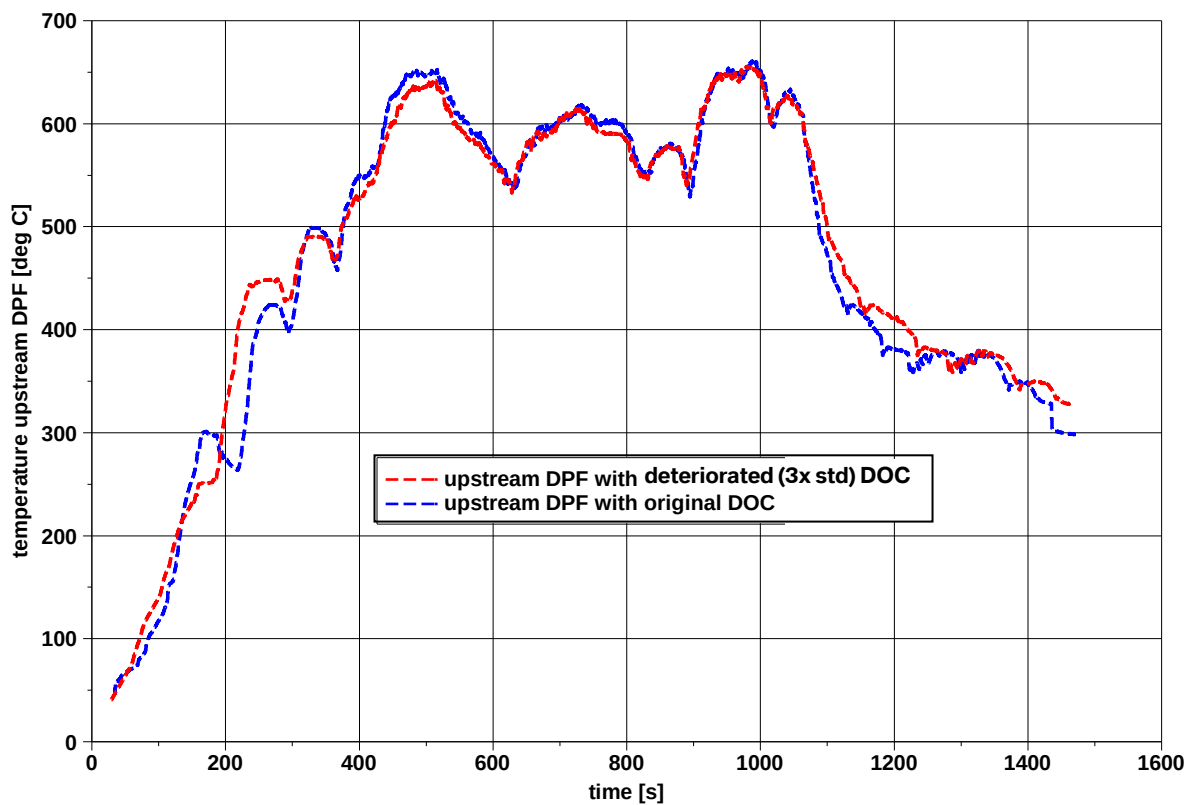


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Temperature DPF upstream during regeneration (CUC)



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1.2 Conversion Efficiency Monitoring (P0420)

General description:

The monitoring concept of the Diesel Oxidation Catalyst (DOC) is based on the evaluation of the HC conversion rate over DOC, indicated by the temperature characteristic during cold start. Due to the current aging status of DOC the measured temperature behaviour is showing significant differences. The characterization is realized by a comparison between the measured and the two simulated borderline temperatures DOC downstream calculated by a temperature model. These limits are on one hand the calculation of the optimal temperature, which is expected during current driving with proper catalyst and on the other hand the temperature which would occur during current operation with an catalyst without any catalytic conversion. Based on these two modelled temperatures in comparison to the measured one the decision regarding proper functionality of the DOC can be established. Therefore a catalyst efficiency factor is introduced. This factor is the ratio between measured exothermic over DOC and expected exothermic which would occur during current operation with a proper DOC. The exothermic level is major effected by the deviation between measured (as well as modelled optimal temp.) in relation to the modelled temperature without any catalytic conversion.

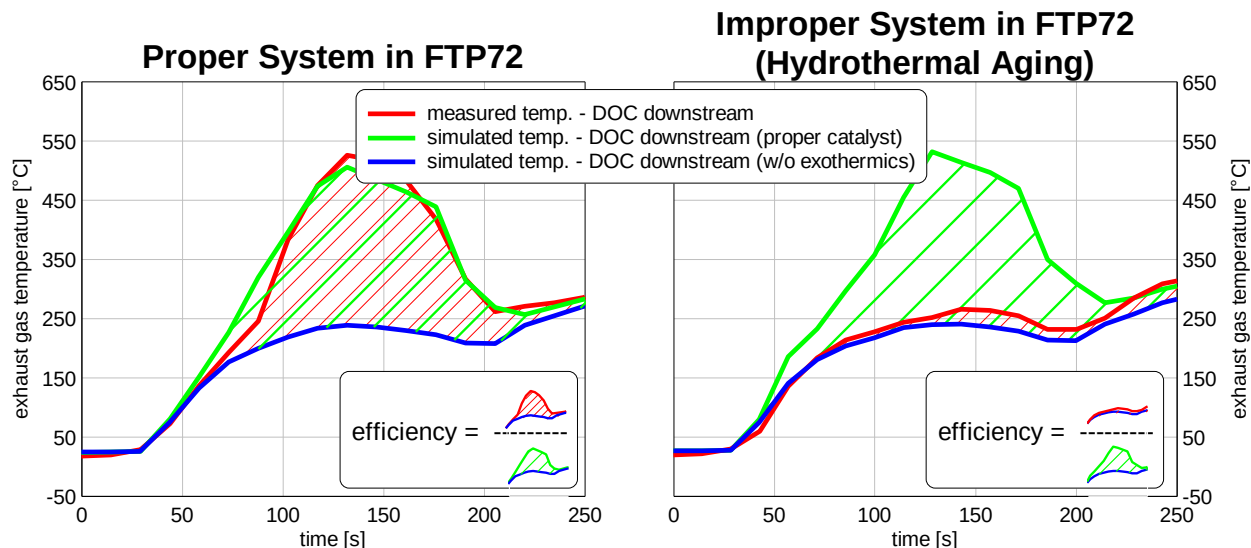


Figure: NMHC catalyst efficiency calculation

A schematic view of efficiency factor calculation is shown in figure above. The DOC monitoring is calculated during each cold start while catalyst warm up. The evaluation of monitoring results will be executed if a sufficient mass of hydrocarbons run through the exhaust line to achieve a certain selectivity level between the measured and the modelled expected exothermic. If these conditions are met and an efficiency lower than a certain threshold is detected a fault is preliminary stored. If this fault is detected after two consecutive DPF-regenerations a DTC is stored and the MIL will be illuminated.

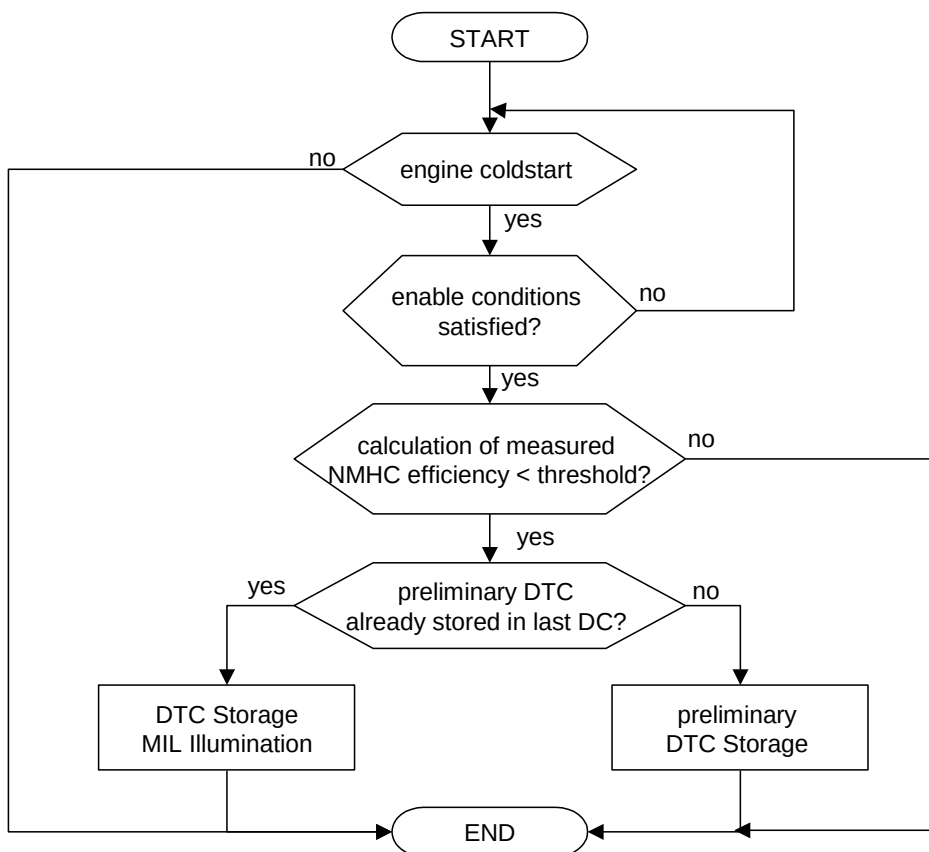


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Flowchart:





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2 NOx Catalyst Monitoring

NOx Catalyst	(f)(2.2.2) NOx Catalyst Efficiency P20EE	(f)(2.2.3)(A) Reductant delivery P20E8, P20E9, P204F	(f)(2.2.3)(B) Reductant tank level P203B, P203A	(f)(2.2.3)(C) Proper Reductant P207F	(f)(2.2.3)(D)(i) Feedback: time to CL P204F	(f)(2.2.3)(D)(ii) Feedback: default/OL see (f)(2.2.3)(F)	(f)(2.2.3)(D)(iii) Feedback: CL limits No closed loop system
--------------	---	--	--	---	--	--	--

2.1 Conversion Efficiency Monitoring (P20EE)

General description:

The conversion efficiency monitoring of the SCR – Catalyst is based on a comparison of the calculated conversion efficiency (through a NOx upstream and a NOx downstream sensor) and threshold value. If the calculated conversion efficiency is lower than the modelled threshold value, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL will be illuminated.

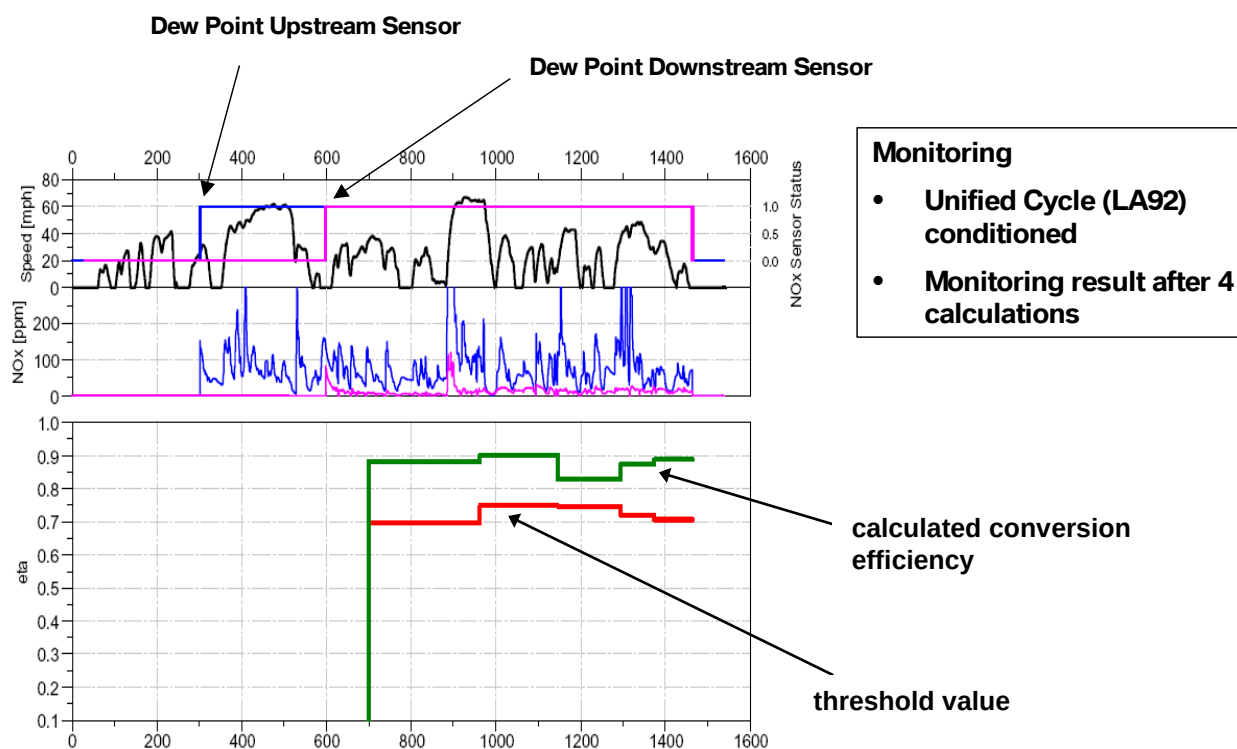


Figure: conversion efficiency monitoring

The following examples the monitor detecting a malfunction versus a nominal system:



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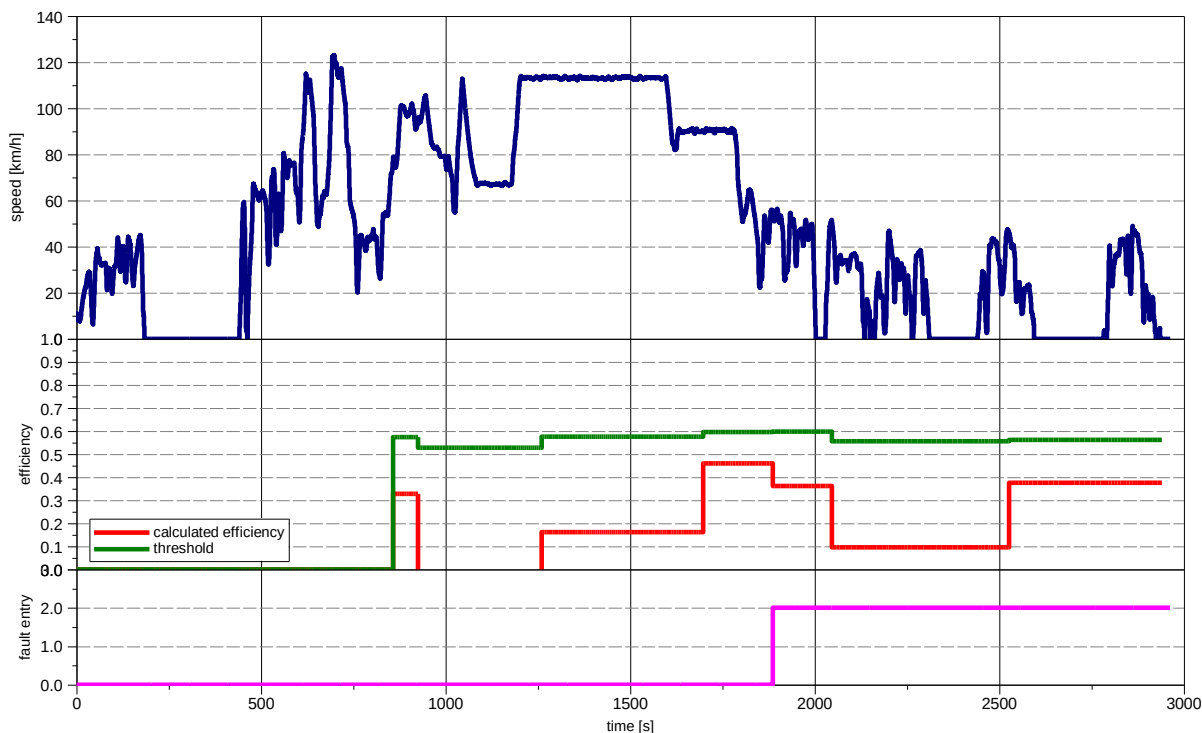


Figure: Detection of an efficiency malfunction in the SCR-System.

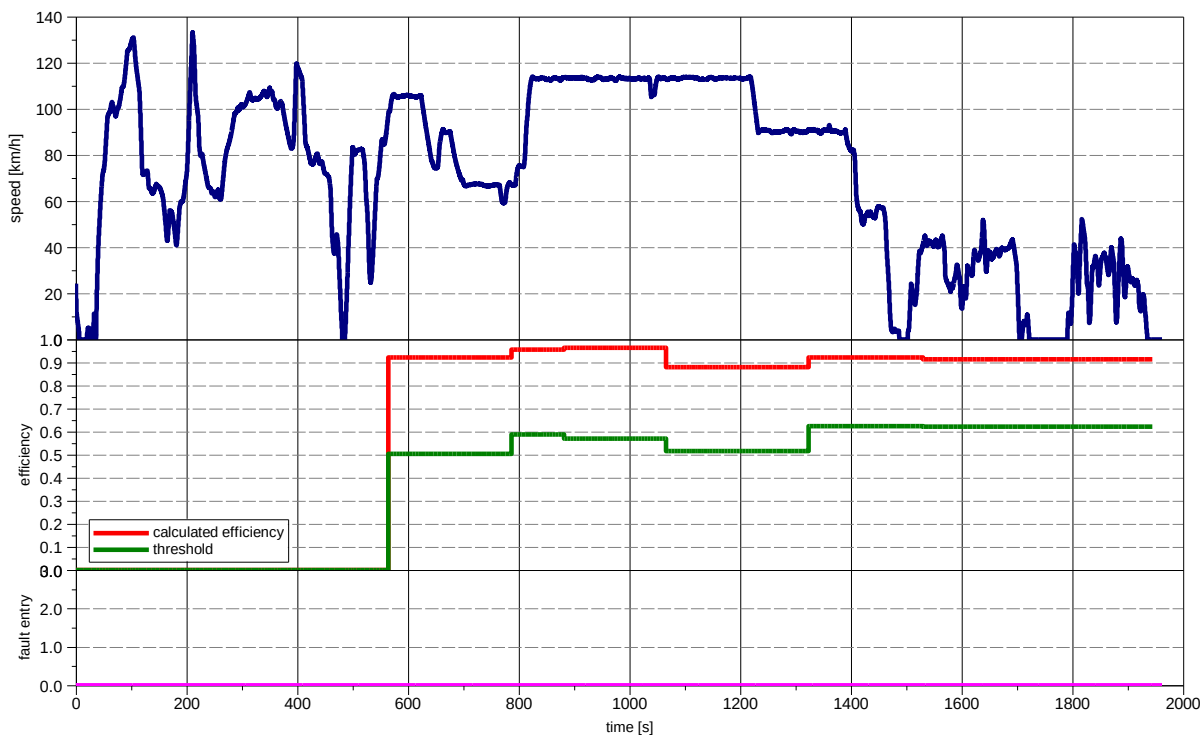


Figure: SCR-System without a malfunction.

For calculation of the conversion efficiency the upstream NO_x mass flow and the downstream NO_x mass flow are used. Therefore the NO_x sensors have to be valid and all enable conditions have to be satisfied. While these conditions are satisfied, the upstream and downstream NO_x mass flow

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are integrated. If the upstream NOx mass reaches an evaluation threshold, a valid actual conversion efficiency is calculated.

For a complete monitoring the calculations have to be done for four times.

The following enable conditions will be checked for Nox mass integration:

- NOx sensors valid
- dosing system is active
- environment temperature
- environment pressure
- regeneration not active
- SCR-exhaust catalyst temperature within a calibrated range
- exhaust gas flow within a calibrated range

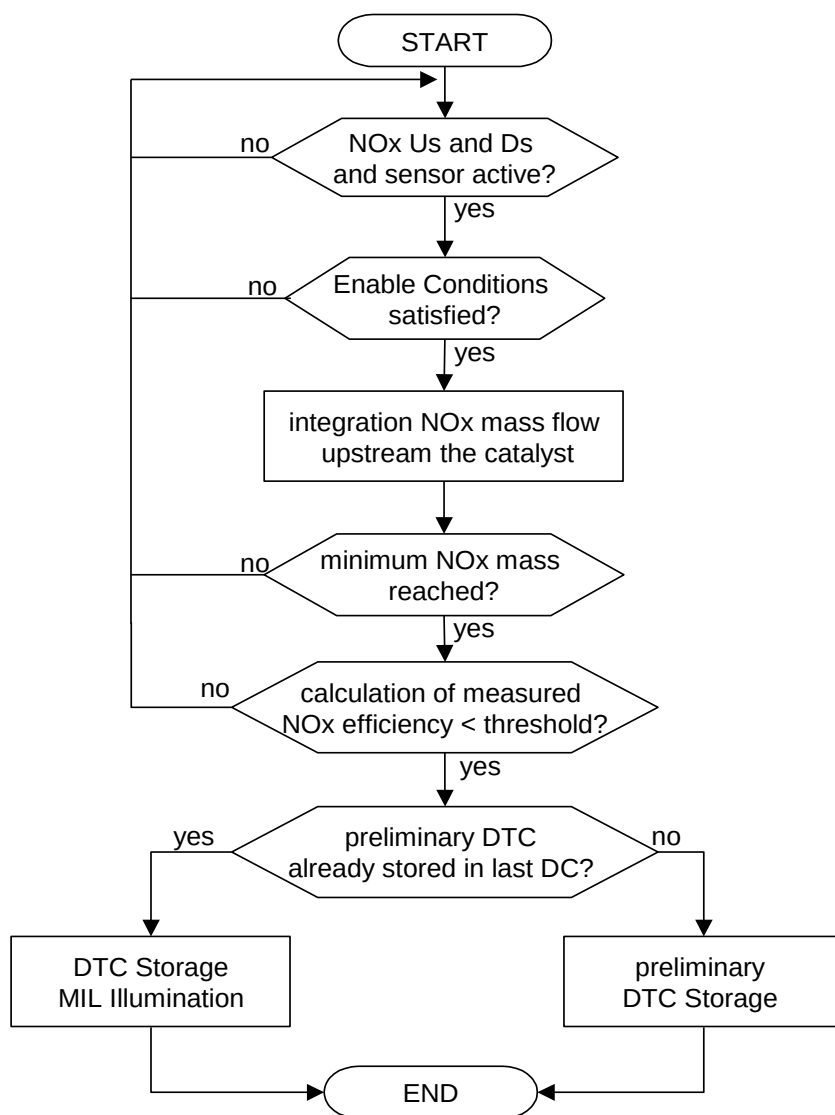


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Flowchart:





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2.2 Long Term Adaptation
(P20EE)

General description:

In a properly working system the adaption doesn't work at all. Tolerances (e.g. wear) in the dosing system can lead to wrong dosing amounts over lifetime. In operation points with a high expected efficiency the adaption is able to detect these deviations and adjusts the correct dosing amount. So the adaptation is a function to guarantee long term efficiency. The learning of the adaption factor is a very slow process (e.g. 7000 miles until it reaches its limits). Short term drifts or failures are always detected by SCR efficiency monitoring.

Therefore the NOx-sensor value downstream SCR is compared to the calculated NOx value downstream SCR. If deviations occur the dosing amount is corrected temporarily. The systematics of the corrections is evaluated and an adaptation factor is applied on the dosing amount. The operation range of the long term adaption is the same as for NOx conversion efficiency monitoring (2.1) in operation points where expected efficiency is higher than 70 %.

If the correction factor exceeds an upper or lower threshold, a fault is detected and a preliminary DTC will be stored. If this fault is detected in two consecutive driving cycles, the MIL will be illuminated.

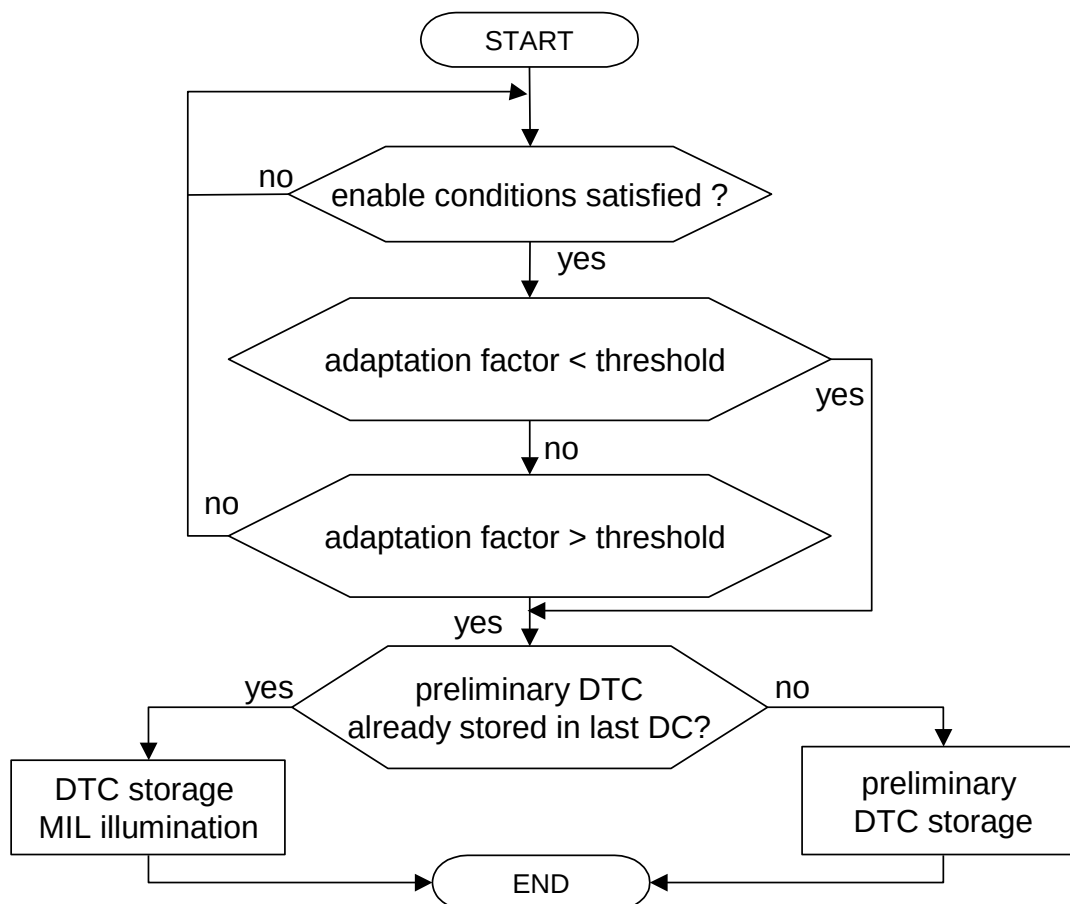


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Flowchart:



2.3 Reductant Delivery Monitoring

2.3.1 Monitoring the Enabling the SCR Reductant Dosing (SCR Time to closed Loop) (P204F)

BMW's system requires that the SCR inlet temperature achieve 190°C in order to enable SCR reductant dosing. To fulfill 2,5x times applicable FTP standard a monitoring function will force the activation.

Description:

1. If modelled exhaust temperature is above a specified temperature threshold a timer will be activated. (wait defect time)
2. After a specified time where the temperatures stays above the temperature threshold the dosing system is checked for activation.
3. If the modeled temperature drops below the temperature threshold before the specified time is reached, the timer is reset to 0 seconds

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4. If the system is not active, the fault code entry will force the activation of the reductant delivery system by switching the release temperature from measured temperature to modelled temperature.
5. If the diagnostic is completed it will not run again in the same driving cycle.

The used modeled temperature does not use exhaust temperature sensors as input.

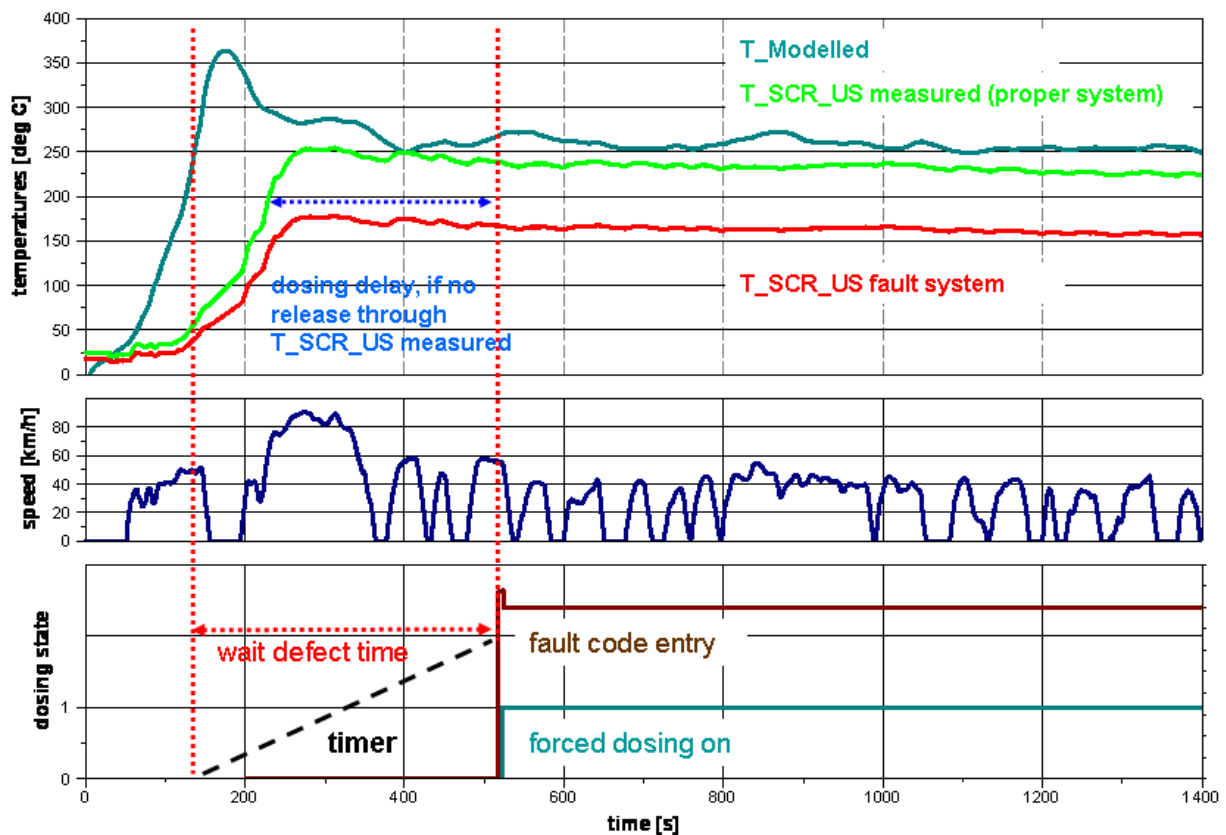


Figure: Proper and improper system in a FTP75



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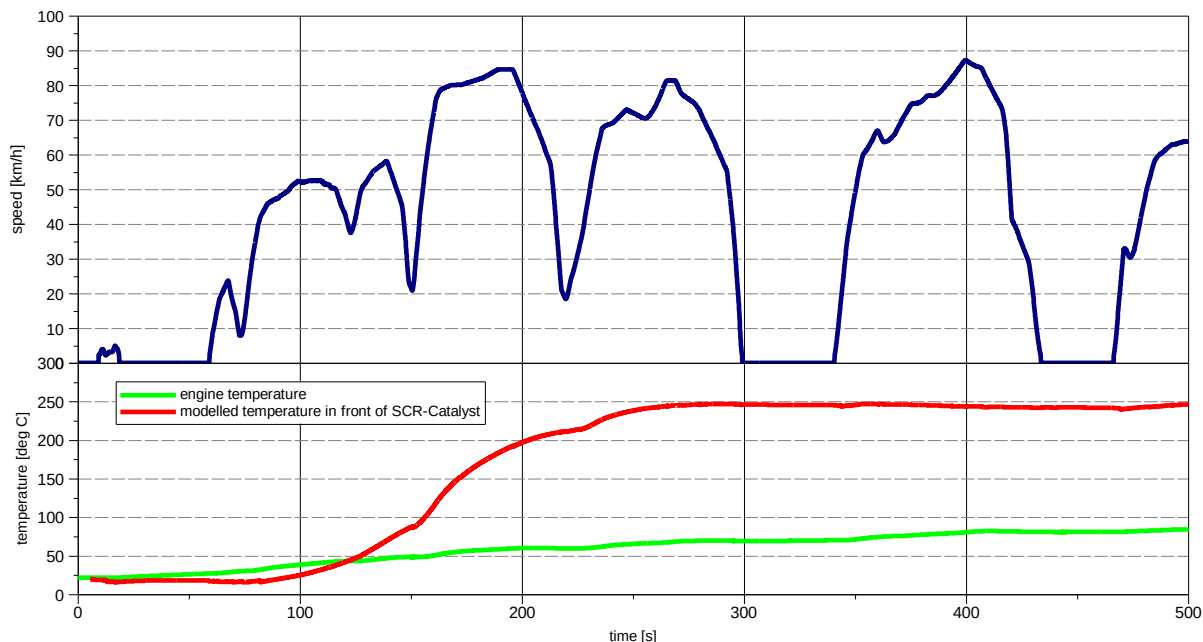


Figure: Warming up of the modeled temperature after cold start driven on road under FTP conditions.

Additional the measured SCR Us temperature is checked with P242A (model plausibility check) continuously.

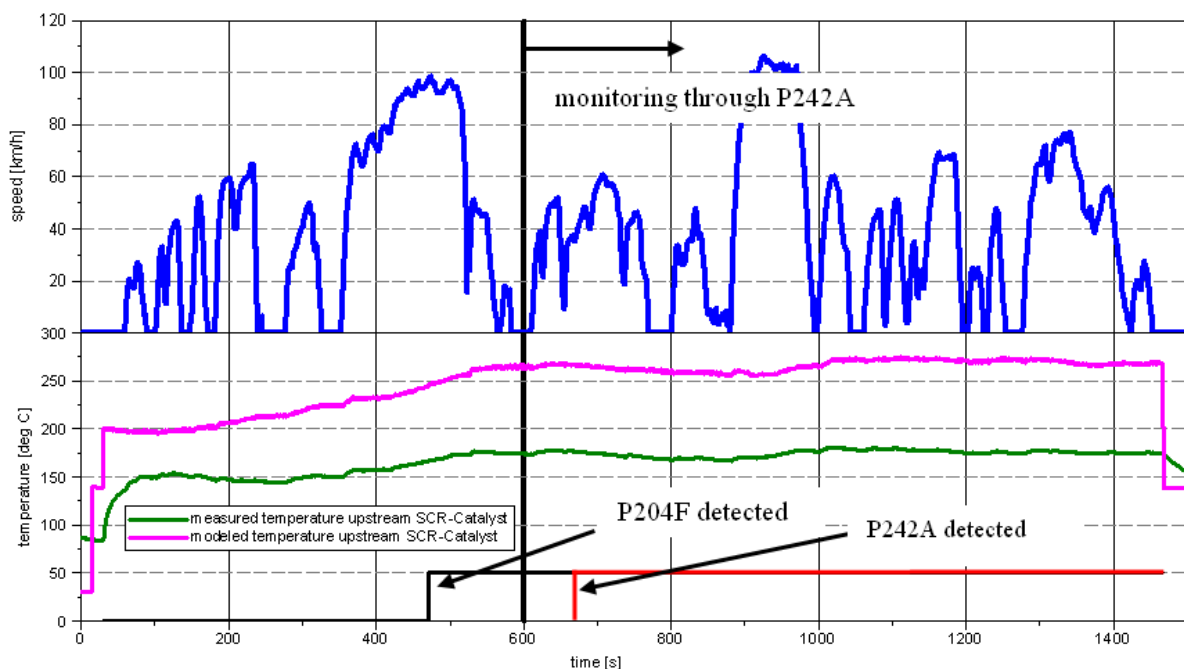


Figure: Continuous Monitoring through P242A with an improper System

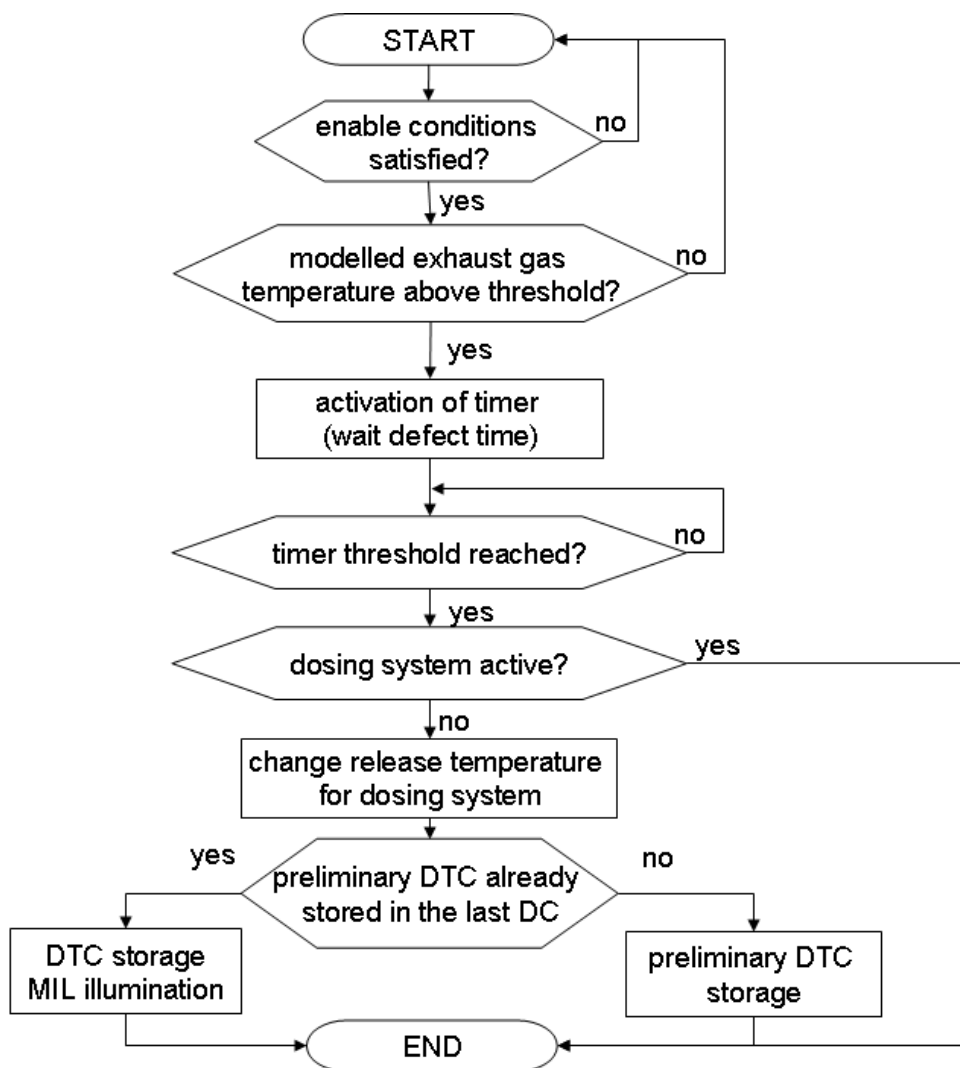


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Flowchart:





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Additionally according to (f)(2.2.3)(F) time to closed loop for SCR system is monitored through single component monitoring. The input (release) components of the SCR system and their monitoring strategies are listed in the following tables:

	Release of dosing dependent on		
	Reductant Delivery System ready for dosing	Engine Speed above threshold	Average SCR Temperature above threshold
		n > 490 rpm	t >= 190 °C
Temperature Sensor Upstream SCR	See separate table below		P242A P204F
Engine Speed Sensor		P0335 P0336	
Reductant Delivery System			

	Reductant delivery system ready for dosing dependent on			
	heater release	SCR exhaust gas temperature above threshold	engine speed above threshold	urea system pressure within range
	Urea-Tank-Temperature < -7°C & Ambient Temperature < -11°C	t >= 80°C	n > 490 rpm	SCR pump pressure > 3100mbar & SCR pump pressure < 6500mbar
Component				
temperature sensor upstream SCR		P242A		
tank temperature sensor	P205B			
urea pressure sensor				P20E8 P20E9 P204B
engine speed sensor			P0335 P0336	
ambient temperature sensor	P0070 P009A			
Pressure Build Up	P20E8			P20E8

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The “time to closed loop” functionality for the heater system is realized by an enforced reductant pressure build-up after a specified time.

The time for the pressure build up depends on:

a) Urea tank temperature:

Urea tank Temperature [°C]	-50	-26,5	-21,5	-16,5	-9,1	-9	-7,5	-7,4
Time [s]	3000	2600	1700	1100	1100	150	150	0

If the pressure build up after the SCR-tank temperature dependent time is not successful an ambient temperature dependent timer is started

b) Ambient temperature:

Ambient Temperature [°C]	-50	-30	-25	-15	-9,6	-9,5	-5,1	-5
Time [s]	2000	1500	1300	800	400	145	145	0

If the pressure build up after this time is not successful the DTC for pressure build up monitoring (P20E8) is set.

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2.3.2 Pressure Build Up Error
(P20E8)

General description:

Due to proper conversion capability of the NOx catalyst, the reductant pressure build up has to be successful. After exceeding a specified catalyst temperature the system tries to build up the pressure of reductant injection system.

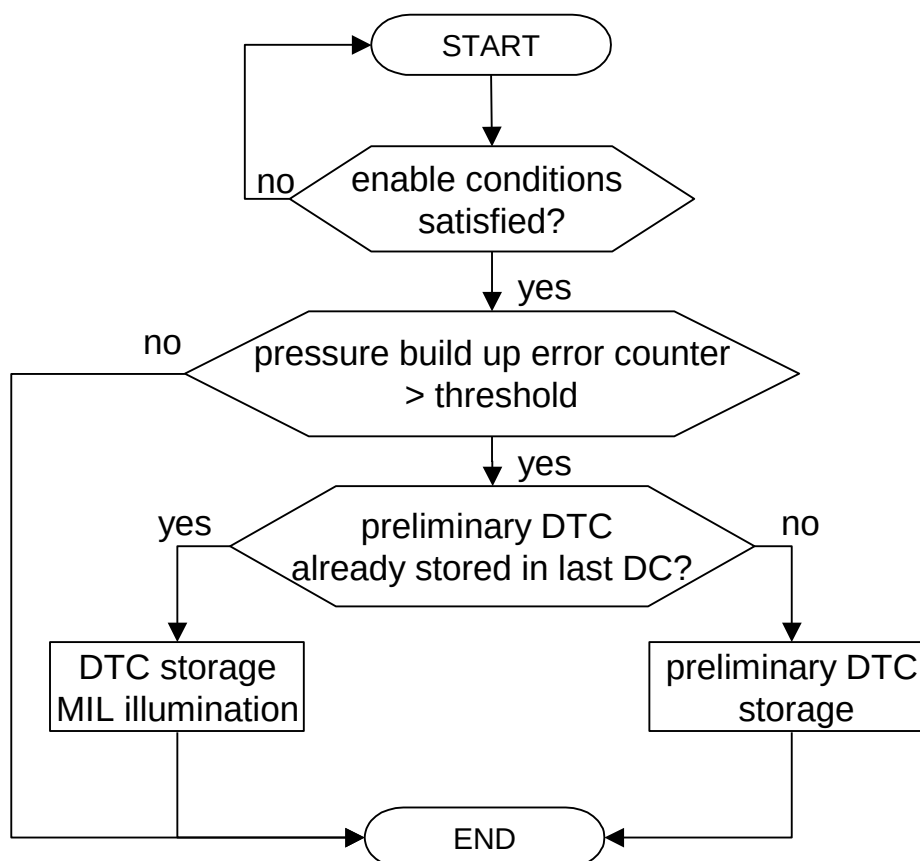
Therefore it is necessary to close the dosing valve and drive the urea metering pump with a specified duty cycle for a calibrated time. If the pressure threshold can not be reached the dosing module will be opened for a specified time to bleed the line. After bleeding the line a counter will be incremented to count the pressure build up cycles.

In one DC three pressure buildup cycles are possible.

A fault is detected if the pressure build up counter exceeds a specified threshold, BMW's application is (3). In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

This monitoring function runs once per drive cycle before urea dosing is active. When this monitor is passed the continuous pressure monitoring is active.

Flowchart:



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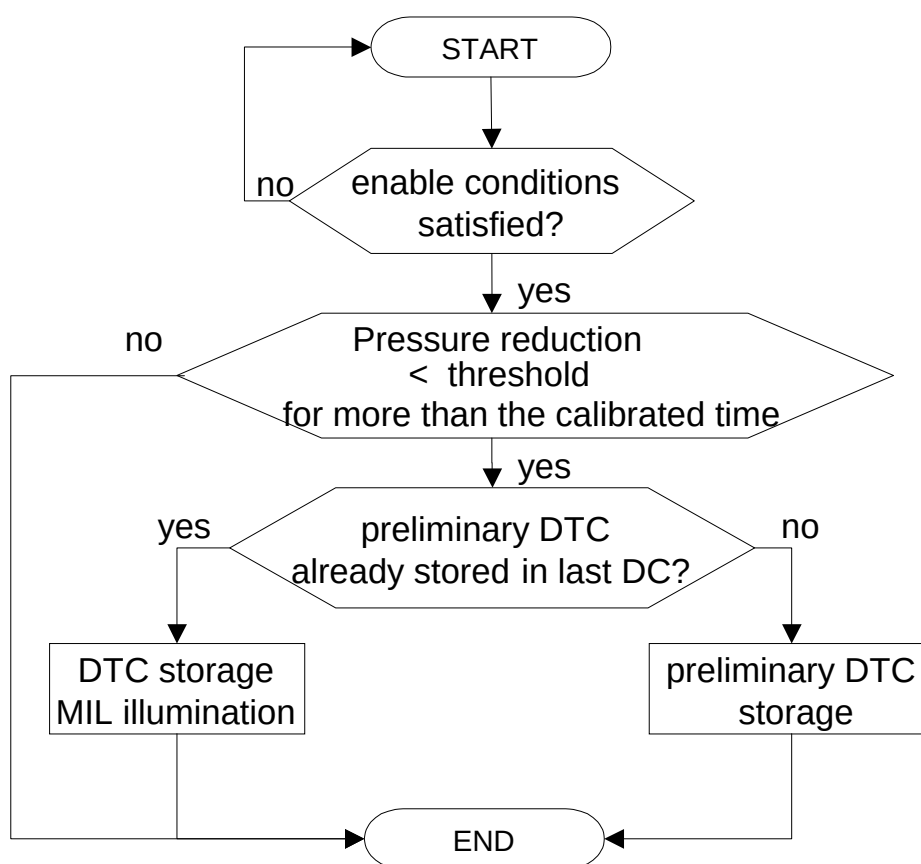
2.3.3 Pressure Reduction Error
(P20A5)

General description:

In this function the opening of the reverse valve is checked. The monitoring checks in engine afterrun an urea pressure reduction. This is only possible, if the reverse control valve opens and the urea pump is running. Therefore a succesfull pressure build up during engine running is necessary.

If the engine is stopped the status pressurereduction is present and the reverse valve opens. If the pressure reduction is smaller than the applicated threshold the pressure reduction error is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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2.3.4 Pressure Control Monitor
(P20E8, P20E9)

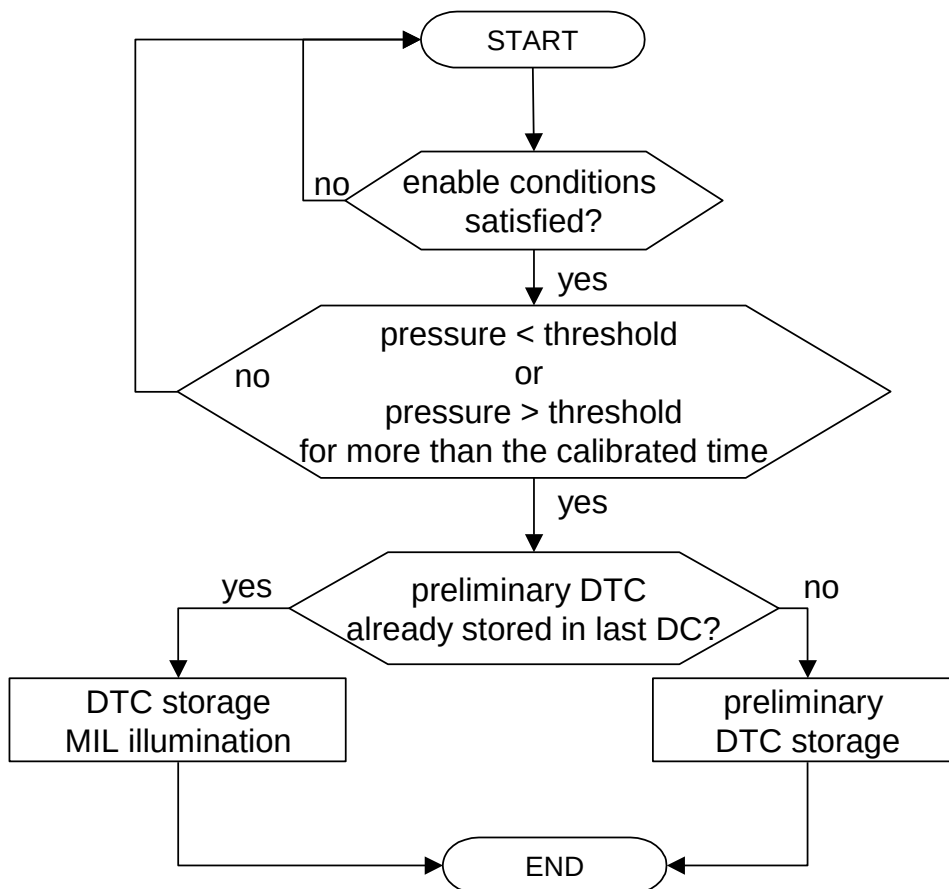
General description:

For proper functionality of the NOx conversion catalyst a constant pressure of the reductant at the dosing valve is necessary. The actual reductant pressure is monitored continuously by comparing with a minimum and a maximum threshold (pressure control deviation). If the actual value exceeds its calibrated limits for more than a specified period of time, a fault is detected and a preliminary DTC will be stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

For monitoring the reductant pressure the urea metering unit has to be active. Therefore following conditions have to be satisfied:

- NOx catalyst temperature above calibrated threshold (guarantees that urea is liquid)
- reductant pressure build up ok (see pressure built up error)

Flowchart:





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2.4 Reductant Tank Level Monitoring

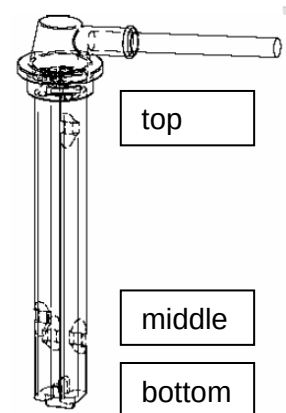
2.4.1 Tank Level Sensor Plausibility Monitoring for Active Tank (P203B)

General description:

For evaluation of the tank level an intelligent sensor with three single level positions is used. The tank level plausibility is monitored by evaluating the PWM signal from the level sensor.

		not moistened		
		bottom	middle	top
moisted	bottom	x	OK	OK
	middle	PF	x	OK
	top	PF	PF	x

PF = plausibility fault



The PWM signal of 30% represents a plausibility fault of the level sensors. This one is detected by evaluating the three single level signals regarding their mounting position in the tank.

e.g. If level 3 (top) gets a fluid signal, level 2 (middle) and level 1 (bottom) are also expected to get a valid fluid level signal, because they are mounted lower in the tank than level 3. Otherwise an error is detected this is indicated by a PWM Signal of 30%.

A fault is detected if the PWM signal of the sensor is lower than a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Following enable condition have to be satisfied for this monitoring:

- tank temperature above a specified threshold

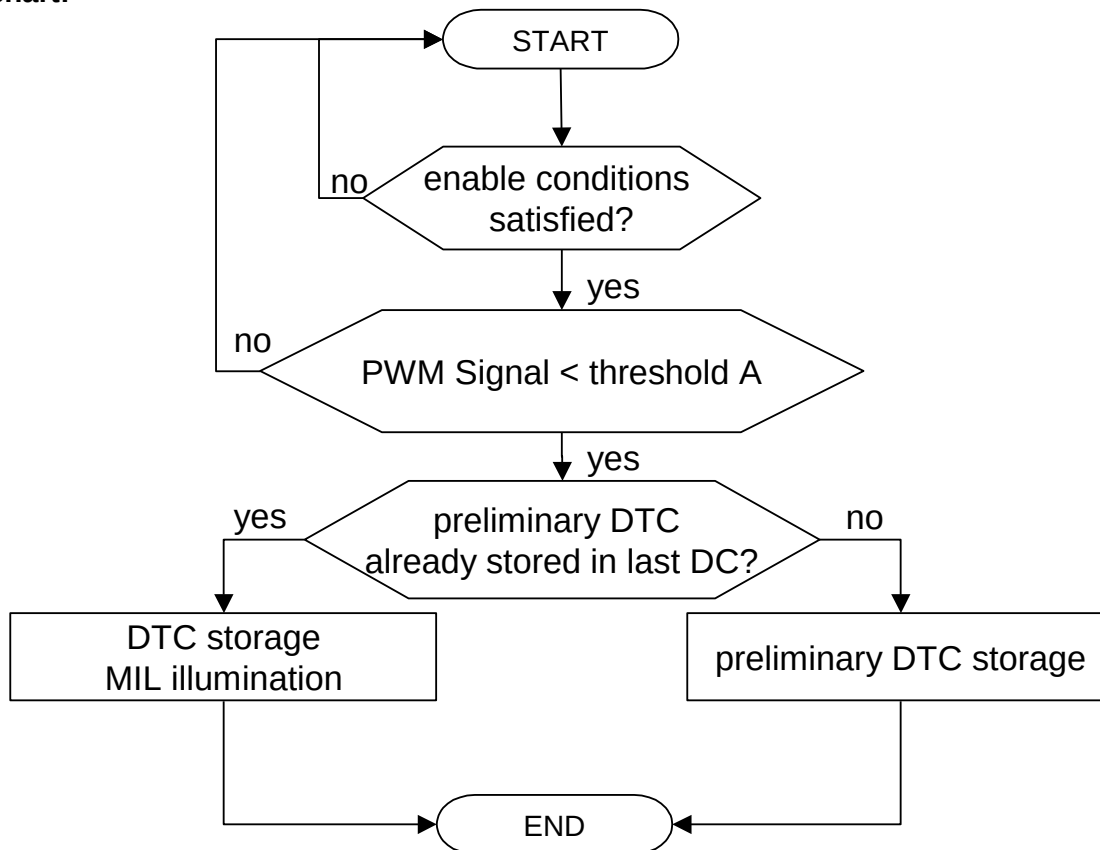


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Flowchart:





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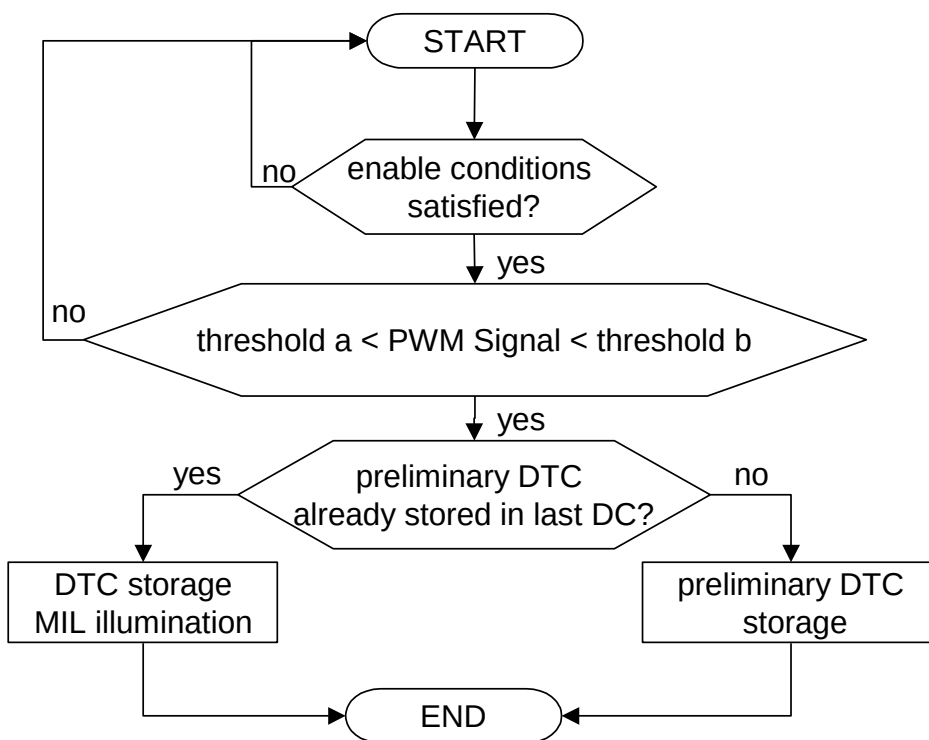
2.4.2 Tank Level Sensor Signal Monitoring for Active Tank
(P203A, P203B)

General description:

For monitoring the electrical signal of the three single level sensors in the tank the PWM signal of the intelligent sensor is evaluated. A PWM signal of 40% represents a circuit continuity fault of at least one of the three single level sensors.

If the PWM signal from the intelligent sensor is within a specified range a fault is detected. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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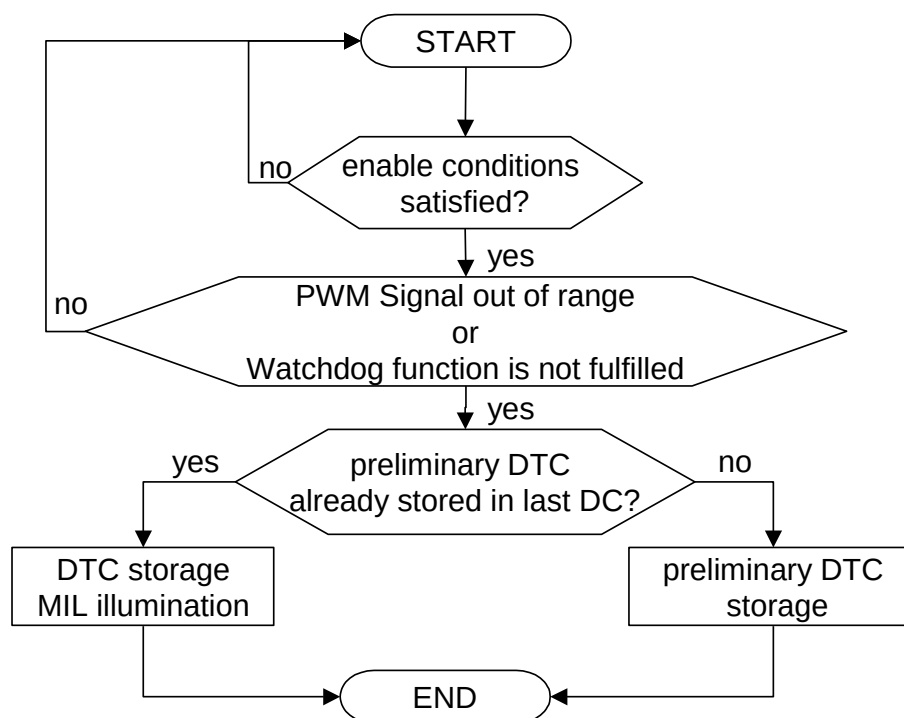
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General description:

For monitoring the tank level sensor its PWM signal is evaluated. The sensor monitoring will detect a fault if the PWM signal is outside a specified range or a watchdog function does not get a specified PWM value in a specified time interval. The tank level sensor sends every 60 sec a defined message for a short time to the ECU (watchdog function). In case of absence of the message in the specified time a fault will set.

In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This fault is the same for active and passive tank.

Flowchart:



Signal description:

The PWM raw signal is transformed into a 'raw sensor signal' via characteristic and then into to the 'sensor signal'.

The PWM 'raw sensor signal' is used for the two P-203A faults, and the PWM 'raw signal' is used for the P203B fault.

The 4 pin level sensor of the active tank (one common pin and three pins with various length) can show four values, 0 %, 33 %, 66 % and 100 %.



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The first fault, the sensor error P-203A shows the defect of an open circuit, shortcut to battery, shortcut to ground or shortcut between the sensors.

This fault is set if the PWM 'raw sensor signal' is between 35 % and 45 %.

The second fault the sensor monitor error (P-203B) shows the defect of the correct transforming (intelligence of the sensor is proofed).

This fault is set if the raw signal value is smaller than 20 % or bigger 90 % or in the second case if the "watchdog function" doesn't work for a time longer then 70 seconds. The watchdog function sends an 'empty to full signal' change one time every 60 seconds, to test the right functionality of the sensor intelligence.

The third fault, the level plaus error (P-203A) indicates a plausibility fault of the sensor, here also the transformed raw sense signal is used.

The fault is set, if the transformed raw sense signal is smaller than 35 %.



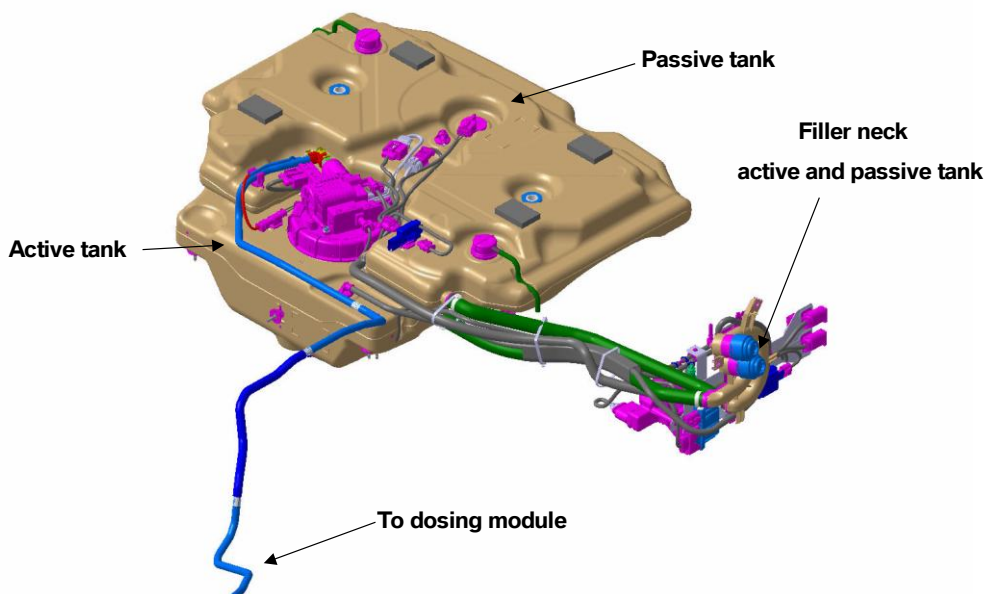
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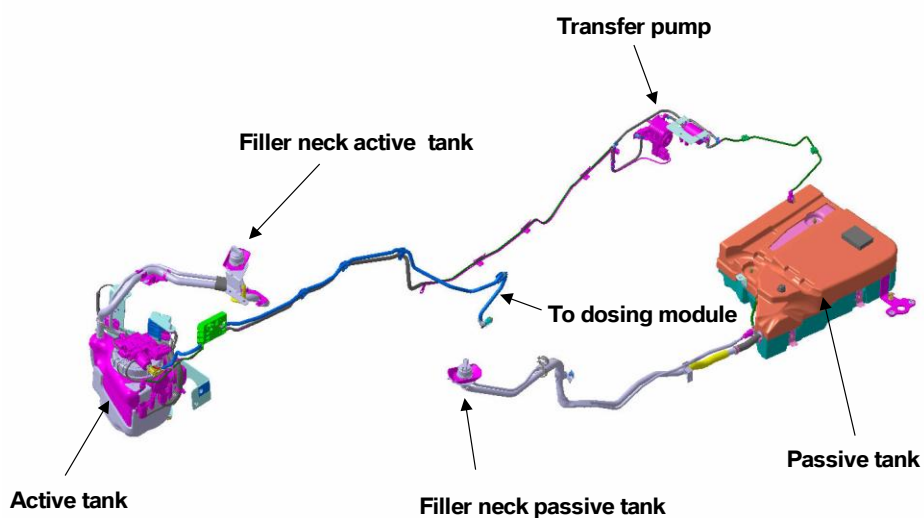
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2.5 Proper Reductant
(P207F)

Adblue System of BMW 3 series:



Adblue System of BMW X5:



see also: 15.36 Reductant Injection System

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General description:

The SCR system is monitored regarding to the quality of the reductant medium, when refilling had taken place.

Refilling active tank:

The wrong medium detection is started if the active tank is refilled from extern through the active tank filler neck (this is only possible, if the passive tank is already empty for a long time, otherwise the active tank is always full) or if the difference between the calculated volume in the tank and the measured volume is more than 0,4 gal (in this case must have been a not detected refilling event before).

The whole volume of the active tank is about 1,5 – 2 gallons.

Refilling without detection (small amounts):

If small amounts e.g. 0,2 gallons are filled in the active tank, the level sensors will not detect this. A wrong medium detection is also started, if some of this small refilling events take place. The injected amount of reductant is calculated and compared with the geometric calculated amount of the reductant between two level sensors. Is the injected calculated amount bigger than the amount between the level sensors the wrong medium detection also starts.

Refilling with detection (big amounts):

If big amounts eg. 1 gallon are filled in the active tank, the level sensors will detect this and the wrong medium detection is started.

If the wrong medium in the active tank is about 70 % of the volume, that means the Ad Blue volume in active tank is 30 % or less, the wrong medium will be detected and the fault code is stored.

If the wrong medium in the active tank is lower than 70 % for example 40 % and the Ad Blue volume is 60 % the quality detection is started, but the quality of emissions will not get bad because more Ad blue/wrong medium is injected to hold the required emission standard.

Refilling passive tank:

The normal transfer pumping event (The transfer pump is started every 150-300 gramm of reductant consumption to keep the active tank always full) is stopped until a larger amount (3000 – 4000 gramm) of volume in the active tank is free. As soon as the large amount is pumped into the active tank the quality detection is started.

The whole volume of the passive tank is about 4 gallons.

Refilling without detection (small amounts):

If small amounts e.g. 0,5 gallons are filled in the passive tank, the level sensors will not detect this. A wrong medium detection is also started, if some of this small refilling events take place.

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The injected amount of reductant is calculated and compared with the geometric calculated amount of the reductant between two level sensors. Is the injected calculated amount bigger than the amount between the level sensors the wrong medium detection also starts.

Refilling with detection (big amounts):

If big amounts eg. 3 gallons are filled in the passive tank, the level sensors will detect this and the wrong medium detection is started.

The normal transfer pumping event from passive to active tank is stopped, until the active tank is before the warning scenario, then a big amount of the wrong medium is pumped in the active tank, and the wrong medium can be detected, the fault code is stored.

Quality detection:

After detecting a refill, the test of the proper reductant will be performed. Therefore the NO_x catalyst conversion capability is monitored for a specified time. This time period depends on the reductant consumption during monitoring interval. If the conversion efficiency falls below a minimum threshold in this interval, a wrong medium fault is detected. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

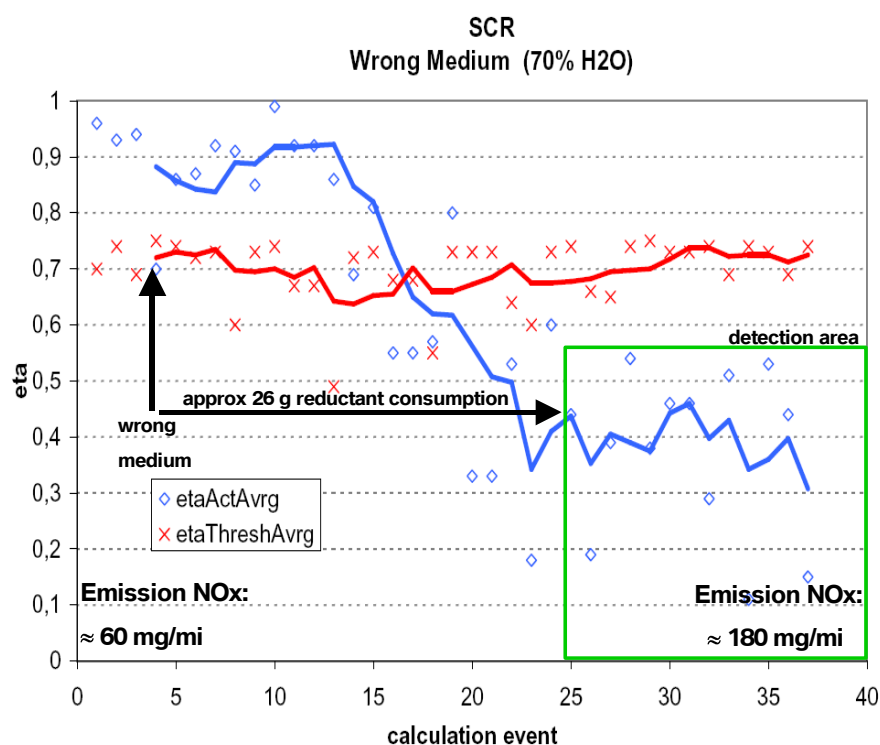


Figure: Monitoring for Proper Reductant



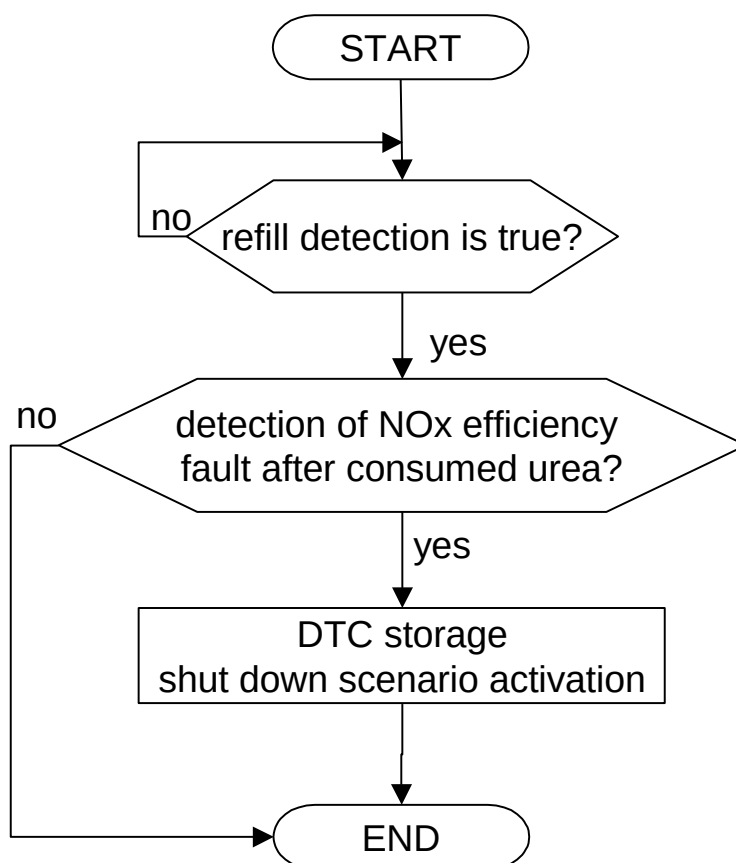
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In case of an incorrect medium detection the warning sequence will be set to the second warning level. This means, the wrong medium has to be replaced within the next 200 mls, otherwise the restart prevention will be activated.

Flowchart:





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3 Misfire Detection

Misfire	(f)(3.2.1) Misfire	(f)(3.2.2) Threshold misfire <i>n/a for MY11</i>
	P0301, P0302, P0303, P0304, P0305, P0306, P0300	

General description:

The misfire monitor detects periodically combustion misfire by evaluating engine (crankshaft) speed fluctuations. If the engine speed increase after ignition top dead center of one cylinder is less than the engine speed increase of the other cylinders, misfire for the particular cylinder is detected. The misfire monitoring starts, if the enable conditions are satisfied. Misfires is detected within the first cumulative 1000 idle revolutions and if the speed increase is less than or equal the minimum speed increase. In this case the misfire counter of the particular cylinder is incremented by one. One testframe includes cumulated 675 rpm. If the misfire counter of one cylinder is above a threshold after a testframe is finished, a preliminary DTC is stored. If a misfire fault is recognized at several cylinders another general DTC is stored preliminary. If one of these two faults is detected in two consecutive driving cycles, the MIL is illuminated.

Enable conditions:

- idle speed
- injection rate
- engine coolant temperature
- vehicle speed
- engine speed
- time since engine running

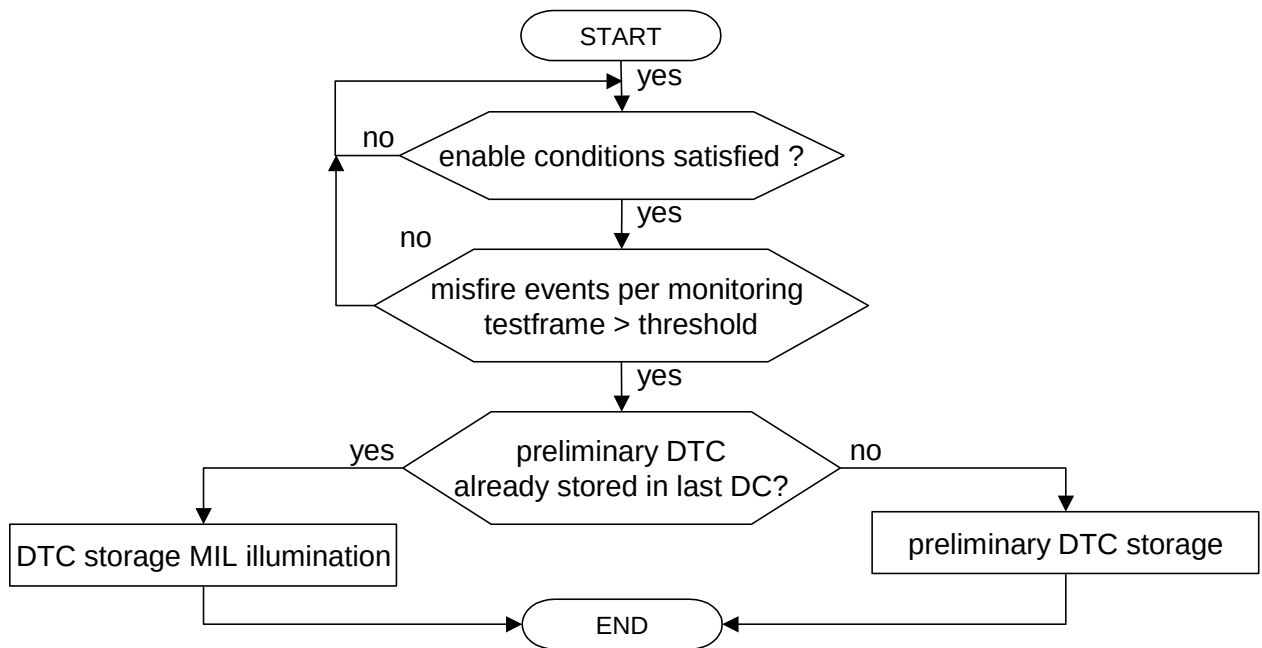


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Flowchart:





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4 Fuel System Monitoring

Fuel System Monitoring	(f)(4.2.1)(A) Pressure Threshold P0087, P0088	(f)(4.2.1)(B) Pressure Functional P0087, P0088	(f)(4.2.2)(A) Quantity Threshold P02CD, P02D5, P02D1, P02D7, P02CF, P02D3, P02CC, P02D4, P02D0, P02D6, P02CE, P02D2, P323F	(f)(4.2.2)(B) Quantity Functional P02CD, P02D5, P02D1, P02D7, P02CF, P02D3, P02CC, P02D4, P02D0, P02D6, P02CE, P02D2, P323F	(f)(4.2.3)(A) Timing Threshold P02CD, P02D5, P02D1, P02D7, P02CF, P02D3, P02CC, P02D4, P02D0, P02D6, P02CE, P02D2, P323F	(f)(4.2.3)(B) Timing Functional P02CD, P02D5, P02D1, P02D7, P02CF, P02D3, P02CC, P02D4, P02D0, P02D6, P02CE, P02D2, P323F
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Fuel System Monitoring	(f)(4.2.4)(A)(i) Feedback: time to CL see (f)(4.2.1)(A) (f)(4.2.1)(B)	(f)(4.2.4)(A)(ii) Feedback: default/OL see (f)(4.2.1)(A) (f)(4.2.1)(B)	(f)(4.2.4)(A)(iii) Feedback: CL limits see (f)(4.2.1)(A) (f)(4.2.1)(B)
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4.1 Rail Pressure Control Loop Monitoring

4.1.1 Rail Pressure Too Low (P0087)

General description:

If the rail pressure is below an engine speed dependent threshold for longer than debounce time, or the rail pressure deviation is above an engine speed dependent threshold (rail pressure lower than demanded) for longer than debounce time, a preliminary DTC is stored immediately. If one of these faults is detected in two consecutive driving cycles, the MIL is illuminated.

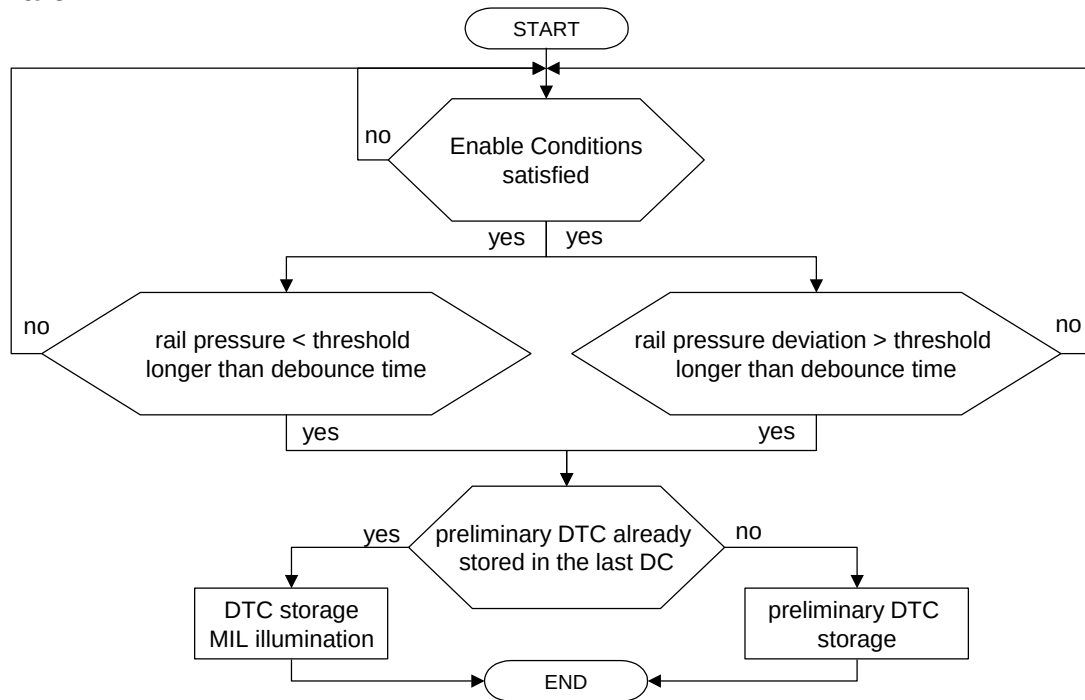


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Flowchart:





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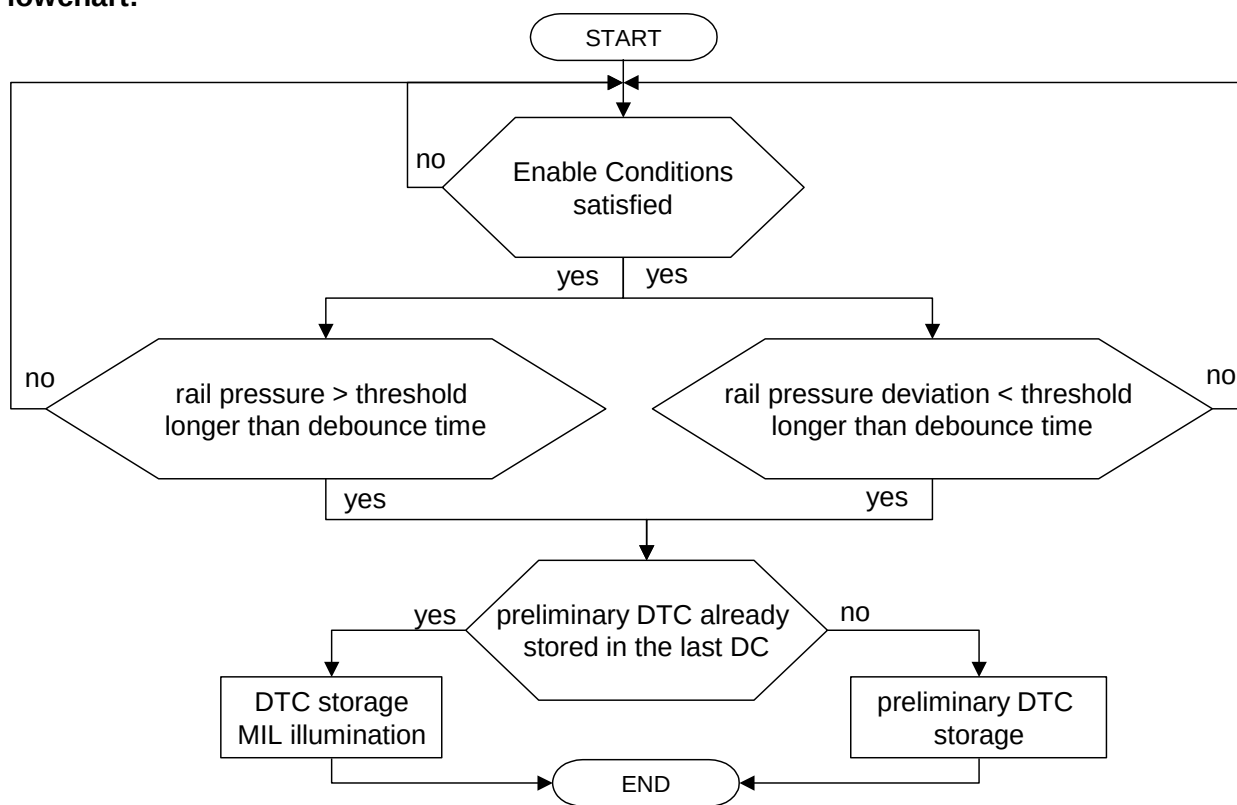
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4.1.2 Rail Pressure Too High
(P0088)

General description:

If the rail pressure is above a fixed threshold or the negative rail pressure deviation is below an engine speed dependent threshold (rail pressure higher than demanded) for longer than debounce time, a preliminary DTC is stored immediately. If one of these faults is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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4.2 Zero Fuel Quantity Calibration

P02CD, P02D5, P02D1, P02D7, P02CF, P02D3;
P02CC, P02D4, P02D0, P02D6, P02CE, P02D2

General description:

The injection quantity of an injector is defined by a certain energizing time at a certain rail pressure. The zero fuel quantity calibration compensates pilot injection drifts to ensure correct injection quantities over lifetime by evaluating corrections for the energizing time for each injector.

During the calibration phase (overrun, injection quantity=0), the zero fuel calibration performs pilot test injections (= "zero fuel quantity") at one single cylinder. The injections cause a speed increase in the crankshaft signal which is evaluated. If the speed increase is above or below a threshold the energizing time of the calibrated injector is decreased or increased until the desired threshold is reached. The evaluated energizing time correction is filtered and written into the ECU EEPROM and so it is considered at the next engine start. This process is executed for every cylinder at several rail pressure calibration points.

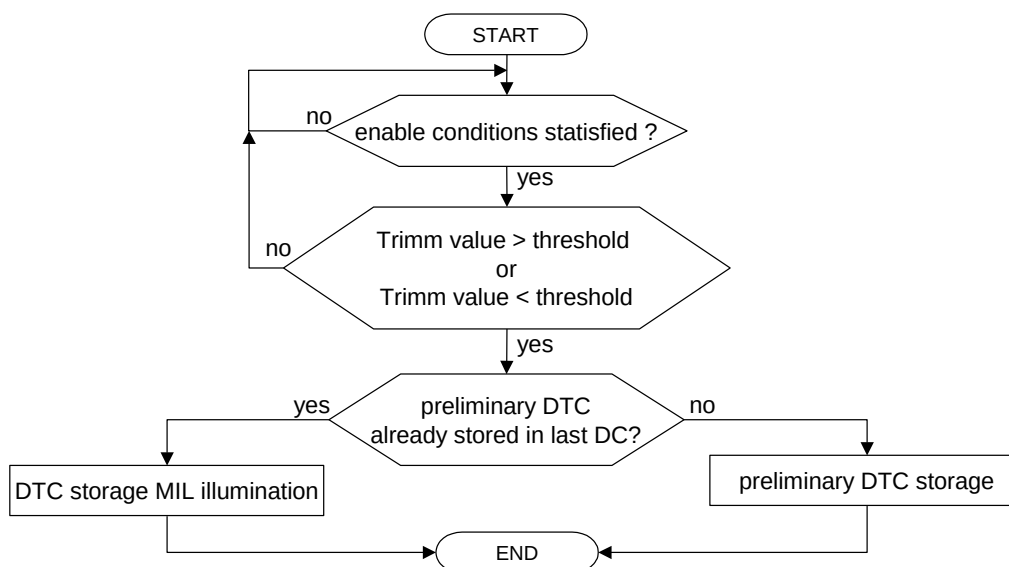
If any of the evaluated energizing time corrections is above or below a threshold a preliminary DTC is stored. If this fault is detected the MIL is illuminated.

The ZFC is enabled only in warm engine conditions to ensure stable combustion of test injections. These conditions are:

- Minimum engine coolant temperature reached
- Limited uninterrupted overrun-time because of cooling down of combustion chamber

Because of noise on the crankshaft signal caused from the drive train in lower gears the ZFC is activated in gear 3 to 6 and in a defined engine speed range.

Flowchart:



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4.2.1 Fuel Mass Observer (FMO)
P323F

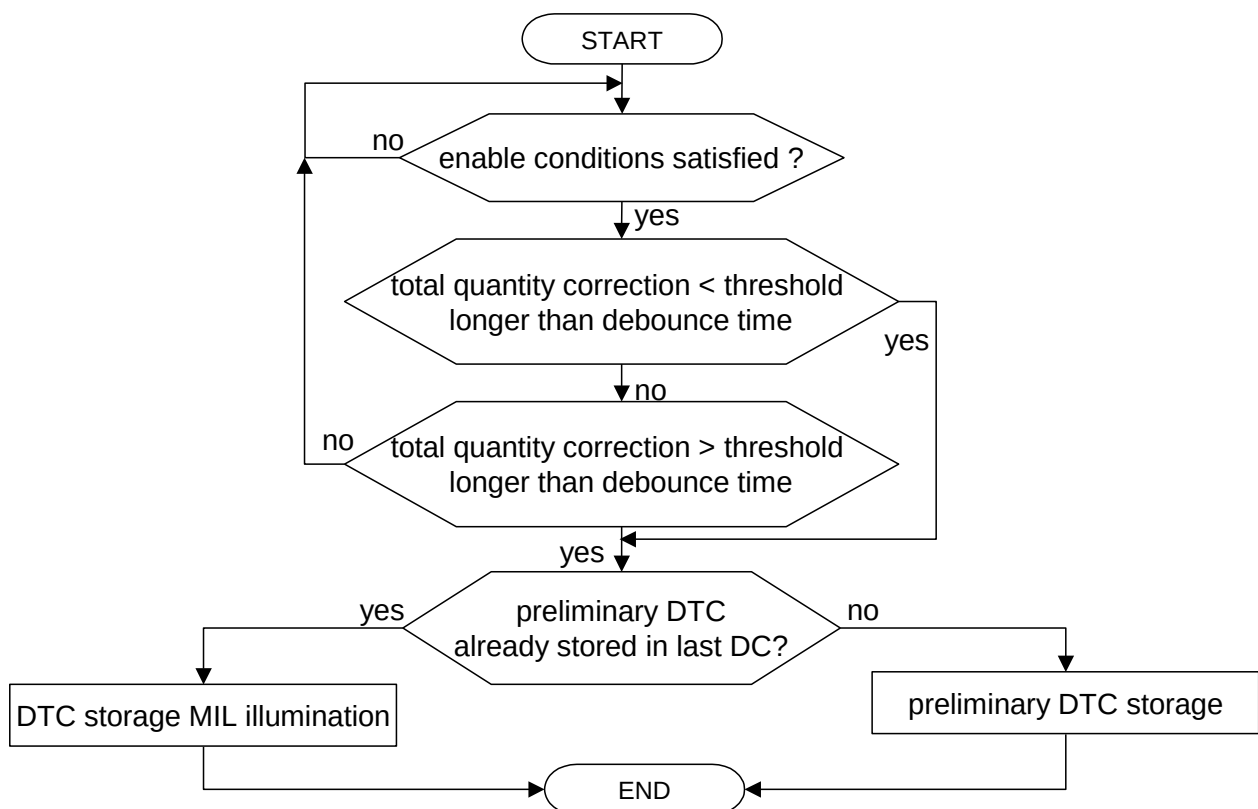
General Description:

The Fuel Mass Observer calculates the difference between

- calculated O₂ concentration (from measured air mass flow and injection quantity)
- and measured O₂ concentration of the exhaust gas by LSU

The O₂ concentration is a criteria for the actual EGR rate. Based on the difference a total quantity correction for exhaust gas recirculation setpoint is calculated. The diagnosis is continuously, after the lambda dew point is reached. A fault is detected, if the correction is below or above the specific thresholds for longer than an allowed time. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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5 Exhaust Gas Sensor Monitoring

Upstream Exhaust Gas Sensor Monitoring (Lambda)	(f)(5.2.1)(A)(i) Emissions threshold	(f)(5.2.1)(A)(ii) Circuit Faults	(f)(5.2.1)(A)(iii) Feedback: default/OL	(f)(5.2.1)(A)(iv) Sufficient for other diagnostics	(f)(5.2.4)(A) Heater Performance	(f)(5.2.4)(B) Heater Circuit Continuity
	P2297, P2A00, P0133	P2243, P2237, P2251, P2238, P2239, P0607, P0132, P0131	see (f)(5.2.1)(A)(i) (f)(5.2.1)(A)(ii)	P3022	P0135	P0032, P0031, P0030

5.1 Lambda Sensor (upstream DOC)

In the BMW Diesel System the LSU is used for the “Fuel Mass Observer” (adaption of the EGR setpoint by comparing measured and simulated lambda). Even with a completely missing LSU the emissions are far below 2x standard.

For all following diagnosis (except P0030, P0032, P0031) the dewpoint of the LSU has to be reached.

The dewpoint detection has to secure that no water is in the exhaust system that can damage the LSU. The duration of the detection of the dew point depends on the following input values:

- model of exhaust pipe temperature at the sensor position
- model of exhaust gas temperature at the sensor position
- air mass flow model at the sensor positions (for gas pulse detection)
- engine temperature at engine start
- environment temperature at engine start
- counter for aborted dew detection cycles



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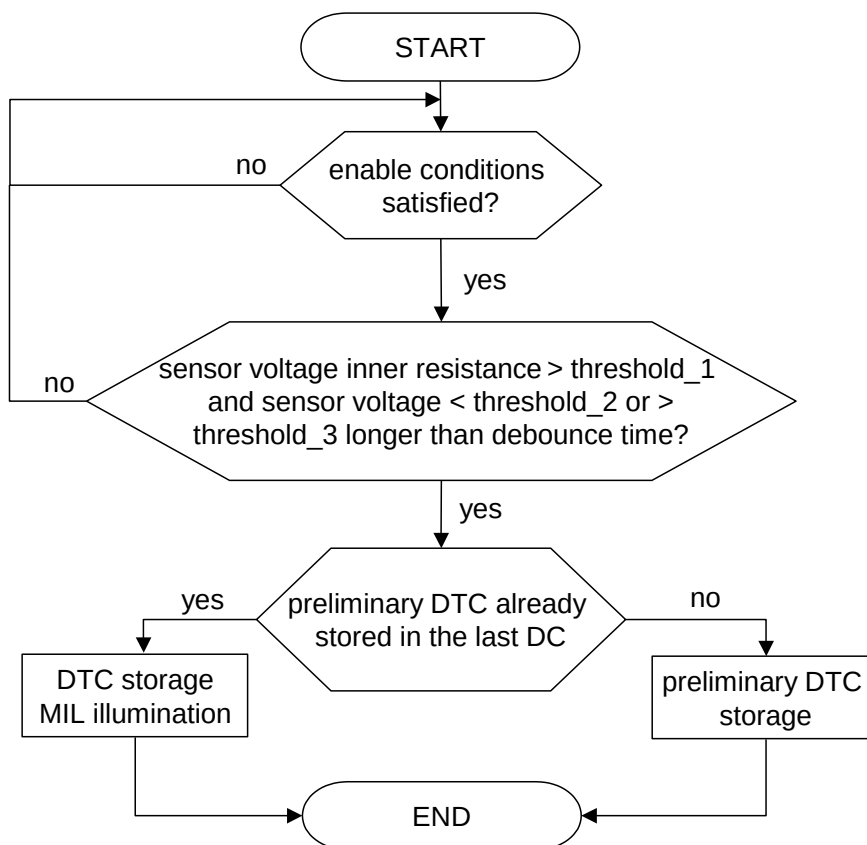
5.1.1 Circuit faults

5.1.1.1 Nernst Cell Open Circuit
(P2243)

General description:

The LSU-Nernst-Wire is monitored for open circuit. A fault is detected, if the LSU signal voltage exceeds the upper threshold or falls below the lower threshold for more than an allowed time while the inner resistance of the LSU is above a threshold for a time. If an error is detected a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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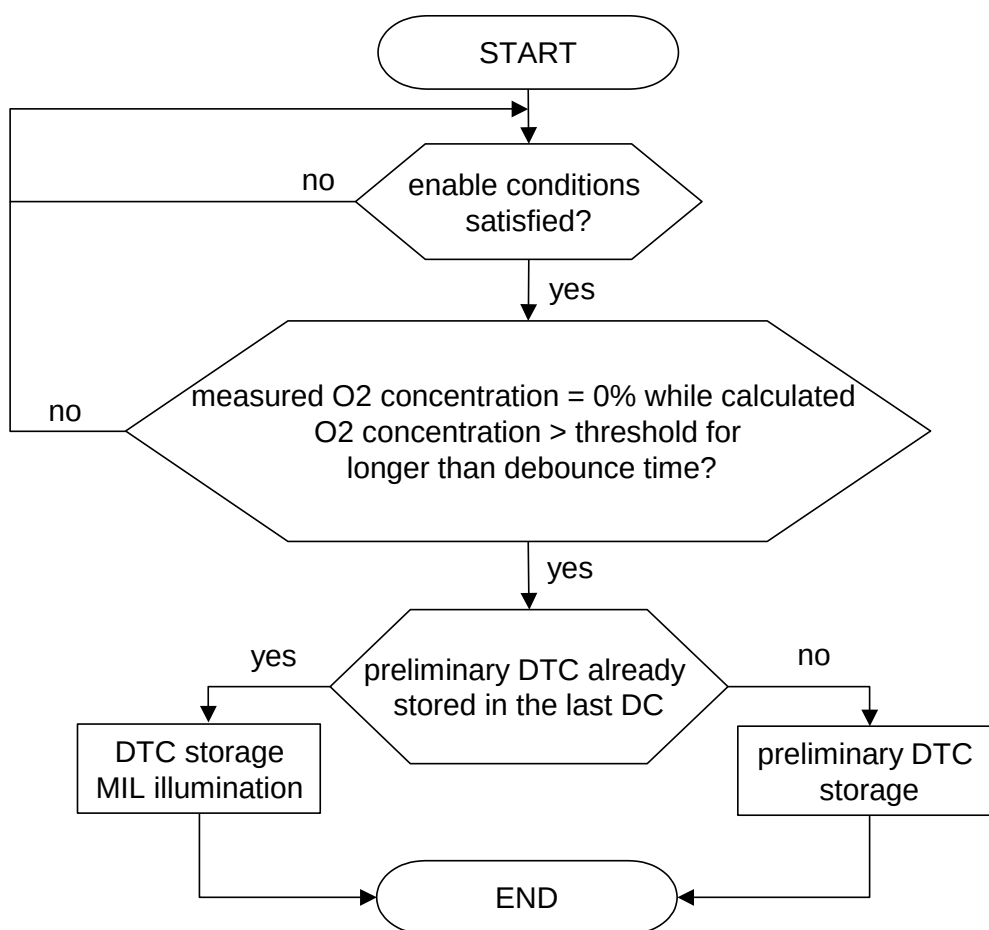
Issued:11/04/09

5.1.1.2 Pump Cell Open Circuit
(P2237)

General Description:

If the pump current wire has an open circuit error, the measured O₂ concentration is closed to 0%. If a calculated O₂ concentration from the air flow and the fuel quantity is above a threshold and the measured O₂ concentration is closed 0% at the same time, an error is detected. If an error is detected a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

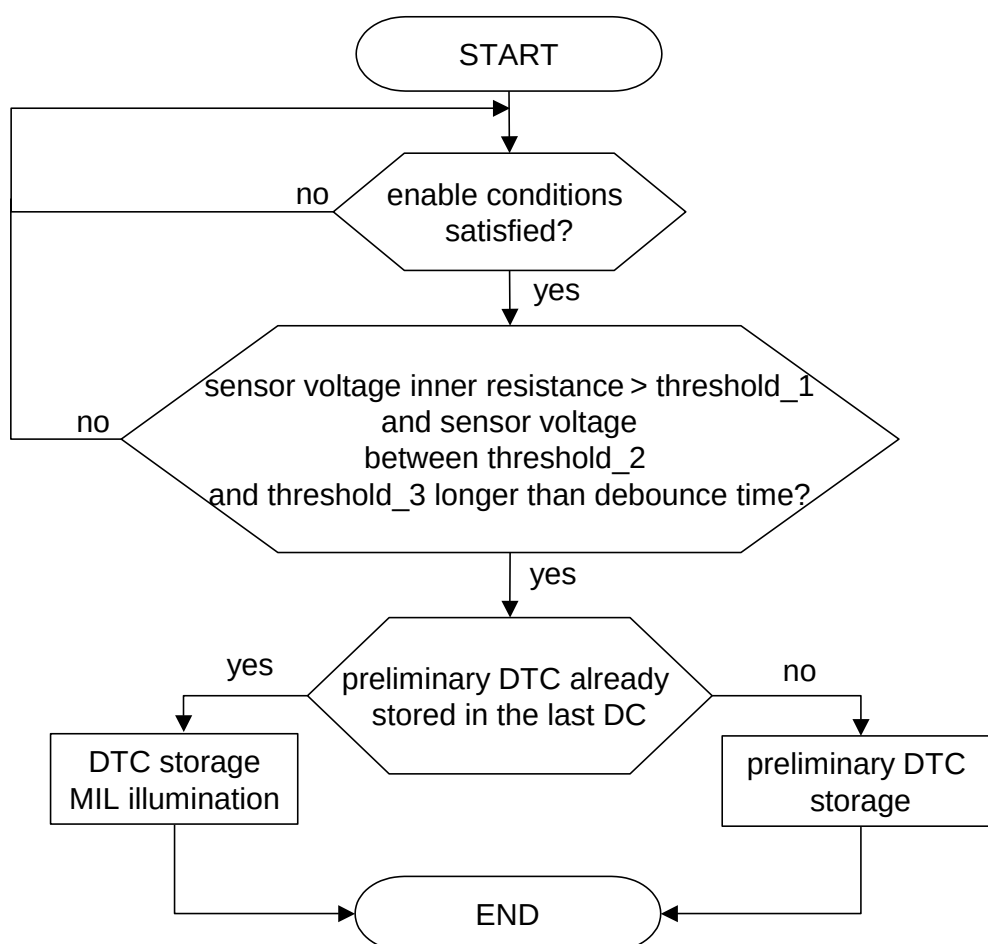
Issued:11/04/09

5.1.1.3 Virtual Ground Open Circuit
(P2251)

General Description:

An open circuit on the virtual ground wire is detected, if the inner resistance of the LSU is above a threshold and the sensor voltage is between a threshold one and a threshold two for a specified time. If an error is detected a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

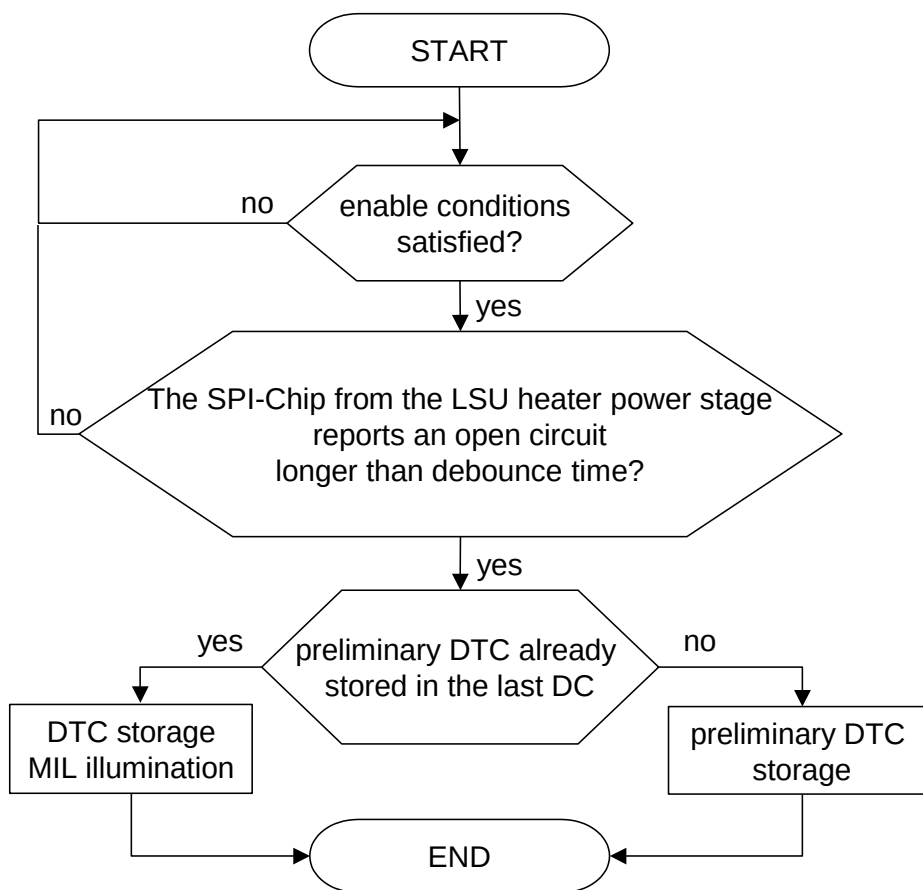
Issued:11/04/09

5.1.1.4 LSU – Sensor Heater Monitoring – Open Circuit
(P0030)

General description:

The LSU sensor driver chip checks continuously and independently whether a load is connected to the heater power stage and reports the result to the processor. If this fault is detected, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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Issued:11/04/09

5.1.1.5 Short Circuit to Battery and Short to Ground

5.1.1.5.1 Short to Ground and short circuit to battery for LSU-Wire
(P2239, P2238)

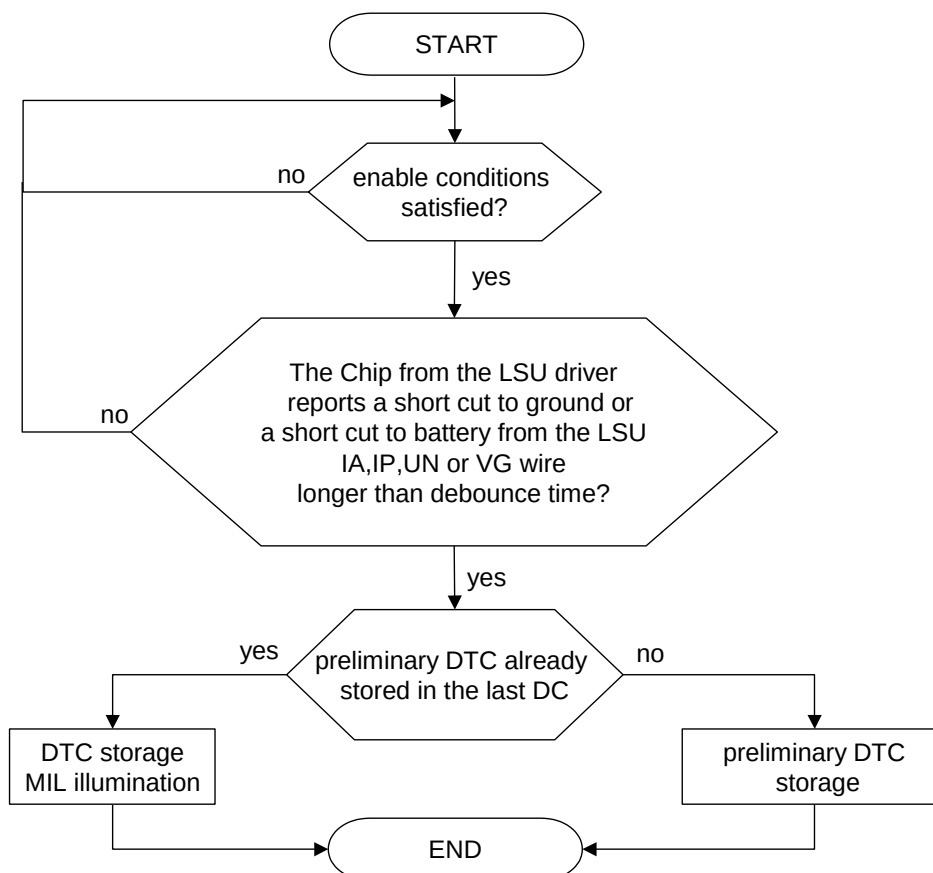
General description:

The LSU sensor driver chip continuously and independently checks for short to ground and short circuit to battery faults and reports these directly to the processor. The monitoring is stopped for the time, while the tests for P2243 and P2251 are performed. The following wires are monitored:

- Nernst Cell (UN)
- Pump Current (IP)
- Virtual Ground (VG)
- Compensation current (IA)

If this fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. If a fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

Issued:11/04/09

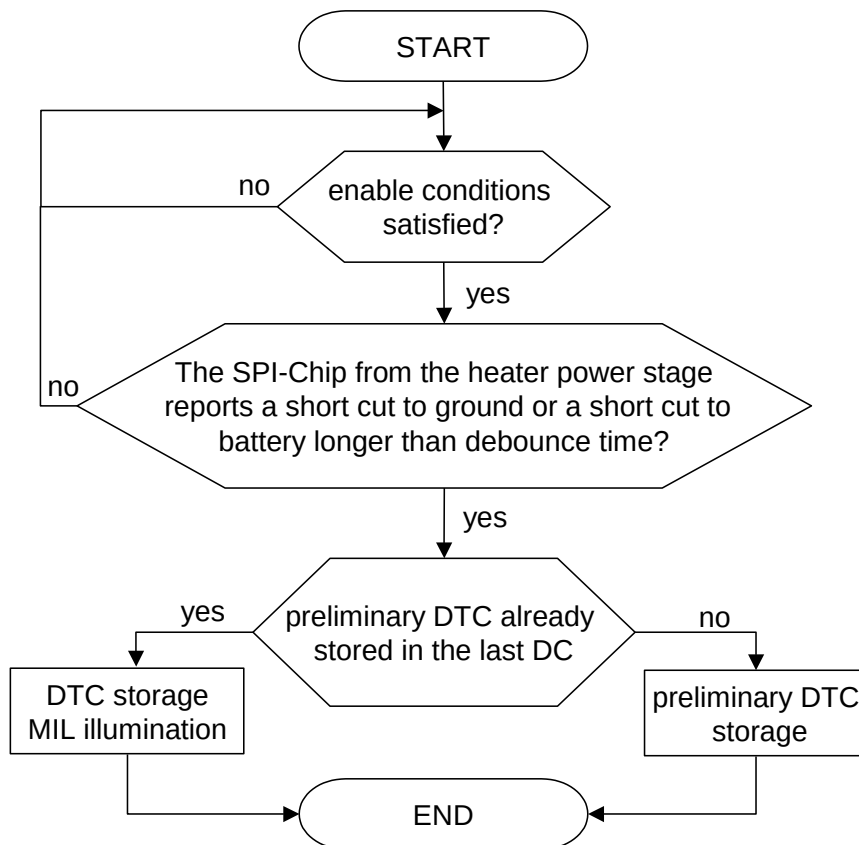
5.1.1.5.2 LSU Sensor Heater Monitoring – Short Circuit to battery and short circuit to ground

(P0031, P0032)

General Description:

The LSU sensor driver chip continuously and independently checks for short circuit to ground and short circuit to battery from the heater power stage and reports these directly to the processor. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

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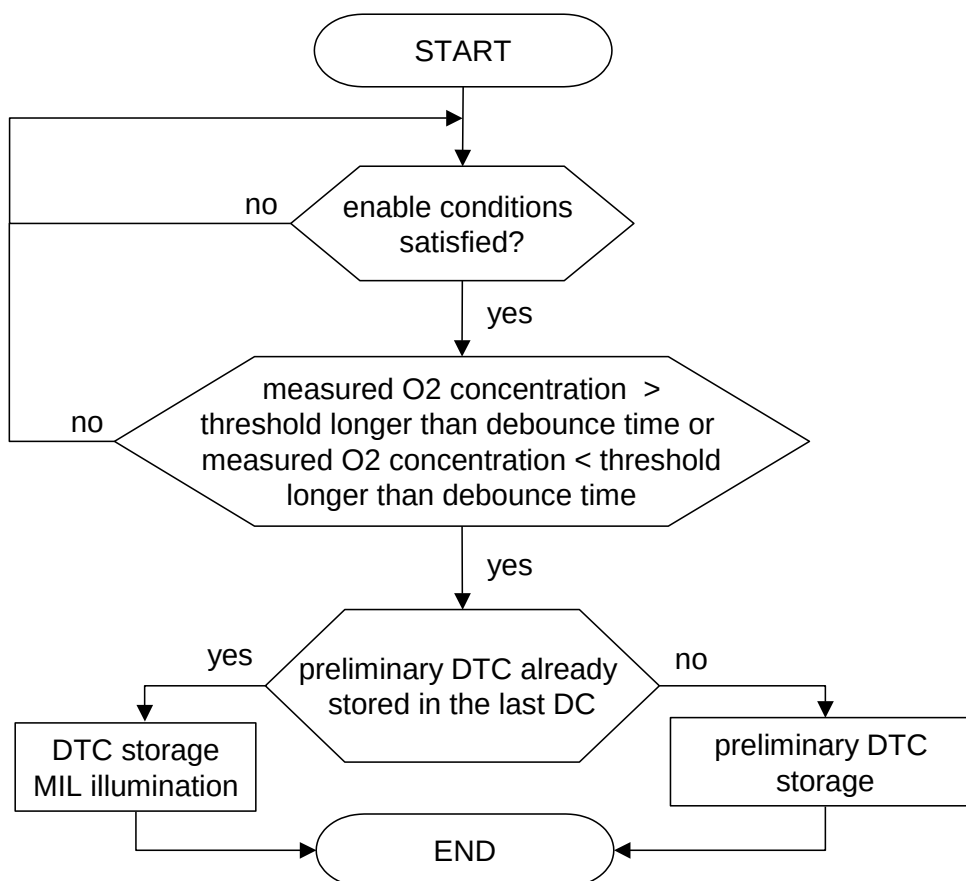
5.1.2 Signal Range Check

5.1.2.1 LSU – Sensor Signal Range Check
(P0132, P0131)

General description:

The measured O2 concentration is monitored for signal range. A fault is detected and a preliminary fault code stored, if the measured O2 concentration exceeds the upper signal range threshold or falls below the lower signal range threshold. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

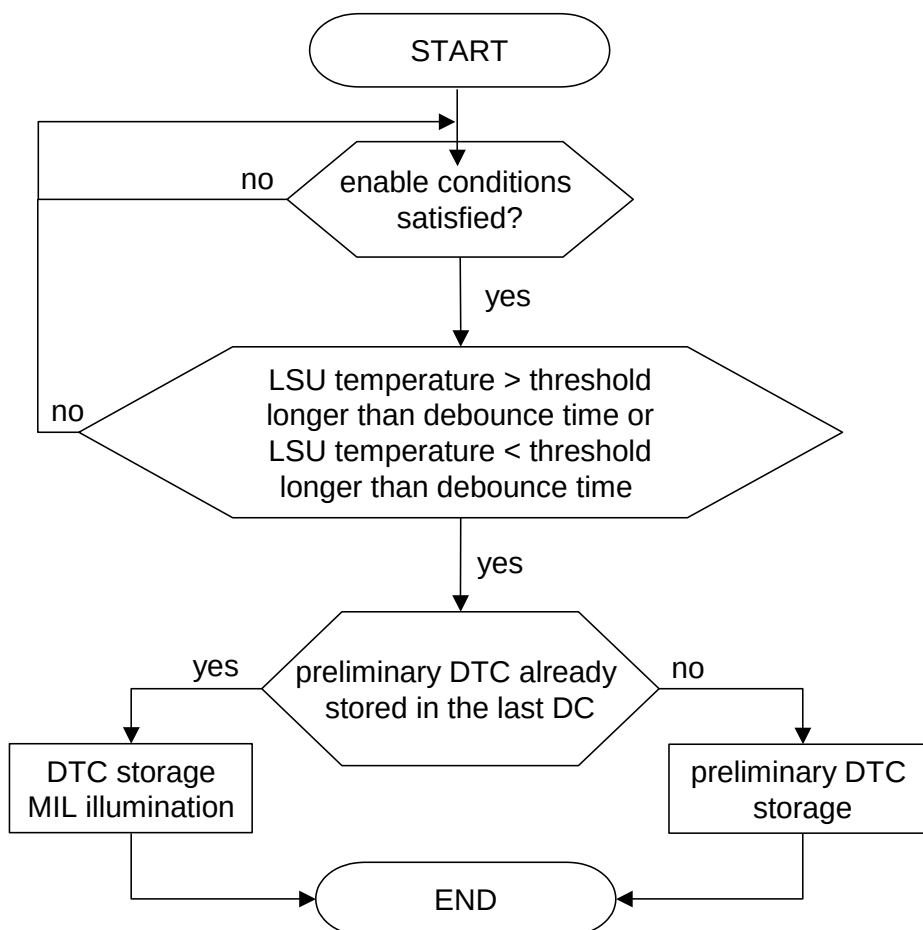
Issued:11/04/09

5.1.2.2 Heater Performance – Signal Range Check
(P0135)

General description:

The LSU sensor heater temperature is monitored for signal range. A fault is detected and a preliminary fault code is stored, if the sensor temperature signal exceeds the upper signal range threshold or falls below the lower signal range threshold for more than an allowed time. If this fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

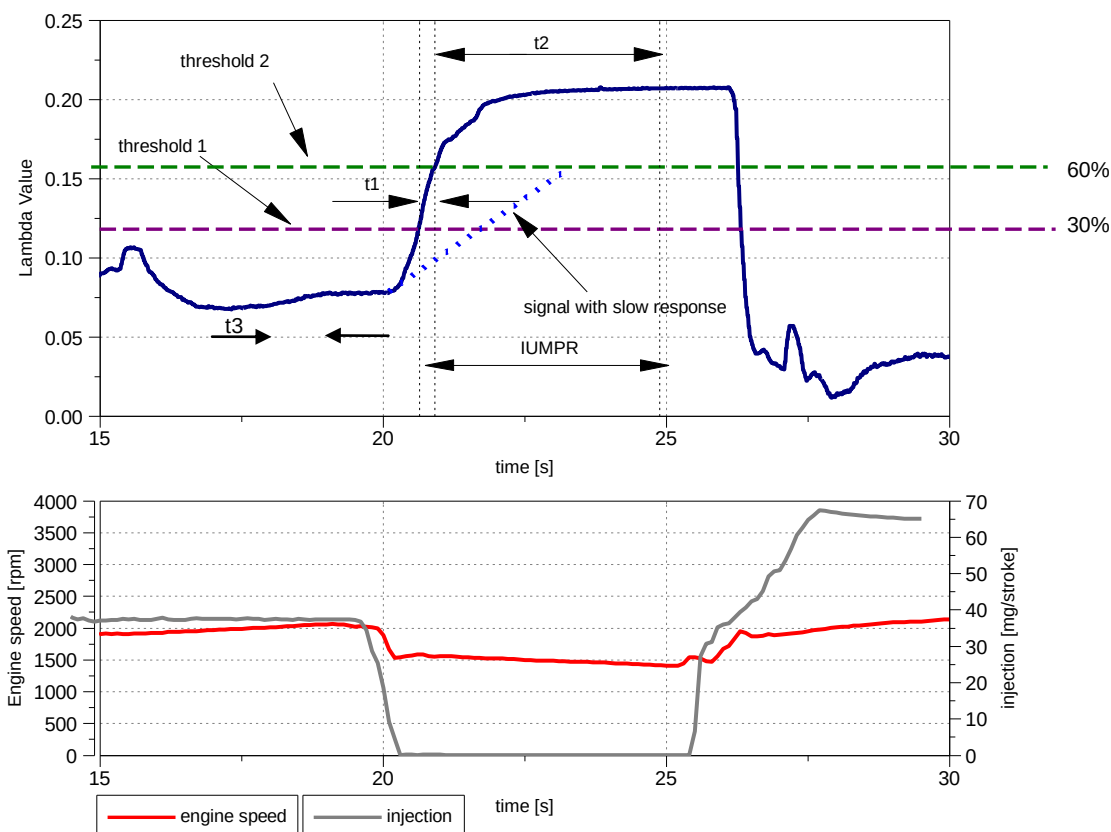
Issued:11/04/09

5.1.2.3 Dynamic test of the LSU Signal in a Load-to-Overrun Transition (P0133)

General description:

In a load to overrun transition the dynamic of the Lambda-Sensor is monitored. Therefore two O₂ thresholds (30% and 60% of the expected O₂ ratio) are used. The base for the calculation of the two thresholds is the Lambda-Value at the time of the load to overrun transition. The load to overrun transition is valid, if several conditions are fulfilled for a time before the load to overrun transition happens (steady state).

If the O₂ Signal does not rise from 30% to 60% of the expected O₂ ratio in a max. time₁ or the O₂ Signal does not reach 60% of the expected O₂ ratio in a max. time₂, or the O₂ signal does not reach 30% of the expected O₂ ration in a max. time₃ an error is detected and a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.



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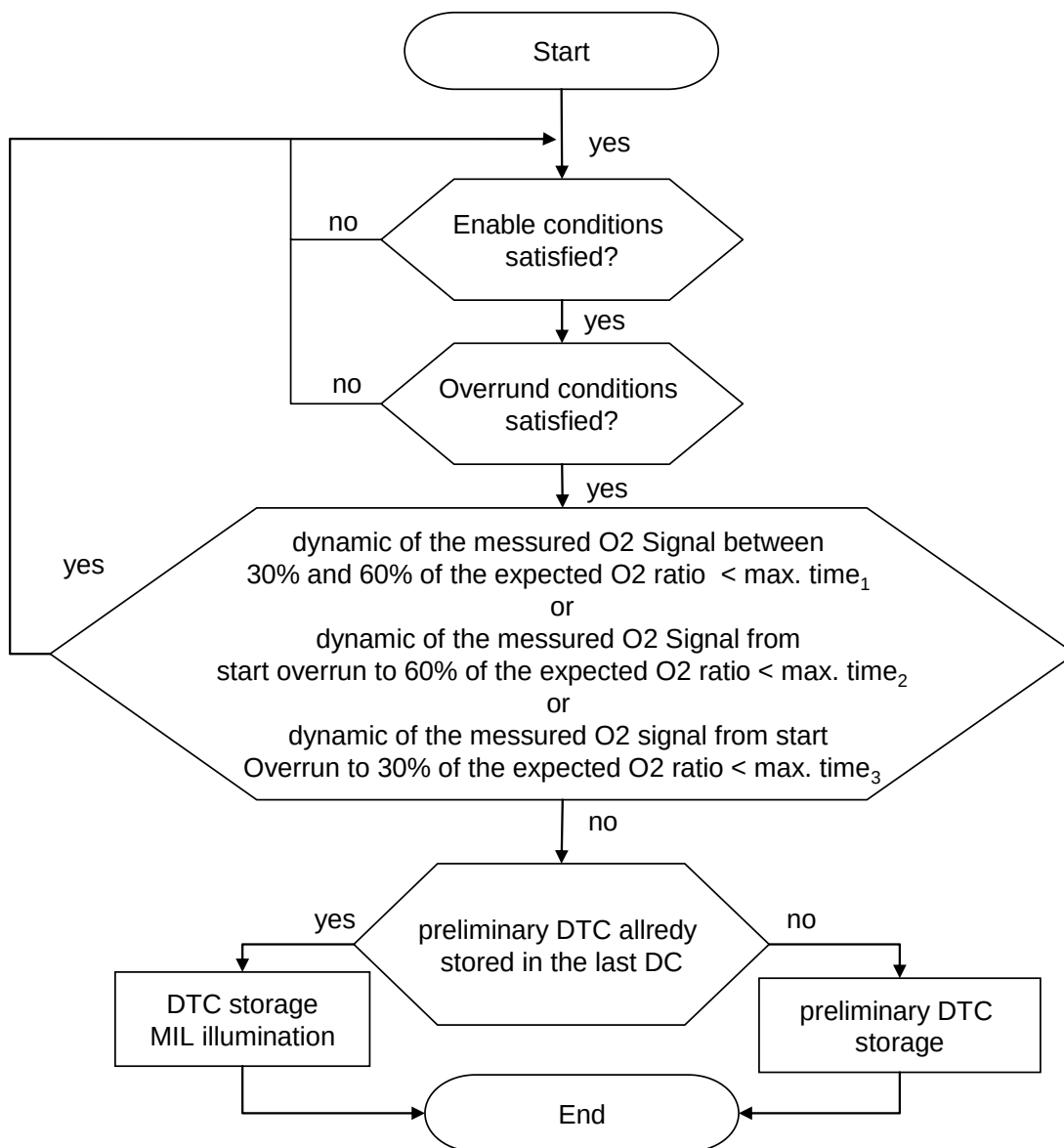


Test Group xxxxx – Tier2 Bin 5 Standard

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Issued:11/04/09

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

Issued:11/04/09

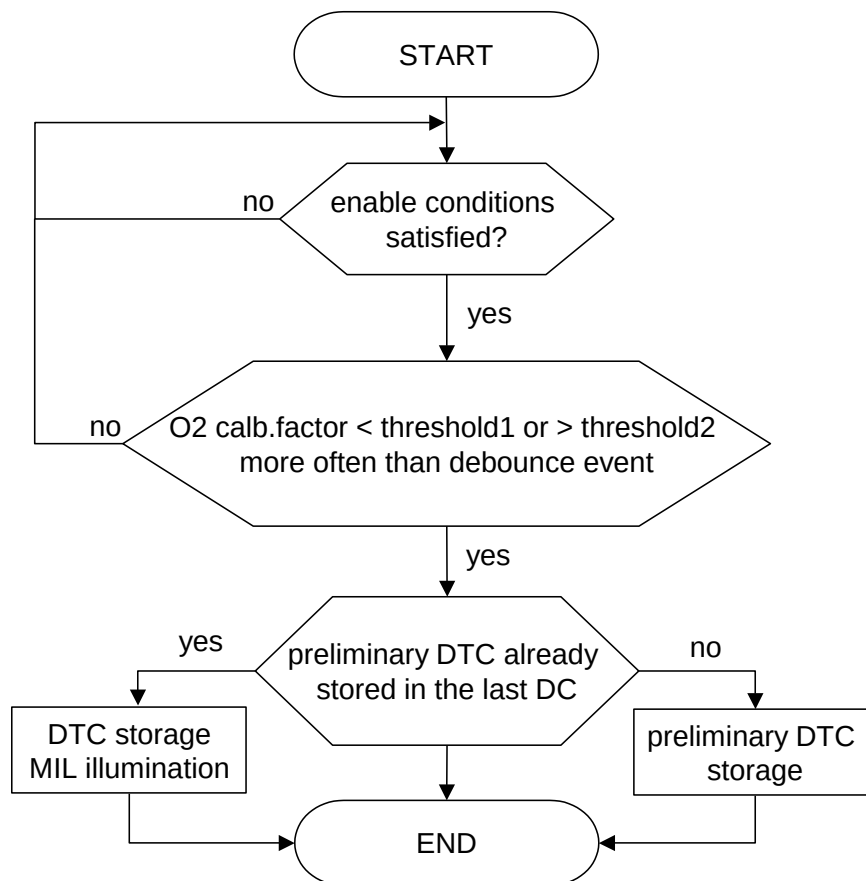
5.1.2.4 Lambda Offset Calibration Value
(P0607)

General description:

To compensate the offset of the LSU signal over lifetime, an offset calibration is triggered by ECU. To detect this offset, the difference between the expected and the actual pump current is determined in the triggered operation points.

This offset of the pump current is monitored for signal range. A fault is detected and a preliminary fault code is stored, if the sensor offset voltage calibration value exceeds the upper signal range threshold or falls below the lower signal range threshold. If this fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

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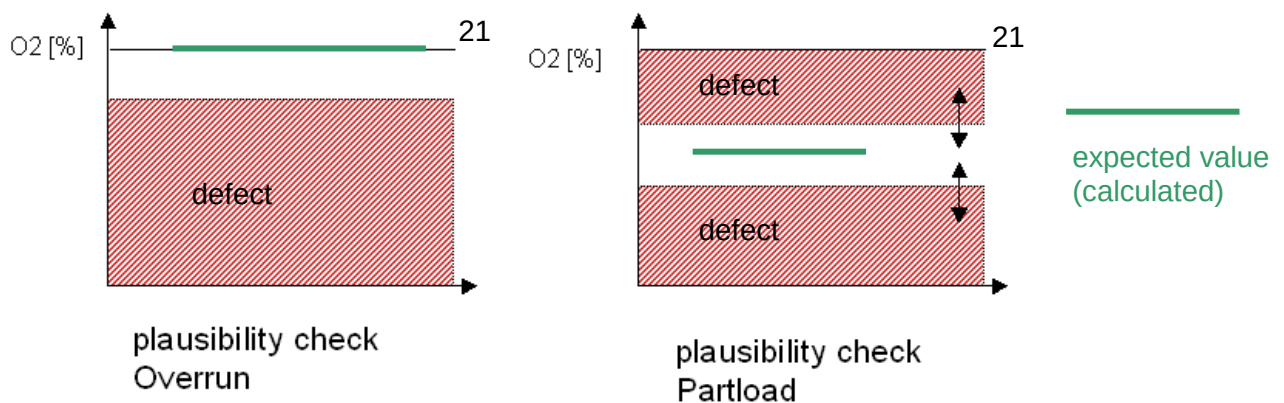
5.1.3 Functional checks

5.1.3.1 Plausibility of the LSU signal in overrun and idle
(P2297, P2A00)

General description:

To detect a stuck signal or a too high adaption of the LSU signal the measured O₂ concentration is monitored in overrun and partload.
Therefore the adapted O₂ signal from the LSU has to be in a range of the calculated O₂ signal (+/- offset). If the measured lambda is above or below this offset a fault is detected and a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

The following cases are monitored:



If the LSU signal is in the monitored region during the operation conditions for the diagnosis are satisfied, a fault is detected.

Secondary Parameters:

- engine speed
- injection quantity
- air mass flow

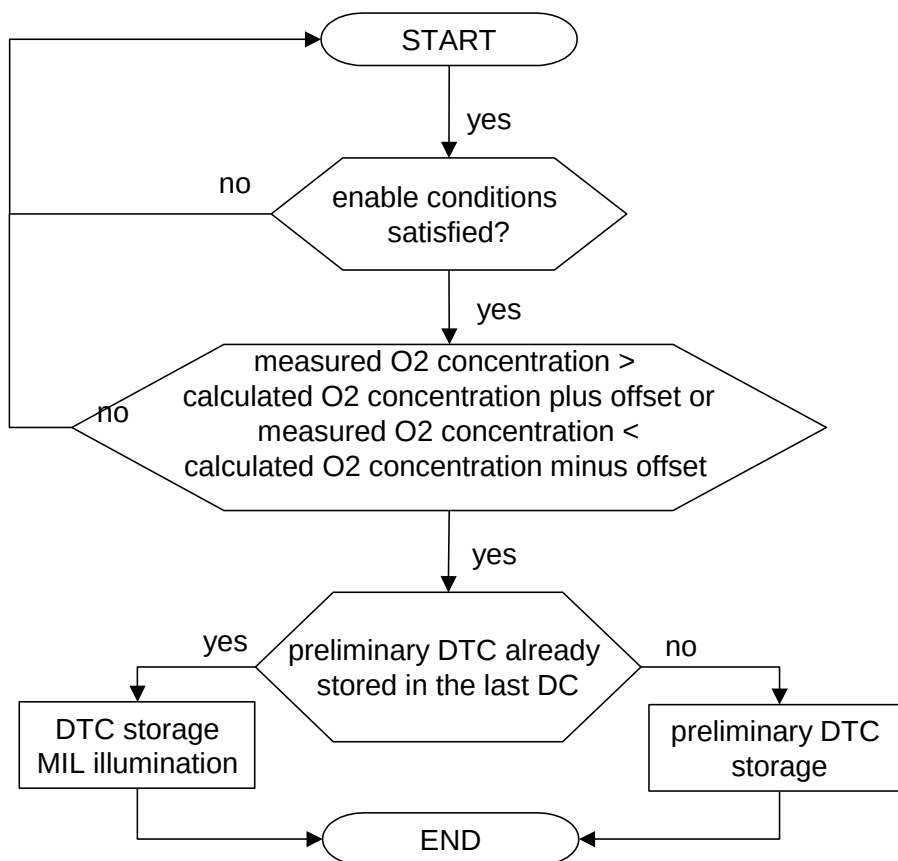


Test Group xxxxx – Tier2 Bin 5 Standard

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Issued:11/04/09

Flowchart:





Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

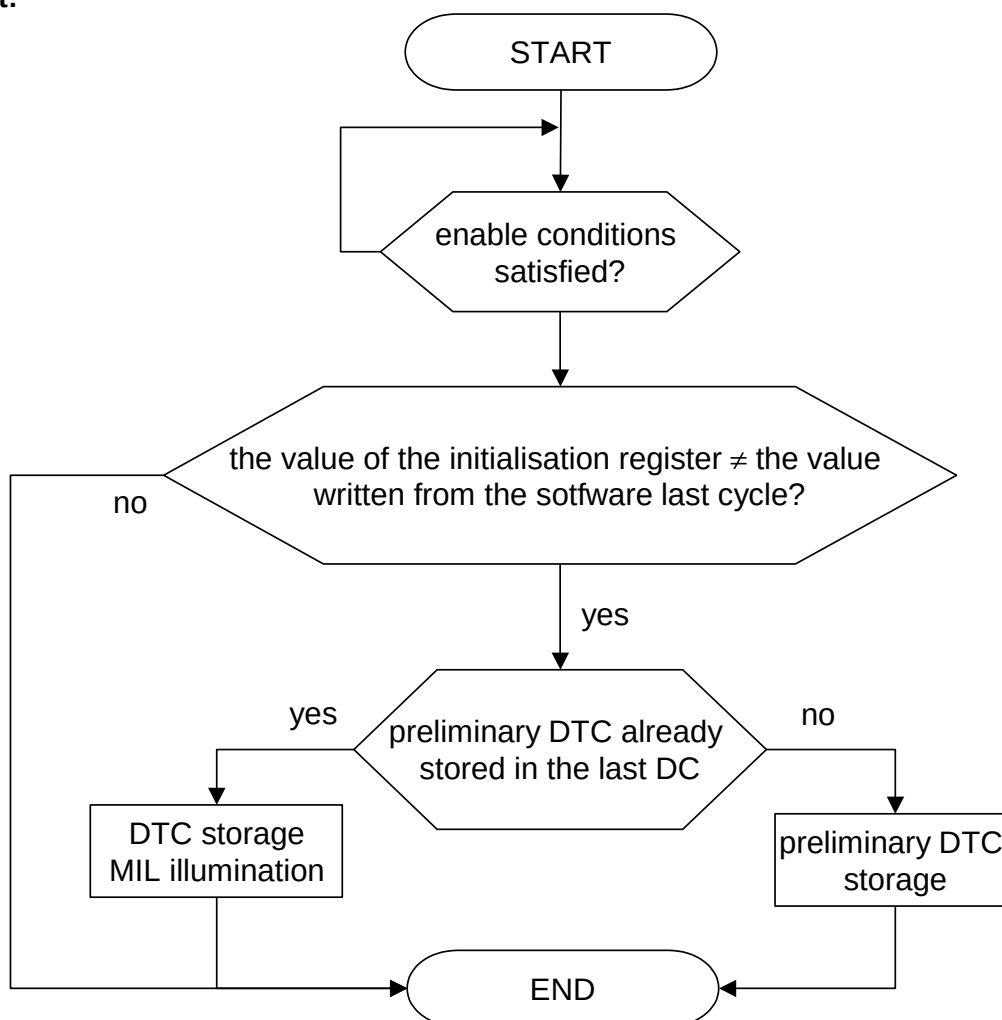
Issued:11/04/09

5.1.4 Disturbed LSU SPI – Signal
(P3022)

General description:

At ECU start the value of the initialization register is compared with the value written in the last driving cycle into the register. If the two values are different an error is detected and a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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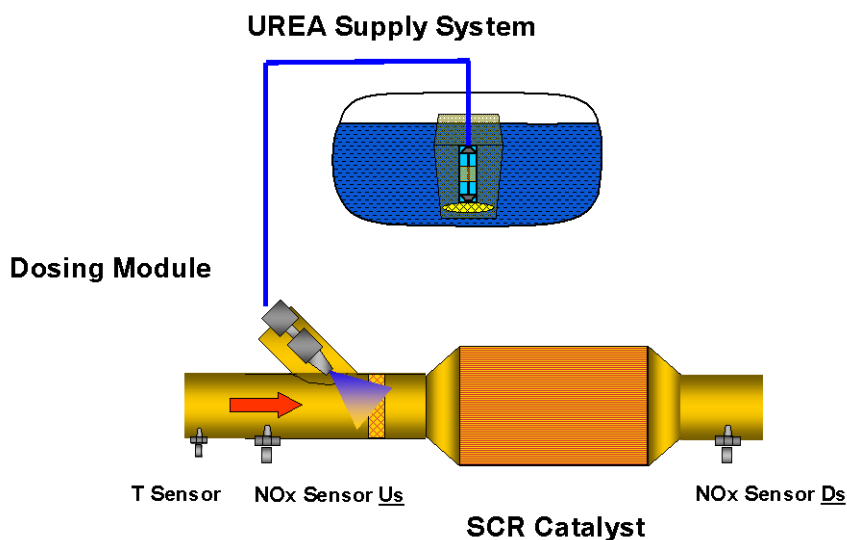
5.2 NOx Sensors (Us and Ds)

NOx and/or PM Sensors (NOx Us and Ds)	(f)(5.2.2)(A) Emissions threshold	(f)(5.2.2)(B) Circuit Faults	(f)(5.2.2)(C) Feedback: default/OL	(f)(5.2.2)(D) Sufficient for other diagnostics	(f)(5.2.4)(A) Heater Performance	(f)(5.2.4)(B) Heater Circuit Continuity
	P229F, P2201	U029D, U029E, P229E, P122D, P22A1, P22A0, P124E, U059F, P2200, P122C, P2203, P2202, P124C, U059E	P124F, P22A7, P124D, P2209	see (f)(5.2.2) (A)-(C)	P124F, P22A7, P124D, P2209	P229E, P122D, P2200, P122C

For all following diagnosis the dewpoint of the NOx sensor has to be reached.
The dewpoint detection has to secure that no water is in the exhaust system that can damage the NOx sensors. The duration of the detection of the dew point depends on the following input values:

- model of exhaust pipe temperature at the sensor position
- model of exhaust gas temperature at the sensor position
- air mass flow model at the sensor positions (for gas pulse detection)
- engine temperature at engine start
- environment temperature at engine start
- counter for aborted dew detection cycles

Overview SCR Catalyst system:



The installed NOx sensors are secured by password against manipulation from third persons and reprogramming of a sensor in service is not intended.



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Part 1

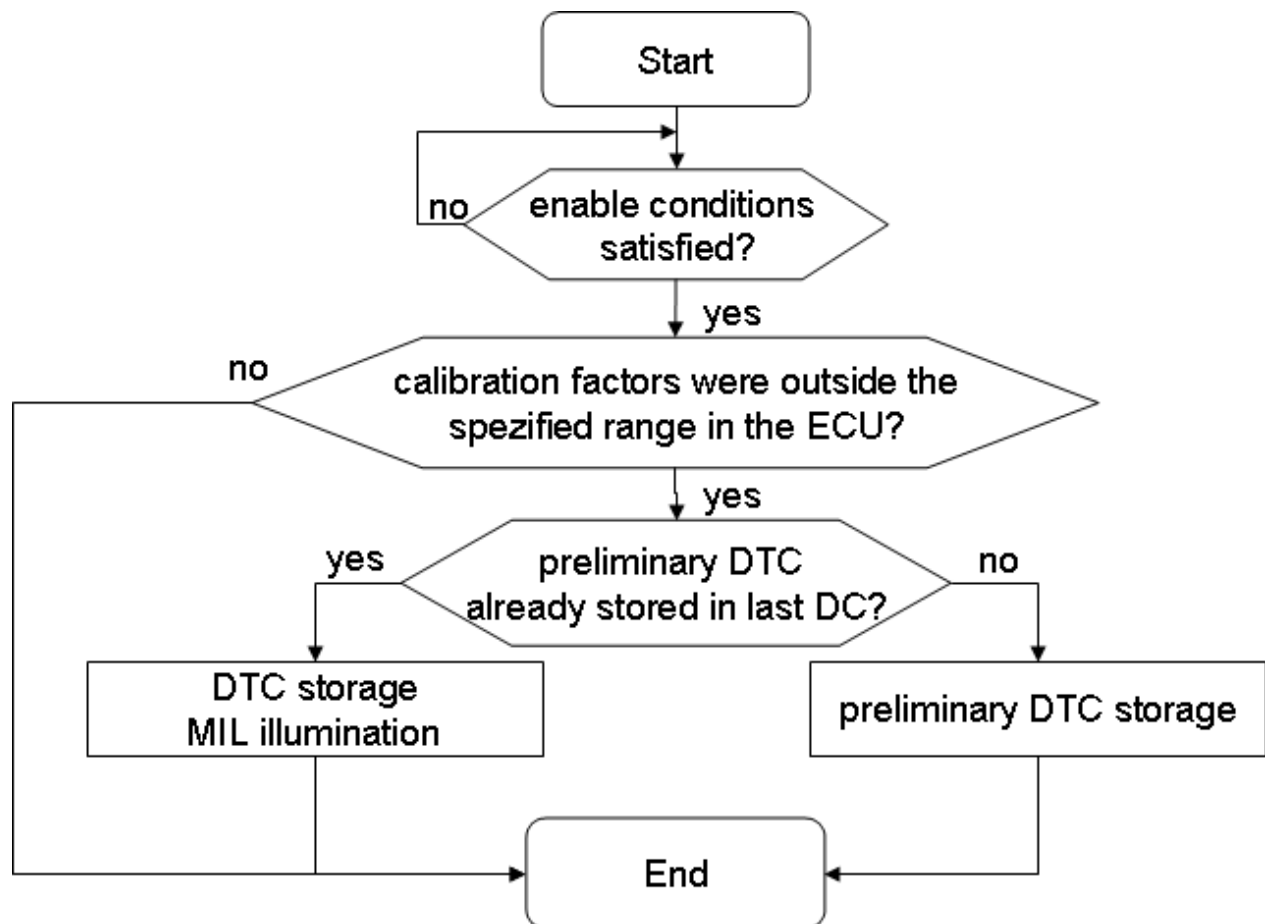
Issued:11/04/09

5.2.1 Sensor Can Feedback (Factor)
(P124C, P124E)

General description:

To secure, that the right NOx sensors are installed in the system, the sensors send a correction factor via CAN bus. If the correction factors are outside the specified range of the expected values, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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Part 1

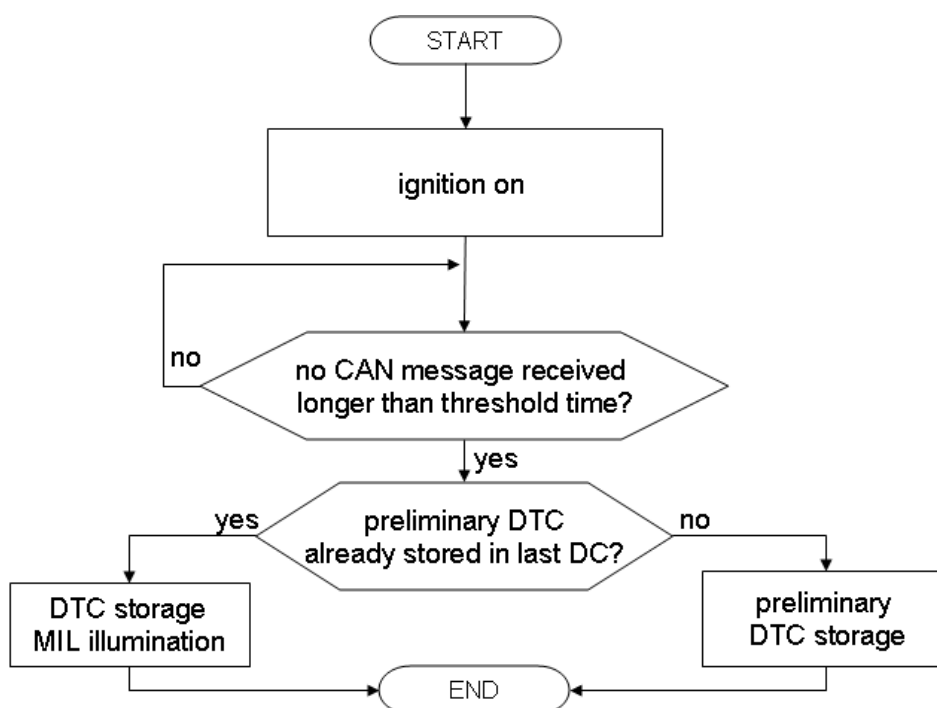
Issued:11/04/09

5.2.2 CAN Message Mode9 Time Out Monitoring
(U059E, U059F, P124C, P124E)

General description:

The monitoring of CAN message works in the same way for both NOx sensors. If there is no correct message send from the sensor to the ECU longer than a threshold time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



5.2.3 Circuit Faults
(P2200, P122C, P229E, P122D)

General description:

If an open load or short circuit error between the sensor and its CAN controller occurs for a defined time the controller sends a CAN message to the ECU. The ECU checks this message and



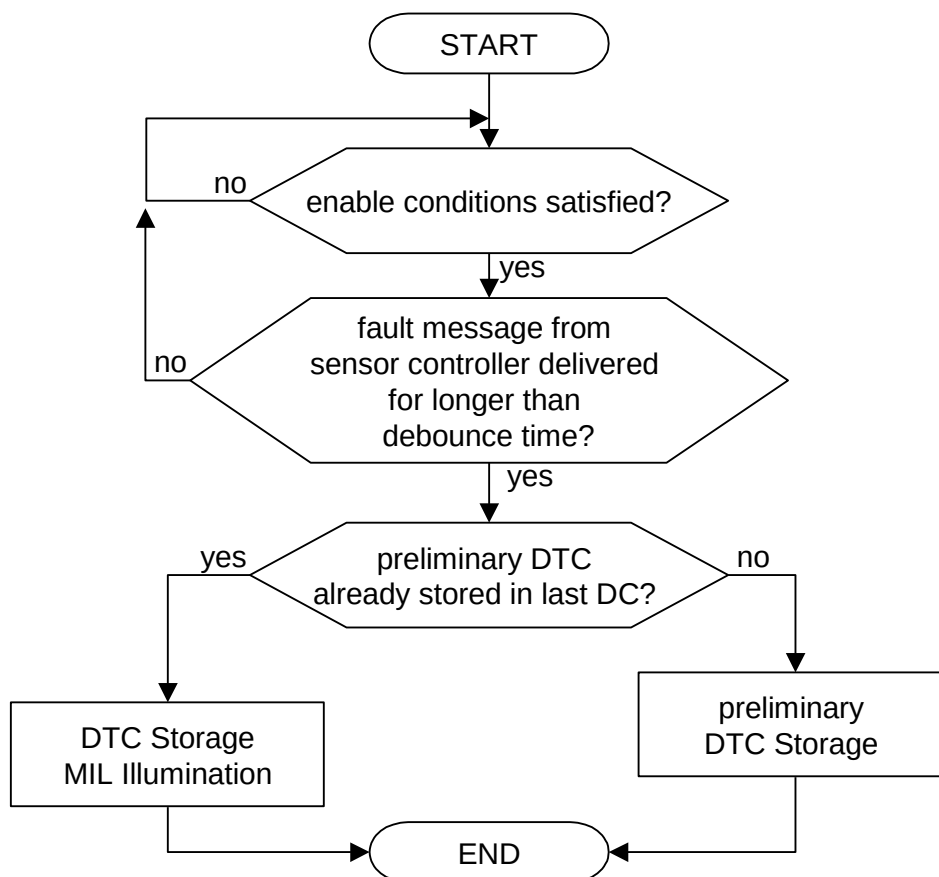
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the CAN communication continuously. In case of an open load, short circuit or CAN communication failure a DTC will be preliminary set. If one of these faults will be detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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5.2.4 Signal Range Check
(P2203, P2202, P22A1, P22A0)

General Description

If the physical range of the NOx sensor is above/below the applicated threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

The total measuring range of the NOx sensors is [-100ppm to 1650ppm]. In some operating conditions these values can occur. The following two figures should support this facts.

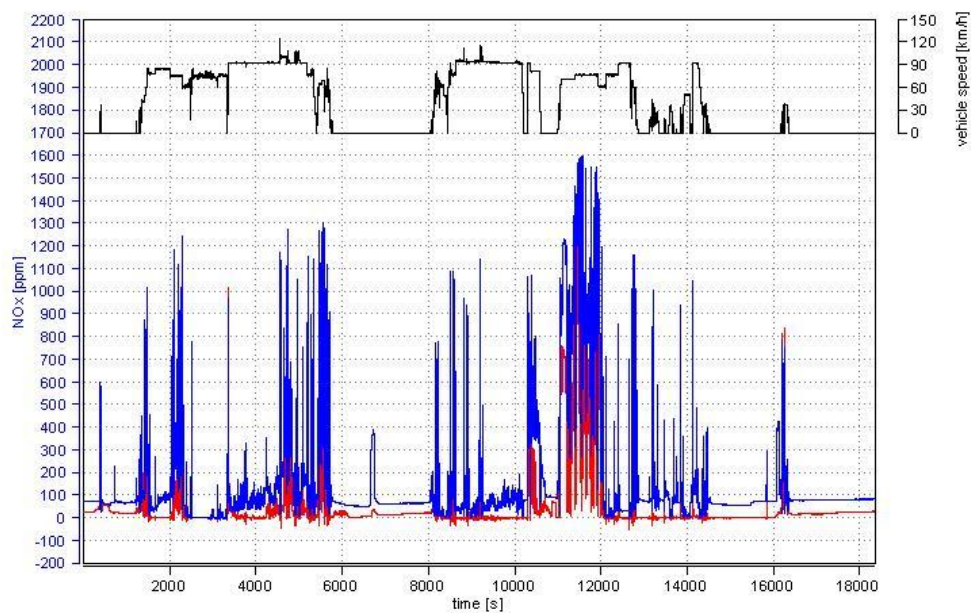


Figure: NOx values up to 1650 ppm



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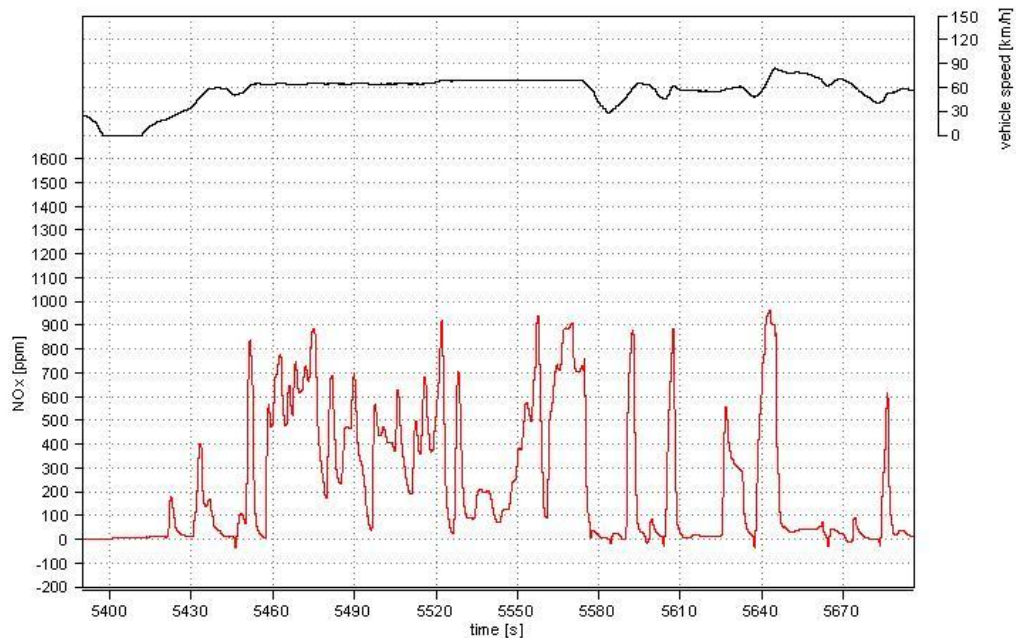
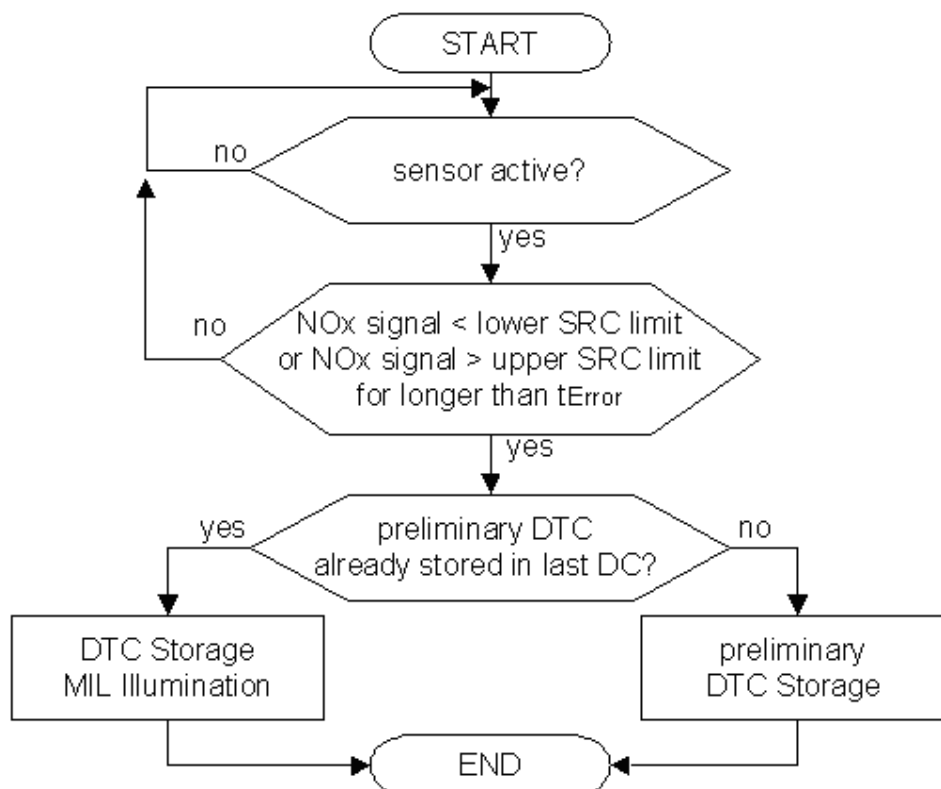


Figure: NOx values down to -50 ppm

Flowchart:



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Application for Certification

Enclosure 1

OBDII Description for Model Year 2011

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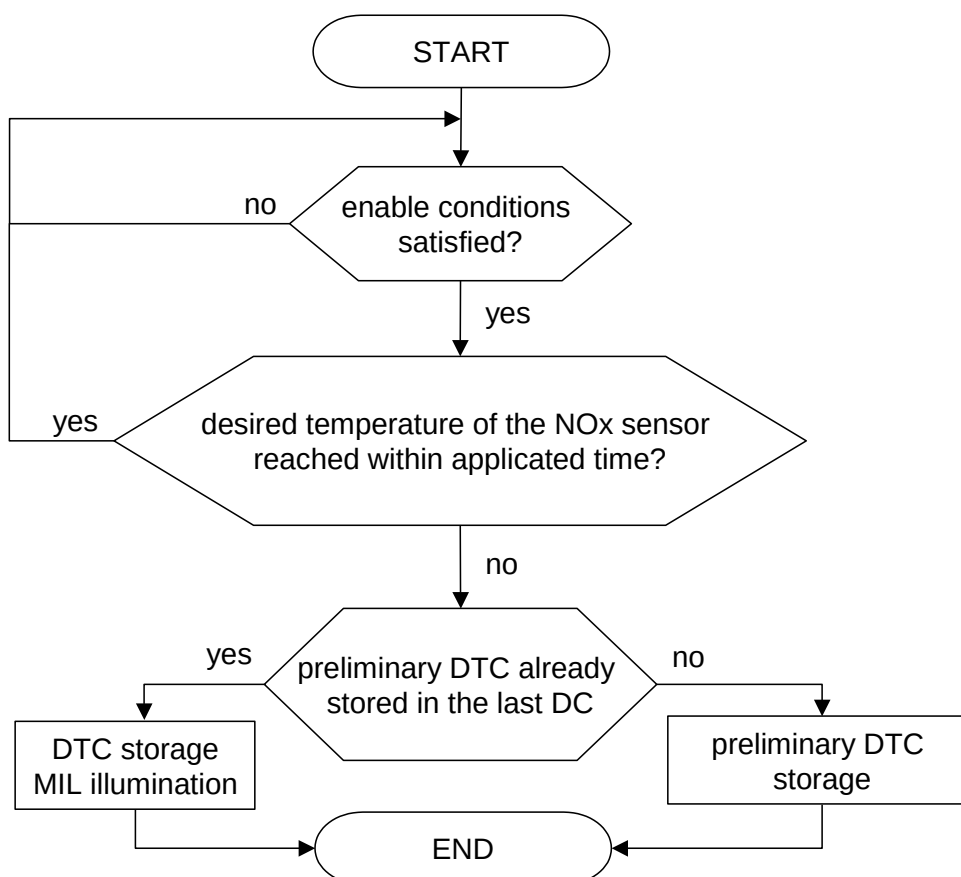
Issued:11/04/09

5.2.5 Heater Performance
(P2209, P22A7)

General Description:

The internal NOx sensor heater temperature has to be in the range 780°C to 820°C to guarantee proper functionality of the sensor (NOx Sensor valid). If the NOx sensors reach there desired temperature, it is communicated to the ECU via CAN. If this message is not received within a applicated time after the heater is switched on (depends on dewpoint of the NOx Sensors and exhaust gas temperature), a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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5.2.6 Feedback Monitoring
(P124D, P124F)

General description:

The valid status of the NOx – signal is used in other monitoring functions. Therefore this status signal is monitored.

For monitoring the NOx – Sensor feedback (valid status) the measured time period of invalid and valid signal status of the NOx signal is evaluated. A ratio of "valid time" and "invalid time" + "valid time" is calculated and compared with a threshold value. If the calculated ratio exceeds its limit a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Therefore following enable conditions have to be satisfied:

- a minimum time for calculating the ratio
- dew point reached
- exhaust gas temperature in specified range
- dynamic detection of Lambda Signal < threshold

For the enable condition "dynamic detection of lambda signal dynamic of O2 signal (difference between measured and filtered O2 signal) < threshold the O2 signal from the oxygen sensor is used.

The purpose of this function is to determine a dynamic driving condition. In such a driving condition the NOx sensor signal can be invalid. To avoid a false fault detection the diagnosis is stopped under these conditions.

The raw O2 sensor voltage is filtered in the ECU, so that a noisy signal from the LSU sensor will be compensated. Therby a inappropriately disabeling of this diagnostic is obviated.

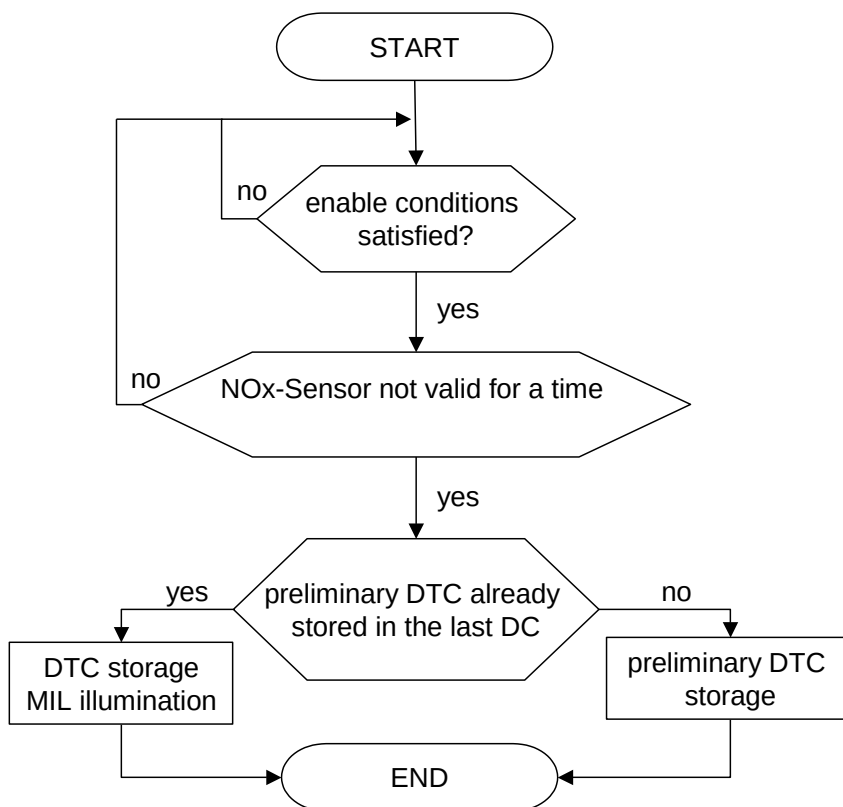


Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

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Flowchart:





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Issued:11/04/09

5.2.7 NOx Offset Test
(P2201, P229F)

General description:

The NOx signal offset works in the same way for both NOx sensors, except the NOx maximum Offset-Test for the downstream sensor. An offset is calculated during overrun when no NOx concentration is expected. The offset value is not valid and is depraved if one of the following conditions are not satisfied:

- the offset test time is shorter than a threshold
- the NOx variation during the offsettest is out of a specified range

If the calculated offset value (difference between actual value and zero point) is out of a specified range a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

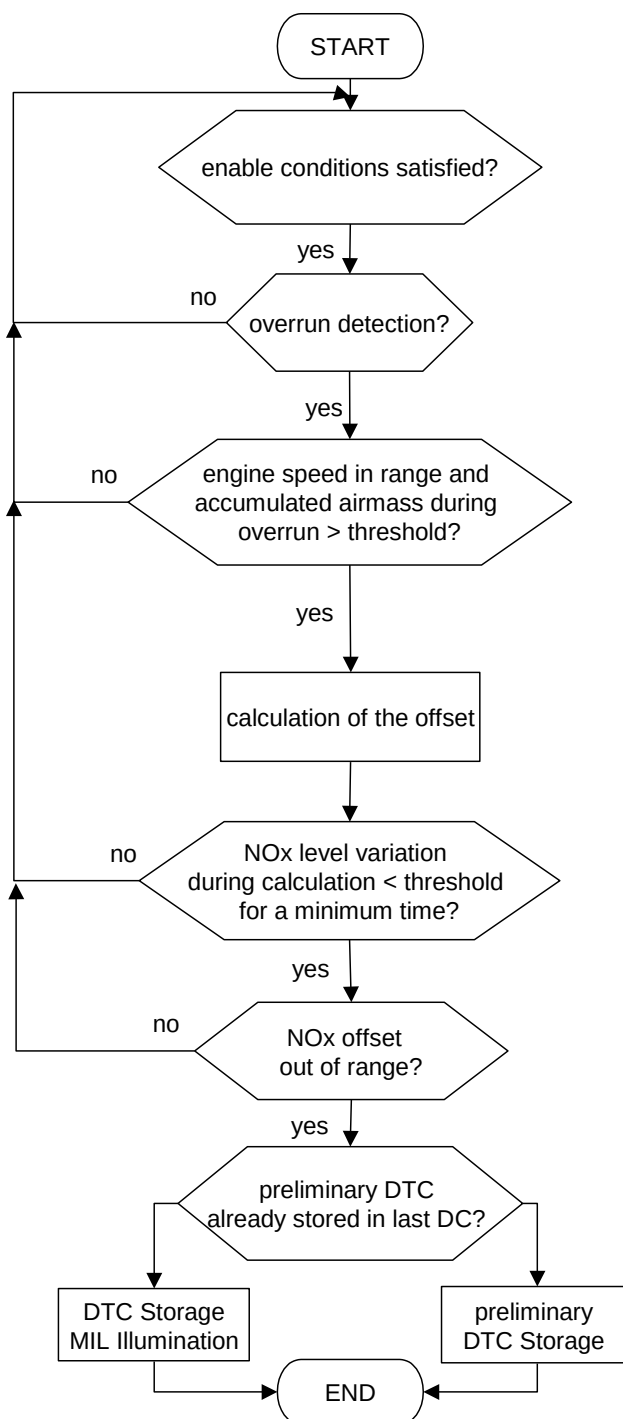


Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

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Flowchart:





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Issued:11/04/09

5.2.8 NOx maximum Offset-Test (only downstream)
(P229F)

General description:

The maximum Offset-Test for the NOx Sensor downstream includes an extended Offset-Test (additional to the basis test → see offset limit min).

Both tests (basis and extended) have to pass through to get a test result.

Both tests have to detect a failure to store a DTC. (see the following table)

basic test	extended test	test result for scantool
OK	OK	OK
OK	fault	OK
fault	OK	OK
fault	fault	fault

In this chapter only the extended Offset-Test will be explained.

The extended Offset-Function works as “minimum-search” of the lowest NOx Ds concentration within a certain time.

The certain time will be calculated if following conditions are true:

- Modelled SCR efficiency above a threshold
- Engine speed within range

After the timer have reached a threshold following enable conditions have to be true:

- Minimum overrun events for a time
- Minimum NOx Mass gradient Us events for a time

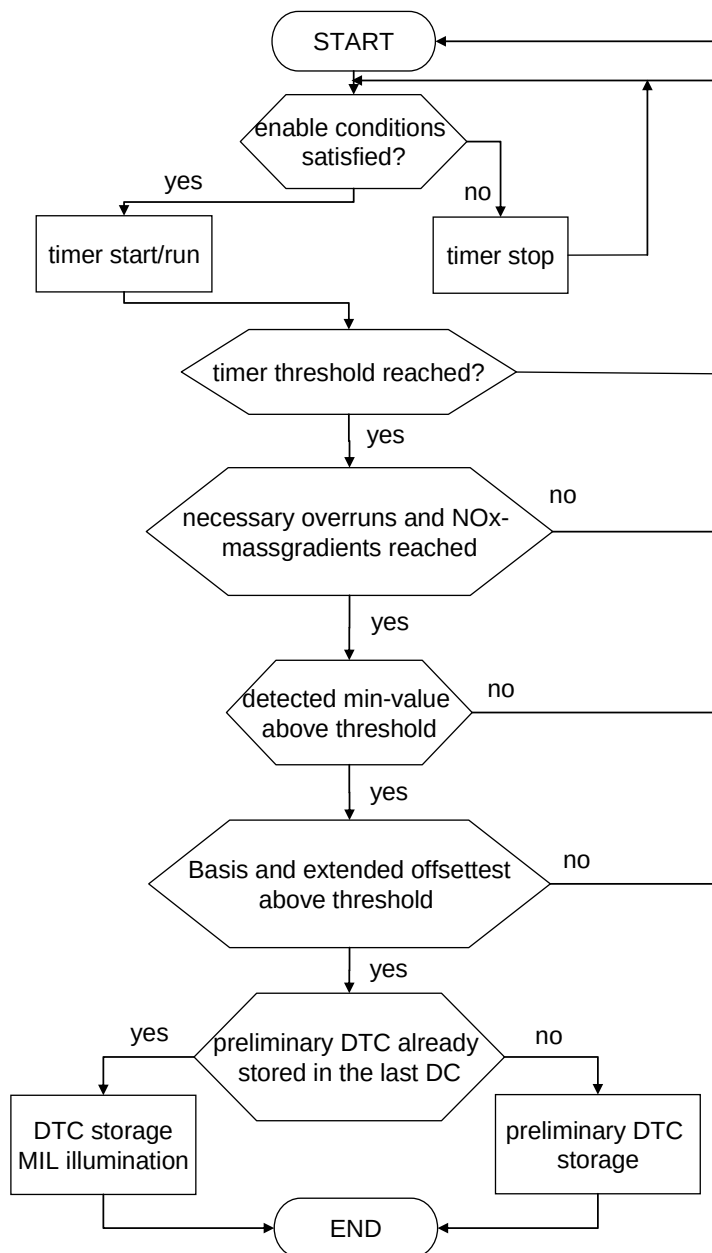


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Flowchart:





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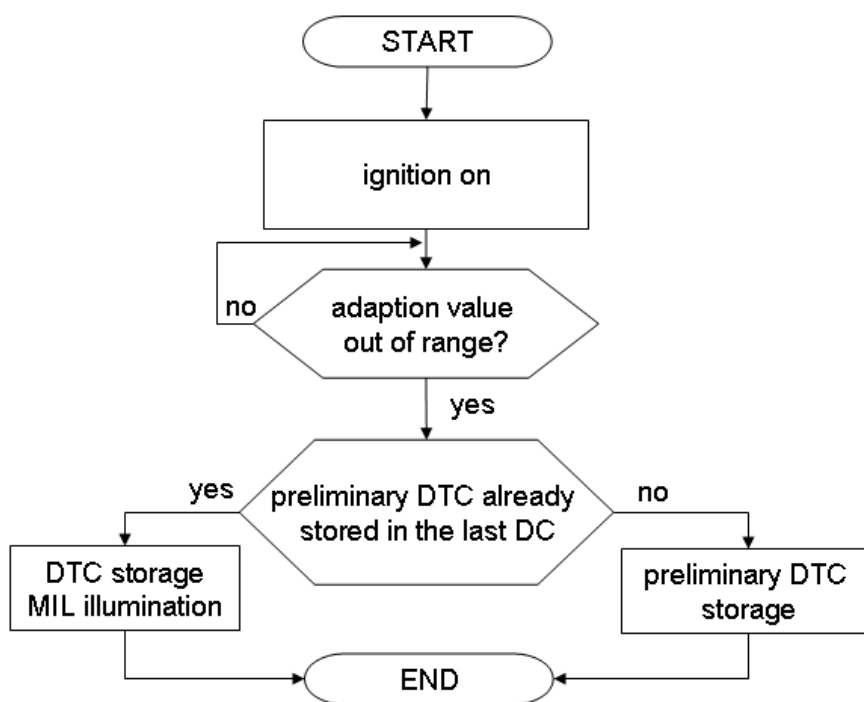
Issued:11/04/09

5.2.9 Signal Adaption Monitoring
(P229F,P2201)

General description:

The NOx signal adaption works in the same way for both NOx sensors. An adaption value is calculated based on the offset value of the NOx Offset Test. This correction offset for the NOx Sensors are monitored continuously. If the calculated offset is out of a range a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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5.3 NOx Sensor Ds Lambda Signal

The relationship between raw value and lambda is in the NOx-Sensor which is connected to the ECU via CAN. The NOx Sensor transmits the 1/Lambda signal to the ECU. This signal is monitored for signal range through P229F. See the tables below for the correct relationships. O2lin or lambda signal of the NOx sensor are not used for any other functionalities in the engine ECU except the monitoring for the DS-NOx sensor.

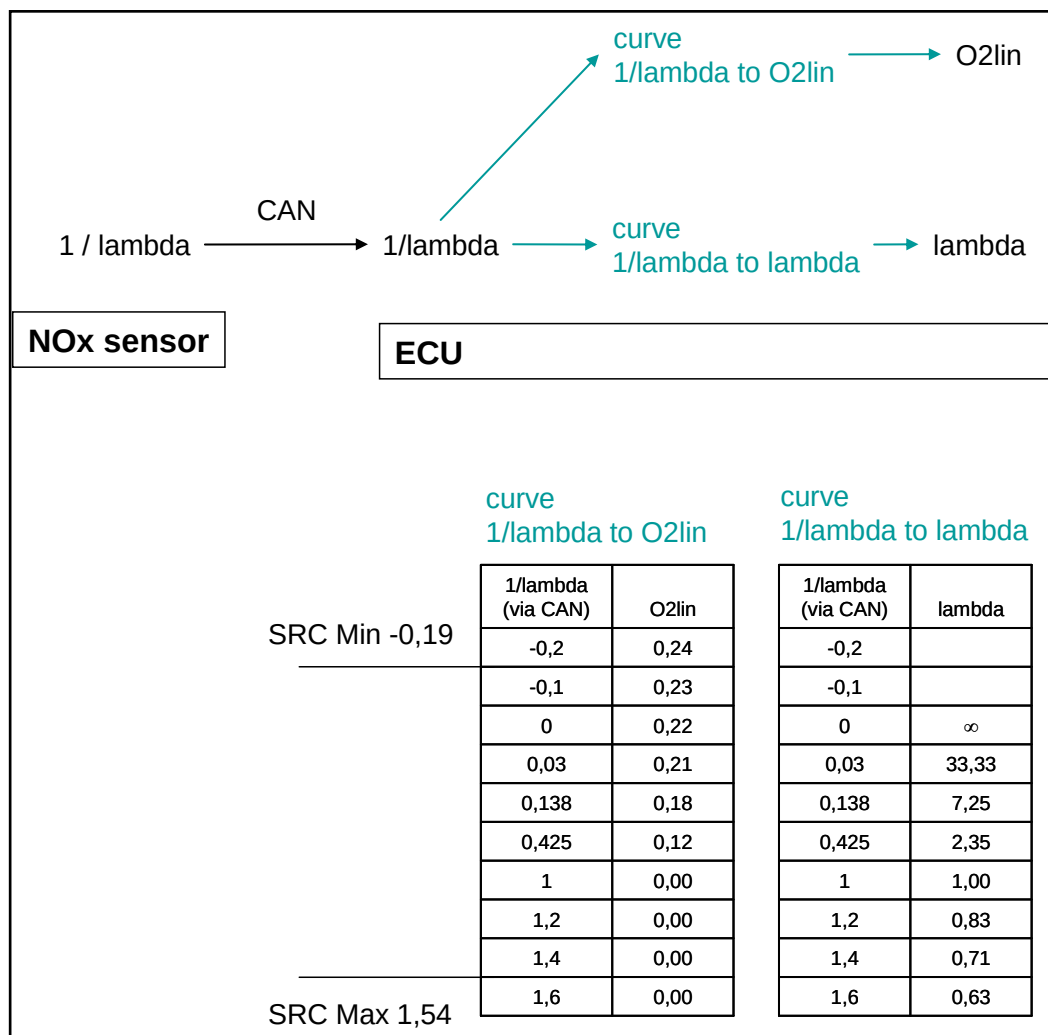


Figure: relationships between the NOx sensor DS signals



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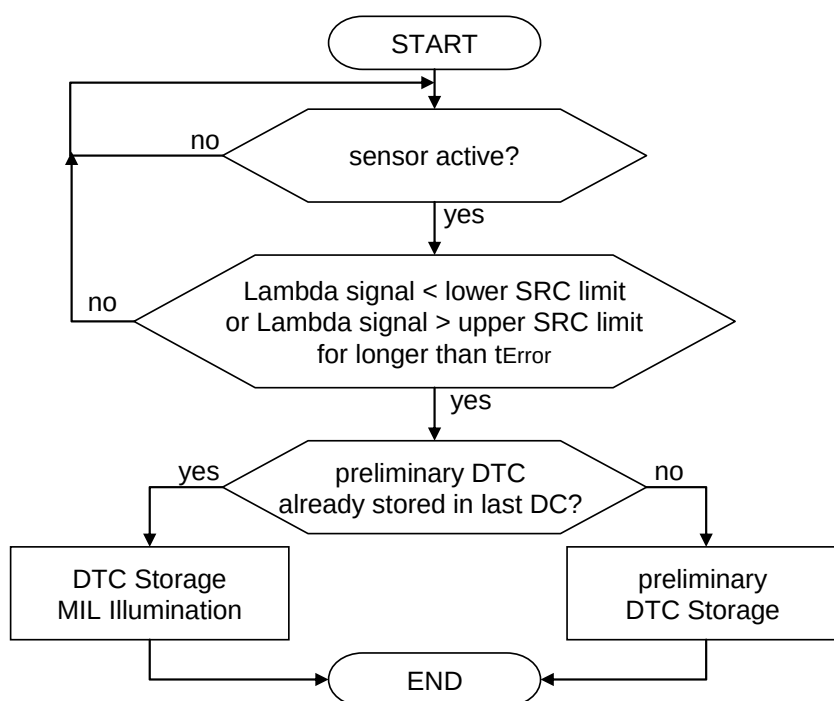
Issued:11/04/09

5.3.1 Lambda Signal Range Check
(P229F)

General description:

If the signal range of the Lambda signal is above/below the applicated threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:





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Part 1

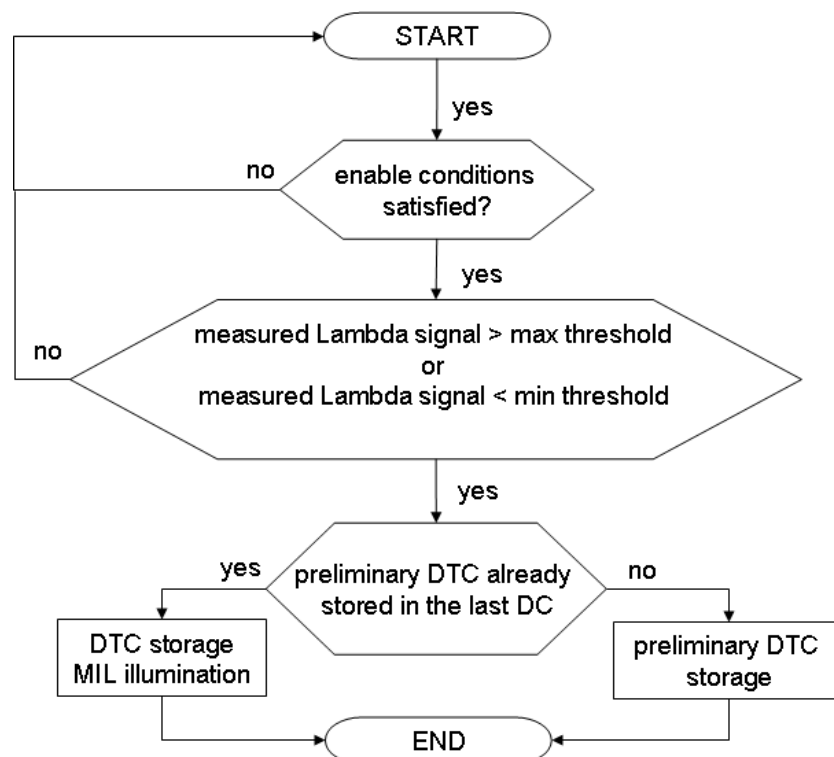
Issued:11/04/09

5.3.2 Lambda Signal Monitoring during overrun
(P229F)

General description:

To eliminate the Nox Sensor monitoring concern, BMW included a diagnostic that monitors the response of the oxygen measuring component of the Nox Ds Sensor during fuel cut. If the test conditions are satisfied the signal is compared with threshold values. If the signal is above/below a threshold a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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5.4 NOx Us Sensor**5.4.1 NOx Us Signal Plausibility Check**
(P2201)**General description:**

The measured NOx concentration of the NOx sensor Us is monitored for plausibility by a NOx model. The diagnosis works in two different operating ranges.

range one: part load

range two: low idle

At first the range one (part load) will be checked and than the range two (low idle).

If the enable conditions for range one are satisfied and the modelled NOx concentration is in steady state, an average of the signal difference will be calculated for a certain time (mean value calculation with timer1).

$$\text{signal difference} = (\text{sensor value} / \text{modelled value}) - 1$$

After finishing the testing in range one, range two will be started.

If the enable conditions for range two are satisfied and the modelled NOx concentration is in steady state, an average of the signal difference will be calculated for a certain time (mean value calculation with timer2).

If the calculated signal difference (in both operating ranges) is out of range a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

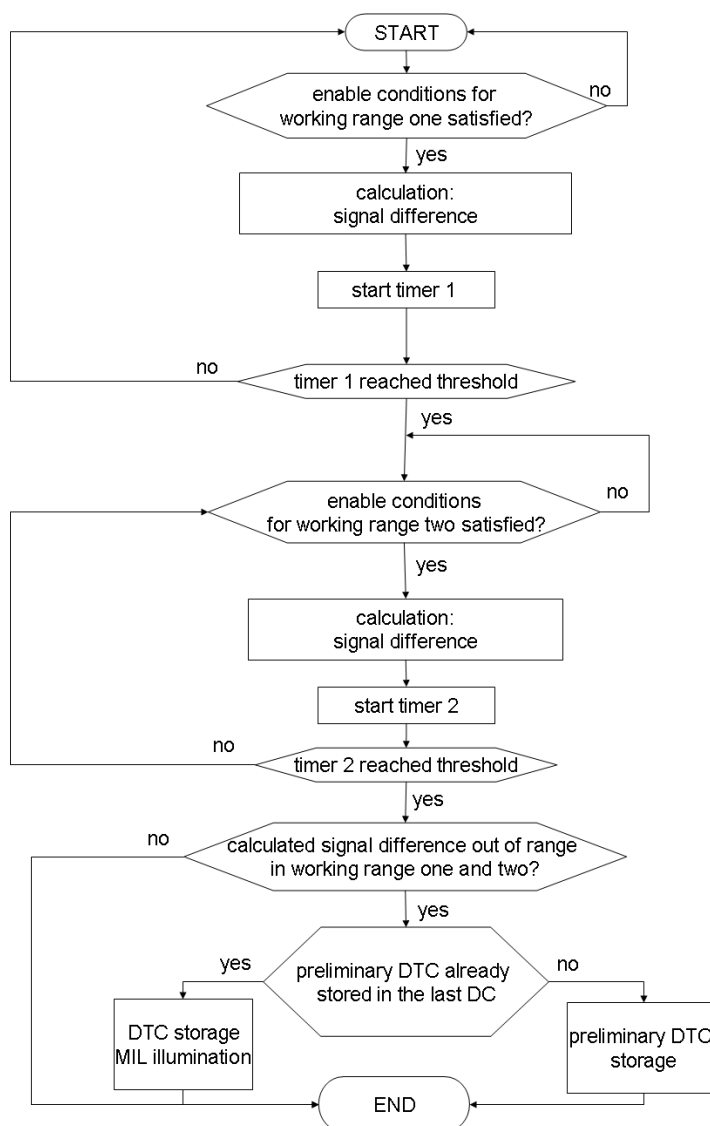


Test Group xxxxx – Tier2 Bin 5 Standard

Part 1

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Flowchart:



5.4.2 Dynamic Test (only upstream)
(P2201)

General description:

In a load to overrun transition the dynamic of the upstream NOx-Sensor is monitored. Therefore the time between two NOx thresholds needed by the Sensor-Signal is measured. The load to overrun transition is valid, if several conditions are fulfilled for a time before the load to overrun transition happens (steady state).



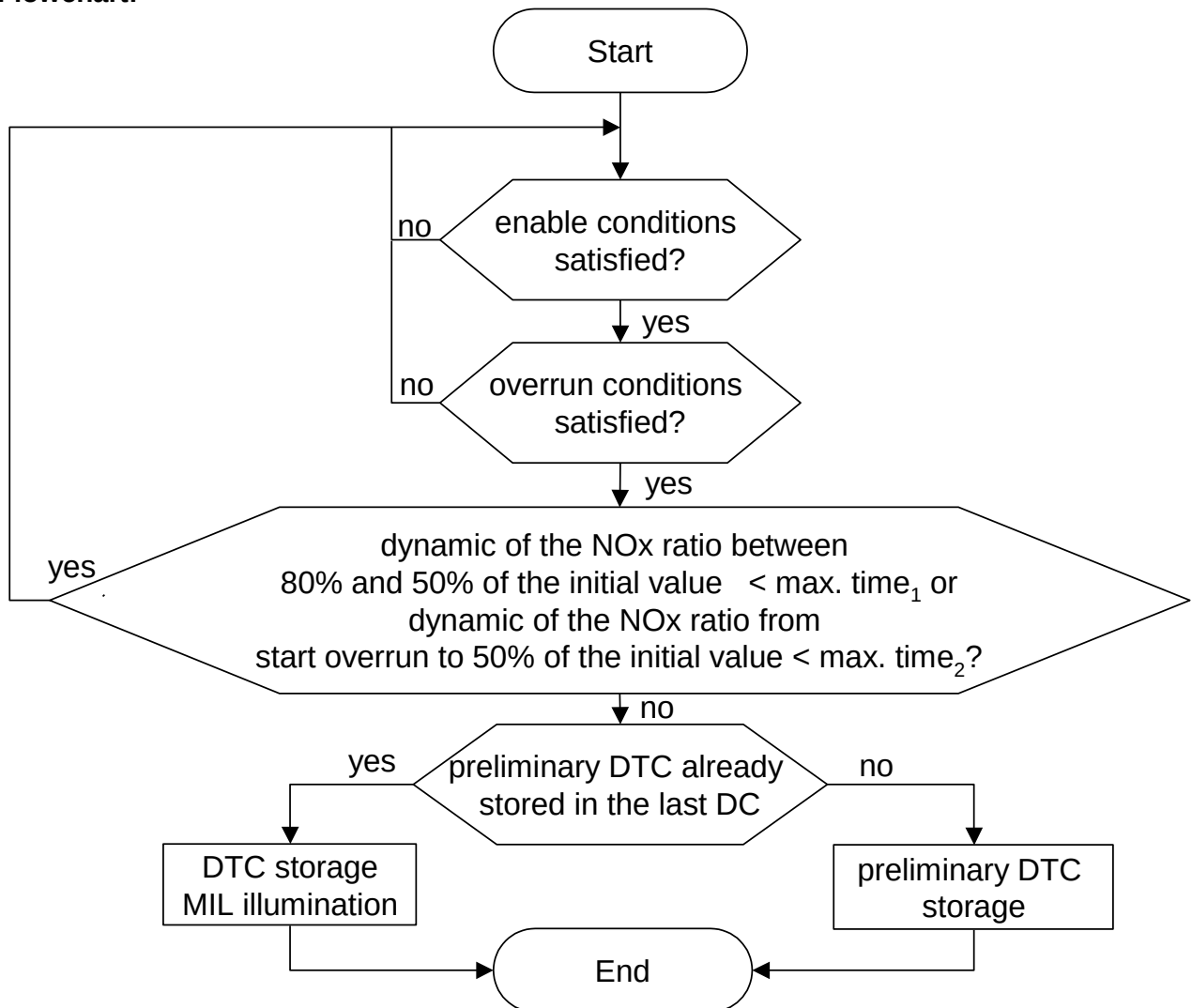
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If the sensor signal does not fall from 80% to 50% from the initial value in a maximum time₁ or the Sensor-Signal does not reach 50% from the initial value in a maximum time₂ an error is detected and a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles (CUC), the MIL is illuminated.

Flowchart:



5.5 NOx Ds Sensor Stuck in Range
(P229F)

General description:

The downstream NOx sensor is monitored for plausibility. The passive monitoring works during active dosing of Urea and will detect a static sensor signal. If a valid high NOx upstream (Us) peak value is detected, an increase of the NOx downstream (Ds) value should occur. If the increasing Nox-signal does not reach a minimum threshold for the NOx downstream (Ds) value, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

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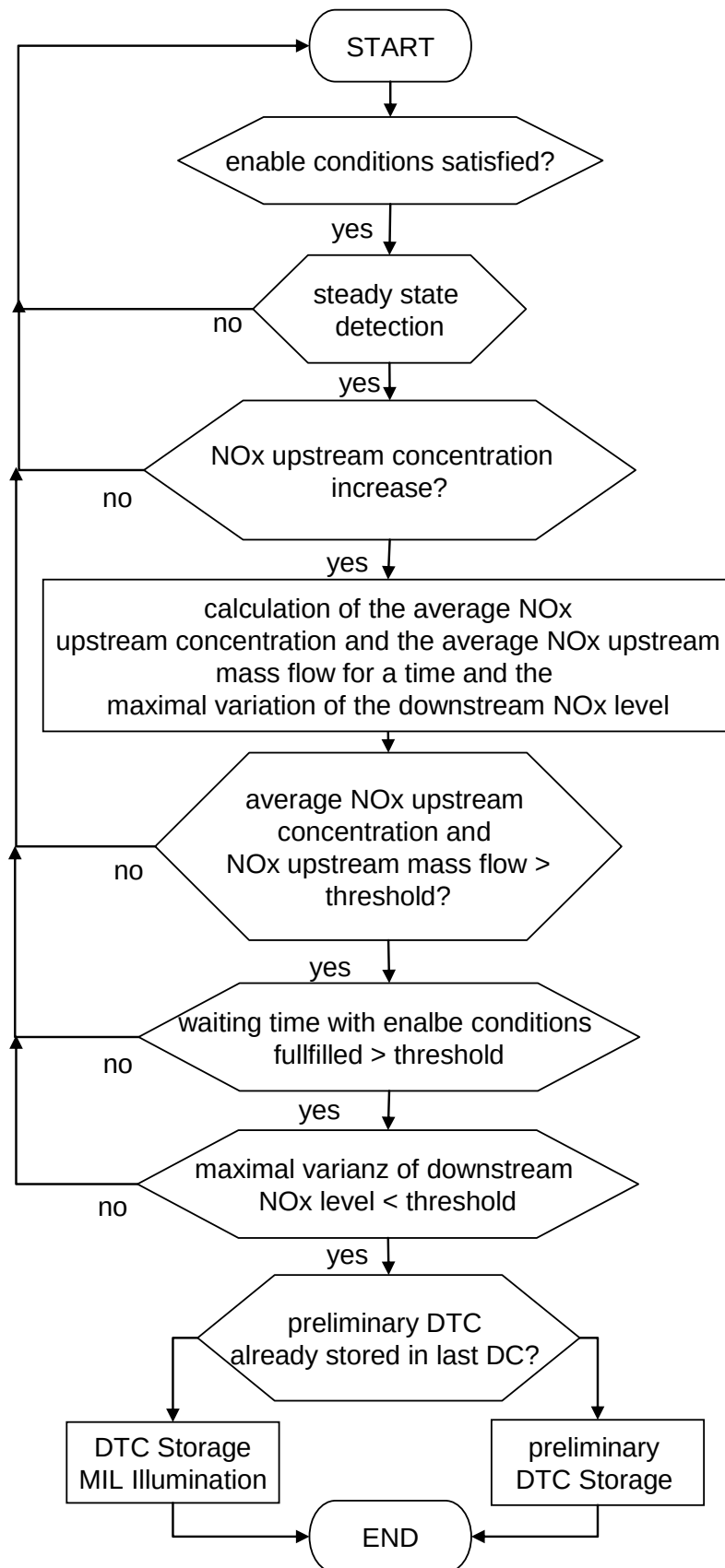


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Flowchart:



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6 Exhaust Gas Recirculation (EGR) System Monitoring

EGR	(f)(6.2.1)(A) Low Flow Threshold	(f)(6.2.1)(B) Low Flow Functional	(f)(6.2.2)(A) Low Flow Threshold	(f)(6.2.2)(B) Low Flow Functional
	P0401	P0401	P0402	P0402

(f)(6.2.3)(A) Slow Response Threshold	(f)(6.2.3)(B) Slow Response Increasing and Decreasing	(f)(6.2.4)(A)(i) Feedback: time to CL	(f)(6.2.4)(A)(ii) Feedback: default/OL	(f)(6.2.4)(A)(iii) Feedback: CL limits
P240F	P240F	see (f)(6.2.4)(C)	see (f)(6.2.4)(C)	P0401, P0402

6.1 EGR Control Loop Monitoring

The EGR control works on an airmass-based control mode in regeneration, and on an EGR-ratio-based control mode outside of regeneration. To prevent the engine from overheating the EGR has to be shut off at a certain engine temperature threshold. (This shut off is also considered by EI-AECD tracking in Scan Tool Mode\$1, PID81).

A stuck ECT sensor (above these temperatures) is detected via the comparison of the Fuel Temperature Sensor (FTS) and the ECT during cold start (see P008F).

6.1.1 Normal mode

Out of DPF regeneration the EGR is controlled based on an EGR-ratio.

$$\text{EGR ratio} = \frac{\text{recirculated exhaust gas}}{\text{recirculated exhaust gas} + \text{fresh air}}$$

**6.1.1.1 EGR Low Flow
(P0401)**

General description:

The controlling of the exhaust gas recirculation (EGR) outside of regeneration is based on the calculated EGR ratio. The monitoring is based on the EGR control deviation between the desired

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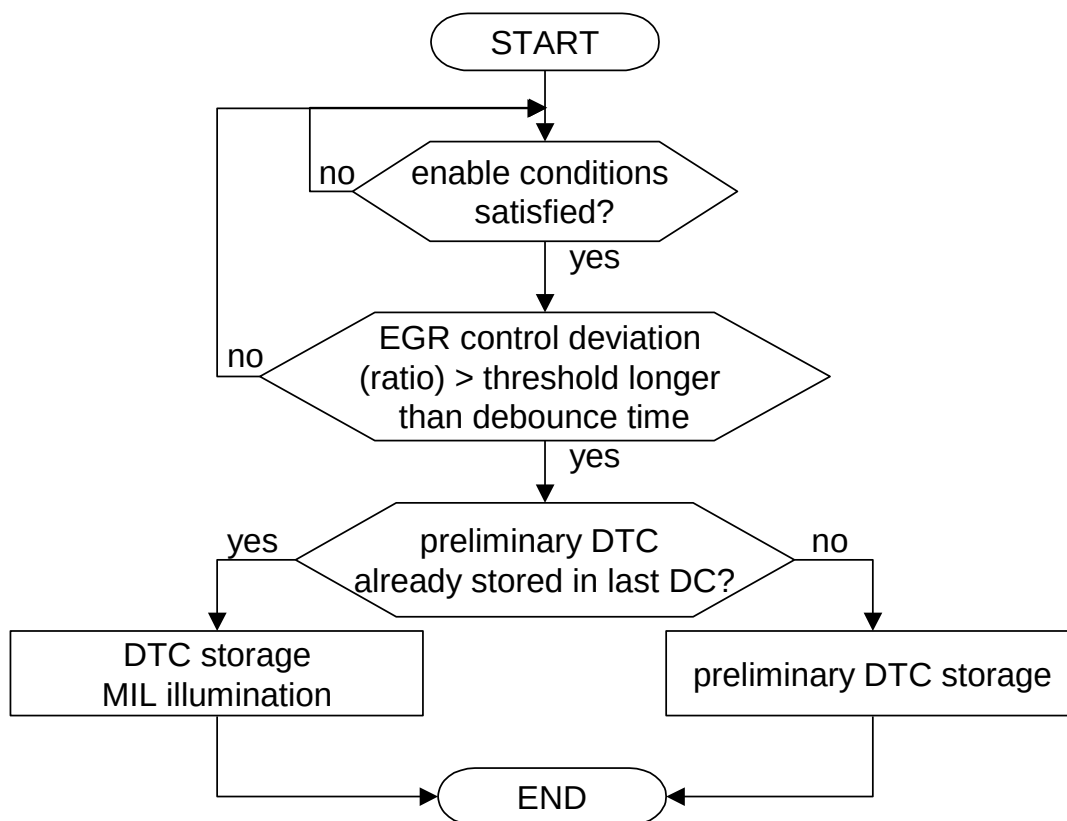
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EGR ratio and the actual EGR ratio. If the EGR control deviation is above the calibrated threshold for longer than the calibrated time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:





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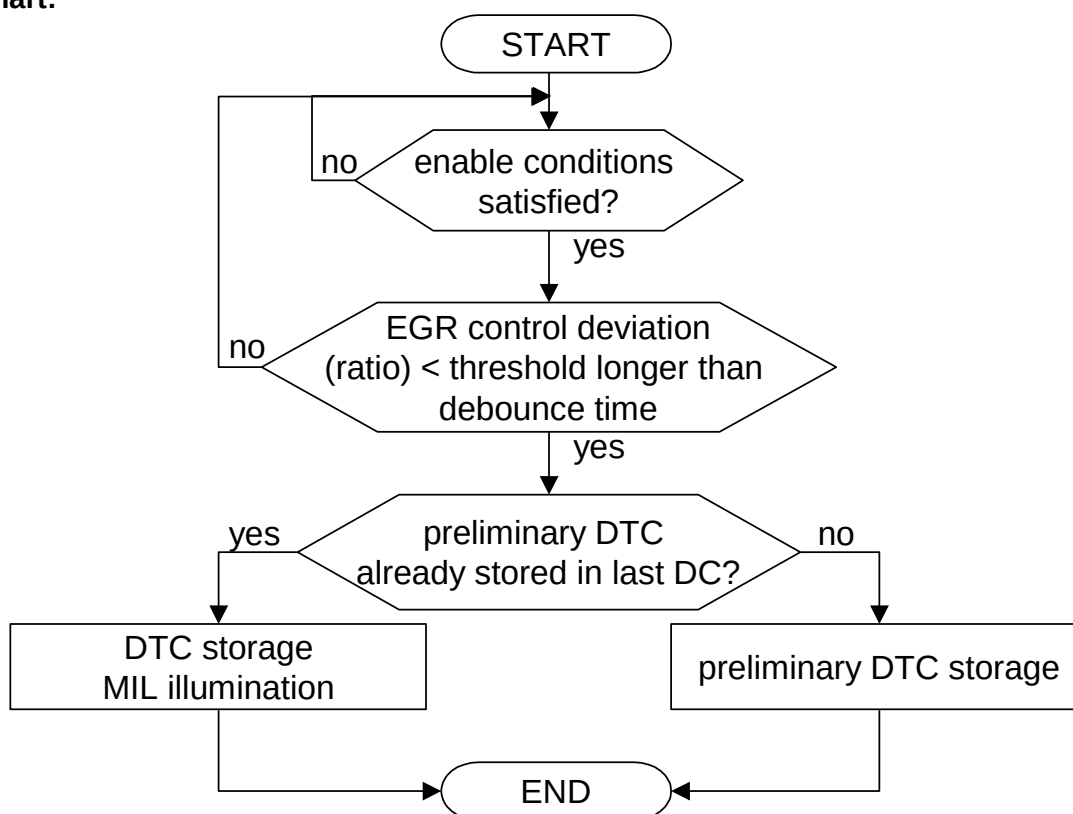
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6.1.1.2 EGR High Flow
(P0402)

General description:

The controlling of the exhaust gas recirculation (EGR) outside regeneration is based on the calculated EGR ratio. The monitoring is based on the EGR control deviation between the desired EGR ratio and the actual EGR ratio. If the EGR control deviation is below the calibrated threshold for longer than the calibrated time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:





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6.1.2 Regeneration

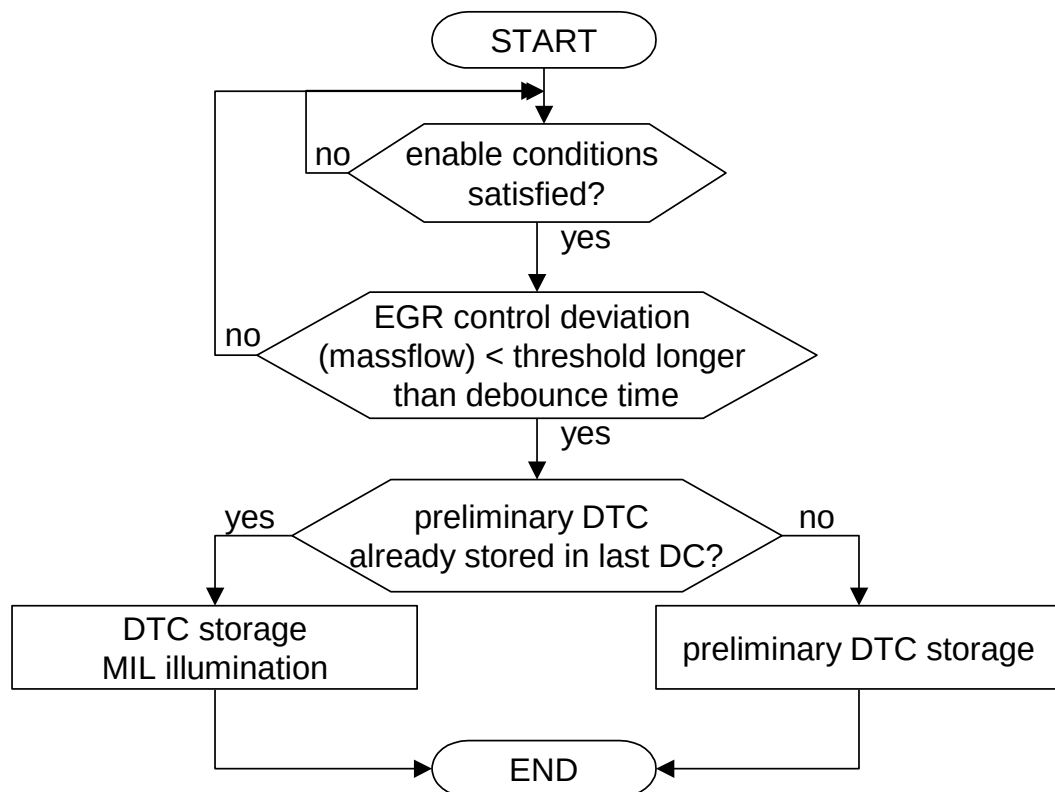
In regeneration mode the EGR is controlled based on airmass.

6.1.2.1 Low Flow
(P0401)

General description:

The controlling of the exhaust gas recirculation (EGR) in regeneration is based on the airmass measured by the air flow sensor. The monitoring is based on the EGR control deviation between this desired airmass and the actual airmass. If the EGR control deviation is below the calibrated threshold for longer than the calibrated time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:





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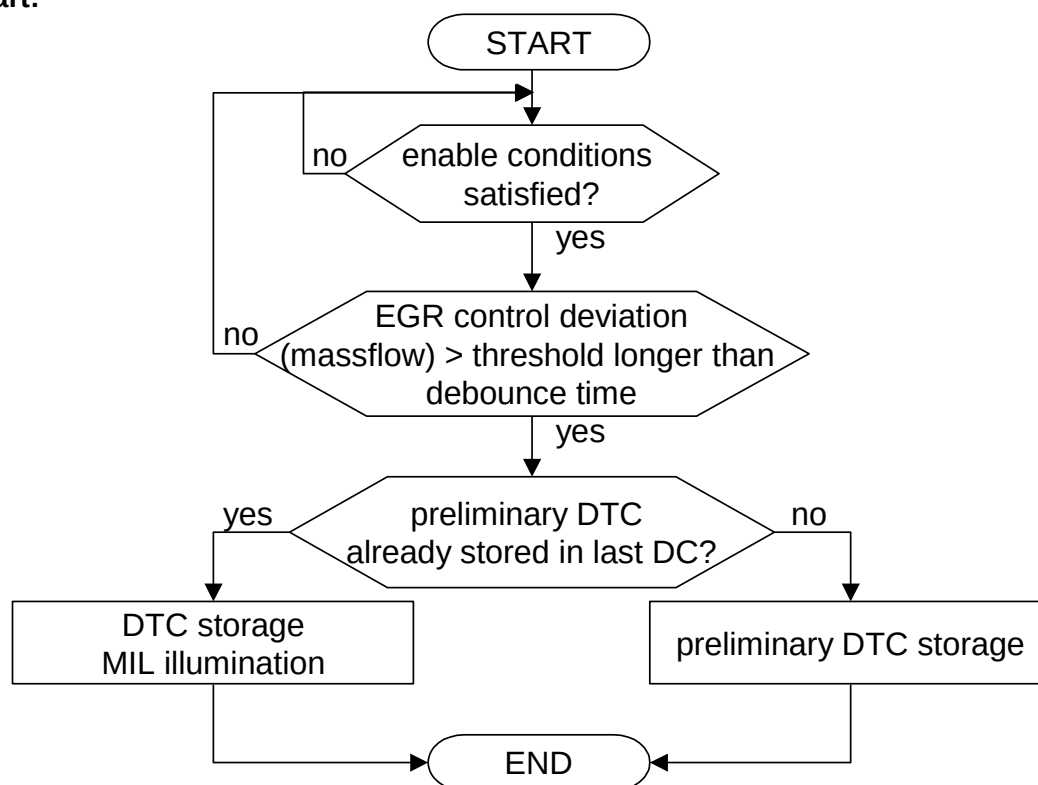
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6.1.2.2 High Flow
(P0402)

General description:

The controlling of the exhaust gas recirculation (EGR) in regeneration is based on the airmass measured by the air flow sensor. The monitoring is based on the EGR control deviation between the desired airmass and the actual airmass. If the EGR control deviation is above the calibrated threshold for longer than the calibrated time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:





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6.2 Feedback / Time to Closed Loop

According to CARB Regulations (f)(6.2.4)(C) time to closed loop for EGR system is monitored through single component monitoring. The input (release) components of the EGR system and their monitoring strategies are listed in the following table:

COMPONENT	ambient air temperature too high ambient air temperature too low ambient pressure too low EGR valve error engine coolant temperature too high engine coolant temperature too low injection quantity too high permanent EGR governor deviation							
Ambient Air Temperature Sensor	P0070	P009A						
Barometric Pressure Sensor			P321E P321F					
Engine Coolant Temperature Sensor					P008F	P0128		
Exhaust Gas Recirculation								P0401 P0402
Exhaust Gas Recirculation Valve				P3043 P042E P042F P045E P045F				
Fuel System							P323F	

6.2.1 EGR Slow Response Threshold

General description:

To detect a slow responding EGR system the characteristic value is monitored. The calculation of the characteristic value is based on the gradient of the commanded value and the EGR governor deviation (see graphic). This characteristic value is determined for each sample point. If the enabling conditions are satisfied the average characteristic value is determined. The characteristic value and the average characteristic value are calculated separately for positive and for negative gradients of the commanded value. While the enabling conditions are satisfied a timer increases. If the timer exceeds its applicable threshold the average characteristic value is compared to an applicable threshold. To enable the diagnosis the gradient of the commanded value has to be above an applicable threshold for a specified time. If the applicable release time is exceeded and the average characteristic value is above the applicable threshold a preliminary fault code is stored. If the fault is detected in two consecutive DC the MIL is illuminated.

Characteristic Value =

$$\frac{\text{EGR ratio governor deviation}^2}{2 * \text{gradient of desired value}}$$

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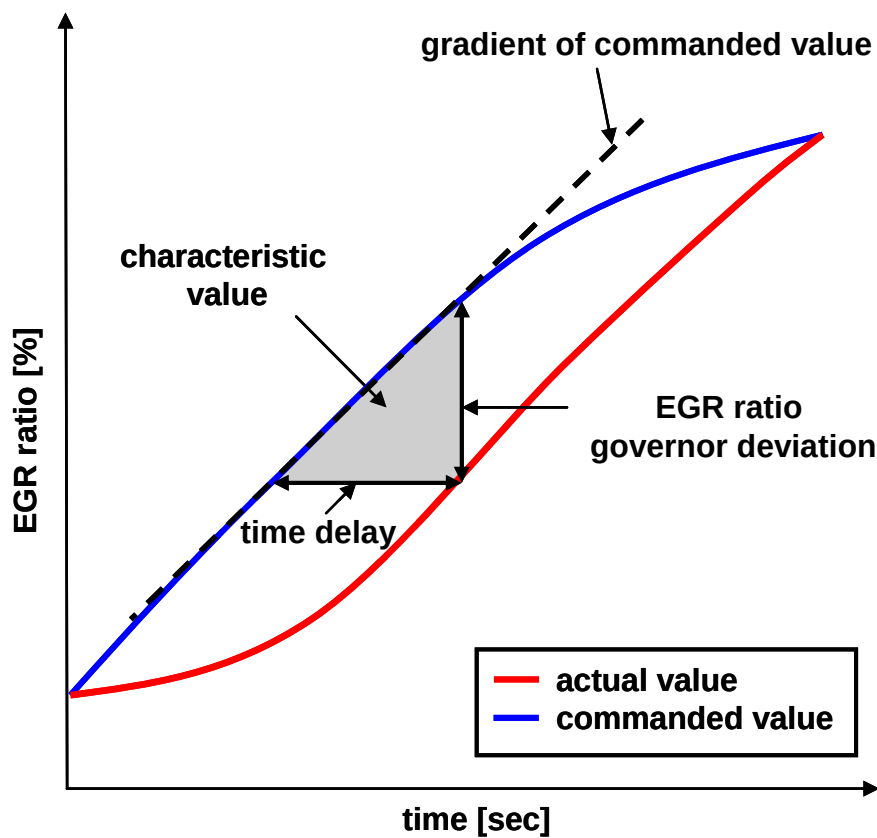


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Graphic:



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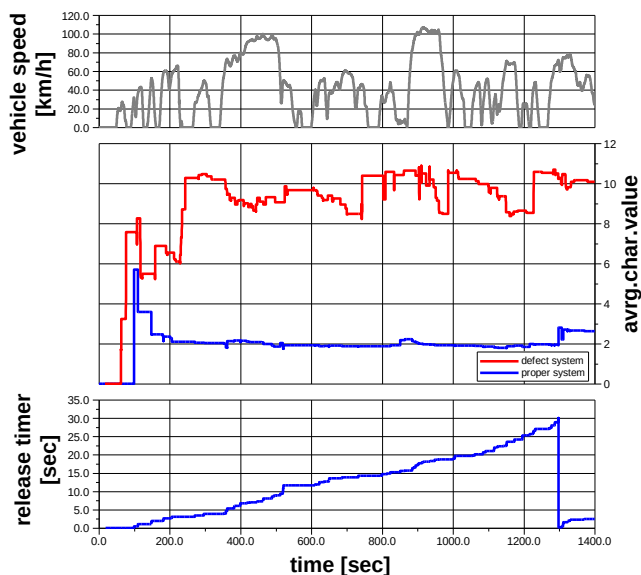


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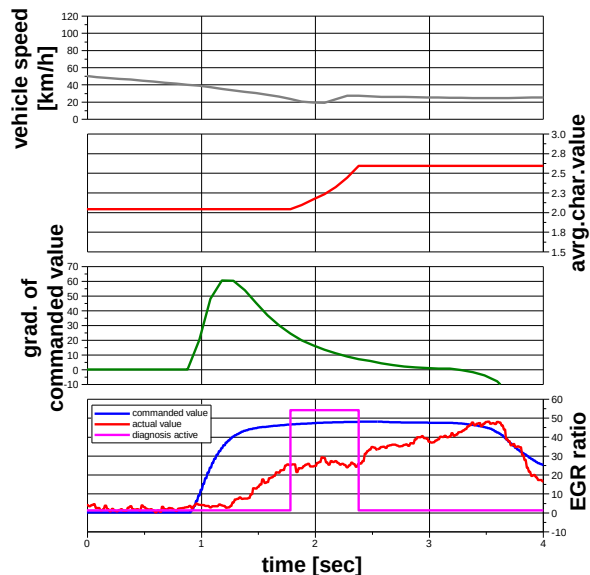
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AirCtl_RatSlwResp in CUC:



example of EGR slow response:



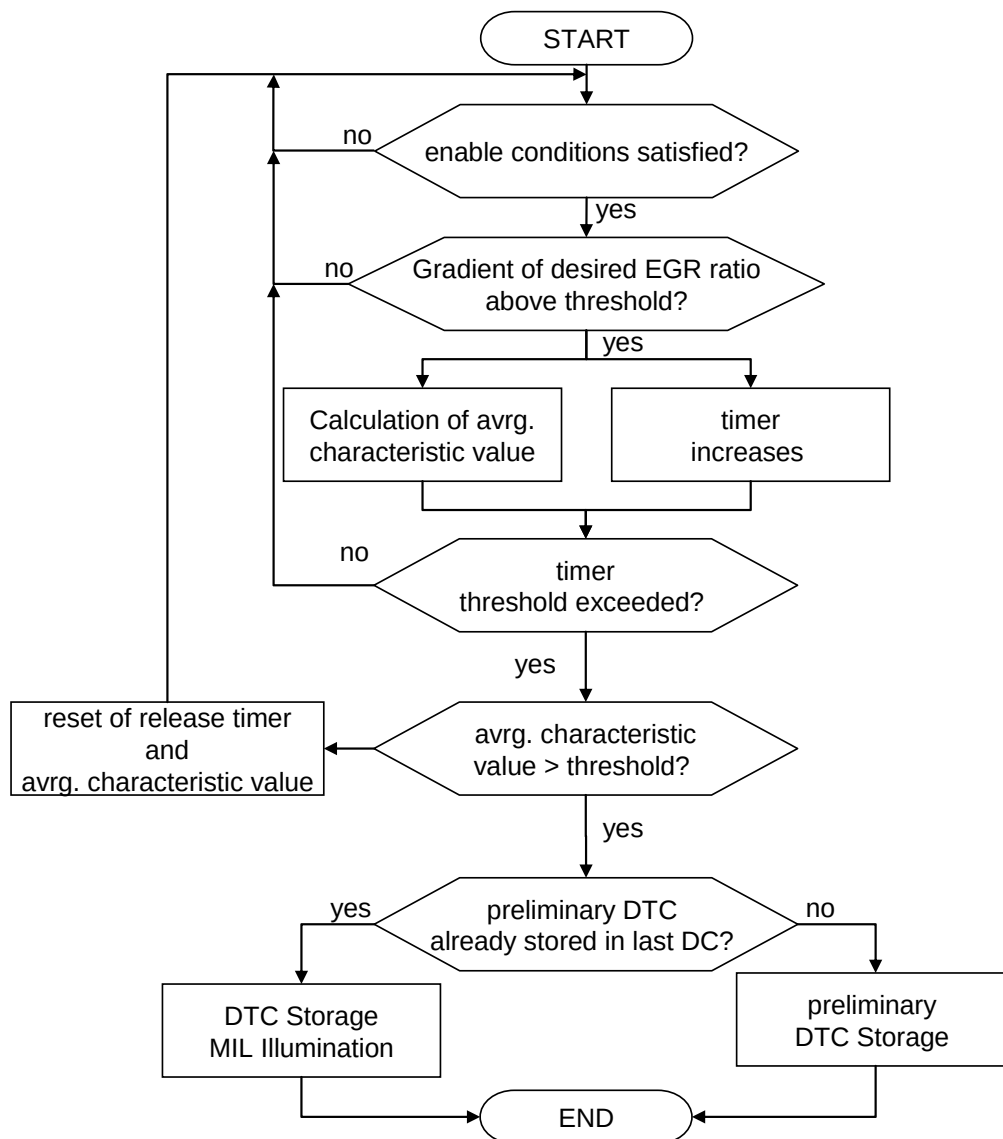


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Flowchart:





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6.3 EGR Target Value Correction – FMO
(P323F)

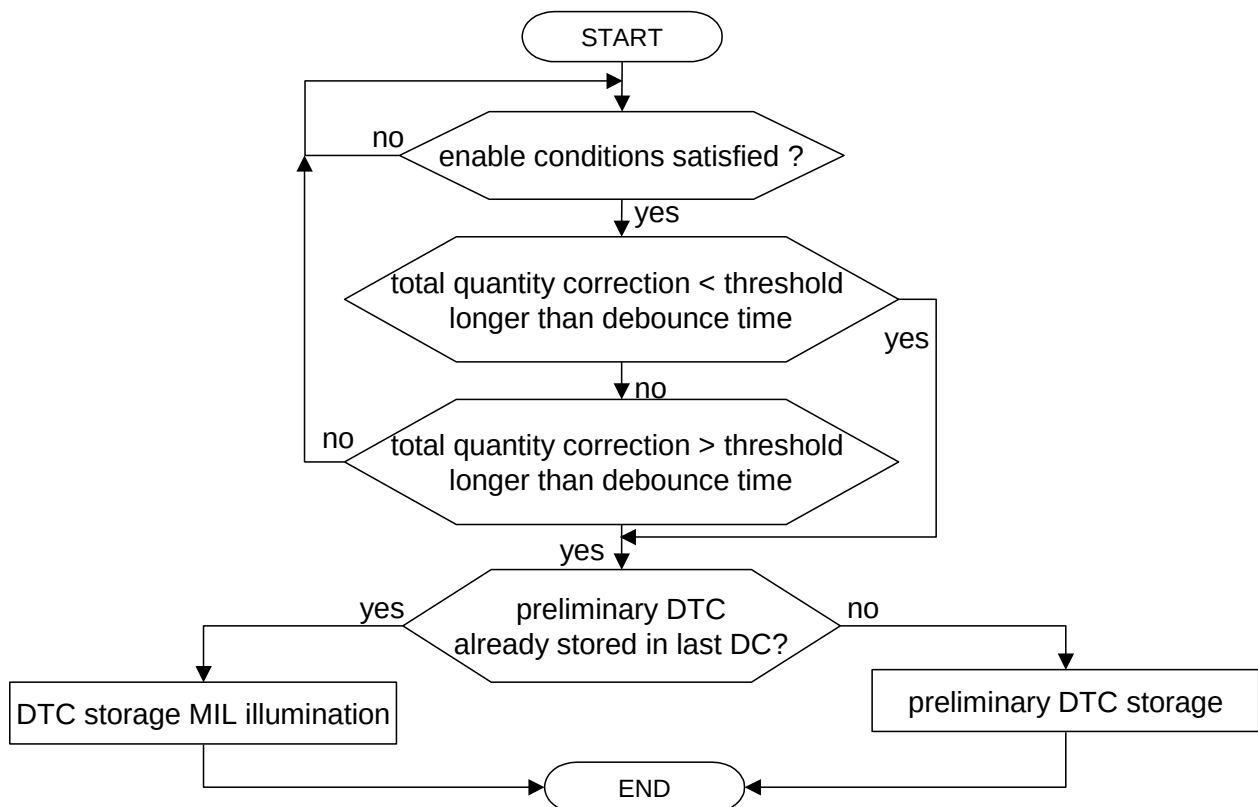
General Description:

The Fuel Mass Observer calculates the difference between

- calculated O2 concentration (from measured air mass flow and injection quantity)
- and measured O2 concentration of the exhaust gas by LSU

The O2 concentration is a criteria for the actual EGR rate. Based on the difference a total quantity correction for exhaust gas recirculation setpoint is calculated. The diagnosis is continuously, after the lambda dew point is reached. A fault is detected, if the correction is below or above the specific thresholds for longer than an allowed time. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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6.4 EGR Cooler Monitoring

EGR Cooler	(f)(6.2.5)(A)	(f)(6.2.5)(B)
	Cooler Threshold	EGR Cooler (HP, LP)
	P2457, P14D0	P2457, P14D0

6.4.1 High Pressure EGR Cooler
(P2457)

General description:

The high pressure exhaust gas recirculation (EGR) cooler is equipped with a bypass valve. This valve is used for monitoring the cooler efficiency.

Therefore, after the enabling conditions are satisfied, the diagnosis is monitoring the temperature change after high pressure exhaust gas recirculation cooler due to the alteration of the EGR cooler bypass valve position (fully opened and fully closed).

If the absolute temperature change is below the calibrated threshold an error is detected and a preliminary DTC is stored. If the error is detected in two consecutive driving cycles the MIL is illuminated.

Because of the highly dynamic exhaust gas temperatures and to minimize the effects of an opened EGR cooler bypass valve (on emission) the diagnosis is active only in idle.

The monitoring is also active only during active exhaust gas recirculation.

Because of the heating up of the EGR cooler during regeneration the diagnosis is inactive during regeneration and a specified time after regeneration to allow the system to cool down.

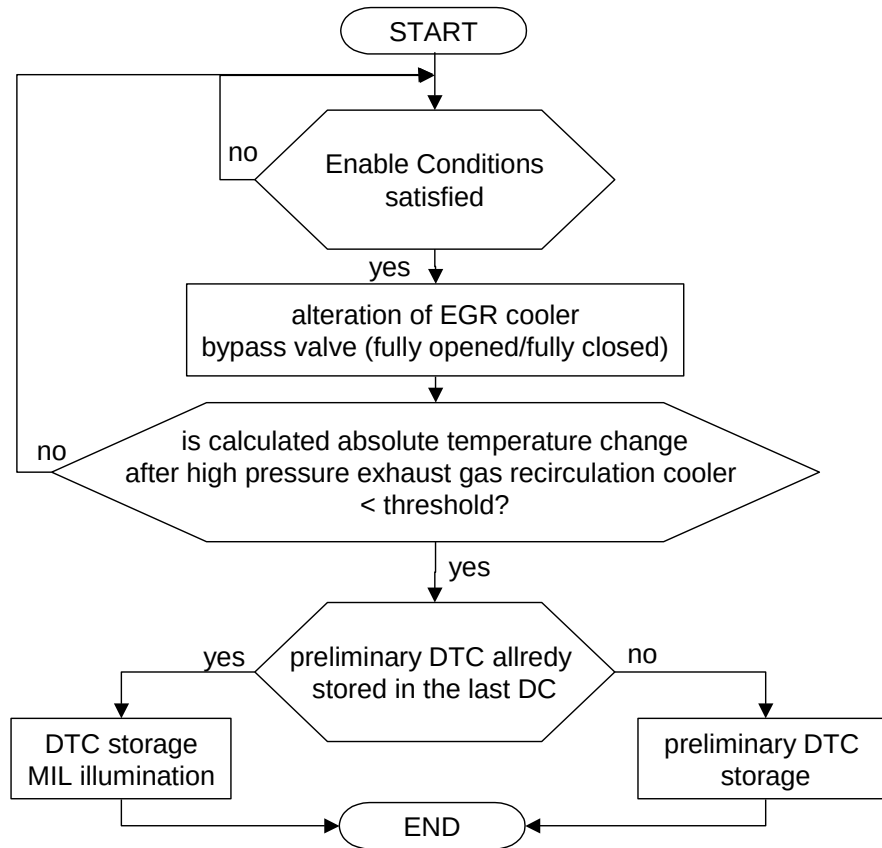


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Flowchart:





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6.4.2 Low Pressure EGR Cooler (only X5 3.0sd)
(P2457)

General description:

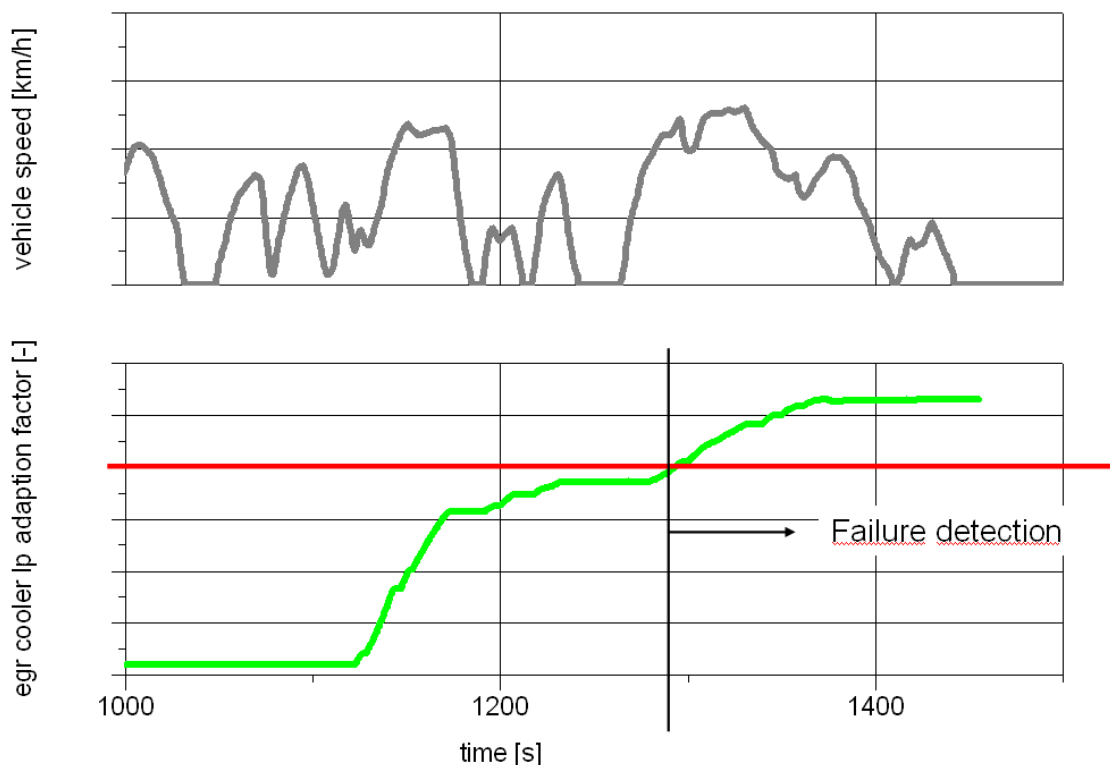
If the enable conditions are satisfied, the difference between the exhaust temperature downstream EGR LP cooler and the modelled exhaust temperature downstream EGR LP cooler is measured and an adaption factor, which is explained in the following, is calculated continuously. The ECU calculates a modeled temperature downstream the EGR cooler based on a cooler efficiency of a proper EGR cooler using data of a proper cooler. If this modeled temperature differs to the measured temperature T_EGR_Ds the cooler efficiency is adapted by an adaption factor in order to equalize measured and modeled temperature downstream the EGR cooler.

The efficiency of the EGR cooler is defined by the temperature difference over the EGR cooler divided by the maximum possible temperature difference:

$$\text{Efficiency} = (T_{\text{EGR_Us}} - T_{\text{EGR_Ds_modelled}}) / (T_{\text{EGR_Us}} - \text{ECT})$$

In a proper system (modeled temperature downstream cooler = measured T_EGR_Ds) the correction factor is = 0.

If the adaption factor exceeds the applicated threshold, an error is detected and a preliminary DTC is stored. If the error is detected in two consecutive driving cycles (CARB Unified Cycle - CUC), the MIL is illuminated.





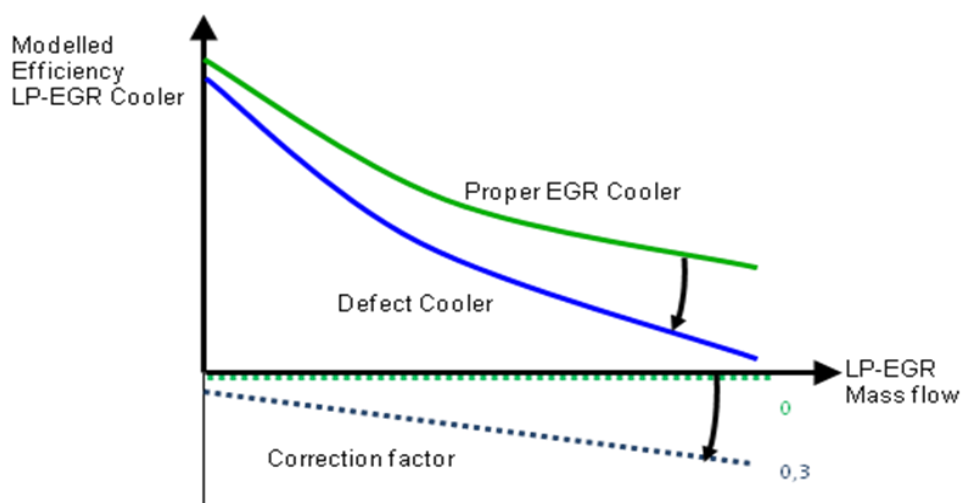
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The correction factor is calculated continuously independent of the temperature difference. Small negative values can occur if the measured temperature is lower than the modeled temperature. This can happen if the EGR cooler has a better efficiency than expected by the data of a proper cooler.

The correction factor is applied to the efficiency map of the cooler:



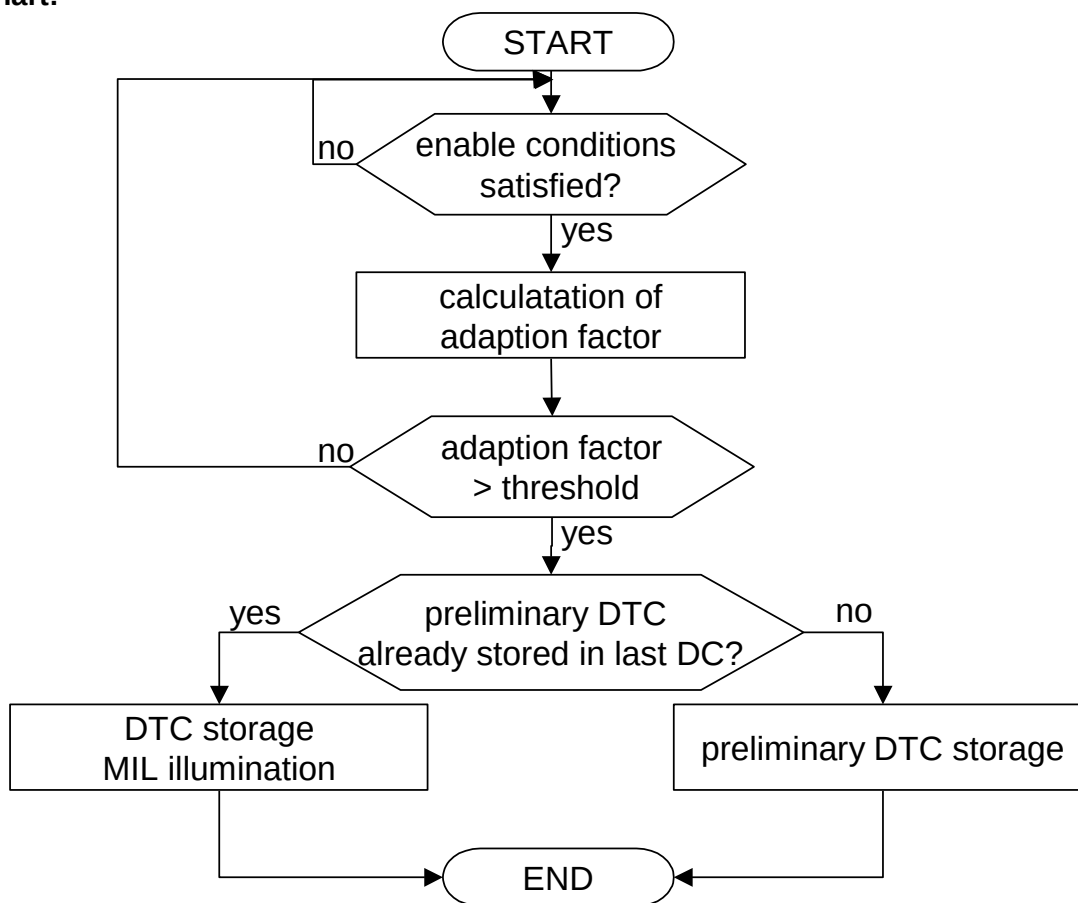


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Flowchart:





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7 Boost Pressure

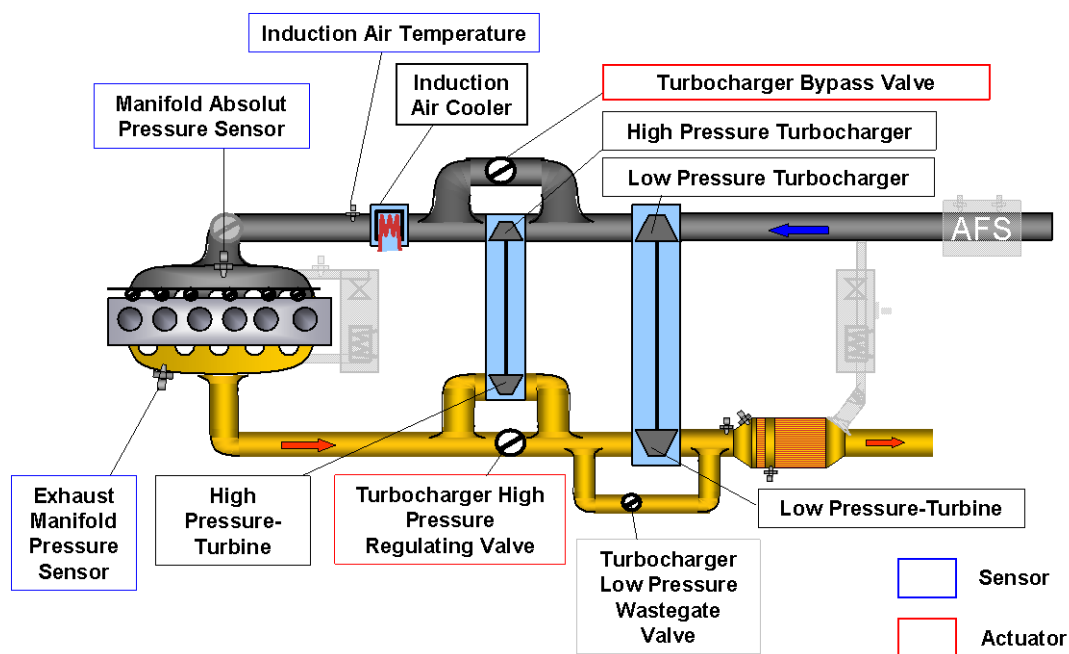
Boost Pressure	(f)(7.2.1)(A) Underboost Threshold	(f)(7.2.1)(B) Underboost Functional	(f)(7.2.2)(A) Overboost Threshold	(f)(7.2.2)(B) Overboost Functional	(f)(7.2.3)(A) VGT Response Threshold	(f)(7.2.3)(B) VGT Response Functional
	P0299	P0299	P0234	P0234	P0234, P0299	P0234, P0299

Boost Pressure	(f)(7.2.4)(A) Charge Air Cooling Threshold P026A	(f)(7.2.4)(B) Charge Air Cooling Functional P026A	(f)(7.2.5)(A)(i) Feedback: time to CL see (f)(7.2.5)(C)	(f)(7.2.5)(A)(ii) Feedback: default/OL see (f)(7.2.5)(C)	(f)(7.2.5)(A)(iii) Feedback: CL limits P0234, P0299
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The monitoring of the boost pressure control regulation (PCR) is based on the control deviation.

Under Boost → positive control deviation P0299
Over Boost → negative control deviation P0234

Boost Pressure System Overview:





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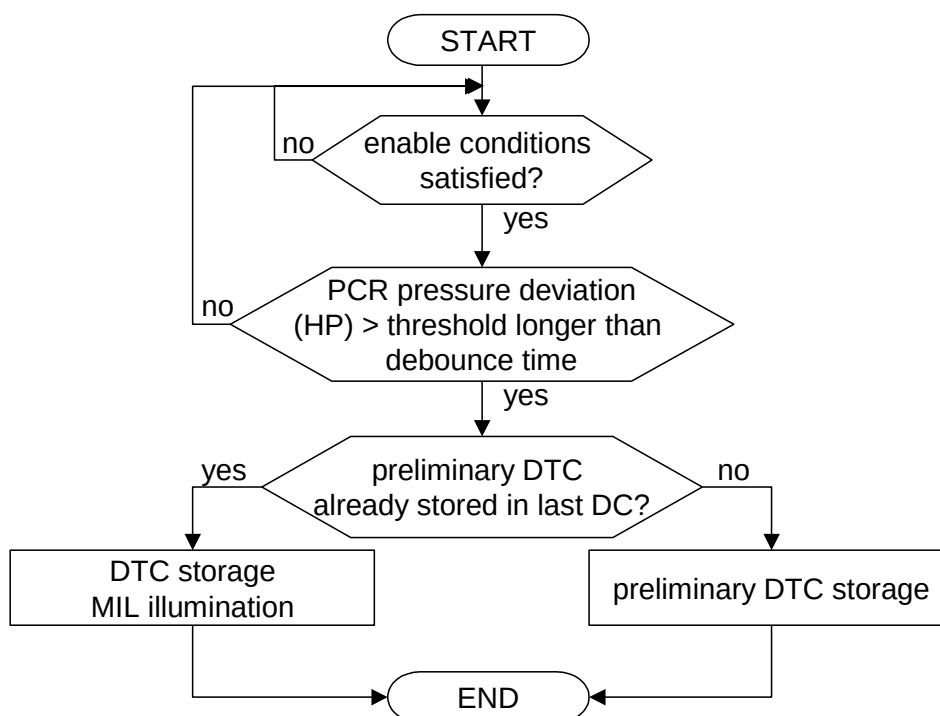
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7.1 Under Boost
(P0299)

General description:

If the PCR (high pressure (HP)) control deviation is above the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:

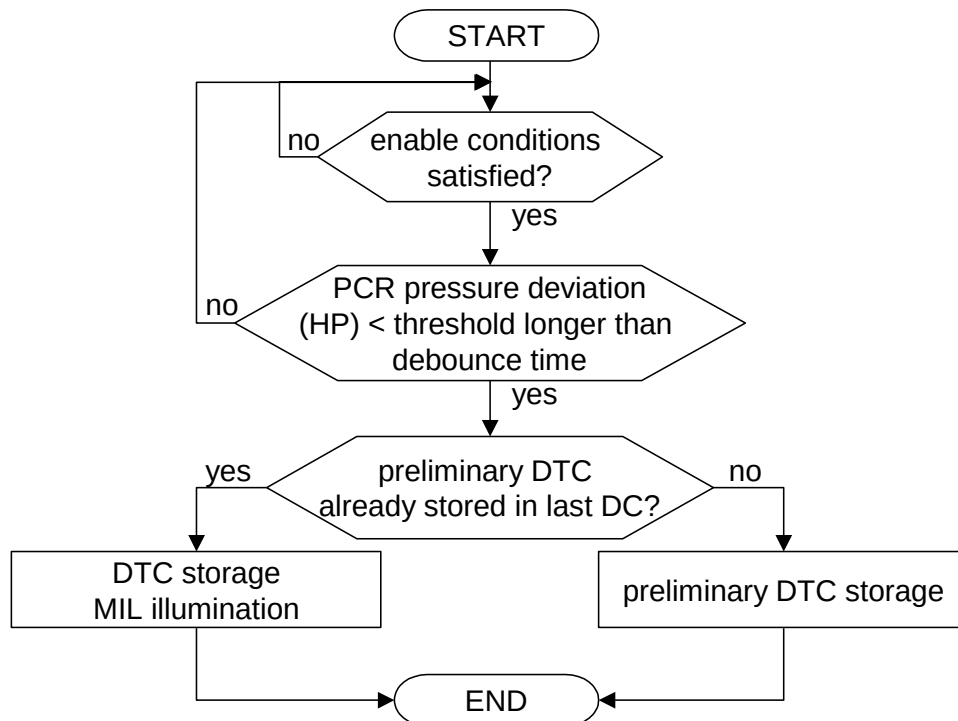


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7.2 Over Boost
(P0234)**General description:**

If the PCR (high pressure (HP)) control deviation is below the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:



7.3 Functional check of Low Pressure Stage (LP)

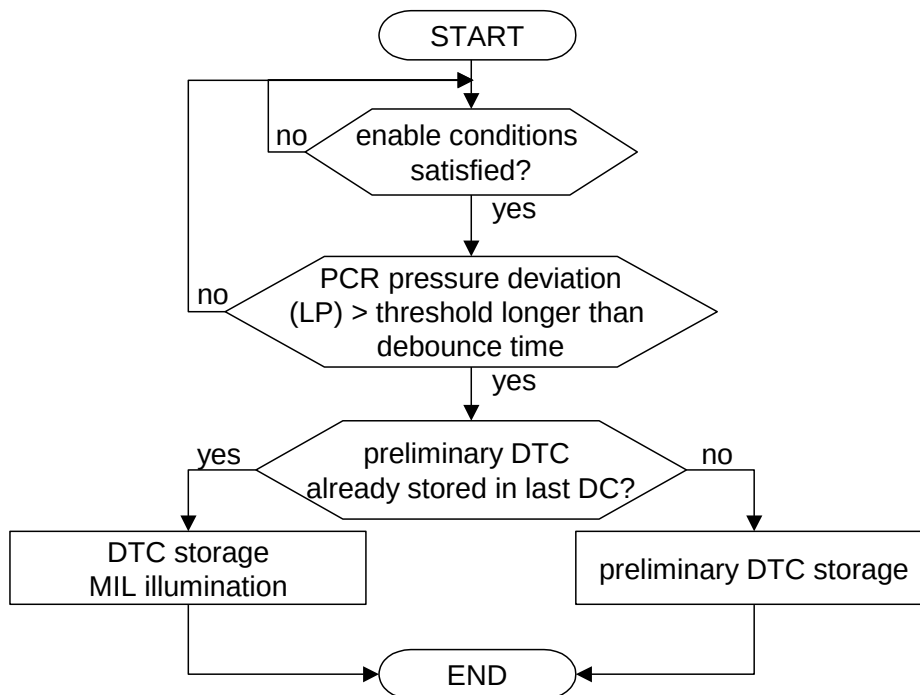
The functional check of the pressure control regulation (PCR) is based on the monitoring of the control deviation.

7.3.1 Boost pressure governor deviation LP (maximum) (P02CB)

General description:

If the PCR (low pressure (LP)) control deviation is above the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:

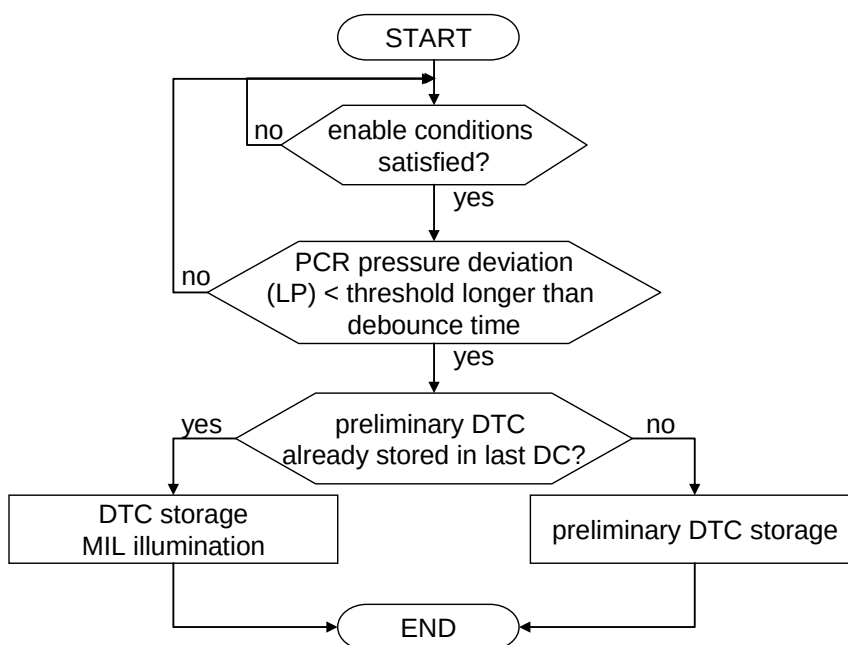


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**7.3.2 Boost pressure governor deviation LP (minimum)
(P02CA)****General description:**

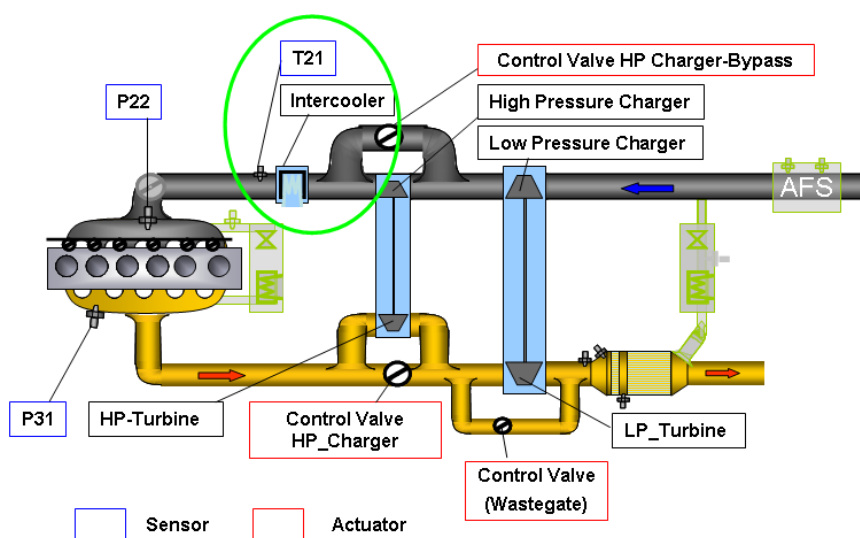
If the PCR (low pressure (LP)) control deviation is below the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:



7.4 Charge Air Cooling Threshold

Schematic Overview of Charge Air Cooling System:



General description:

The charged air cooler (CAC) is monitored by a calculated efficiency based on the temperature difference between the measured temperature downstream of the intercooler CAC (T21) and the modelled temperature value upstream of the CAC.

The basic CAC efficiency is calculated via the temperature difference over the CAC (modeled temperature upstream CAC – measured temperature downstream CAC) divided by the maximum possible temperature difference (modeled temperature upstream CAC – measured ambient air temperature):

$$\text{CAC efficiency} = \frac{\text{modeled temperature upstream CAC} - \text{measured temperature downstream CAC}}{\text{modeled temperature upstream CAC} - \text{measured ambient air temperature}}$$

If the enabling conditions are satisfied a timer increases and the efficiency is calculated. When the timer exceeds its applicable threshold and the calculated CAC efficiency is below the applied threshold a preliminary fault code is stored. If the fault is detected in two consecutive driving cycles the MIL is illuminated.



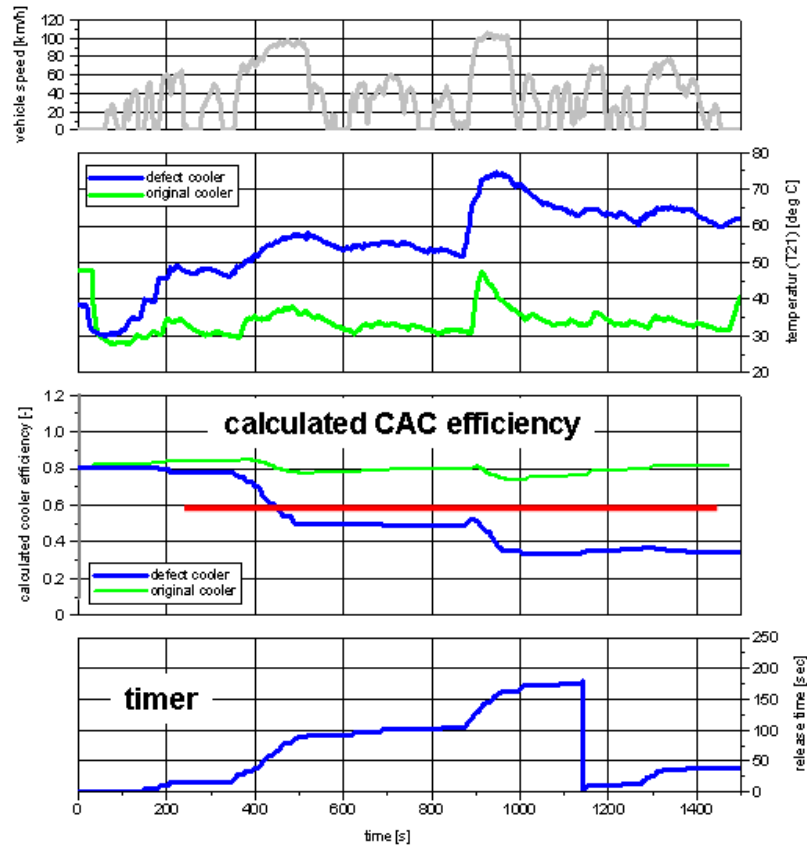
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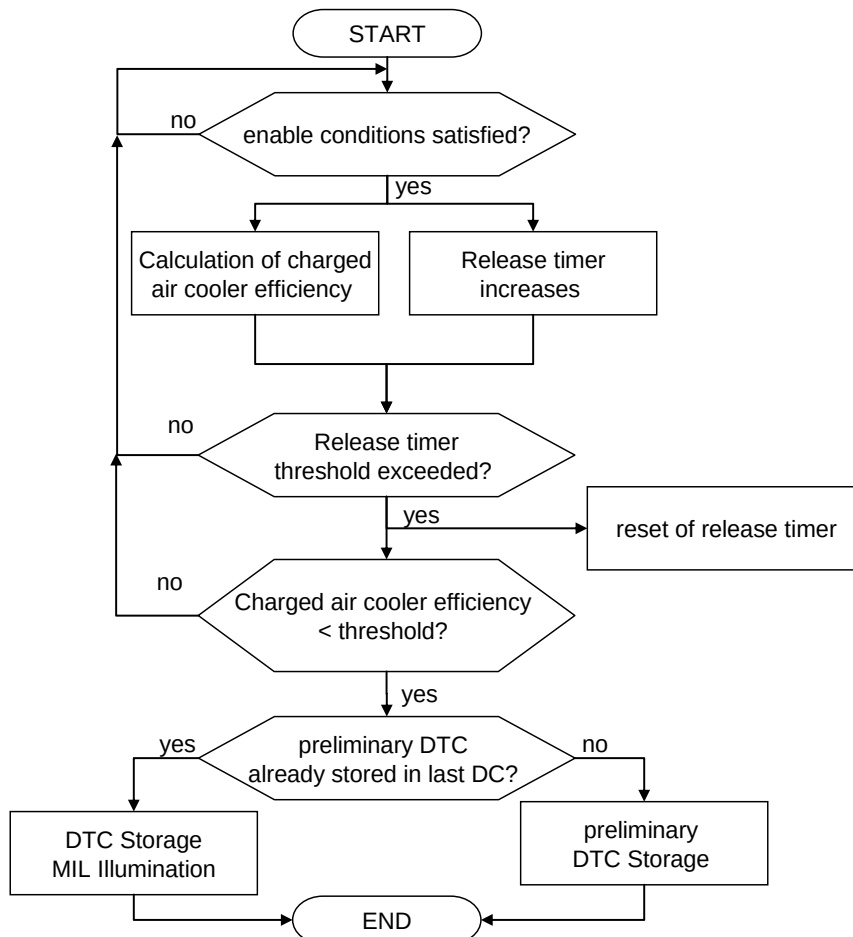
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Flowchart:



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7.5 Slow Response

Regulation:

f7.2.3 (B) For vehicles in which no failure or deterioration of the VGT system response could result in a vehicle's emissions exceeding the malfunction criteria specified in section (f)(7.2.3)(A), the OBD II system shall detect a malfunction of the VGT system when proper functional response of the system to computer commands does not occur.

According to (f)(7.2.3)(B) BMW detects a slow response system with the P0299.

Hints:

- Emission test with jammed open HP control valve below 2 x std
- Wastegate for LP Charger is opened only at full load to avoid overboost (see discussion and measurements for MY2009)
- HP Charger Bypass Valve is closed during FTP / CUC boost pressure levels. The valve opens at high loads. Only two positions (open / closed) are possible
- For closed loop control the HP control valve is used. As BMW system doesn't use VGT we see the HP control valve as the corresponding actor.
- Malfunction of a blocked control valve is detected by f 7.2.1 „underboost“

7.6 Feedback control

According to regulation f7.2.5 C) time to closed loop is monitored by the individual input components of the boost pressure controller:

Release components	Activation threshold	Monitoring threshold
Engine speed	> 1250 rpm	Desired value
Injection quantity	> 6 mg / stroke	Desired value

8 NOx Adsorber --> N/A

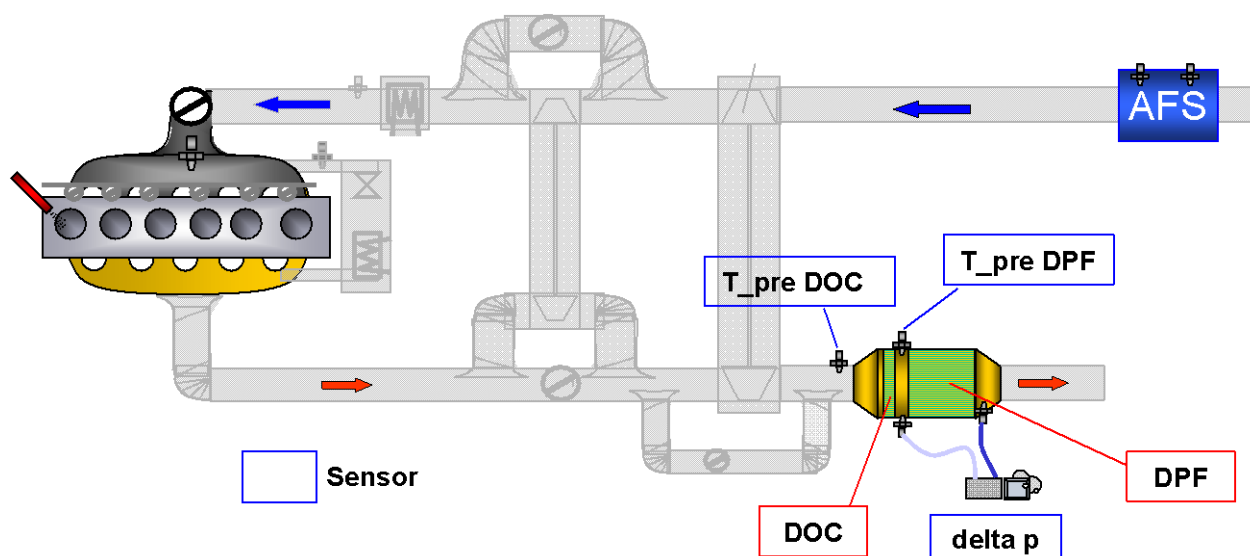


9 PM Filter

PM Filter	(f)(9.2.1)(A) Threshold	(f)(9.2.1)(B) Functional	(f)(9.2.2)(A) Regen frequency: Mfr spec P2459	(f)(9.2.2)(B) Regen frequency: Threshold P2459	(f)(9.2.2)(C) Regen frequency: Functional P2459
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(f)(9.2.3) Incomplete Regen	(f)(9.2.4) NMHC Conversion	(f)(9.2.5) Missing Substrate	(f)(9.2.6) Failure to achieve regeneration P14A7	(f)(9.2.7)(A)(i) Feedback: time to CL P244C	(f)(9.2.7)(A)(ii) Feedback: default/OL P244C	(f)(9.2.7)(A)(iii) Feedback: CL limits P244C, P244D
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9.1 System Overview

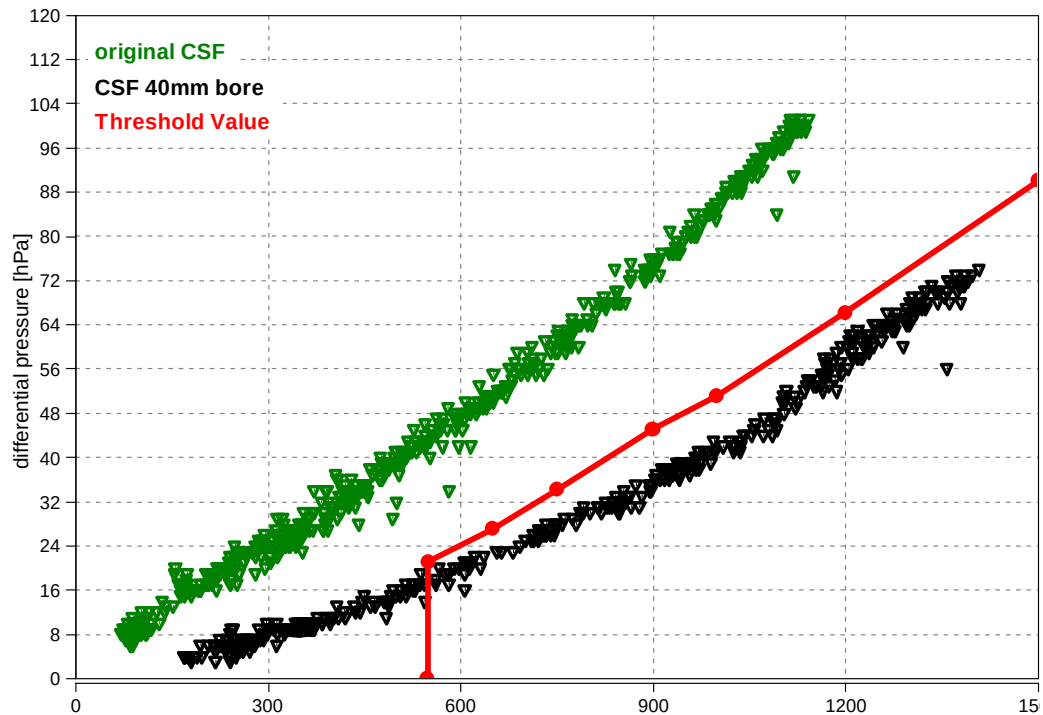


9.2 Efficiency (P2002)

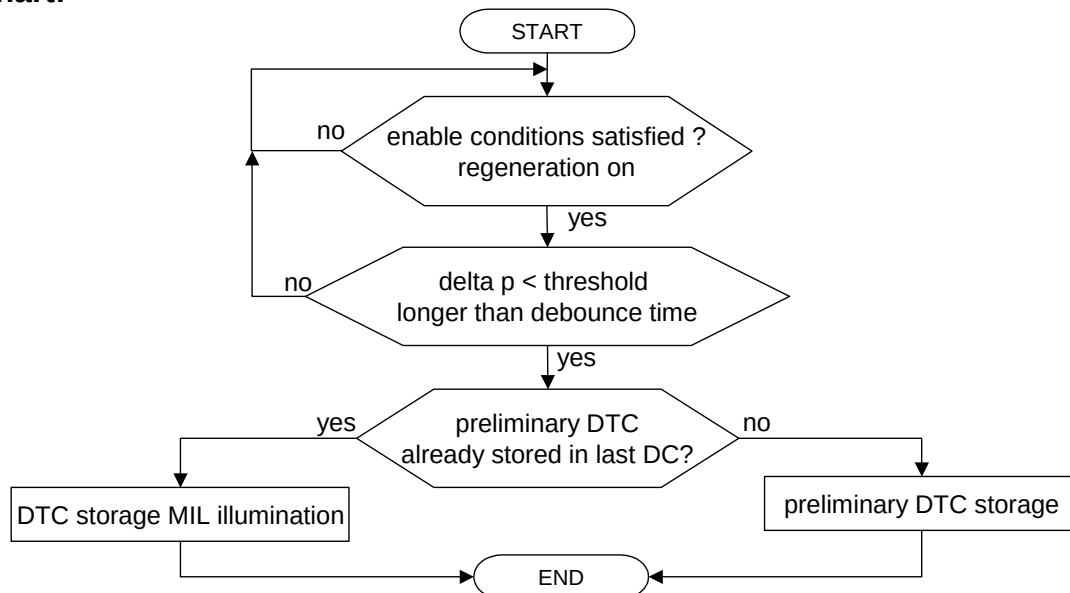
General Description:

The particulate filter is monitored through pressure decrease of the destroyed particulate filter. The differential pressure of a destroyed particulate filter is smaller than the differential pressure of an original particulate filter. Due to that the monitoring concept of the particulate filter efficiency is based on the comparison of differential pressure sensor signal with a threshold. The diagnosis starts if the exhaust flow and the particulate filter temperature are above thresholds.

A fault is detected, if the differential pressure falls below the threshold for more than an allowed time. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC in regeneration), the MIL is illuminated.



Flowchart:



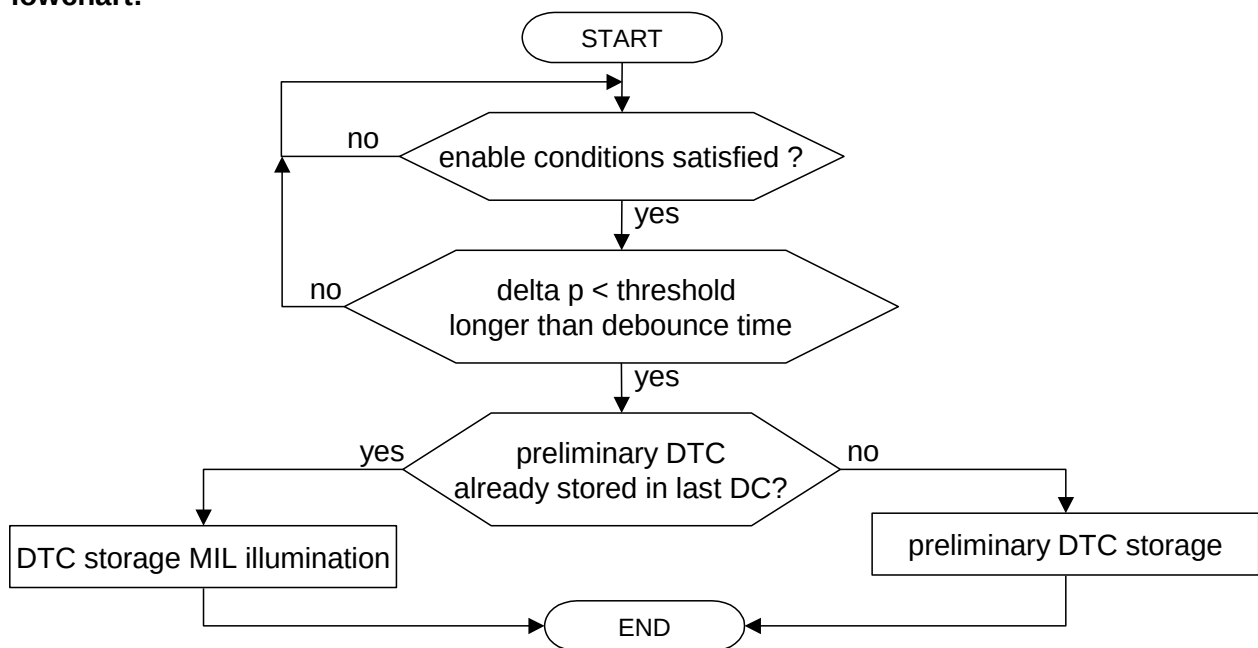


9.3 Missing Substrate (P14A6)

General description:

The particulate filter is monitored through pressure decrease of the "empty" can. The monitoring concept of missing particulate filter substrate is based on the comparison of differential-pressure to threshold. This concept is a continuous diagnosis. The maximum exhaust flow occurred during whole driving cycle is saved with the related differential pressure in engine afterrun. The saving process depends on the detection of a stationary operation point. The evaluation of the saved values is done in afterrun. Therefore the values are compared with a defect limit (curve with exhaust flow and related differential pressure). After a successful exhaust differential pressure monitoring (Offsettest --> P14A3) the missing substrate is released and the fault is detected. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





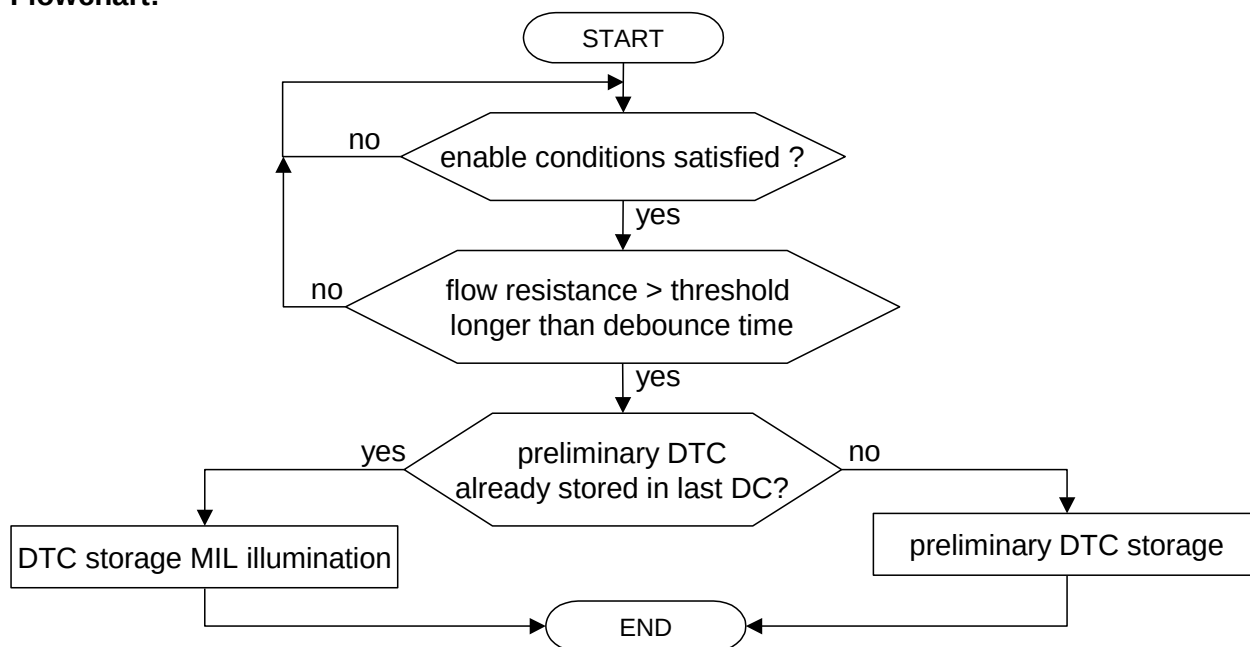
9.4 Overload Detection

(P14A7)

General description:

To detect an overloaded particulate filter, the particulate filter is continuously monitored for maximum flow resistance. A fault is detected, if the flow resistance is above the threshold for more than an allowed time. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





9.5 Frequent Regeneration

General description:

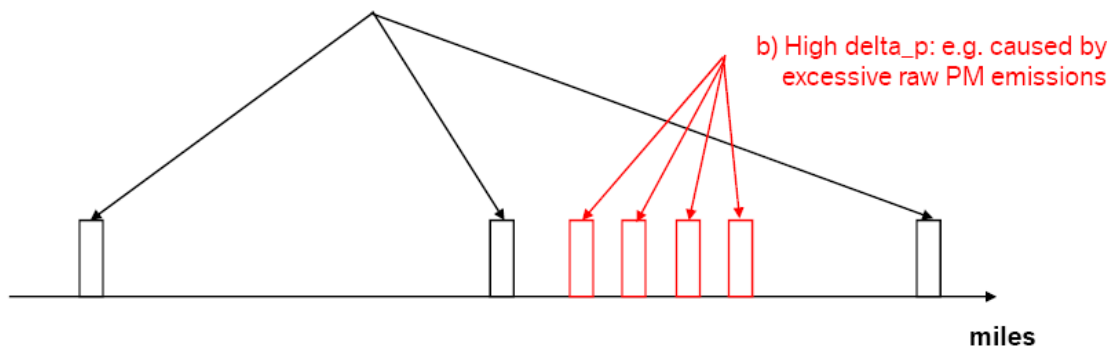
The diagnosis monitors the ratio between the time in normal engine operation and the time in regeneration (triggered through high delta_p) continuously.

The threshold of the ratio is based on a normal regeneration interval in UDC. If the calculated ratio is above the threshold and the exhaust temperature controller is active, a fault will be detected. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive regenerations, the MIL is illuminated.

slide 1

Regenerations are triggered by:

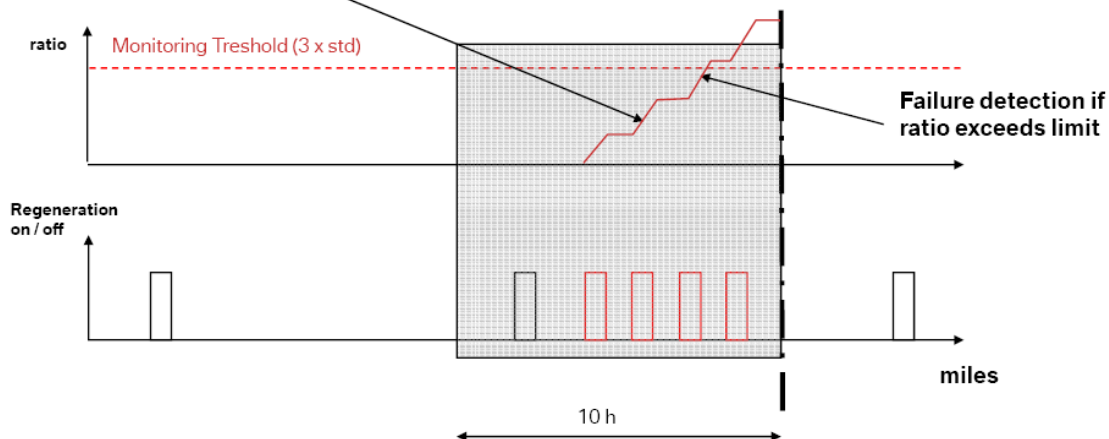
a) Mileage (standard): e.g. every 530 miles during UDC conditions



slide 2

Monitoring concept:

Calculation of a ratio:
$$\frac{\text{Cumulative time with delta}_p \text{ regenerations active in time window (e.g. last 10h)}}{\text{Total time (e.g. 10 hours)}}$$
 (continuously)





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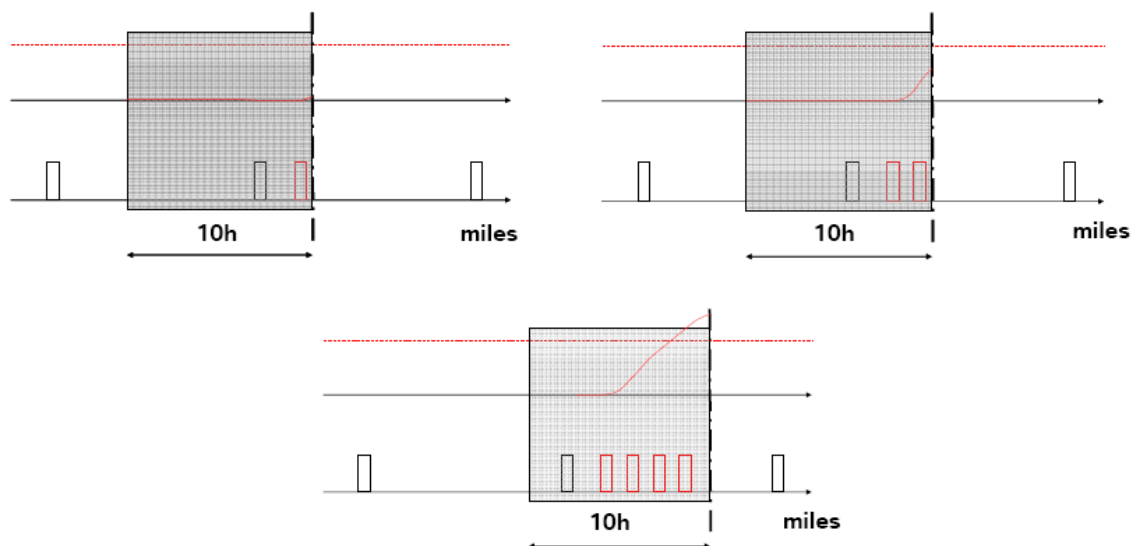
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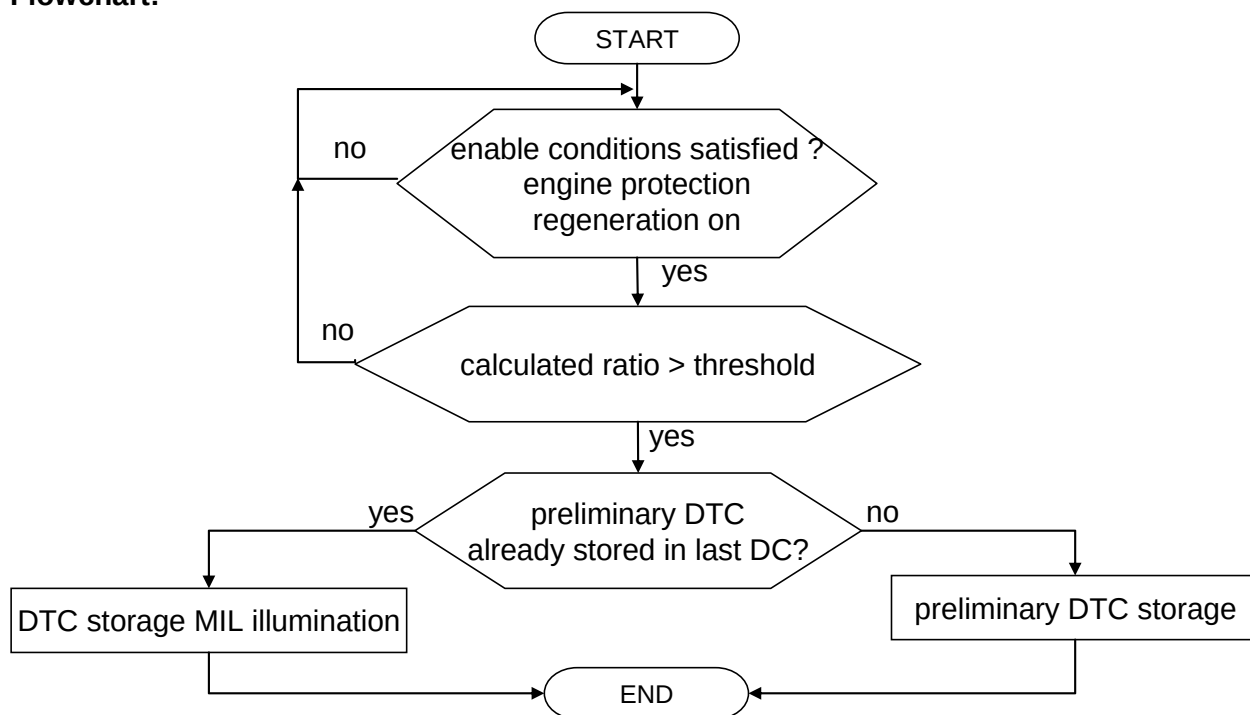
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„Sliding window“



Flowchart:



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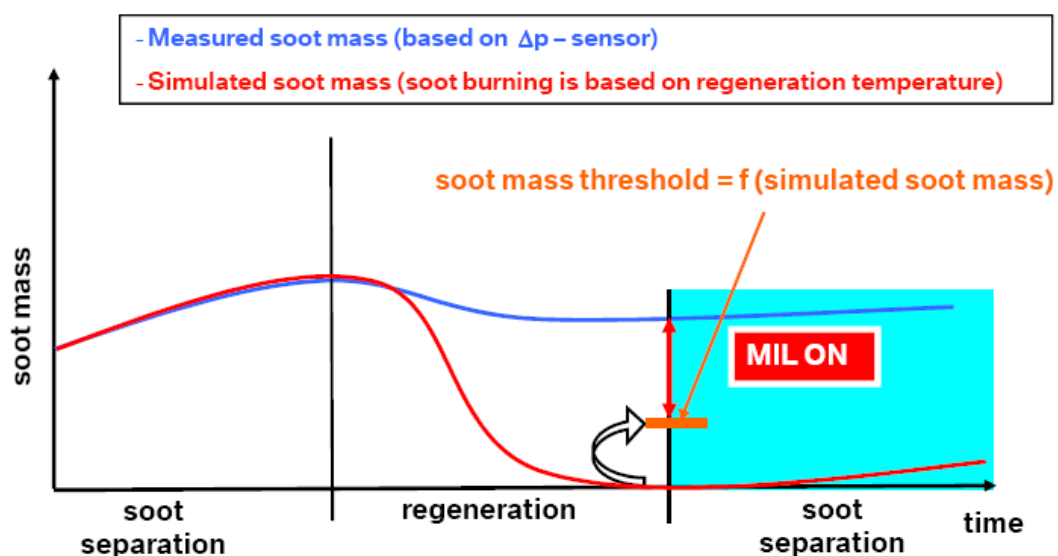


9.6 Incomplete Regeneration

General description:

The monitoring of incomplete regeneration is based on the measured and simulated soot mass after a regeneration event.

After a normal regeneration the measured soot mass (based on delta-p-sensor) will be small. The diagnosis compares after the regeneration the measured soot mass with a threshold, which is calculated from the simulated soot mass (during regeneration). If the regeneration is incomplete the measured soot mass is above the calculated threshold and a fault code will be detected. In this case, a preliminary DTC is stored. If this fault is detected after two consecutive regenerations, the MIL is illuminated.





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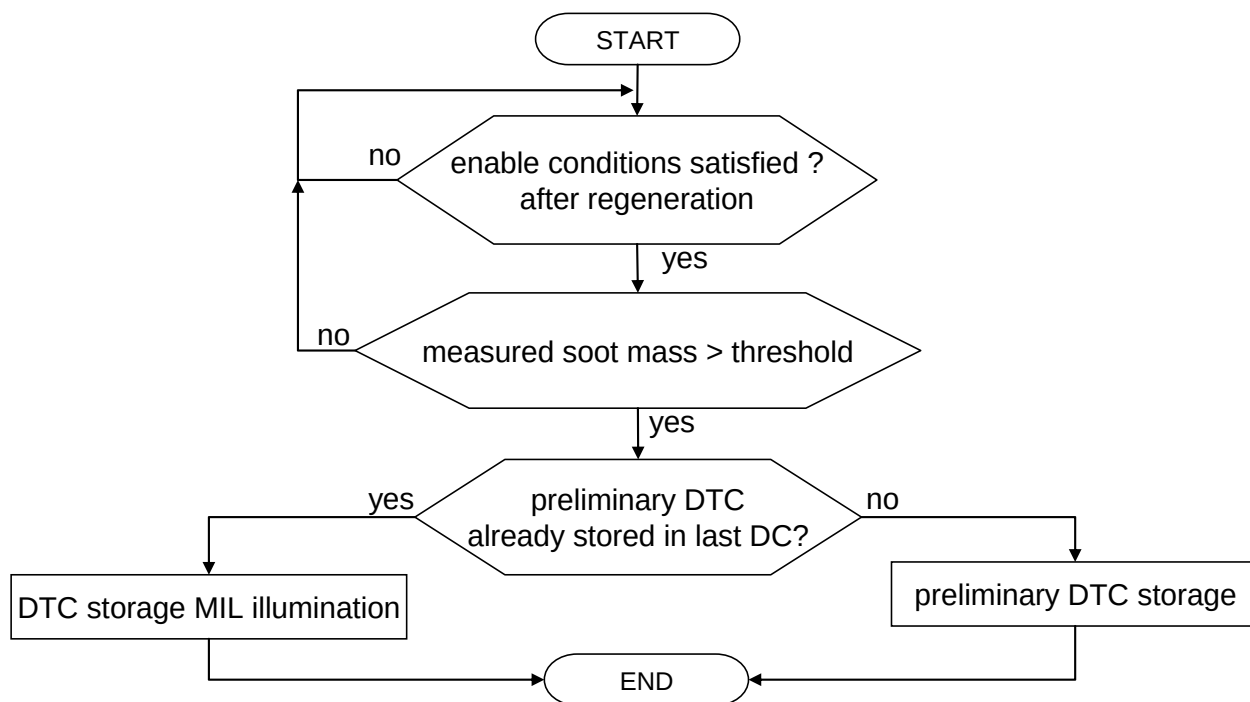
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Flowchart:



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9.7 Regeneration Temperature Monitoring

9.7.1 Response Time (P244C)

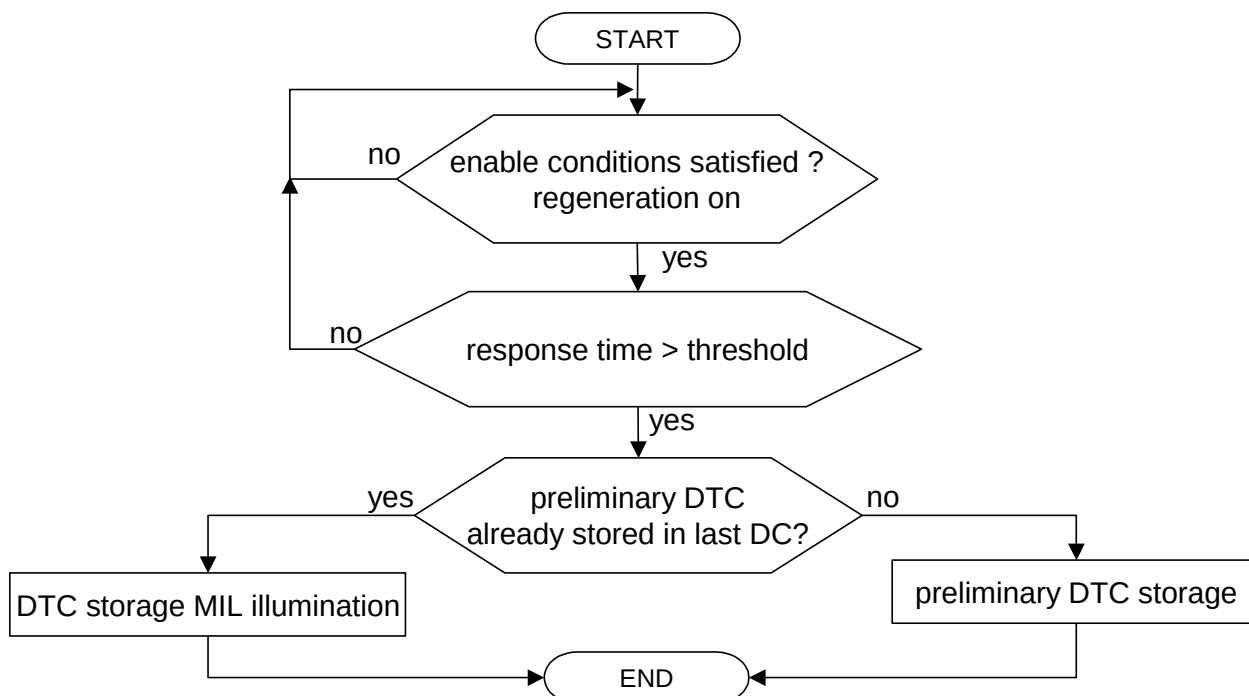
General description:

To detect nonactivation of the exhaust temperature controller, the maximum applicated activation time must be exceeded. This time is dependant on the measured exhaust temperature upstream the DPF at the moment when DPF regeneration is requested.

In the first phase of the regeneration the temperature is increased to 250°C. If the temperature upstream DPF has reached 250°C the controller is activated. The time until 250°C is reached is monitored by this time to closed loop monitor.

The diagnosis starts, when the regeneration is requested. With the beginning of the diagnosis a timer starts to count. If the engine operation mode changes to idle or overrun the timer resets. A fault code is detected, if the timer exceeds the maximum activation time and the exhaust temperature controller is not activated. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC in regeneration), the MIL is illuminated.

Flowchart:





9.7.2 Temperature Controller Deviation (RGN temperature too low) (P244C)

General description:

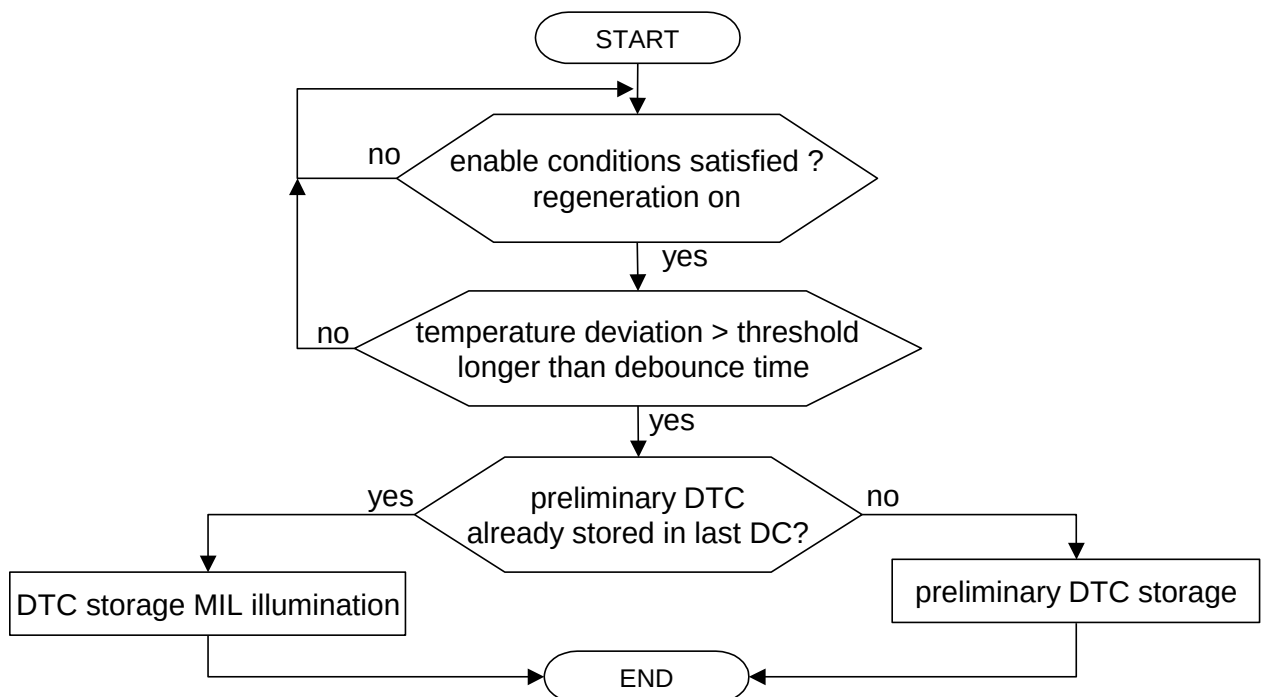
The DPF temperature controller is monitored for controller deviation during regeneration. At the start of a regeneration the temperature deviation between the desired temperature and the actual measured value is very high due to the PT1 behaviour of the measured exhaust temperature sensor upstream DPF. For this reason the threshold is also PT1 filtered. The absolute threshold is calculated from this PT1 filtered value +/- offset.

After a time (depending on initial temperature) the thresholds are only the offset.

The diagnosis is dependent on injection rate (no idle and overrun).

A fault code is detected, if the temperature deviation is above a threshold (PT1 filtered expected temperature behaviour + offset) and the controller output has reached its limits. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC in regeneration), the MIL is illuminated.

Flowchart:

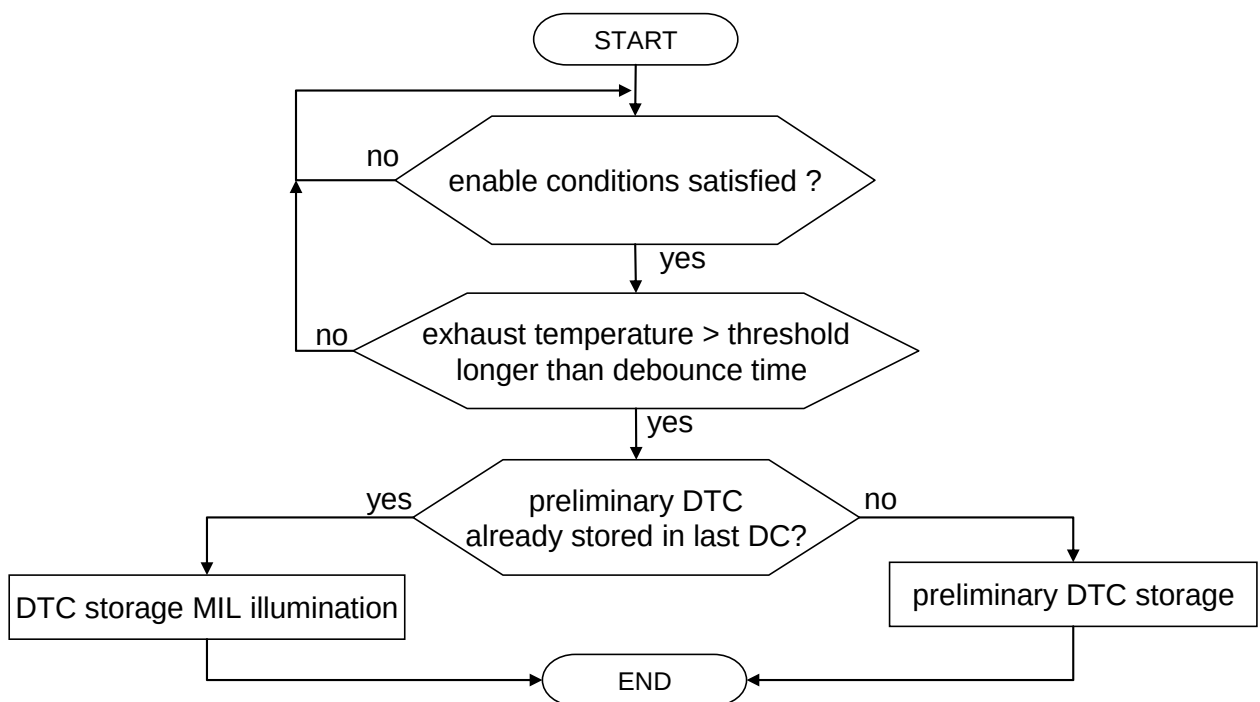


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**9.7.3 Temperature Controller Deviation (RGN temperature too high)
(P244C)****General description:**

Monitoring of maximum exhaust gas temperature during regeneration starts if the exhaust gas temperature controller is active and tested (Response Time - P244C). After this trigger the diagnosis compares continuously the exhaust gas temperature with a maximum threshold. If the regeneration temperature is above this applied threshold, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive regenerations, the MIL is illuminated.

Flowchart:



10 Crankcase Ventilation (CV)

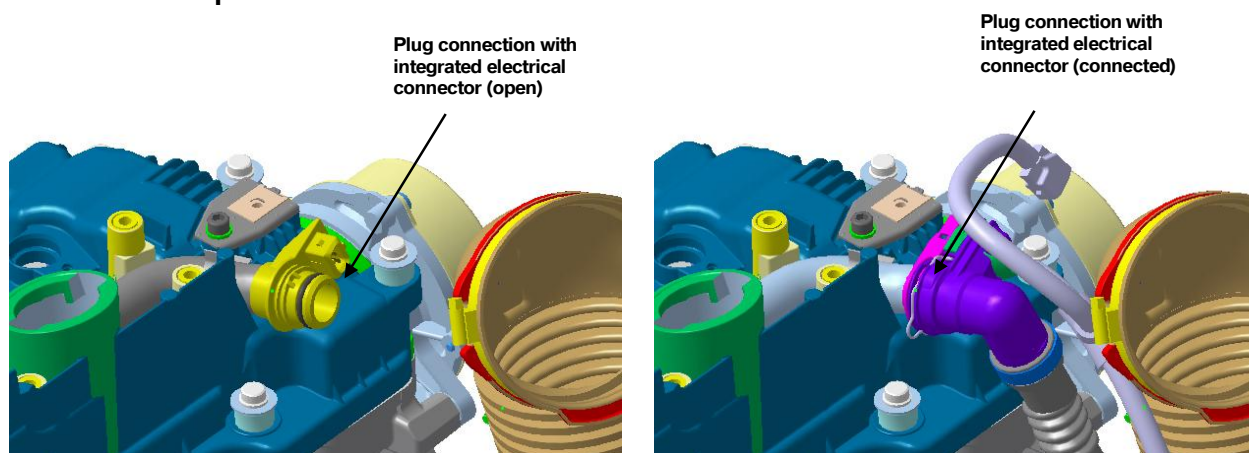
Crankcase Ventilation	(f)(10.2.2) Disconnection P053A, P120D, P053C, P053B
-----------------------	---

General:

The disconnection of the crankcase ventilation hose ((f)(10.2.2)) is detected through the crankcase ventilation heater power stage monitoring (Open Circuit).

10.1.1 Circuit continuity

General description:



The crankcase ventilation heater is monitored to detect a seceded blowby-hoseline. The pictures above show the monitoring principle. The electrical connectors are integrated in the ends of the blowby hose plug. If the blowby hoseline seceded, the electrical connectors are disconnected. The crankcase ventilation heater power stage driver checks continuously and independently for short to ground, short circuit to battery, open circuit and overtemperature faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

The overtemperature-, short circuit to battery- and short circuit to ground faults are only OBD relevant, because they will lock the open circuit fault if they appear. The diagnosis is disabled at high engine coolant temperature, because in this operating points there can be false diagnosis (high resistance of heater will cause in open circuit error).



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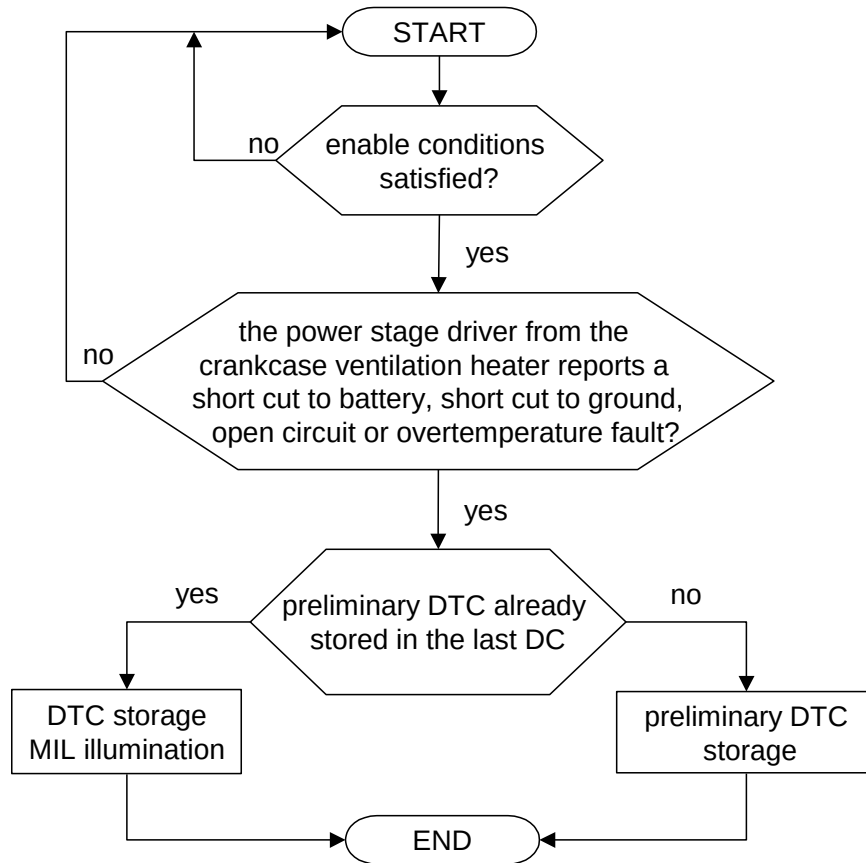
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Flowchart:



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11 Engine Cooling System Monitoring

	(f)(11.2.1)(A)	(f)(11.2.2)(A)	(f)(11.2.2)(A)	(f)(11.2.2)(A)	(f)(11.2.1)(B)	(f)(11.2.1)(C)	(f)(11.2.1)(D)
	Threshold temperature P0128	OOR high P0118	OOR low P0117	Circuit continuity P0117, P0118	Time to reach closed loop P0128	Stuck below the highest minimum enable temp P008F	Stuck above the lowest maximum enable temp P0128
Engine Cooling System							

11.1 Circuit continuity check**General Description:**

If the raw voltage signal of the coolant engine temperature sensor is above/below the applied threshold or the interpreted raw voltage signal (°C) is out of the applied range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:**21.1.1 Circuit continuity****11.2 Rationality checks**

Referring to (11.2.1)(A)(ii) either with a stuck-open thermostat and/or an engine coolant temperature below the applied threshold (60°C for MY2011) the emission does not increase more than 50%. Due to that the monitored fault criteria is an engine coolant temperature that does not reach the highest enabling temperature for the OBDII system (11.2.1)(A)(i)

Emission increase from proper system to stuck-open thermostat (with maximum cooling) at ambient temperature of -7°C and engine coolant temperature < 60°C:

	HC	CO	NOx	CO2
Emission increase	16%	6%	-12%	7%

11.2.1 Stuck Below the Highest Minimum Enable Temperature**General description:**

To monitor the engine coolant temperature sensor the engine coolant temperature is modelled. The model calculates the in- and decrease of the engine coolant system.

This model is calculated with the following input values:

- injection quantity
- inner torque
- engine speed
- number of cylinders
- vehicle speed
- temperature difference between the engine coolant temperature and the ambient air temperature

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A fault is detected if the modelled temperature is above the applicated threshold (incl. safety margin) while the real coolant temperature is still below a different applicated threshold. In this case a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.



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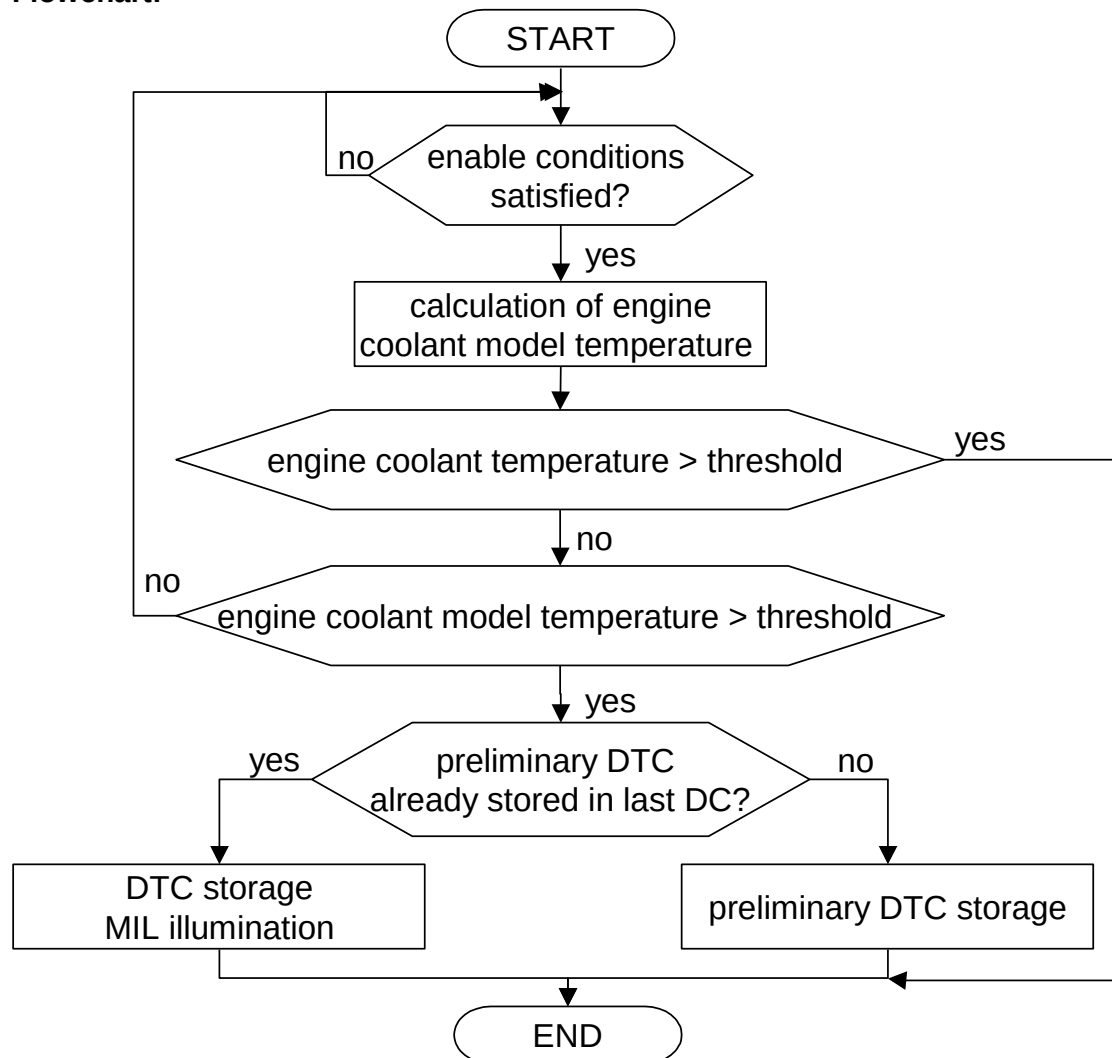
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Flowchart:



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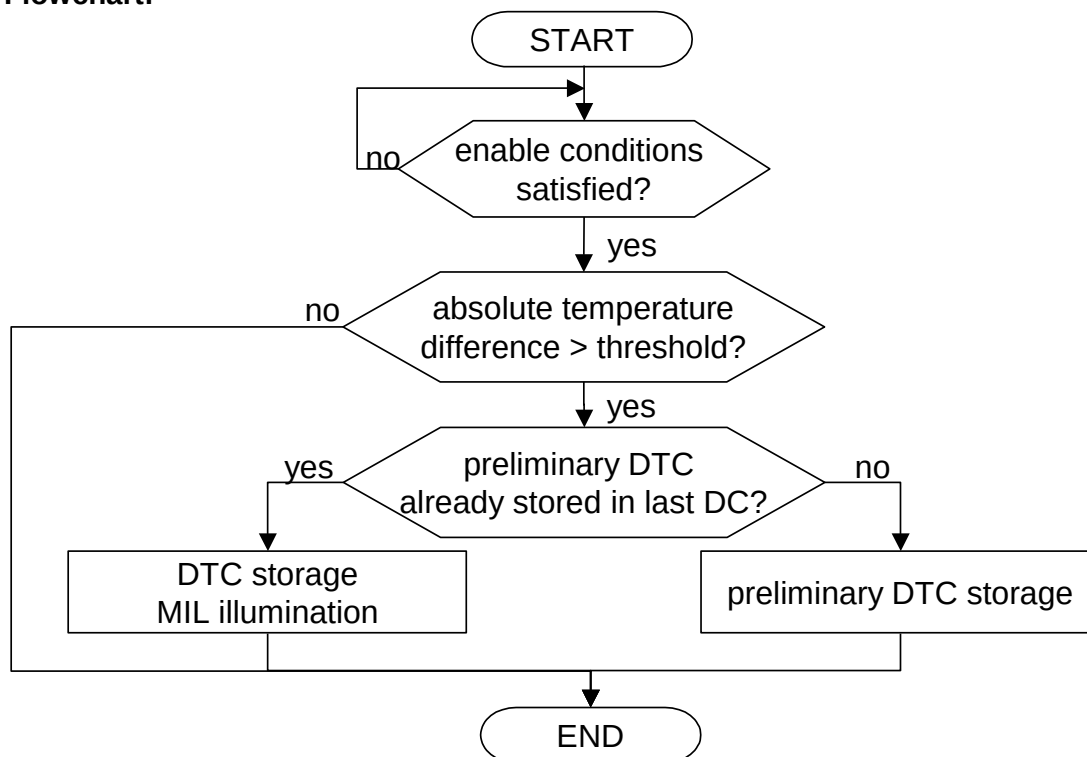
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11.2.2 Stuck Above the Lowest Maximum Enable Temperature

General description:

If an appicated minimum engine-off time is exceeded the absolute difference between the engine coolant temperature sensor value and the fuel temperature sensor value is calibrated. If the difference is above the appicated threshold, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:



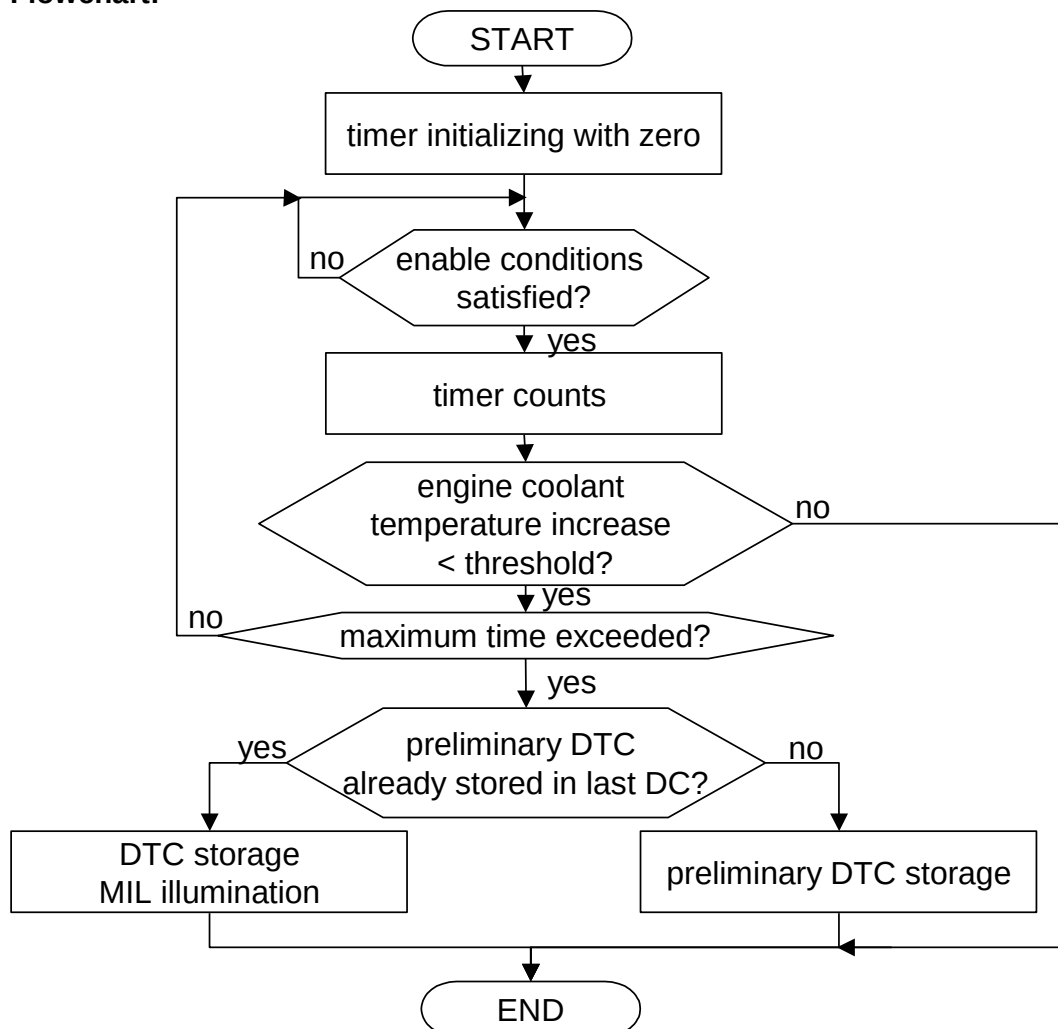
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11.2.3 Stuck Check ECT**General description:**

The engine coolant temperature sensor is monitored for stuck error. Therefore the engine coolant temperature at engine start is saved. If the enable conditions are satisfied a timer counts (initialized with zero at engine start). The engine coolant temperature's increase is monitored by the difference between the actual engine coolant temperature and the coolant starting temperature saved at start. If the temperature increase does not reach an applicated threshold until the timer expires, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:

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12 COLD START EMISSION REDUCTION STRATEGY MONITORING

Cold start strategy	(f)(12.2.1) Single element functional fail P026B	(f)(12.2.2) Threshold monitor P026B
---------------------	---	--

General description:

BMW uses RHU („Rapid Heat Up“) to increase exhaust temperature for quick SCR system operation after coldstart. The monitoring strategy is to check the commanded elements.

The post injection is the most important parameter for RHU. It is the primary commanded element to increase the exhaust temperature.

The other variations of the commanded elements (airmass flow, fuel rail pressure and swirl valve) during RHU are necessary to guarantee proper combustion in operating modes using post injections. These secondary commanded elements are not necessary to increase the exhaust temperature directly.

Emission influence:

Tests X5 3.0sd:	NMHC [g/mi]	CO [g/mi]	NOx [g/mi]	PM [g/mi]
FTP75 with RHU active	0,023	0,206	0,058	-
FTP75 with RHU active and without primary commanded elements (post injections turned off)	0,025	0,166	0,087	-
FTP75 with RHU inactive	0,019	0,116	0,089	-

A system without RHU is still far below 2,5 standard emission limits. The increased NOx-Emission is caused by the delayed warming up of the SCR system.

The comparison of the emissions of a system without RHU (neither primary nor secondary parameters active) and a system without primary elements (only secondary parameters active) shows that there is no emission impact caused by the secondary parameters.

This points out that the post injection is the most important parameter of the RHU to warm up the SCR system and therefore to reduce the emissions.



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Overview:

- Monitoring of commanded elements (regulation f 12.2)

Primary Commanded Element	Monitor	active	DTC
Injection timing post injection Injection quantity post injection	Monitoring of final commanded value of post injection timing and quantity	during RHU	P026B
	Exothermal reaction during RHU	during RHU	P0420

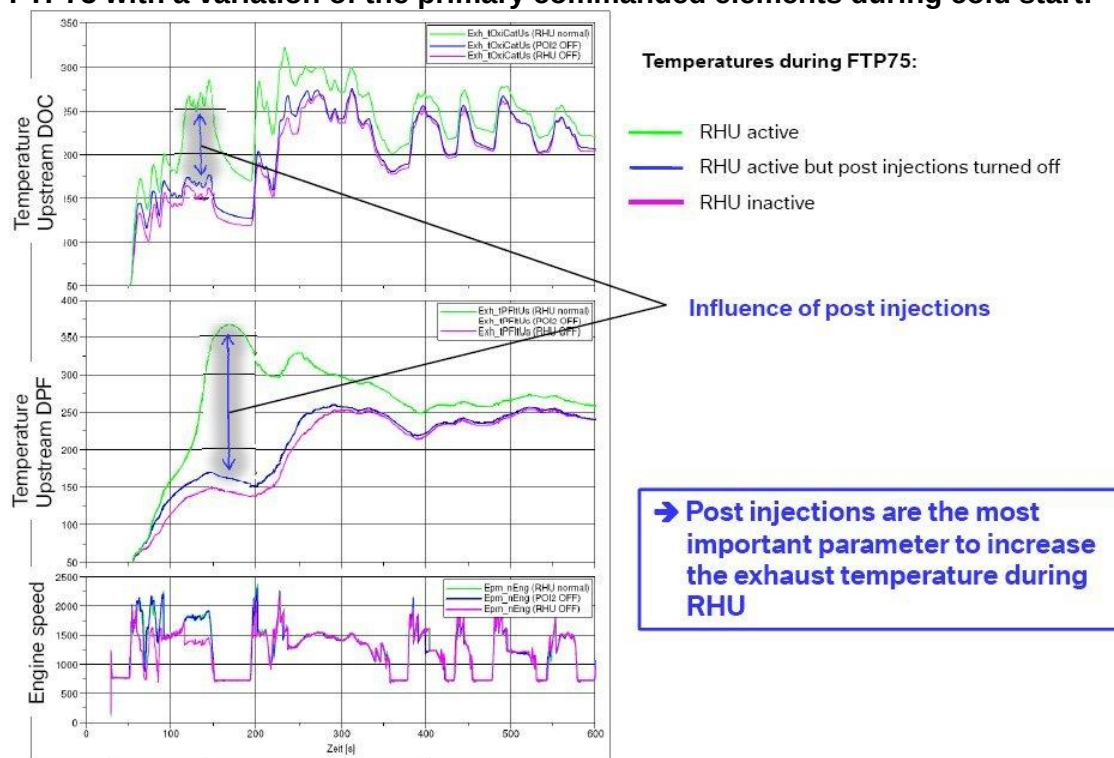
Secondary Commanded Element	Monitor	active	DTC
Air Mass Flow	EGR High Flow / Low Flow	continuous	P0401/P0402
Fuel Rail Pressure	Overpressure / Underpressure	continuous	P0087/P0088
Swirl Valve	Position Sensor	continuous	P200A
Transmission Shift	CAN Monitoring	continuous	U0101

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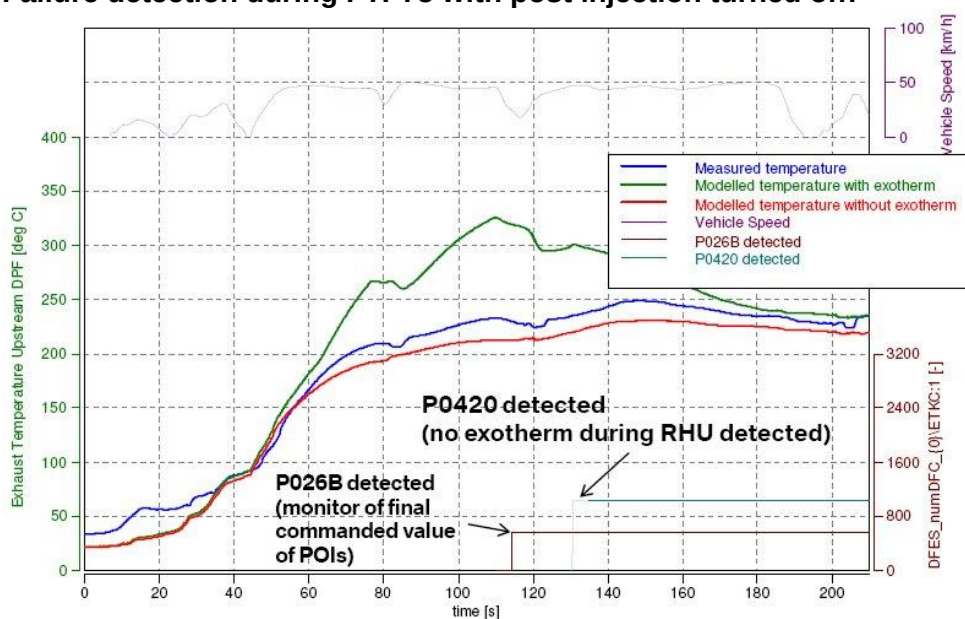


12.1 Primary Commanded Elements

FTP75 with a variation of the primary commanded elements during cold start:



Failure detection during FTP75 with post injection turned off:



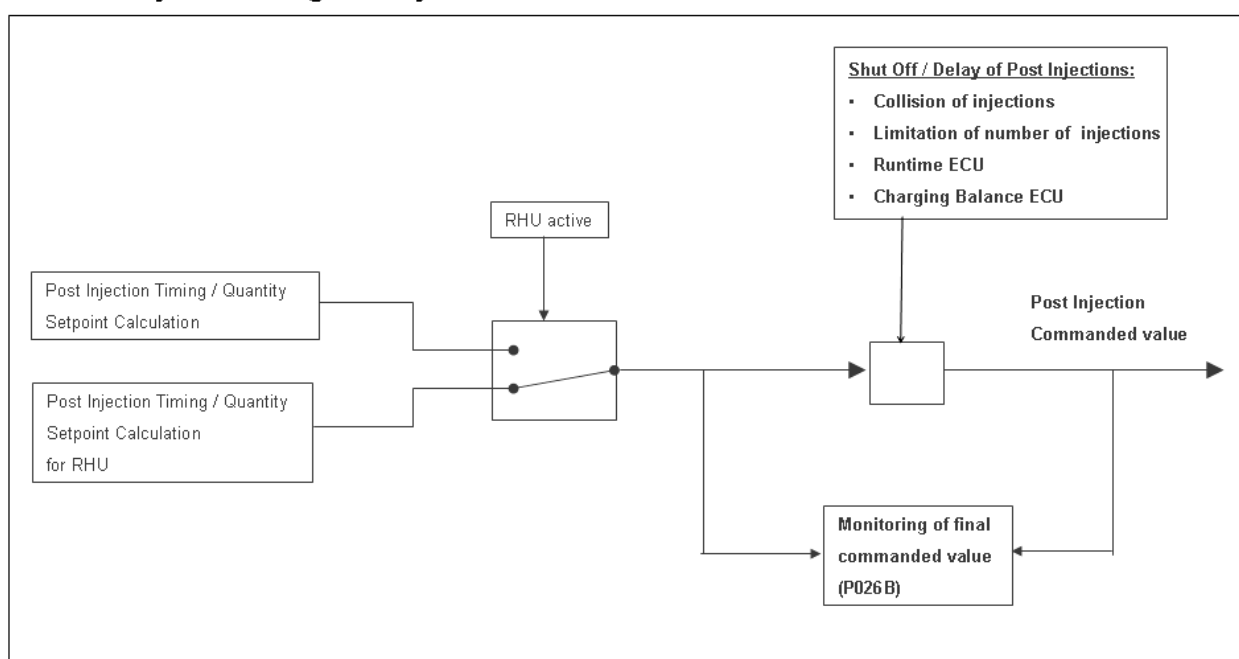


12.1.1 Post Injection Timing/Quantity (P026B)

General Description

The diagnosis for limited post injection monitors continuously during cold starts. It monitors if the commanded values for post injection timing and quantity are carried out by the power stage in the correct way.

- **Post injection Timing/Quantity : Commanded Value for RHU**



The monitoring of post injections consists in:

Monitoring of injection timing (begin of injection (BOI)):

Comparison of the commanded angle of BOI with the actual angle of BOI.
The actual angle of BOI is determined by the start of charging of the power stage.
If the absolute value of the deviation is above a threshold an error is detected.

Monitoring of injection quantity (energizing time of the injection (ET)):

Comparison of the commanded ET with the actual ET of the injection
The actual ET is determined by the time difference between charging and discharging of the power stage.
If the absolute value of the deviation is above a threshold an error is detected.

Monitoring feedback signal of the power stage (discharge time):

The actual discharge time of the power stage is measured and checked. If the actual discharge time is above/below a threshold an error is detected.
This monitor ensures that the energizing of the post injection was carried out in a correct way.

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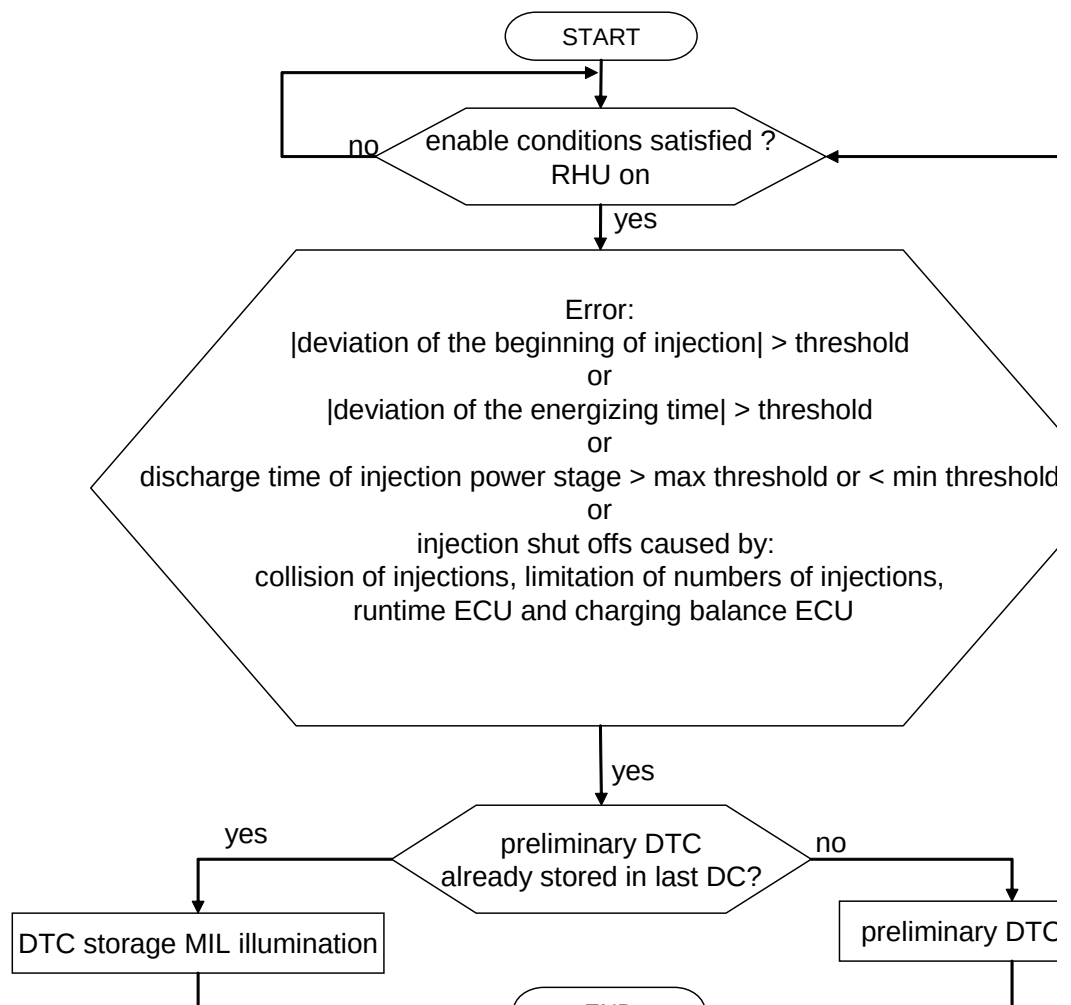
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Monitoring of shut off of injections:

The ECU has implemented an internal priority management for injections. Post injections have the lowest priority and they will be the first injection which are shut off in the following failure cases:

- collision of injections with other injections (e.g. poor calibration)
- limitation of numbers of injection (e.g. poor calibration)
- runtime ECU
- charging balance ECU

If the monitoring detects one of the failures above, a fault will be detected. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles with cold start conditions, the MIL is illuminated.

Flowchart**CONFIDENTIAL**



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12.1.2 Exothermal reaction during RHU
(P0420)

For the monitoring of the exothermal reaction during RHU see chapter “1 NMHC Catalyst Monitoring”.



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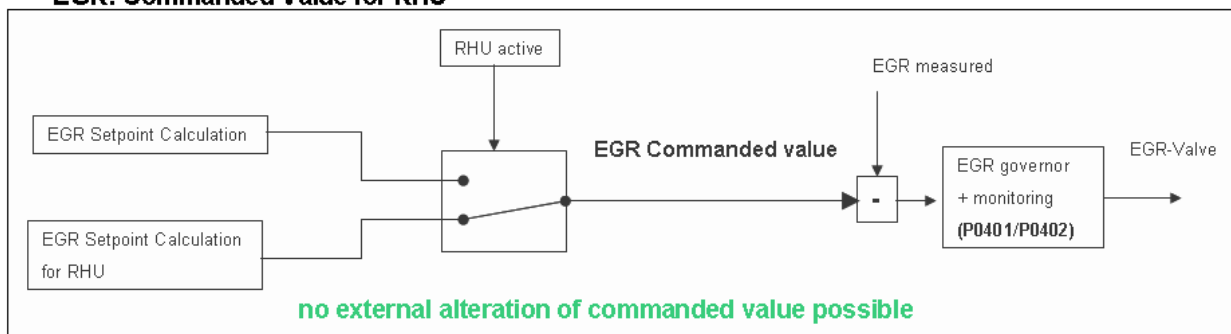
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12.2 Secondary Commanded Elements

12.2.1 EGR High Flow / Low Flow (P0401/P0402)

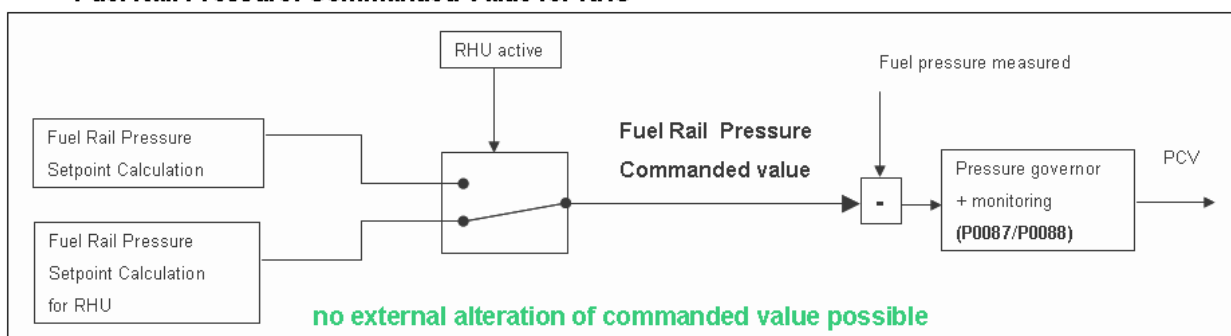
- EGR: Commanded Value for RHU**



For details about the monitor see chapter “6.1 EGR Control Loop Monitoring”.

12.2.2 Fuel Rail Overpressure/Underpressure (P0087/P0088)

- Fuel Rail Pressure: Commanded Value for RHU**



For details about the monitor see chapter “4.1 Rail Pressure Control Loop Monitoring”

12.2.3 Swirl Valve Position Sensor (P200A)

See “15.39 Swirl Valve”

12.2.4 Transmission Shift CAN Monitoring (U0101)

See “15.5 CAN Communication Transmission Control Module”

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- 13 VARIABLE VALVE TIMING AND/OR CONTROL (VVT) SYSTEM MONITORING
 --> N/A**
- 14 RESERVED --> N/A**



15 Comprehensive Component Monitoring

15.1 Ambient Air Temperature Sensor

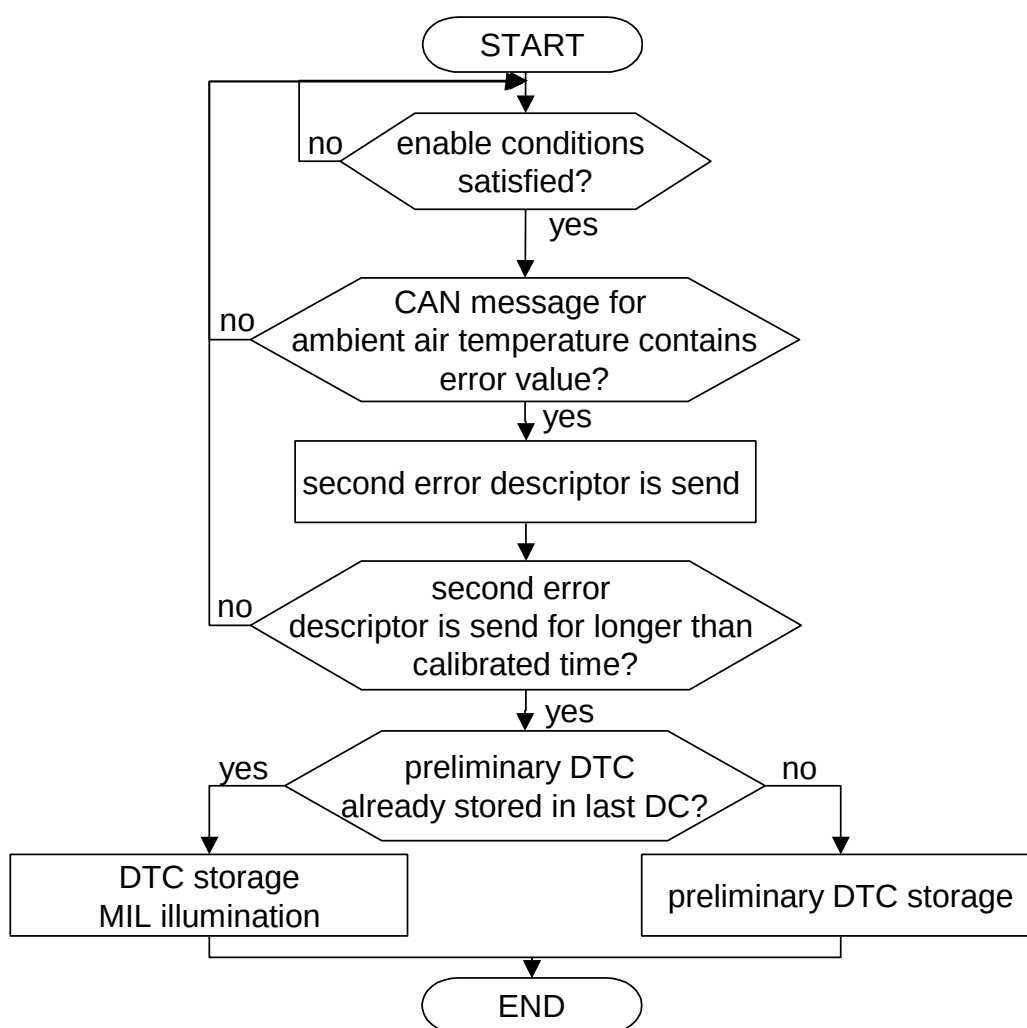
15.1.1 Circuit continuity

(P0072/P0073)

General description:

The ambient air temperature sensor signal is provided by the instrument cluster via CAN. If the CAN message contains an error value a second error descriptor message is sent. If this second error descriptor message is sent for longer than the calibrated time a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:



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15.1.2 Rationality Check**15.1.2.1 Cross-Check
(P0070)****General description:**

A fault is detected via the cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

To enable the diagnosis the applicated engine off time has to be exceeded.

The ambient air temperature sensor is compared to:

- induction air temperature sensor
- exhaust temperature sensor downstream EGR cooler
- exhaust temperature sensor downstream EGR LP cooler(only X5 3.0sd)

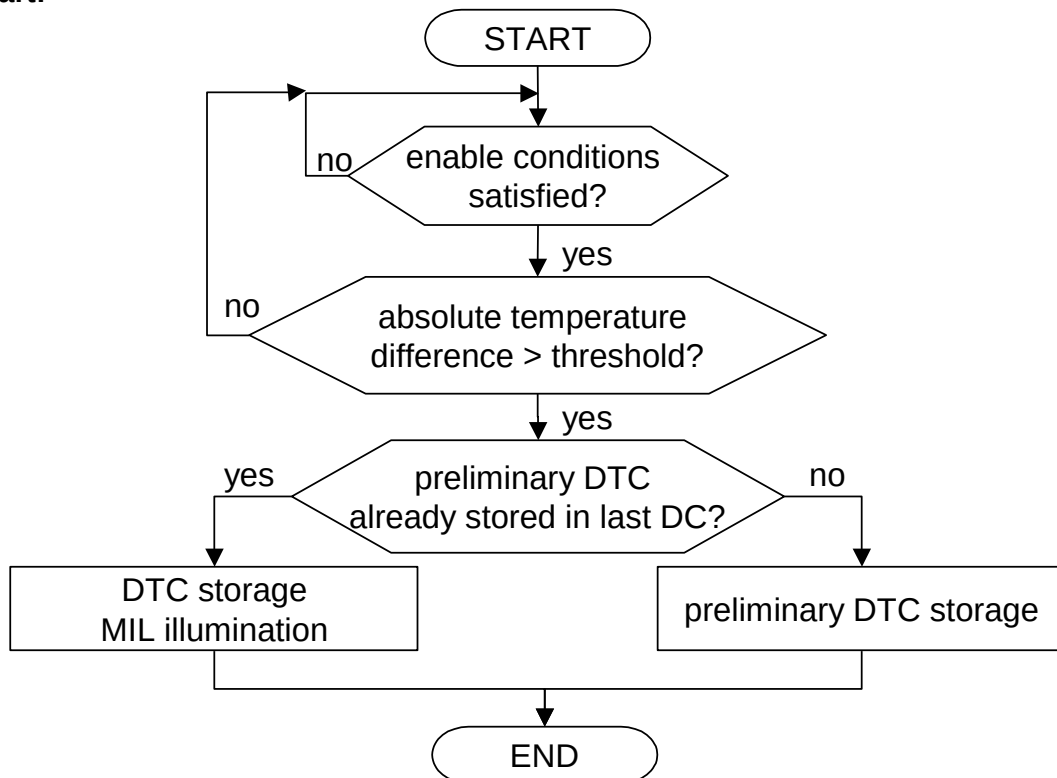
Flowchart:

see 21.1.2 Cross-check of Temperature Sensors

**15.1.2.2 Other
(P009A)****General description:**

The diagnosis compares the mass airflow temperature value with the ambient air temperature value. If the absolute difference between the mass airflow temperature value and the ambient air temperature value is above the applicated threshold, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

To enable the diagnosis an applicated amount of calculated airmass must have passed the mass airflow temperature sensor.

**Flowchart:****15.1.3 Functional check****15.1.3.1 CAN Signal Fault
(U0155/P110F)****General description:**

If one of the received CAN messages (ambient air temperature sensor, second error descriptor) is not plausible for longer than the applied time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

see 21.1.4 CAN Signal Fault



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15.1.3.2 CAN Timeout Fault
(U0155/P110F)

General description

If no CAN message (ambient air temperature sensor, second error descriptor) is received longer than the applicated time a fault is detected. This diagnosis is performed continuously.

Flowchart:

see 21.1.5 CAN Timeout Fault



15.2 Barometric Pressure Sensor

15.2.1 Circuit continuity (P2228, P2229)

General description:

If the raw voltage signal of the barometric pressure sensor is above/below the applicated threshold or the interpreted raw voltage signal (hPa) is out of the applicated range longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

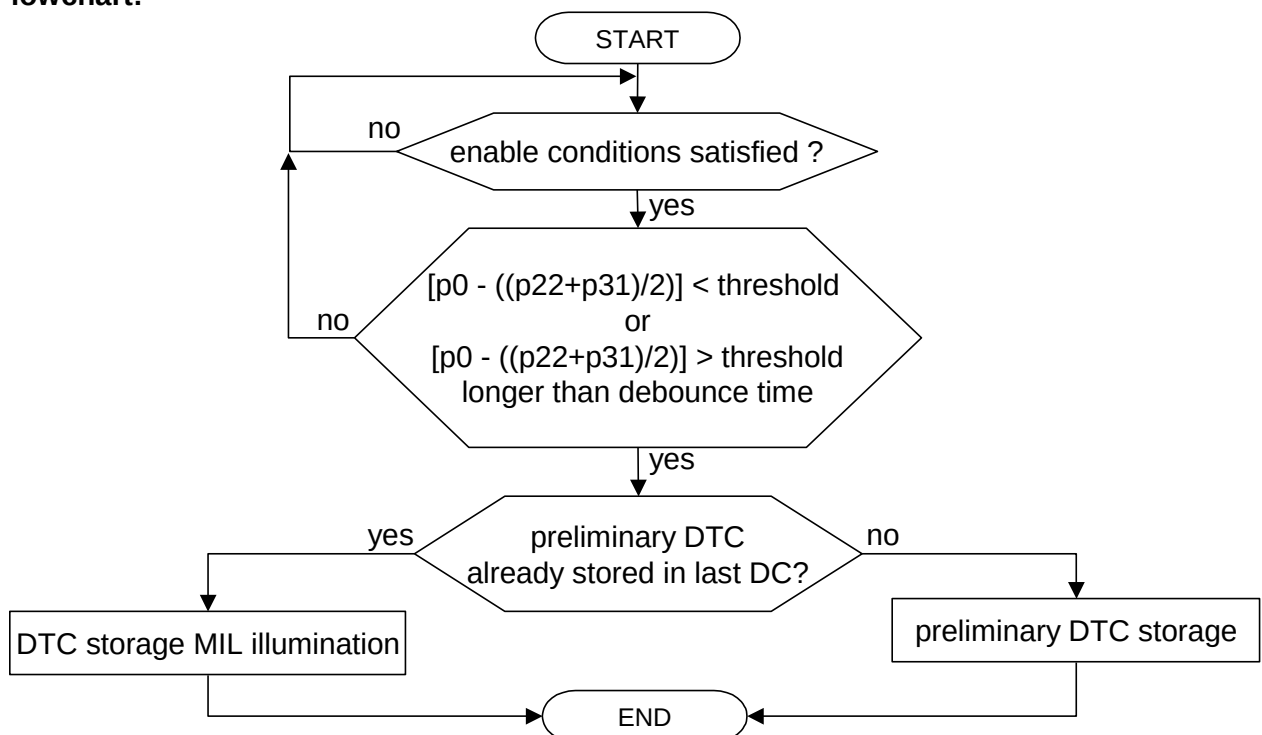
see 21.1.1 Circuit continuity

15.2.2 Rationality check (P321E, P321F)

General description:

The plausibility check is executed with ignition on or in the afterrun. During the plausibility check of the barometric pressure sensor (p0) a deviation between the barometric pressure sensor and the other sensors (manifold absolute pressure sensor (p22), exhaust manifold pressure sensor (p31)) is calculated. If the calculated deviation is above/below the specific threshold longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:





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15.3 Camshaft Position

**15.3.1 Rationality check
(P0016)**

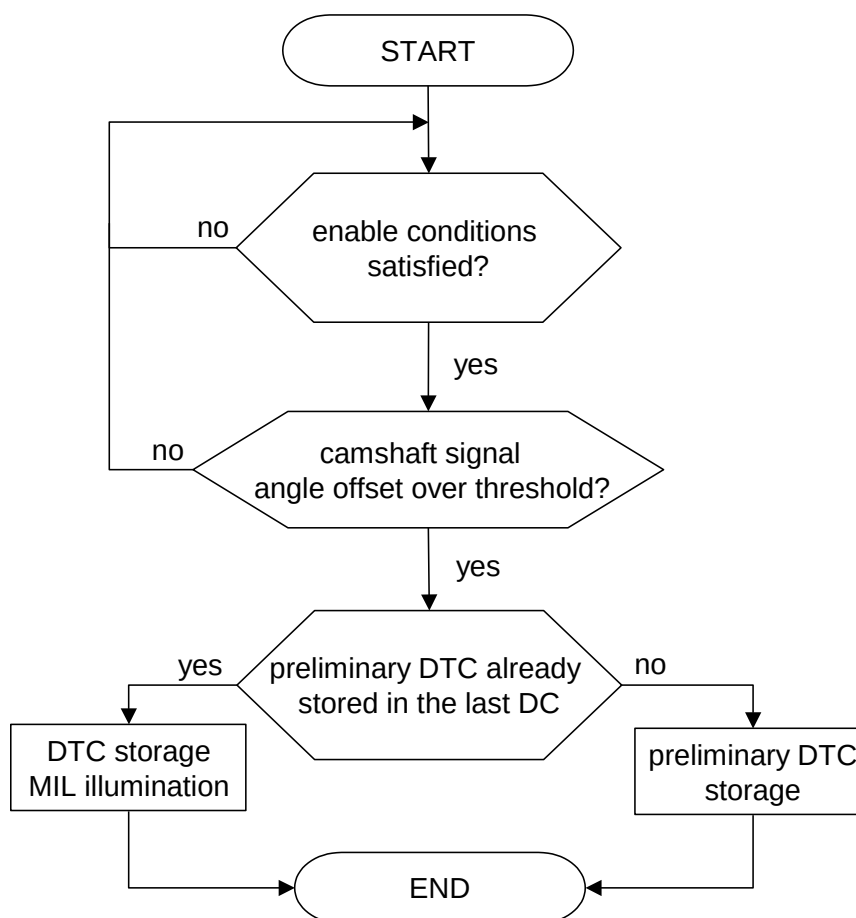
General description:

The Camshaft Position Sensor (CMP) is a Hall-Sensor. An error of the CMP is detected in the following case:

- camshaft offset: camshaft angle exceeded relating to crankshaft

If an error is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.4 CAN Communication System

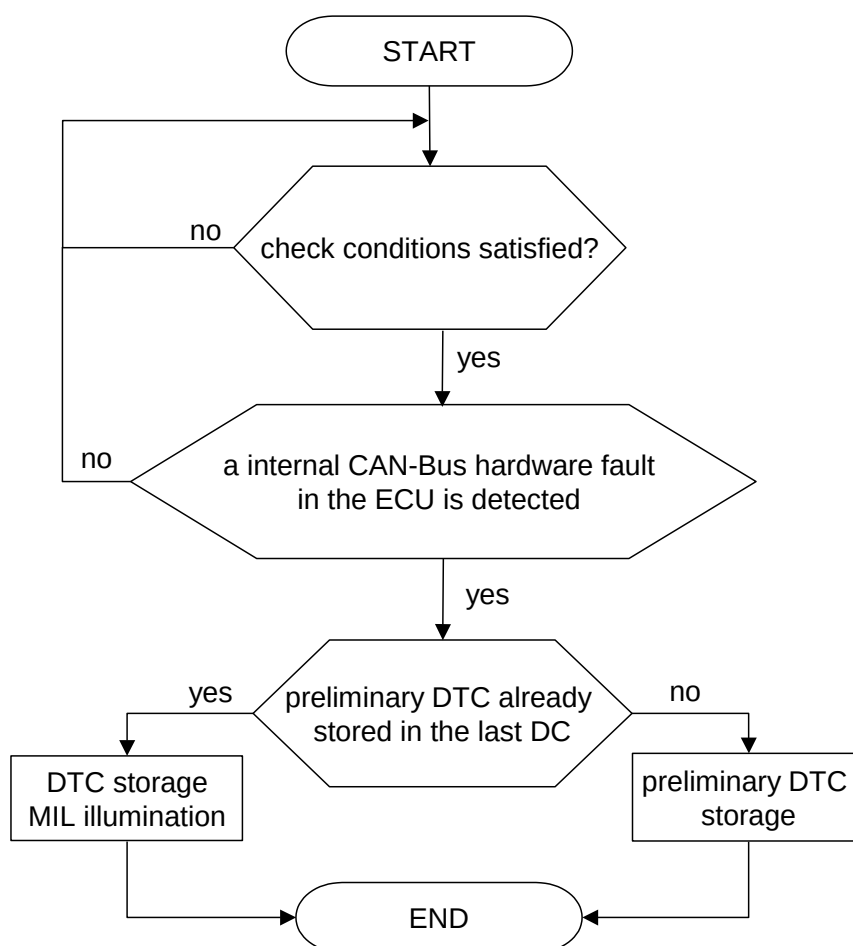
15.4.1 Functional check

15.4.1.1 ECU Internal CAN-Bus Error (P3200)

General description:

The CAN driver chip monitors the bus function. If an internal hardware fault in the ECU is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.4.1.2 ECU External CAN-Bus Error
(U0028, U029D)

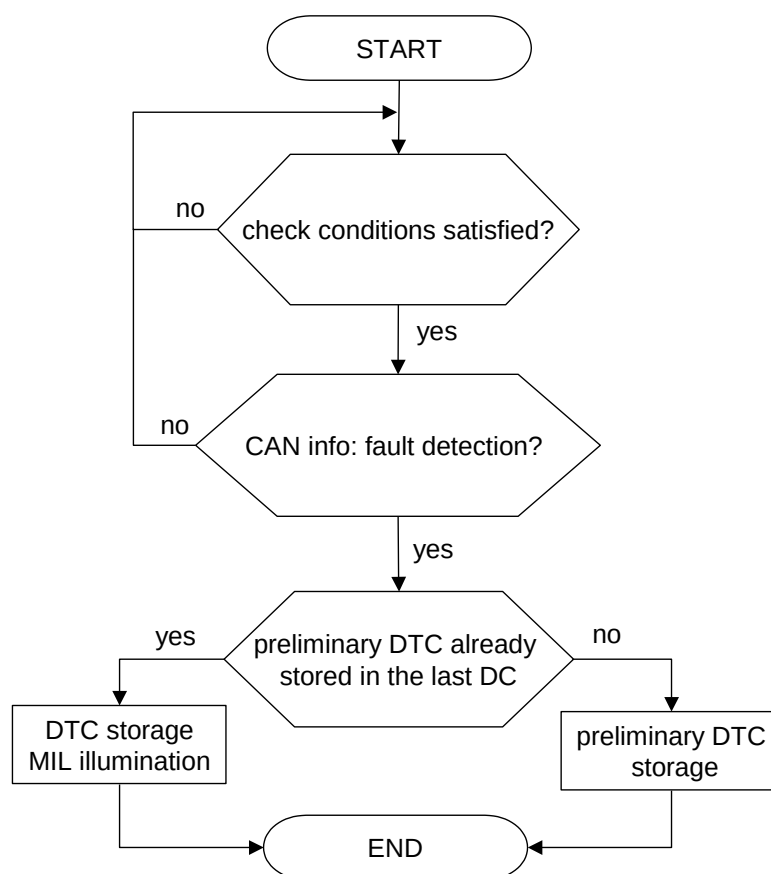
General description:

The CAN driver chip monitors the bus function. Faults are detected for the following cases:

- Bus Off
- Stuff Error
- Form Error
- Acknowledge Error

If an error is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.5 CAN Communication Transmission Control Module

15.5.1 Functional check
(U0101)

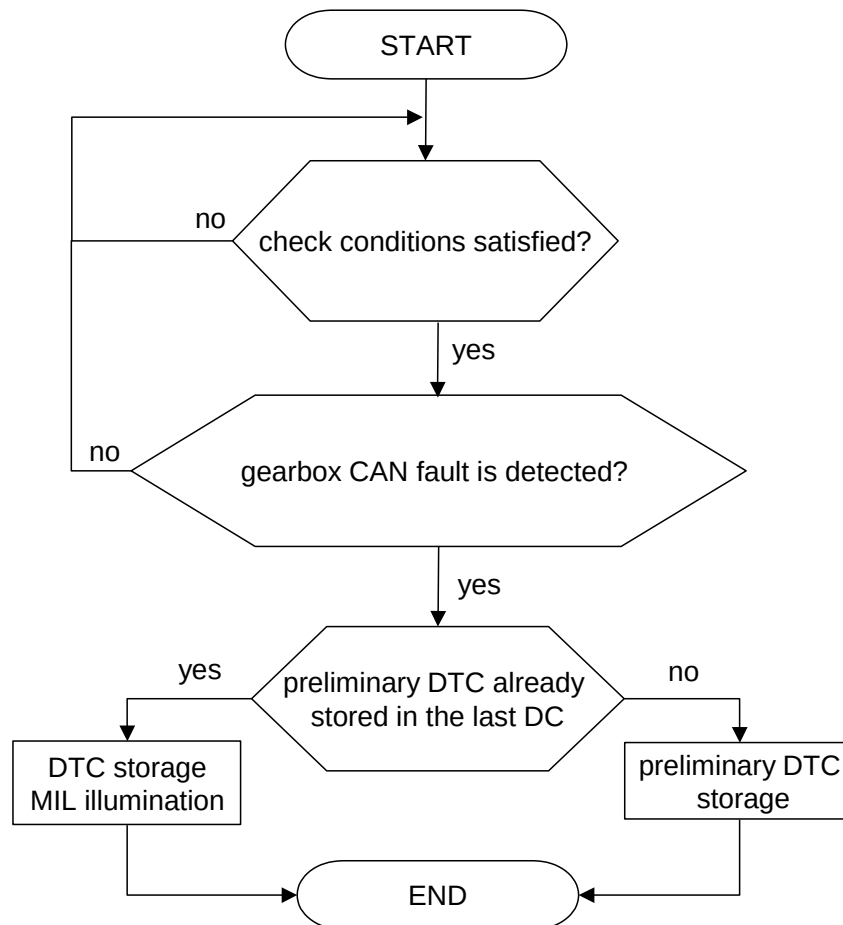
General description:

The CAN driver chip monitors the CAN-messages from the transmission control unit. Faults are detected for the following cases:

- Timeout of the CAN-Signal
- Checksum
- Alive Counter
- Signal fault

If an error is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.6 Crankshaft Position Sensor (P0336 & P0335)

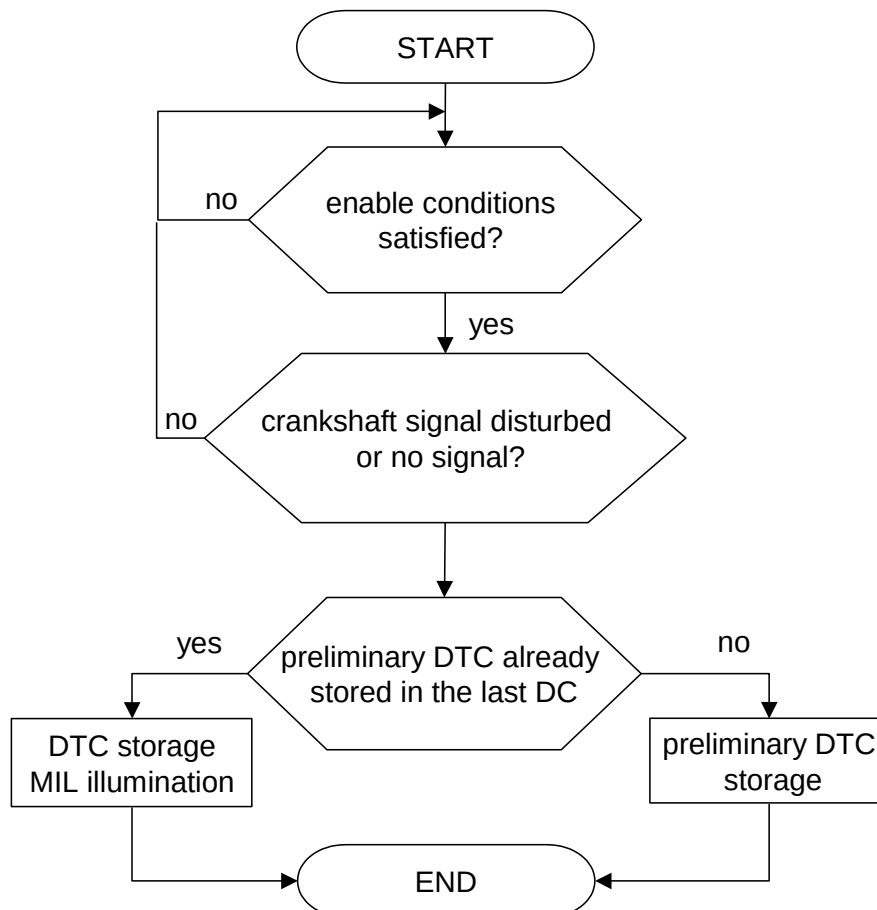
General description:

The Crankshaft Position Sensor (CRP) is a Hall-Sensor. An error of the CRP is detected for the following cases:

- disturbed signal --> number or position of the camshaft ramp not plausible
- no signal --> no cam ramp change has been detected for a certain angle

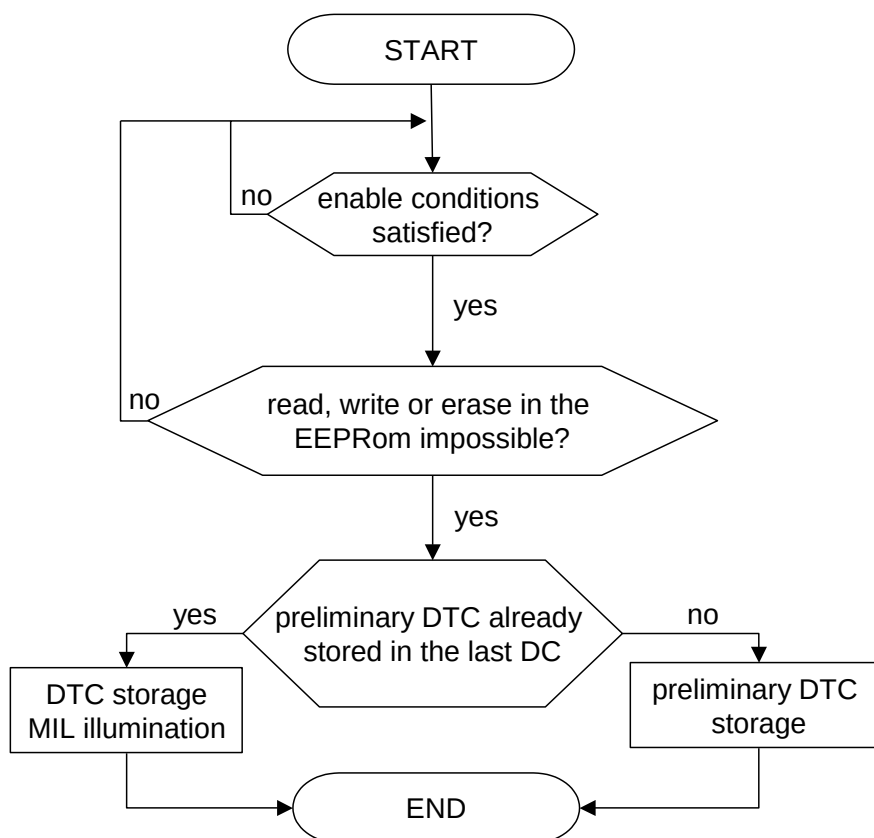
If an error is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



**15.7 Engine Control Module****15.7.1 Functional check****15.7.1.1 EEPROM Error
(P062F)****General description:**

If it is not possible to read, write or erase in the EEPROM a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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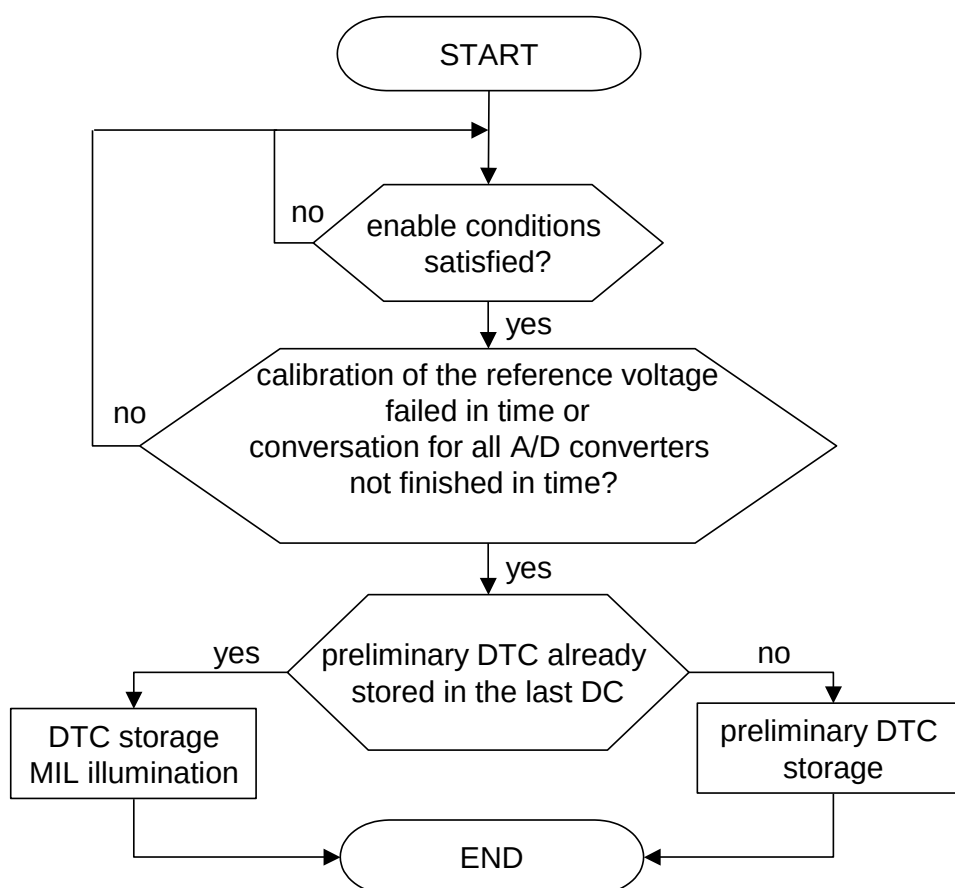
15.8 Engine Control Module Analog Digital Converter

**15.8.1 Rationality check
(P0607)**

General description:

The A/D converter is monitored for malfunction. If the calibration of the reference voltage or the conversation of all A/D converters are not finished in time, a fault is detected. If this malfunction is detected during two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.9 Engine Control Module

15.9.1 Functional check

15.9.1.1 SPI-Bus-Monitoring (P0607)

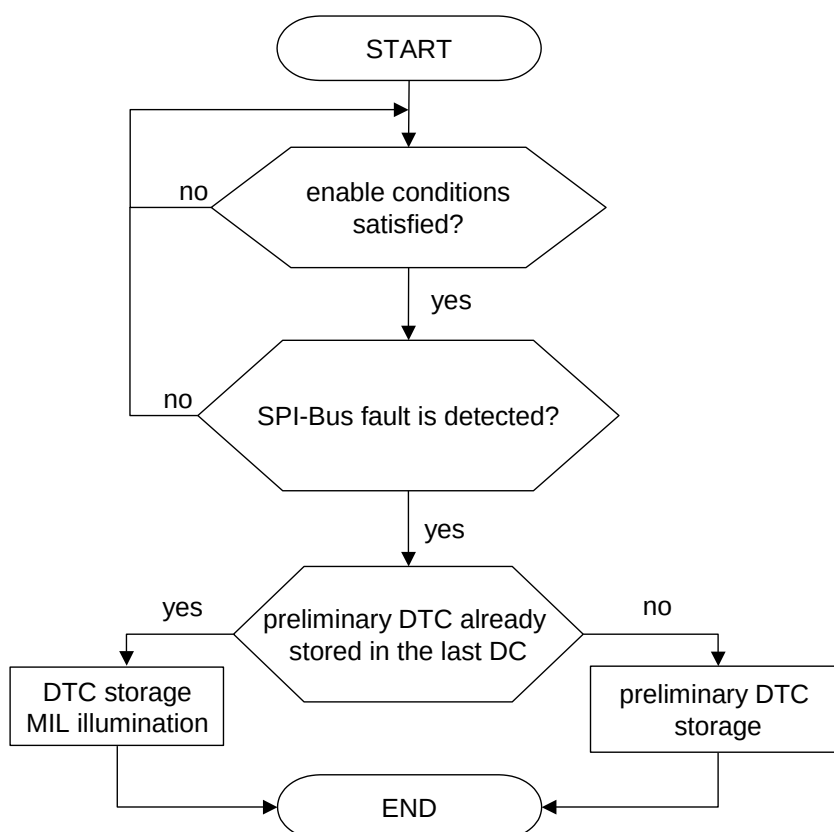
General description:

The SPI-Communication is monitored by a controller chip. Faults are detected for the following cases:

- checksum error
- transmission not possible

If an error is detected, a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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15.10 Engine Coolant Temperature Sensor

15.10.1 Circuit continuity
(P0117,P0118)

General Description

If the raw voltage signal of the engine coolant temperature sensor is above/below the applicated threshold or the interpreted raw voltage signal (°C) is out of the applicated range longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

see 21.1.1 Circuit continuity

15.10.2 Rationality check

see 11.2 Rationality Checks

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15.11 Engine Off Timer

The engine-off time (time since the last engine shut-off event) is calculated with the continuously counting timer information of the instrument cluster coming via CAN.

15.11.1 Rationality check

(P1515)

General description:

The engine off time is calculated via the internal instrument cluster timer.

At time of engine shut-off the actual value of the internal instrument cluster timer is saved ("time A"; see below) and the engine off time starts increasing based on the ECU-timer (with a starting value of 0) until ECU shut-off. The post drive is this time between engine shut-off and ECU shut-off.

The internal instrument cluster timer is continuously increasing even if ECU is off.

When ECU is turned on again the actual engine off time is calculated by subtracting the previously saved "time A" off the actual value of the internal instrument cluster timer (the result is "time C"). With this calculated "time C" as starting value the engine off timer increases based on the ECU-timer. The EOT diagnostic consists of two checking conditions. Only one of these conditions has to fail to detect a fault.

The first condition checks min and max deviation of the CAN-timer (internal instrument cluster timer) to the internal ECU-timer from ECU on until ECU off. If the maximum calculated deviation allowed is exceeded (for example if one timer stops, or runs too slow/too fast) a fault is detected. The second condition compares the calculated engine off time of last DC ("time B", calculated in the post drive and saved prior to ECU shut-off) with the newly calculated engine off time at ECU start of the actual DC ("time C"). If "time C" is below "time B" a fault is detected.



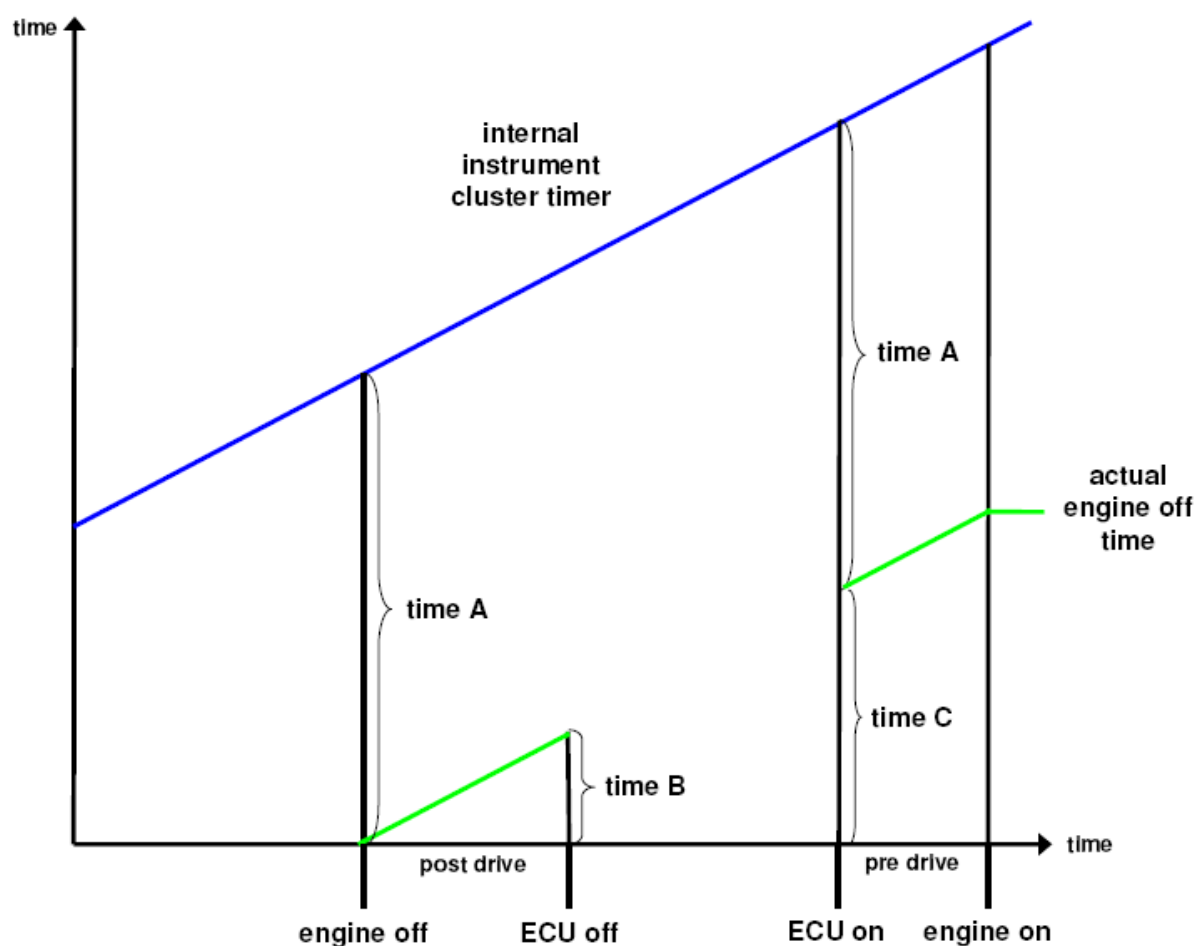
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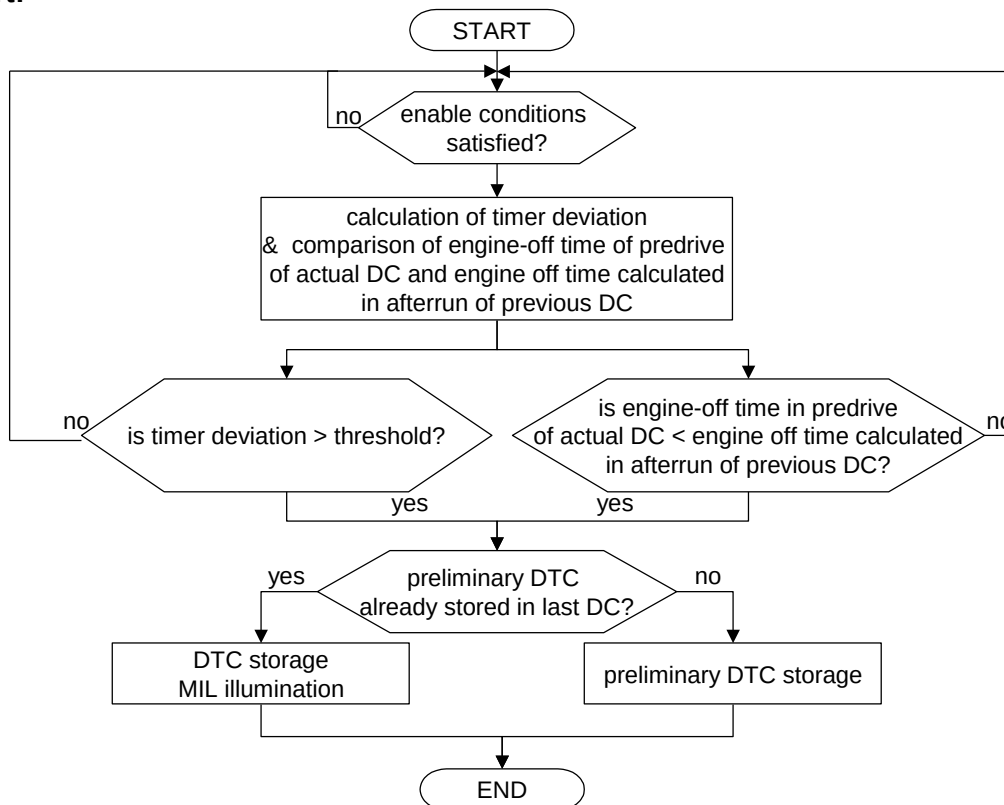
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Flowchart:



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15.11.2 Other functional check

To monitor the timer information via CAN a signal and a timeout check is done.

**15.11.2.1 CAN Signal Fault
(U0155)**

General description:

If the received CAN message (instrument panel timer) is not plausible for longer than the applicated time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

see 21.1.4 CAN Signal Fault

**15.11.2.2 CAN Timeout Fault
(U0155)**

General description:

If no CAN message (instrument panel timer) is received for longer than the applicated time, a fault is detected and a preliminary DTC is stored. This diagnosis is performed continuously.

Flowchart:

see 21.1.5 CAN Timeout Fault



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15.12 Engine Speed

15.12.1 Functional check

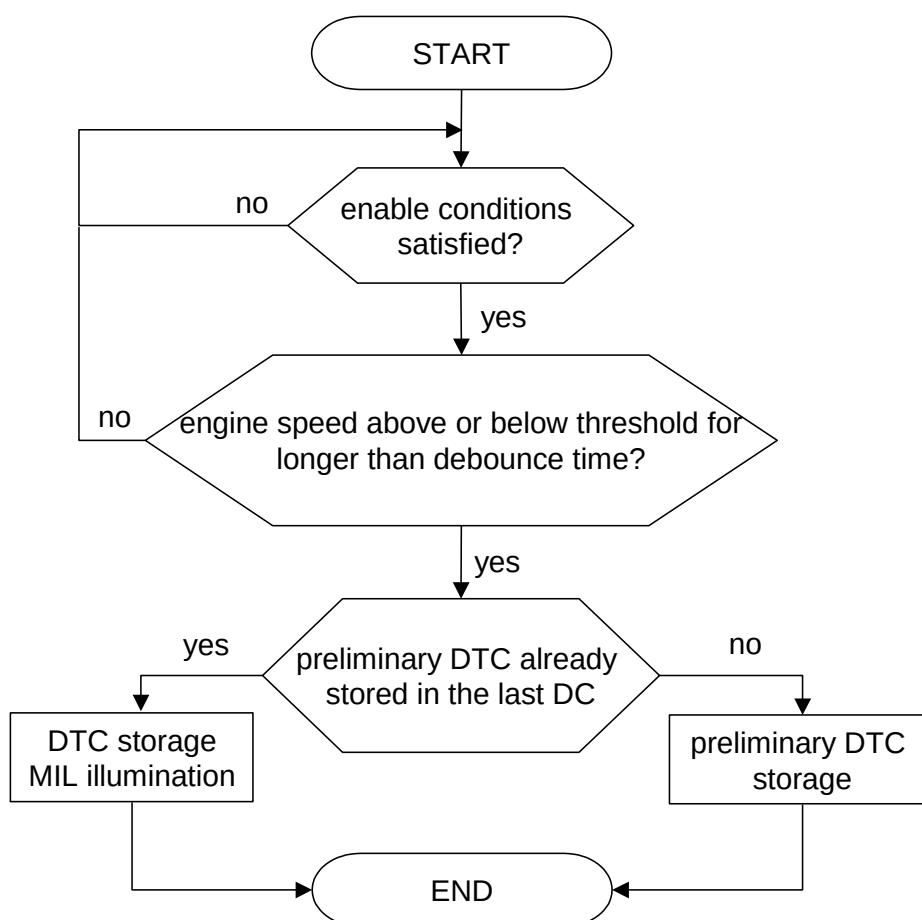
15.12.1.1 Idle Speed Monitoring
(P0507 & P0506)

General description:

The engine speed is monitored in the idle state. If the engine speed falls below a threshold for a certain time or reaches an upper threshold for a certain time a fault is detected and a preliminary DTC is stored.

The minimum and maximum threshold is the desired idle speed multiplied with a constant factor. If a fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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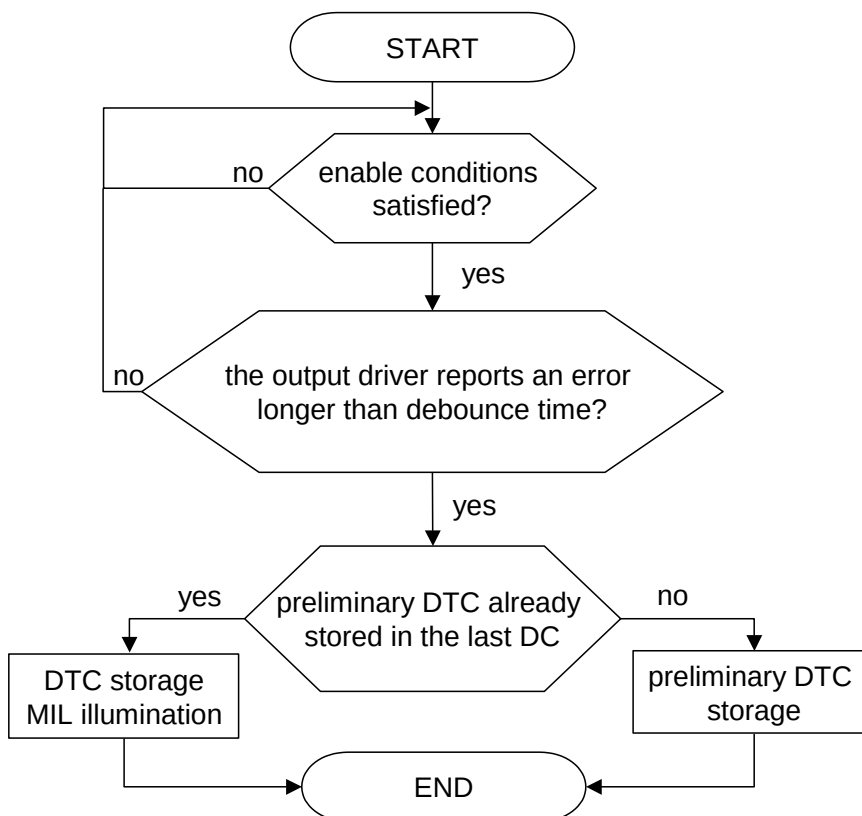
15.13 Exhaust Gas Recirculation Cooler Bypass Valve

15.13.1 Circuit continuity
(P245A, P245B, P245C, P245D)

General description:

The exhaust gas recirculation cooler bypass valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature of the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.14 Exhaust Gas Recirculation Valve

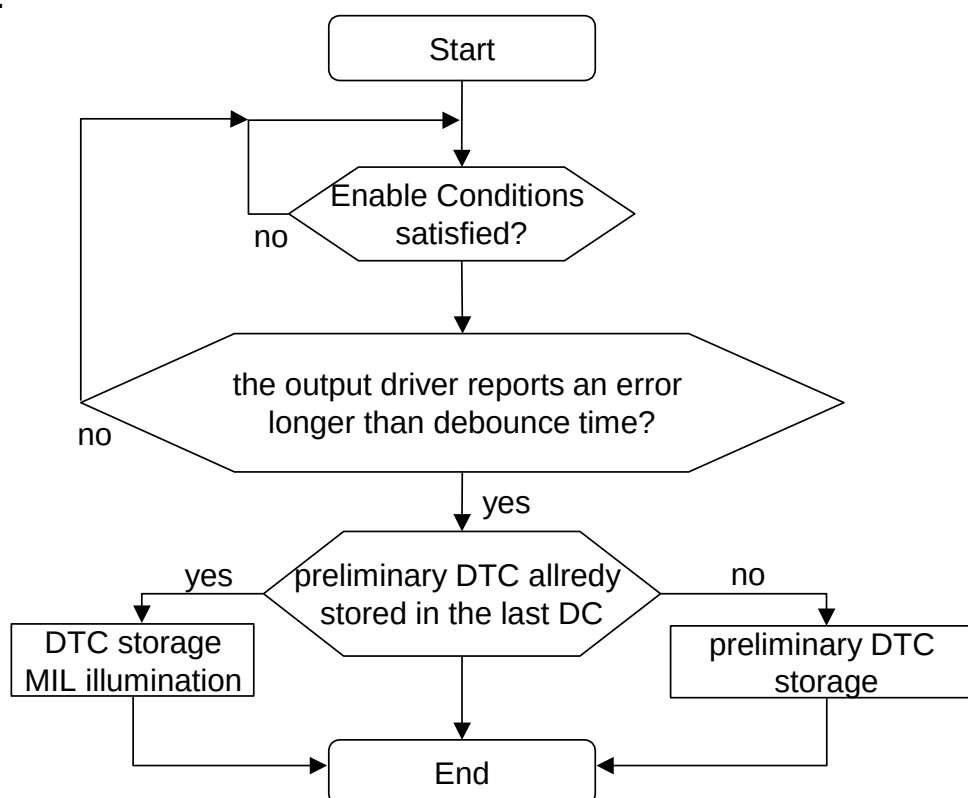
15.14.1 Circuit continuity

15.14.1.1 Self Diagnostic (P1286, P1269, P1212, P1213)

General description:

The EGR valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature from the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



15.14.1.2 Other (P046C)

General description:

If the raw voltage signal of the high pressure exhaust gas recirculation valve sensor is above/below the applicated threshold, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

see 21.1.1.1 Sensor Voltage

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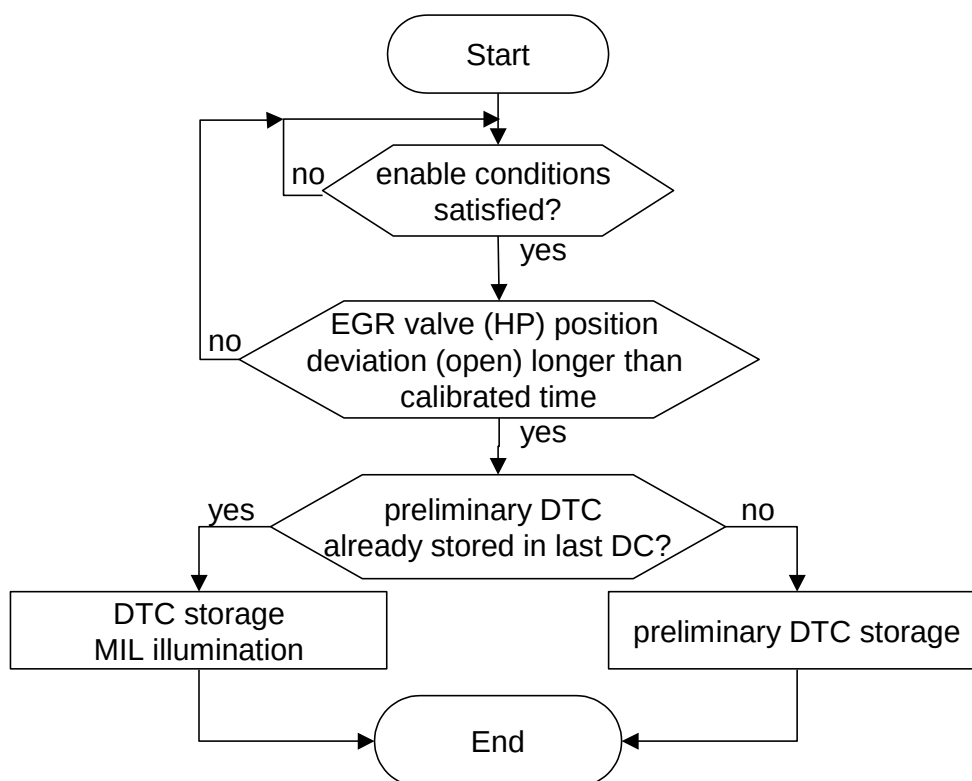
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15.14.2 Functional check**15.14.2.1 Jammed Valve****15.14.2.1.1 Jammed Open
(P042E)****General description:**

Due to a stuck EGR valve this diagnosis detects an open EGR valve under certain conditions. The enable conditions are:

- position deviation for open EGR Valve
- actual EGR valve position
- desired EGR valve position

An error is detected if this position deviation lasts longer than the calibrated time. If an error is detected a preliminary fault code is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

Flowchart:

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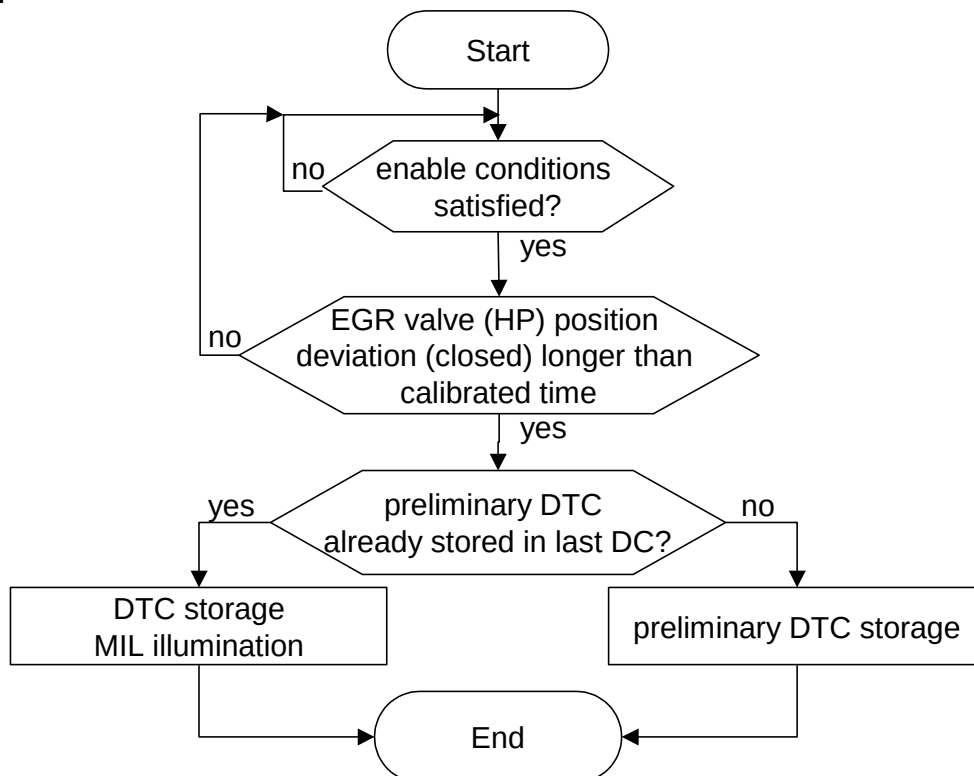
15.14.2.1.2 Jammed Closed
(P042F)**General description:**

To avoid damage due to a stuck EGR valve this diagnosis detects a closed EGR valve under certain conditions.

The enable conditions are:

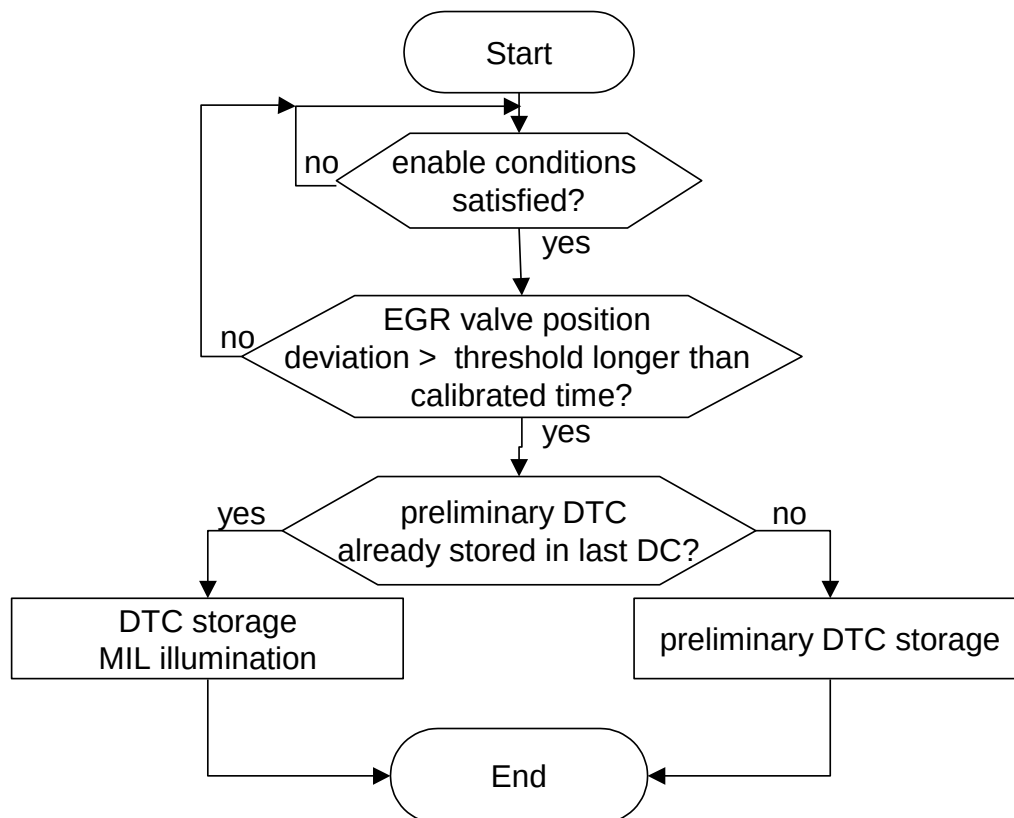
- position deviation for closed EGR Valve
- actual EGR valve position
- desired EGR valve position

An error is detected if this position deviation lasts longer than the calibrated time. If an error is detected, a preliminary fault code is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

Flowchart:

**15.14.2.1.3 Governor Position Deviation****Maximum Deviation****General description:**

If the position deviation of the high pressure EGR valve is above the calibrated threshold, an error is detected and a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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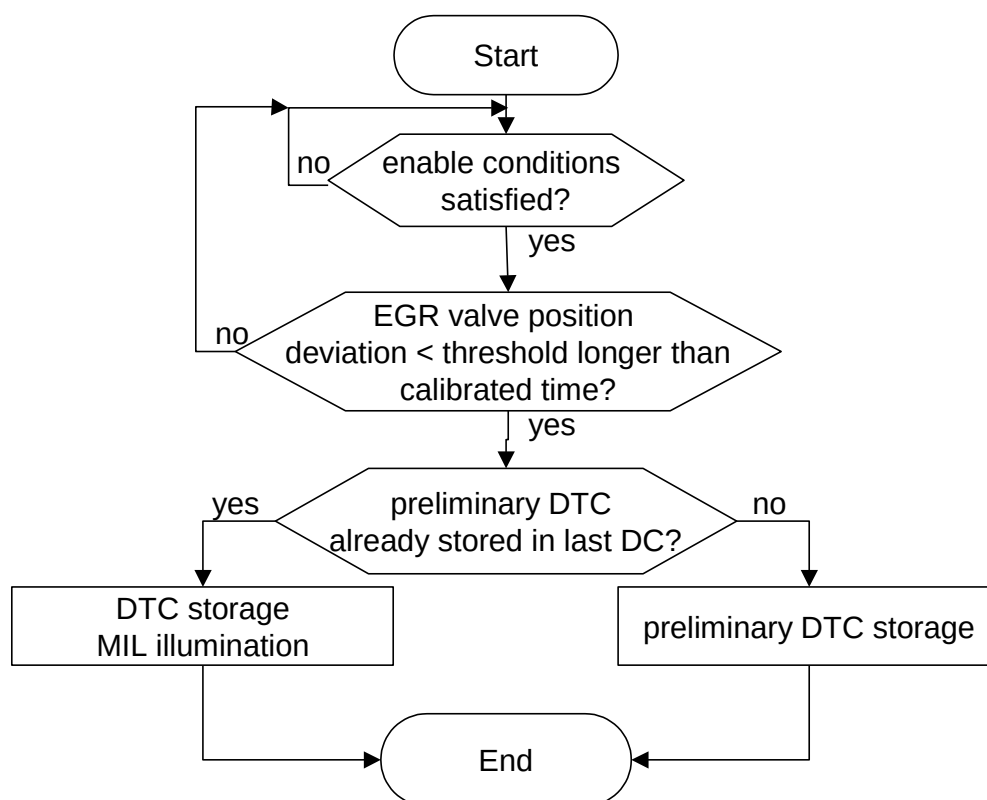
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Minimum Deviation

General description:

If the position deviation of the high pressure EGR valve is below the calibrated threshold an error is detected and a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.15 Exhaust Manifold Pressure Sensor**15.15.1 Circuit continuity**

(P0472, P0473)

General description:

If the raw voltage signal of the exhaust manifold pressure sensor (p31) is above/below the applicated threshold or the interpreted raw voltage signal (hPa) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

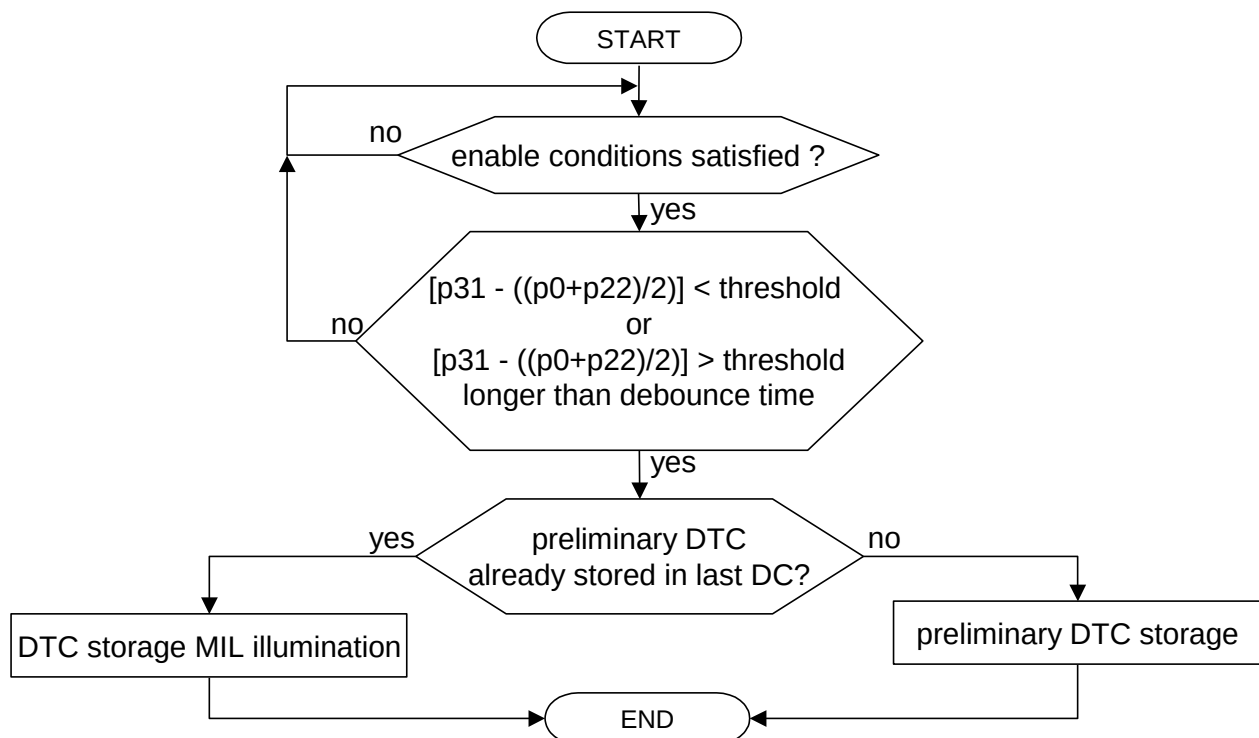
see 21.1.1 Circuit continuity

15.15.2 Rationality check

(P0471)

General description:

The plausibility check is executed by ignition on or in the afterrun. During the plausibility check of the exhaust manifold pressure sensor (p31) is calculated a deviation between the turbine pressure sensor and the other sensors (barometric pressure sensor (p0), manifold absolute pressure sensor (p22)). If the calculated deviation is above/below the threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:**CONFIDENTIAL**

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15.16 Exhaust Temperature Sensor Downstream EGR Cooler**15.16.1 Circuit continuity**

(P040D, P040C)

General Description:

If the raw voltage signal of the exhaust temperature sensor downstream EGR cooler is above/below the applicated threshold or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This is a continuous diagnosis.

Flowchart:

see 21.1.1 Circuit continuity

15.16.2 Rationality check

(P040B)

General description:

A fault is detected via the cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared with every other sensor value (within the function) and a difference to each sensor value is calculated. If one Sensor value is above the applicated thresholds compared to all other sensor values within the funktion, a fault is detected and a preliminary DTC is stored.

To enable this diagnosis the applicated engine off time has to be exceeded.

The exhaust temperature sensor downstream EGR cooler is compared to:

- ambient air temperature sensor
- induction air temperature sensor
- exhaust temperature sensor downstream EGR LP cooler(only X5 3.0sd)

Flowchart:

see 21.1.3 Rationality Check Low



15.17 Fuel Injector

15.17.1 Circuit continuity / Functional check

(P0201, P0202, P0203, P0204, P0205, P0206, P0261, P0262, P0264, P0265, P0267, P0268, P0270, P0271, P0273, P0274, P0276, P0277, P0611, P3148, P3151, P3154, P3157, P3160, P3163)

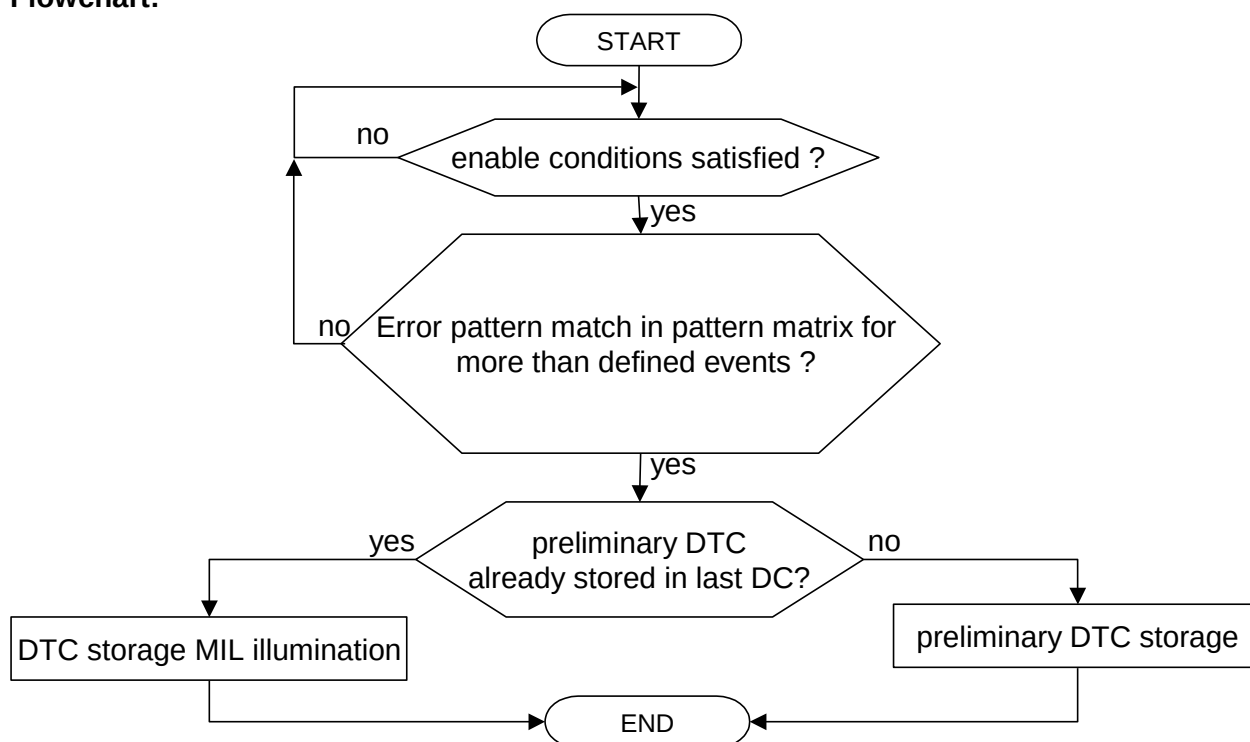
General description:

This is an internal function, which detects injector faults and faults between the ECU and the injectors. To detect faults of the injector-control the check byte and the status of plausibility check of the actuator-voltage are evaluated by the diagnosis-register. The injectors and the power stage are controlled by sample-detection in every injection event. Sample-detection means, a number of events that describe a possible physical fault. The sample-detection detects the following possible faults:

- Short circuit between high-side and low-side
- Open circuit (broken cable)
- Short circuit to ground and to battery
- Short circuit bank
- Short circuit charge-switch
- Chip-error

If one of this faults is detected, a preliminary DTC for the particular cylinder/bank/chip is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





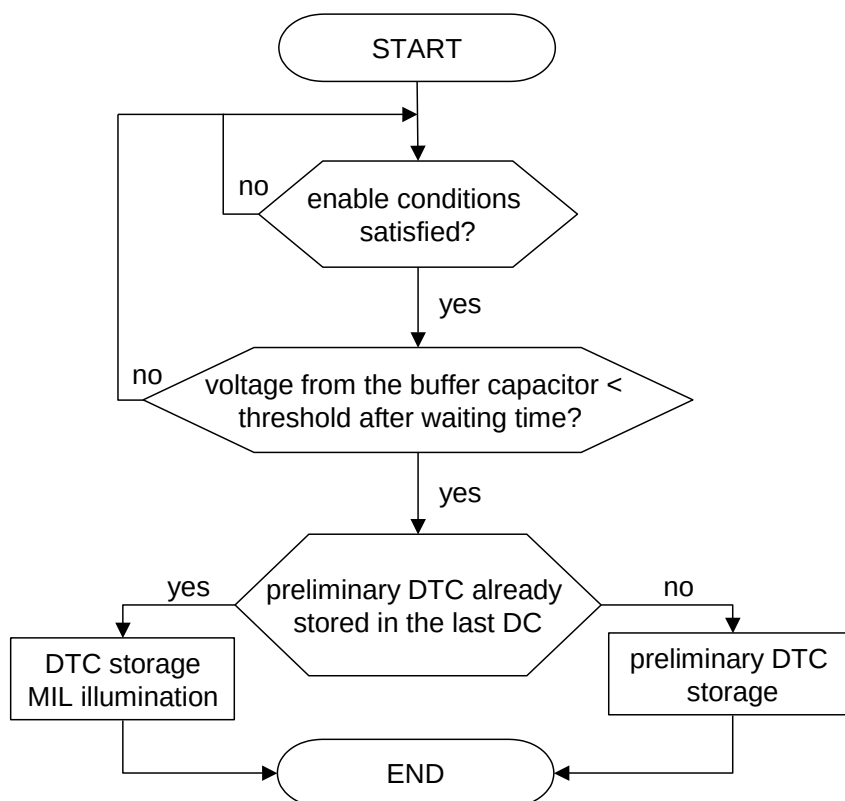
15.18 Fuel Injector System

15.18.1 Rationality check (P0611)

General description:

The high voltage for the piezo injectors is generated by a DC/DC converter and stored in a buffer capacitor. This buffer capacitor is monitored for undercharge. If the charge is under a threshold after a defined waiting-time, a fault is detected. If this malfunction is detected during two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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15.19 Fuel Metering Unit

15.19.1 Circuit continuity
(P0001, P0003, P0004)

General Description:

The metering unit power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:

see 21.1.1.3 Power Stage



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15.20 Fuel Rail Pressure Sensor

15.20.1 Circuit continuity
(P0192, P0193)

General description:

If the raw voltage signal of the rail pressure sensor is above/below the applicated threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

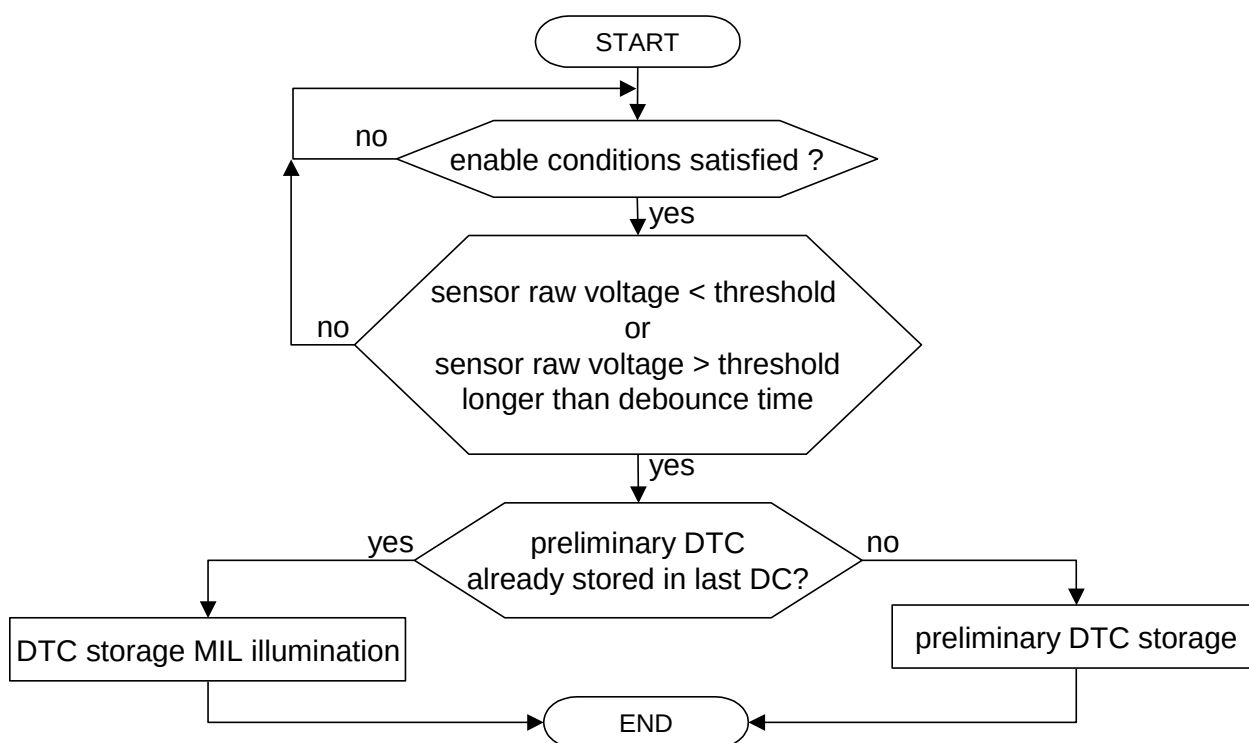
see 21.1.1 Circuit continuity

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15.20.2 Rationality check
(P3000, P3001)**General description:**

The diagnosis starts in the afterrun. After the engine has stopped the system must wait a fixed time period until the rail pressure is equal to the ambient pressure. During the afterrun the ECU compares the raw voltage of the rail pressure sensor with a threshold. If the raw value of the rail pressure sensor is above/below the threshold for longer than a allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



15.21 Fuel Rail Pressure Control Valve

15.21.1 Circuit continuity (P0090, P0091, P0092)

General description:

The rail pressure control valve power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

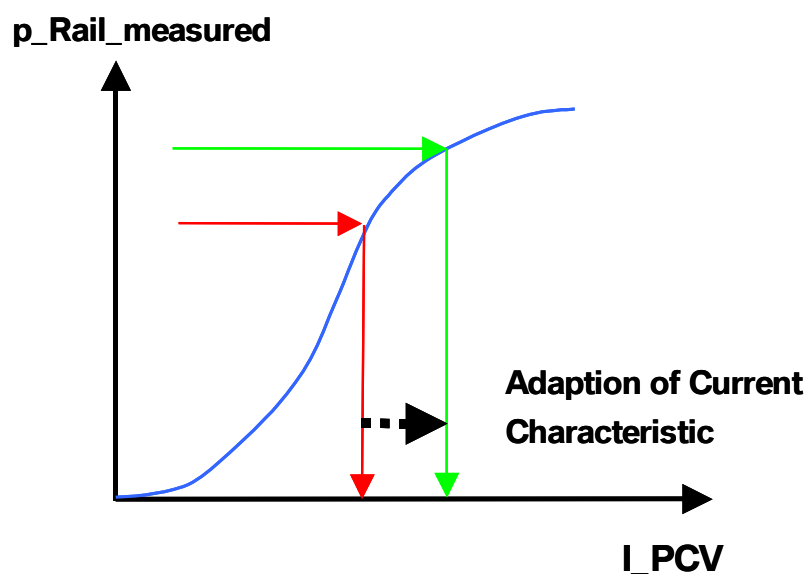
see 21.1.1.3 Power Stage

15.21.2 Rationality check (Adaption of Pressure Control Valve) (P228F,P228E)

General description:

The adaption function executes a plausibility check of the measured current of the pressure control valve with the measured rail pressure. In pressure control mode, the adaption function is calculating an adaption factor via the calculated deviation between measured pressure and the actual value of the current curve. If the adaption factor is above or below the specific threshold, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Characteristic of PCV





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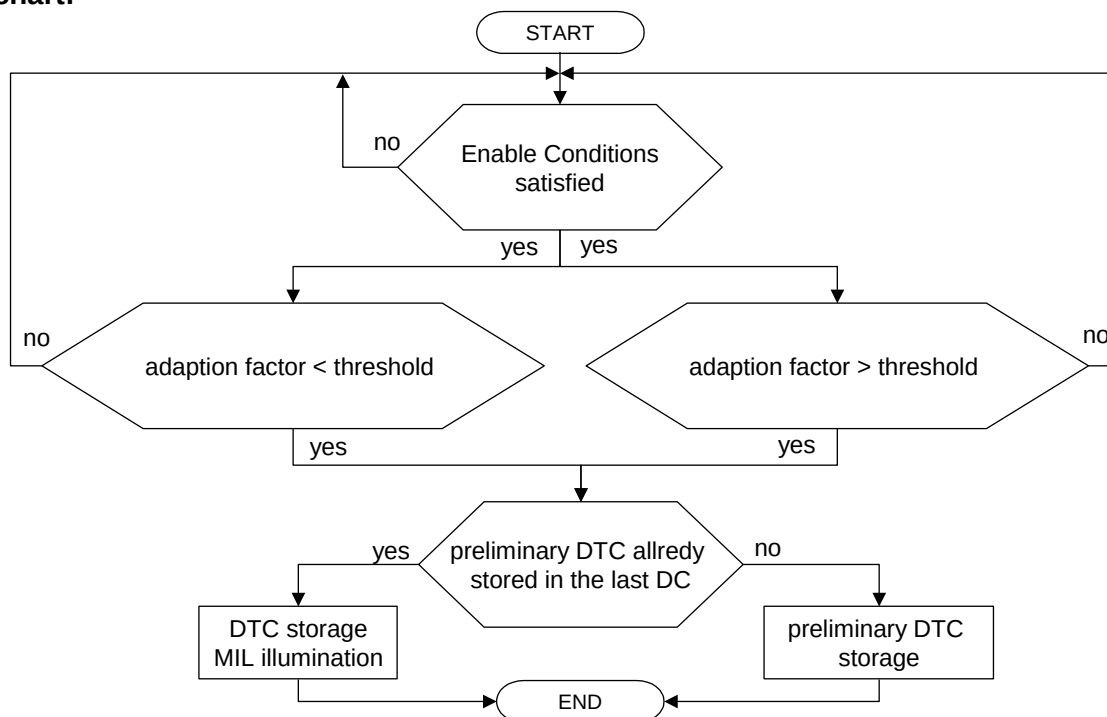
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Flowchart:



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15.22 Fuel Temperature Sensor

15.22.1 Circuit continuity (P0182, P0182)

General description:

If the raw voltage signal of the fuel temperature sensor is above/below the applicated threshold or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

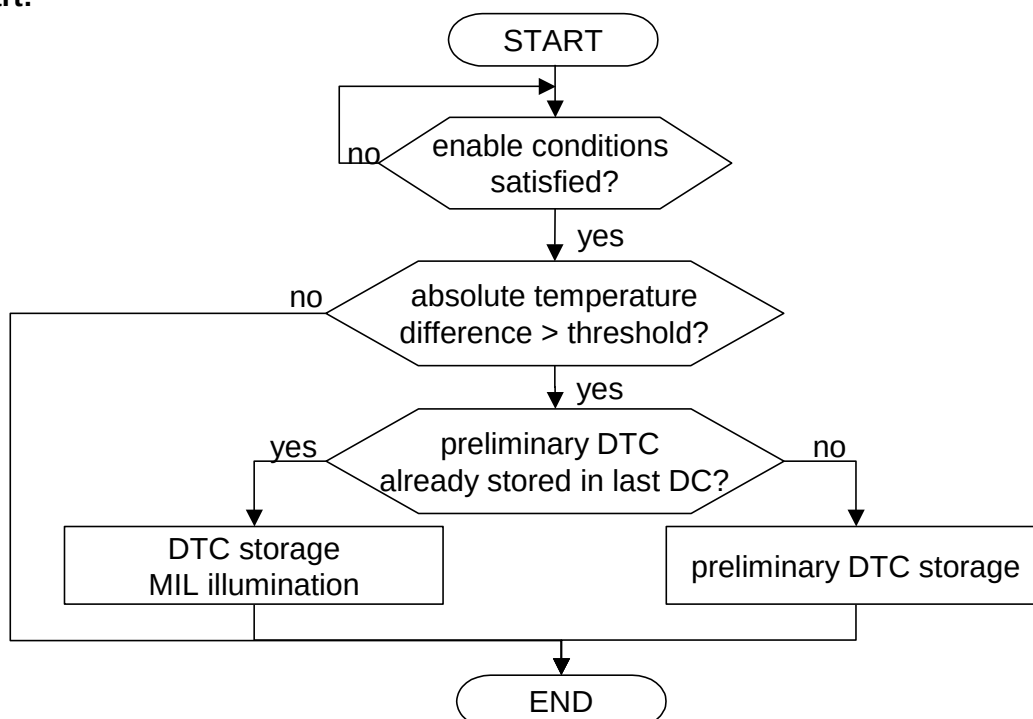
see 21.1.1 Circuit continuity

15.22.2 Rationality check (P008F)

General description:

If an applicated minimum engine-off time is exceeded the absolute difference between the engine coolant temperature sensor value and the fuel temperature sensor value is calibrated. If the difference is above the applicated threshold, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:





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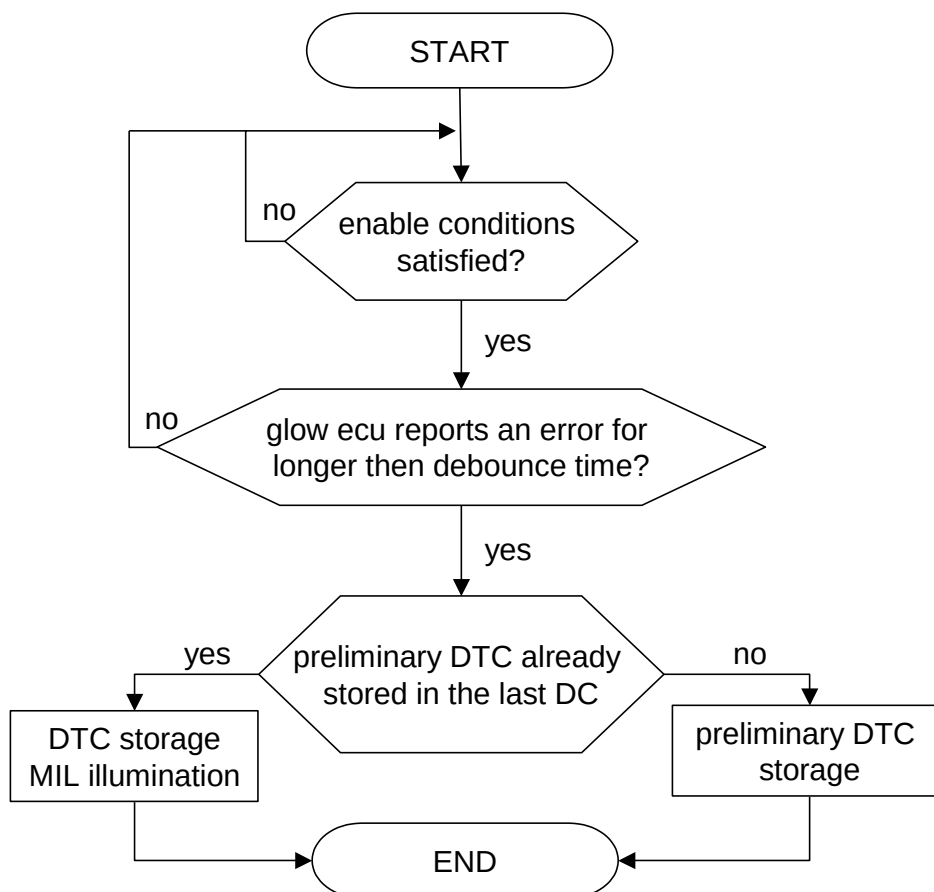
15.23 Glow Plug

15.23.1 Circuit continuity

(P0671, P0672, P0673, P0674, P0675, P0676, P066A, P066C, P066E, P067A, P067C, P067E)

General description:

The specifications of the glow plugs for high mileage performance are defined by the resistance of the glow plugs. Each glow plug has a wire that leads to a connector on the glow ECU. The glow ECU is connected via CAN to the engine ECU. All glow plugs are monitored for low or high resistance, short circuits, open loop and permanent switch on/off by the glow ECU. If the glow ECU sends a fault to the ECU longer than debounce time, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



15.24 Glow Plug System Control Module

15.24.1 Functional check

(P0670), (P064C), (U0106),

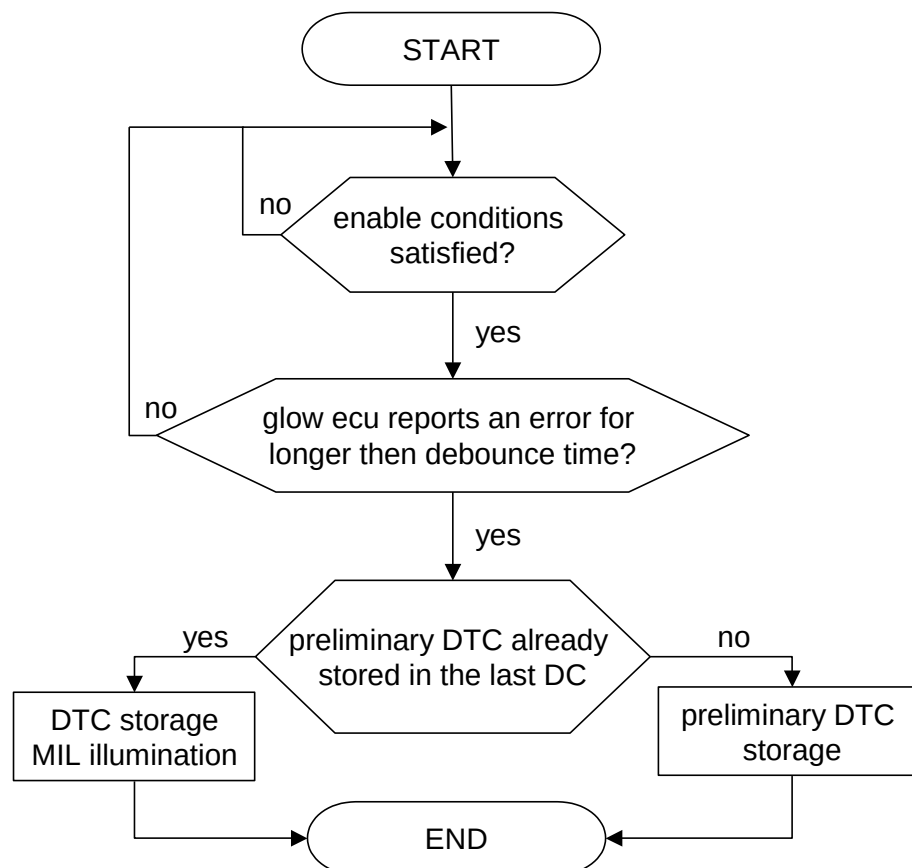
General description:

The glow ECU monitors itself for several malfunctions. These malfunctions are:

- EEPROM Error
- supply power loss
- internal hardware error
- temperature exceeds
- ground error

If the glow ECU sends a fault to the ECU longer then debounce time, a fault is detected and a preliminary fault code is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:





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15.24.2 Glow control Unit LIN Bus
(P0670, P064C)

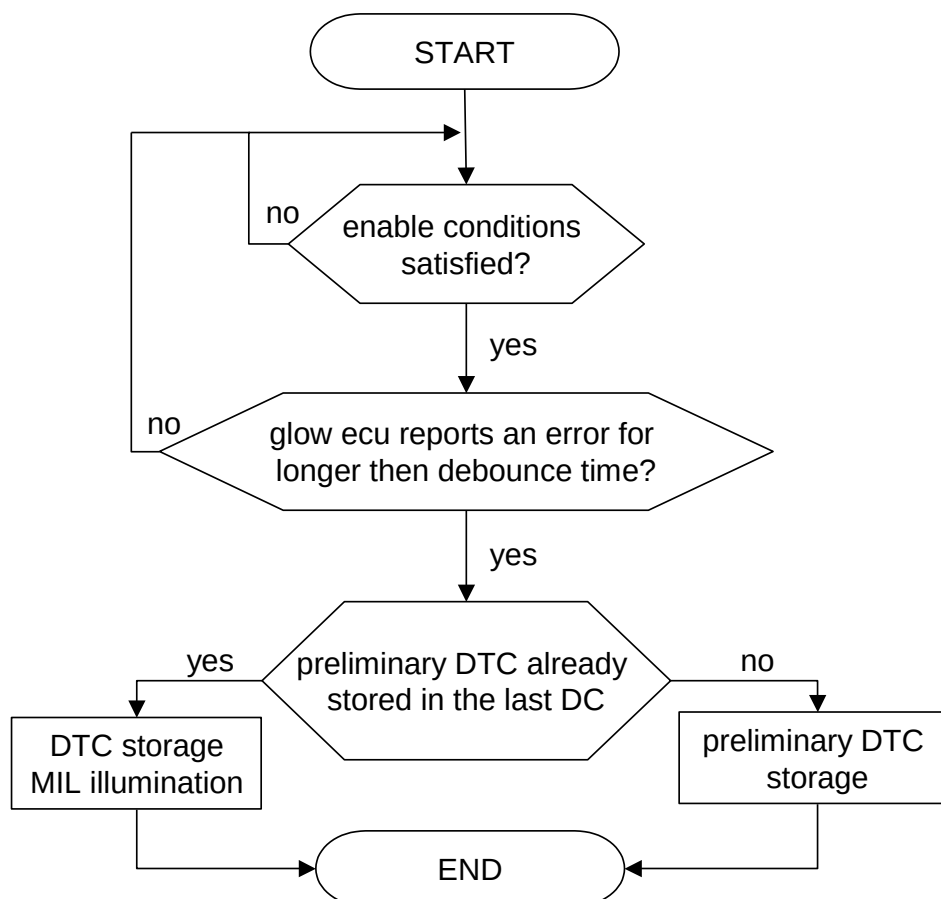
General description:

The ECU monitors the LIN-Bus from the glow ECU for several faults. The faults are:

- checksum error
- message timeout
- voltage difference system voltage – LIN-signal too high

If a fault is detected in two consecutive driving cycles, the MIL is illuminated.

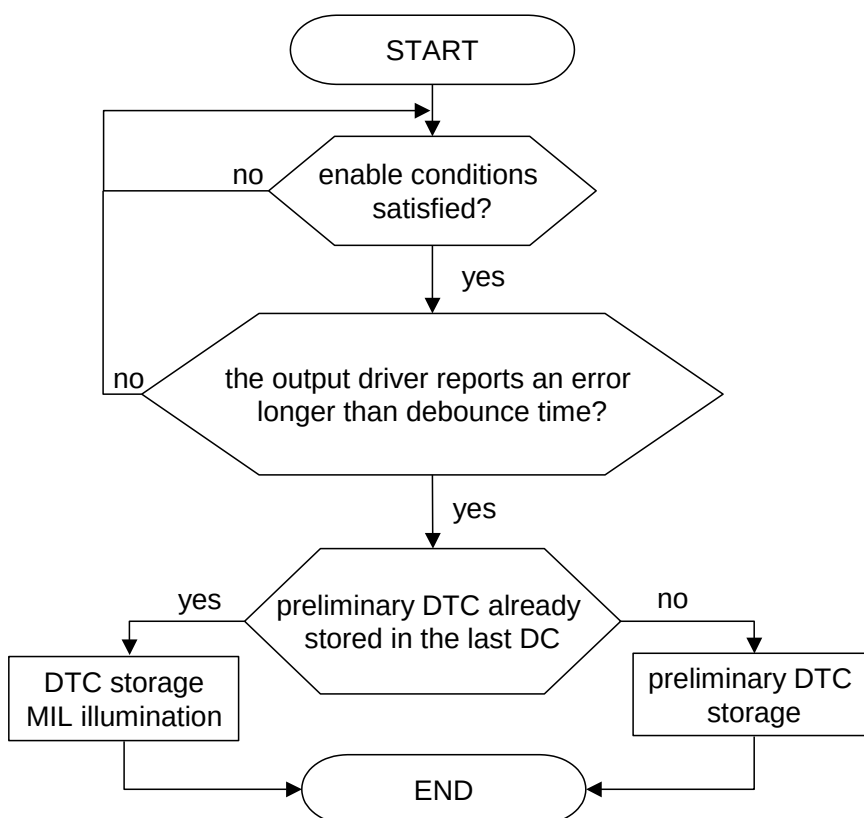
Flowchart:



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**15.25 Low Pressure Exhaust Gas Recirculation Valve (only X5 3.0sd)****15.25.1 Circuit continuity****15.25.1.1 Self diagnostic**
(P045B, P045C, P045D)**General description:**

The low pressure EGR valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature of the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:

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**15.25.1.2 Other
(P046E)****General description:**

If the raw voltage signal of the low pressure exhaust gas recirculation valve sensor is above/below the applied thresholds, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

see 21.1.1 Circuit continuity

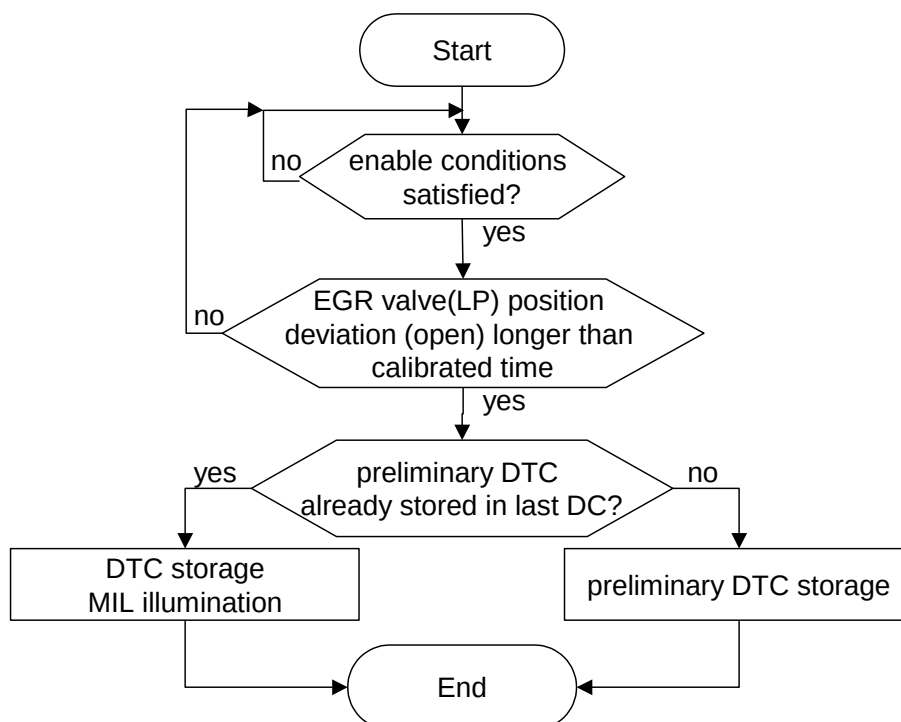
15.25.2 Functional check**15.25.2.1 Jammed Valve****15.25.2.1.1 Jammed Open****(P045E)****General description:**

To avoid damages due to a stucked low pressure EGR valve this diagnosis detects an opened low pressure EGR valve under certain conditions.

The enable conditions are:

- position deviation for opened low pressure EGR Valve
- actual low pressure EGR valve position
- desired low pressure EGR valve position

An error is detected if this position deviation lasts longer than the calibrated time. If an error is detected a preliminary fault code is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

Flowchart:**CONFIDENTIAL**

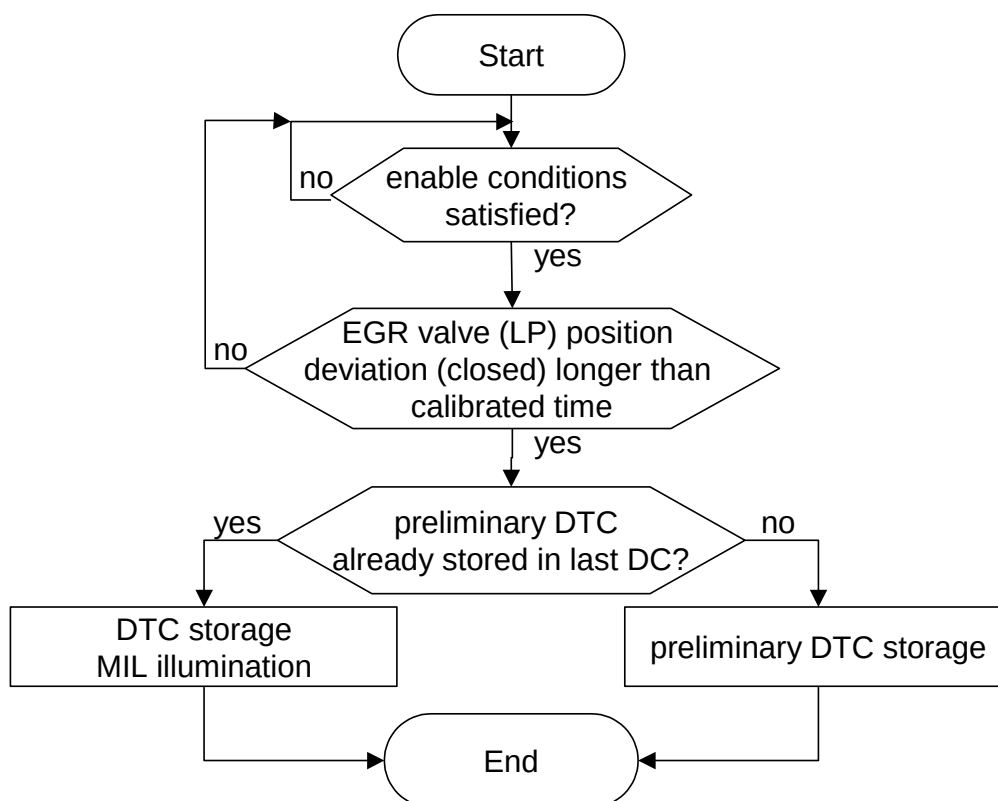
**15.25.2.1.2 Jammed Close**
(P045F)**General description:**

To avoid damages due to a stucked low pressure EGR valve this diagnosis detects a closed low pressure EGR valve under certain conditions.

The enable conditions are:

- position deviation for closed low pressure EGR Valve
- actual low pressure EGR valve position
- desired low pressure EGR valve position

An error is detected if this position deviation lasts longer than the calibrated time. If an error is detected a preliminary fault code is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

Flowchart:



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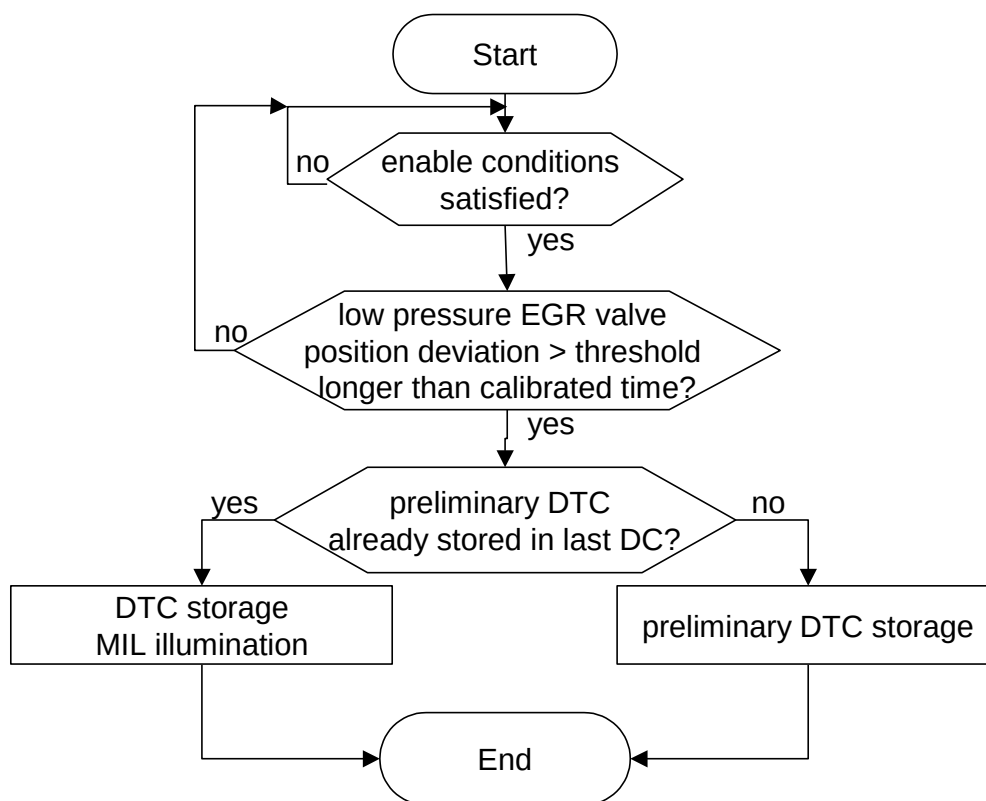
15.25.2.1.3 Governor Position Deviation

Maximum Deviation

General description:

If the position deviation of the low pressure EGR valve is above the calibrated threshold an error is detected and a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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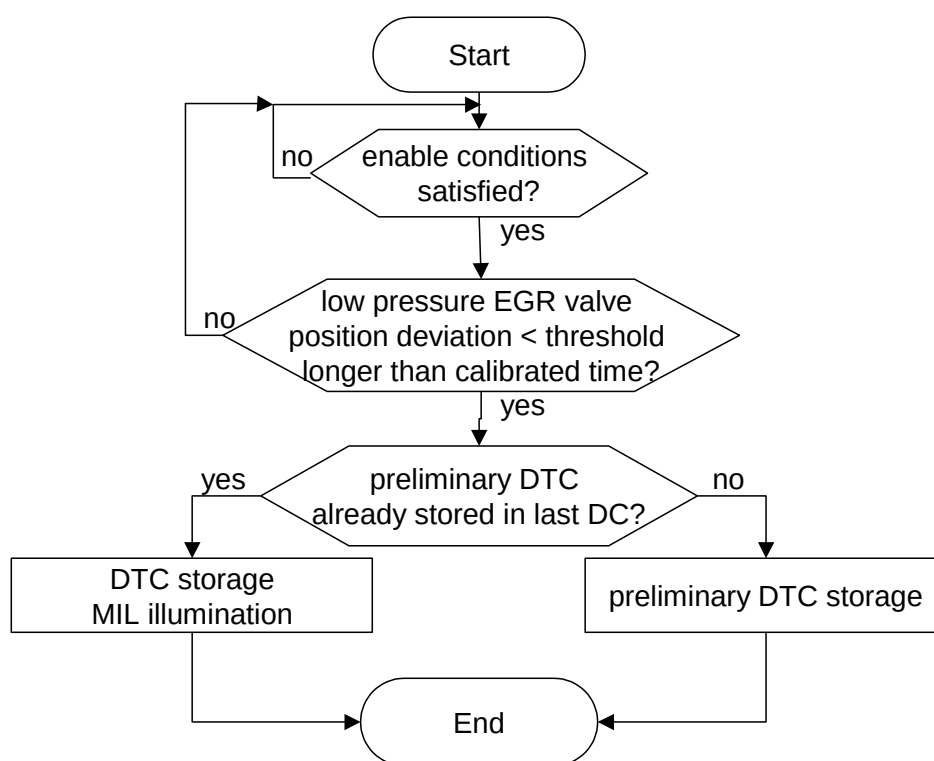
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Minimum deviation

General description:

If the position deviation of the low pressure EGR valve is below the calibrated threshold an error is detected and a preliminary fault code is stored. If an error is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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15.26 Exhaust Temperature Sensor Downstream EGR LP Cooler(only X5 3.0sd)**15.26.1 Circuit continuity**
(P041D)**General description:**

If the raw voltage signal of the exhaust temperature sensor downstream EGR LP cooler is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

21.1.1 Circuit continuity

15.26.2 Rationality check
(P041B)**General description:**

A fault is detected via the cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

To enable the diagnosis the applicated engine off time has to be exceeded.

The exhaust temperature sensor downstream EGR LP cooler is compared to:

- ambient air temperature sensor
- induction air temperature sensor
- exhaust temperature sensor downstream EGR cooler

Flowchart:

see 21.1.2 Cross-check of Temperature Sensors



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15.27 Main Relay

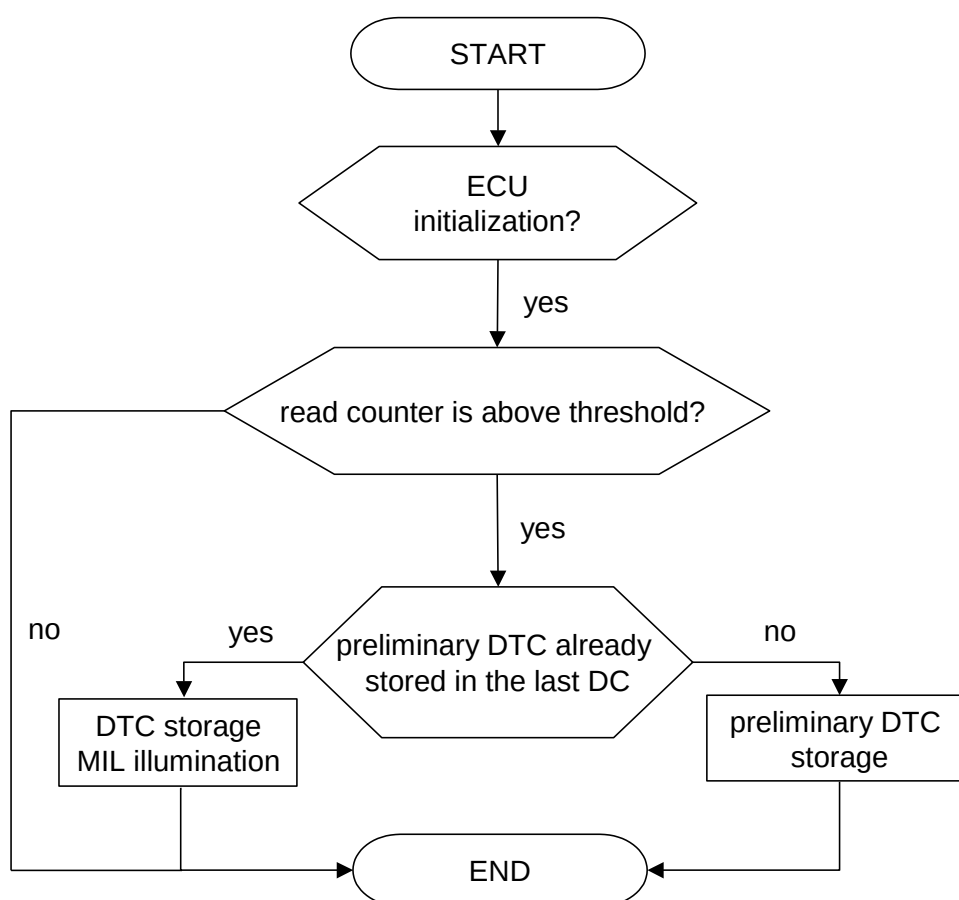
15.27.1 Functional check

**15.27.1.1 Main relay early shut off detection
(P0685)**

General description:

After key off, a counter is incremented in the ECU. After all processes in the ECU afterrun are finished, the counter is forced to zero. At ECU start the counter is compared with a threshold. If the counter is greater than the threshold, a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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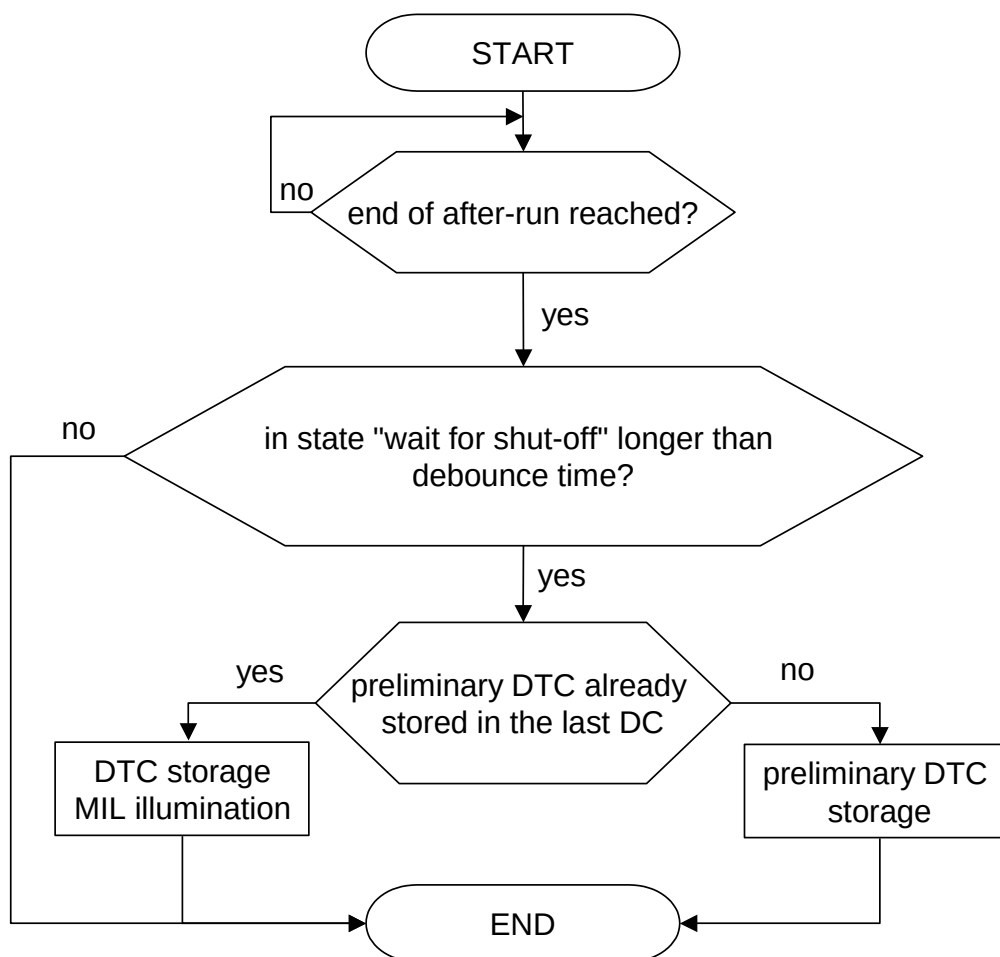
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**15.27.1.2 Main relay late shut off detection
(P0685)**

General description:

After key off, the main relay is kept shut by the ECU until all processes in the afterrun were finished. After that, the ECU waits to be shut off. If the ECU is not shut off in a certain time a fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles, the corresponding fault code is stored and the MIL is illuminated.

Flowchart:



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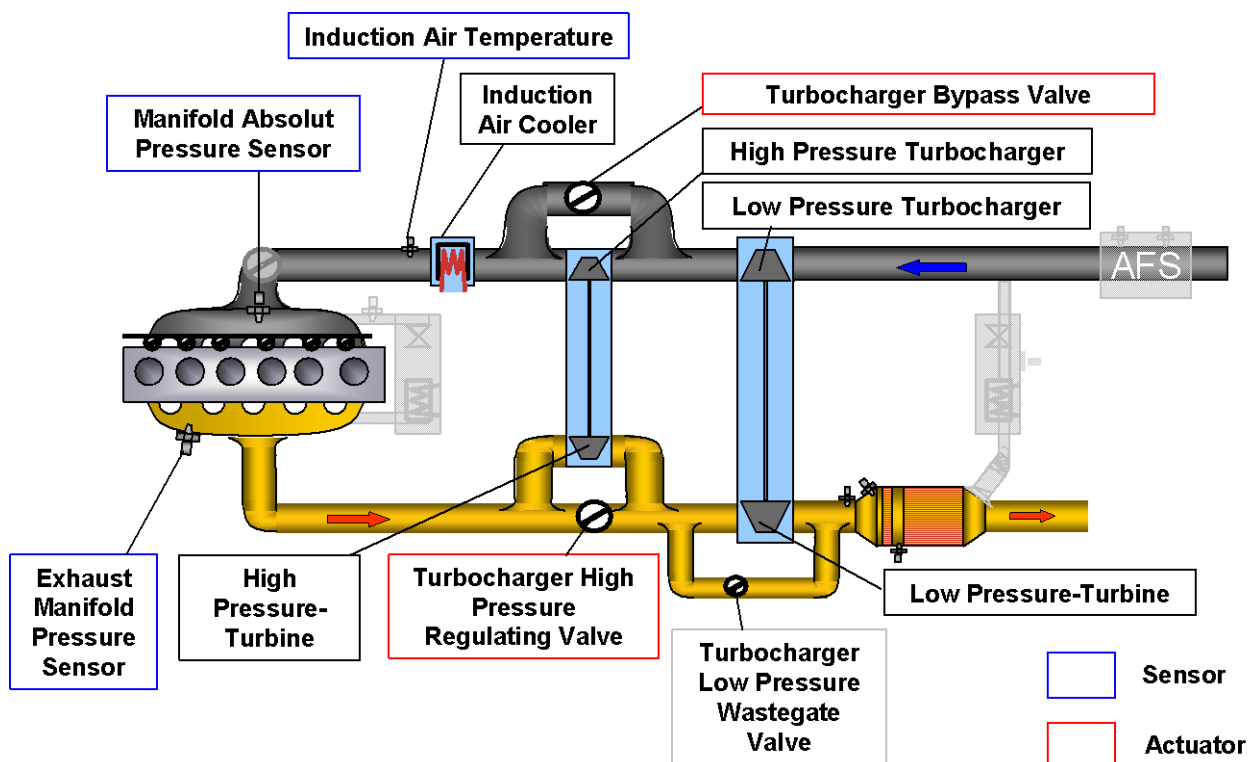


15.28 Boost Pressure Control System

15.28.1 System Overview

Regarding the boost pressure control system the following comprehensive components are monitored:

- Electrical monitoring of all solenoid valves
- Functional response check of control flaps
 - Control flap high pressure charger: Failure recognition with boost pressure governor deviation
 - Control flap high pressure bypass: Failure recognition with boost pressure governor deviation
- Control flap wastegate (not emission relevant)

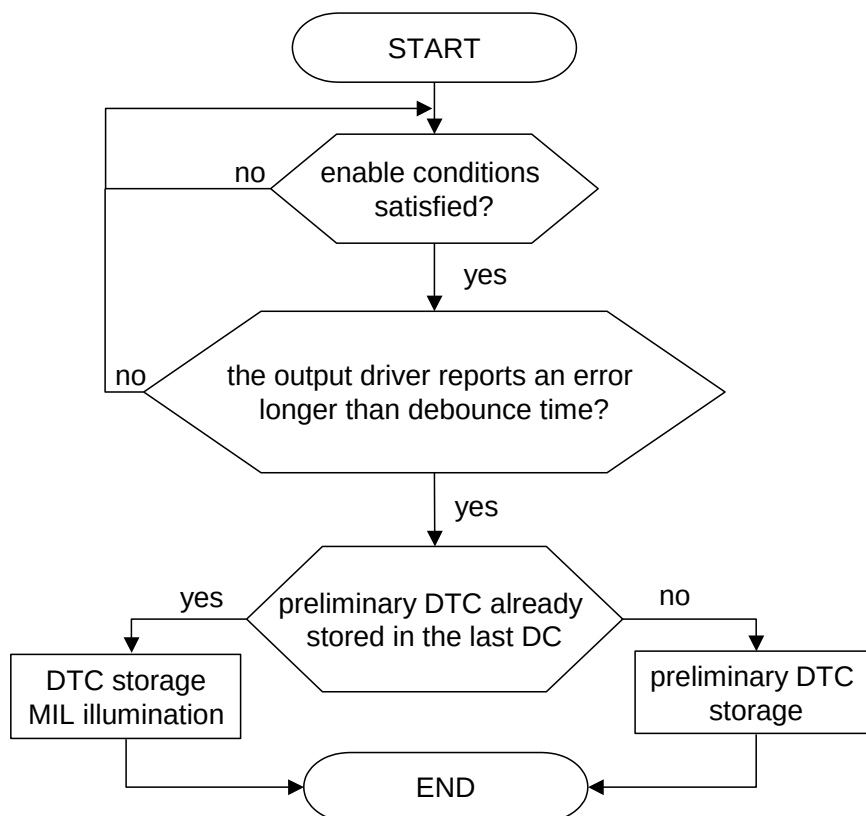


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15.28.2 Turbocharger Bypass Valve**15.28.2.1 Circuit continuity**
(P0033, P0035, P0034, P0039)**General description:**

The turbo charger bypass valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature of the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

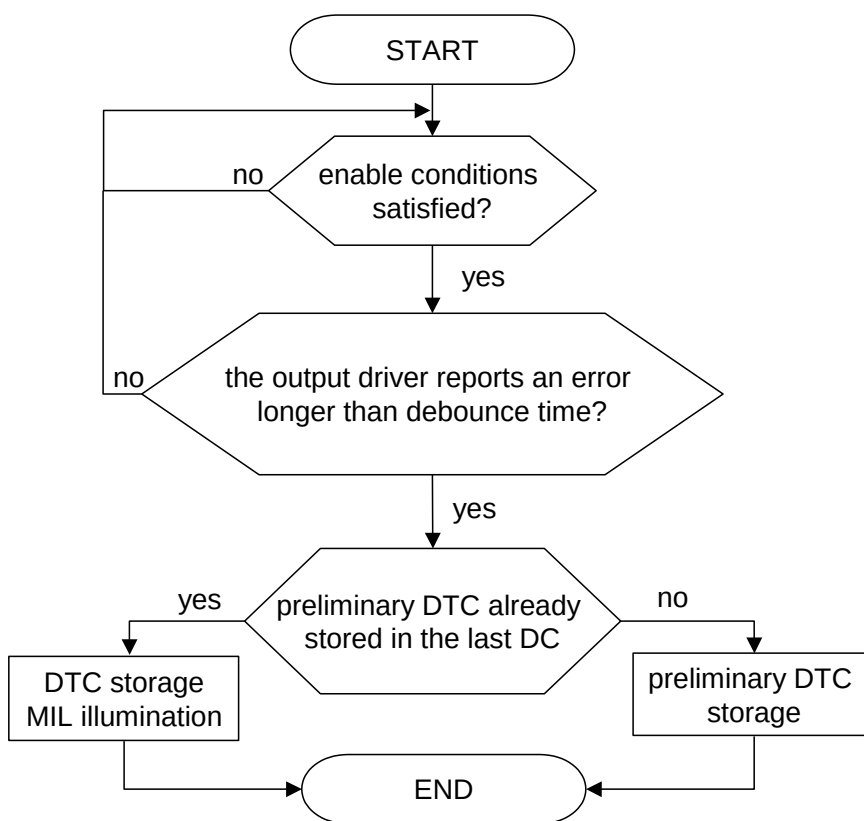
Flowchart:

**15.28.3 Turbocharger High Pressure Regulating Valve****15.28.3.1 Circuit continuity**

(P0045, P1254, P0047, P0048)

General description:

The turbo charger high pressure regulating valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature of the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

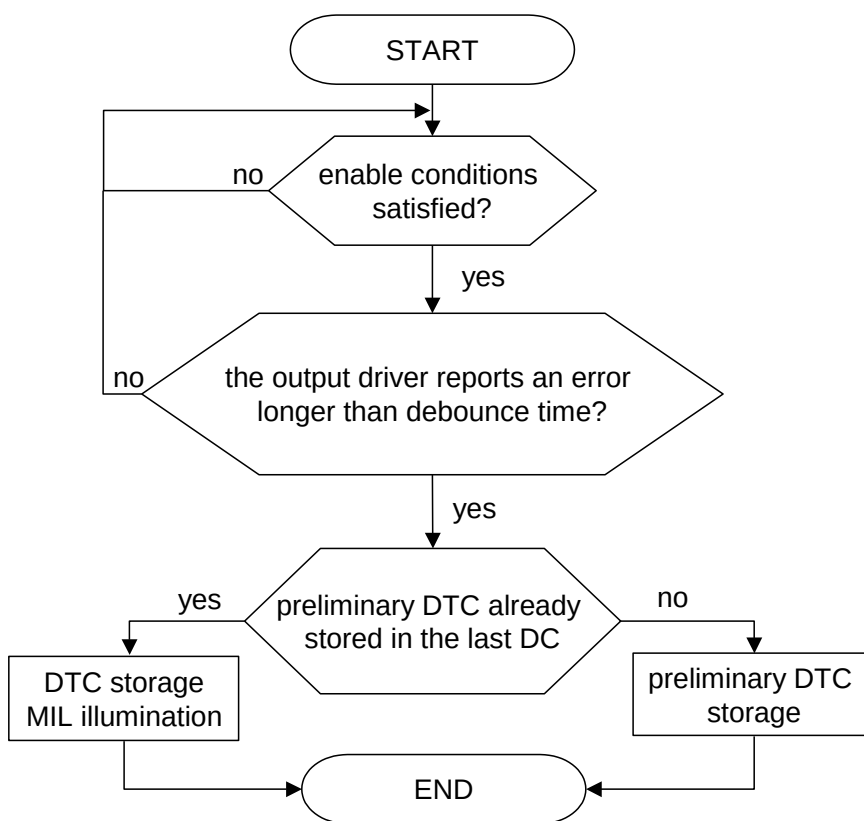
Flowchart :

**15.28.4 Turbocharger Low Pressure Wastegate Valve****15.28.4.1 Circuit continuity**

(P0243, P0244, P0245, P0246)

General Description:

The turbo charger low pressure wastegate valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature of the power stage. If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

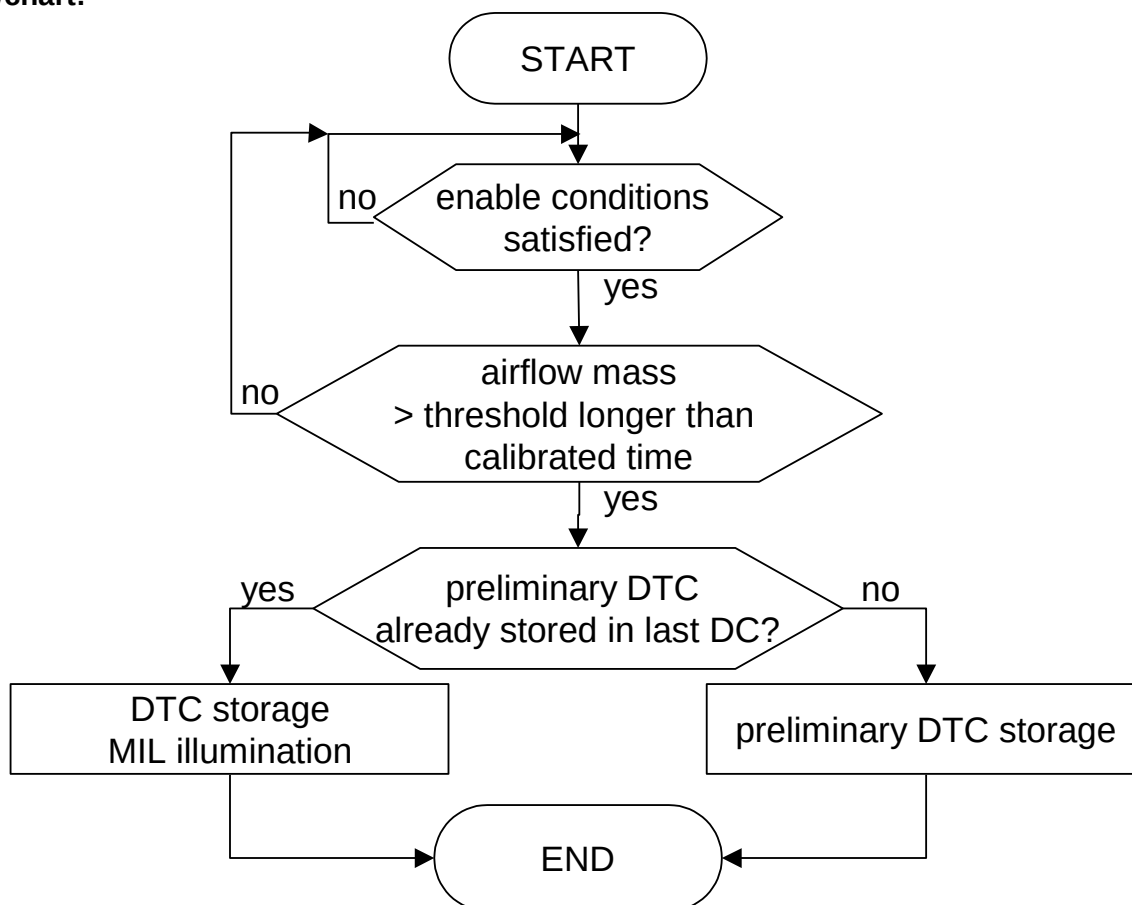
Flowchart:

**15.28.5 Manifold Absolute Pressure Regulation (Functional Response)****15.28.5.1 Rationality check****15.28.5.1.1 Static
(P2279)****General description:**

To avoid high soot emissions for DPF protection it is necessary to detect a disconnected charge air tube. This diagnosis compares the airflow mass with a threshold under certain conditions. The enable conditions are:

- engine speed
- injection quantity

An error is detected if the airflow mass is above a threshold for longer than the calibrated time. If an error is detected a preliminary DTC is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

Flowchart:

**15.28.5.1.2** **Dynamic**
(P2279)**General description:**

To avoid high soot emissions for DPF protection it is necessary to detect a disconnected charge air tube. This diagnosis compares the manifold pressure control deviation with a corrected threshold under certain conditions.

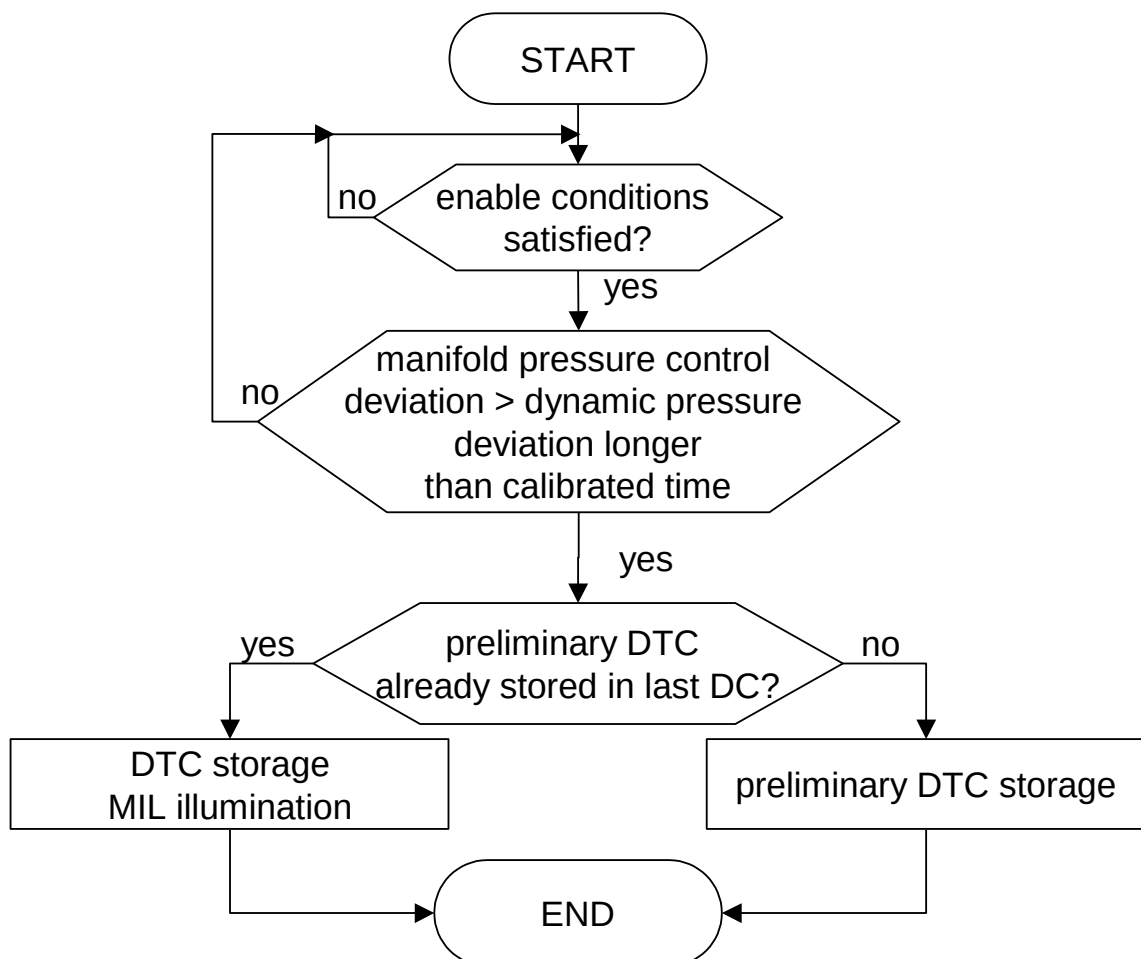
The enable condition is:

- dynamic of injection quantity

The corrections are due to:

- barometric pressure
- engine coolant temperature

An error is detected if the manifold pressure control deviation is above a threshold for longer than the calibrated time. If an error is detected a preliminary DTC is stored. The MIL is illuminated, if this fault is detected in two consecutive driving cycles.

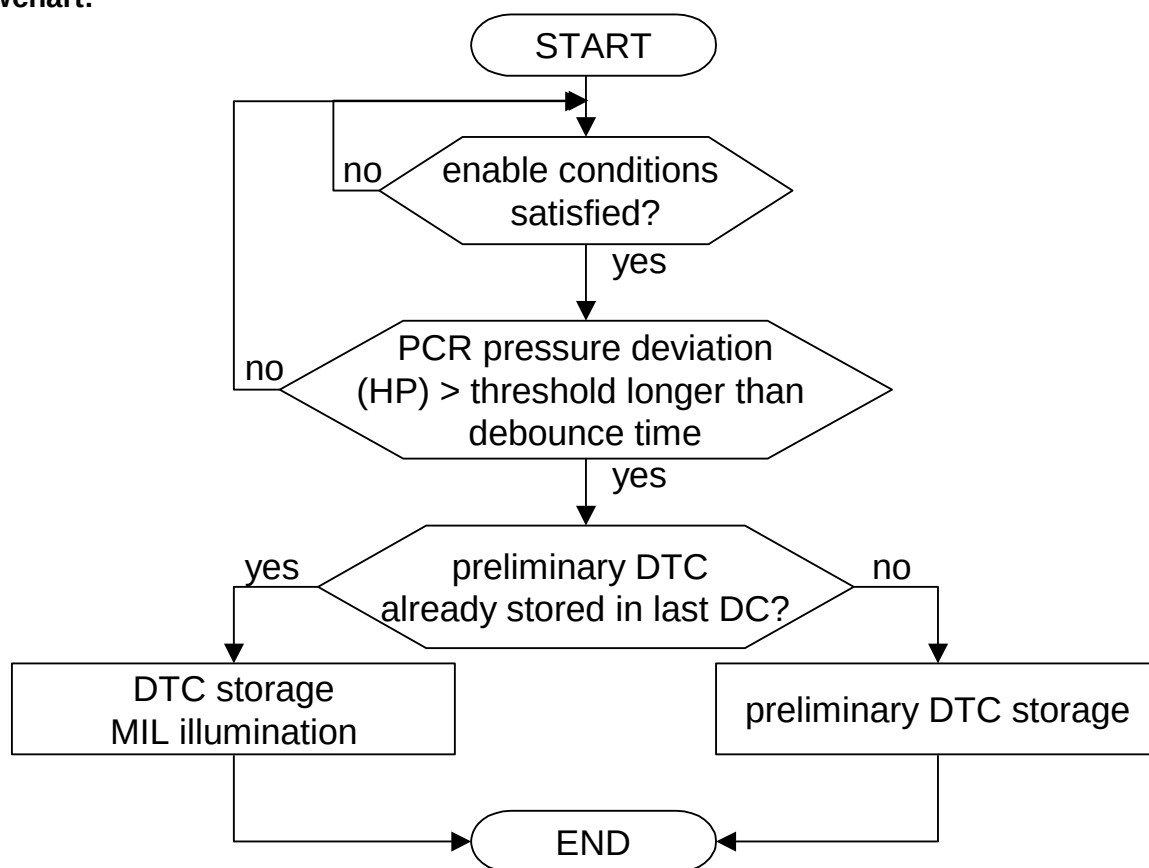
Flowchart:

**15.28.5.2 Functional check**

The functional check of the pressure control regulation (PCR) is based on the monitoring of the control deviation.

**15.28.5.2.1 Boost pressure governor deviation (maximum)
(P2099)****General description:**

If the PCR (high pressure (HP)) control deviation is above the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

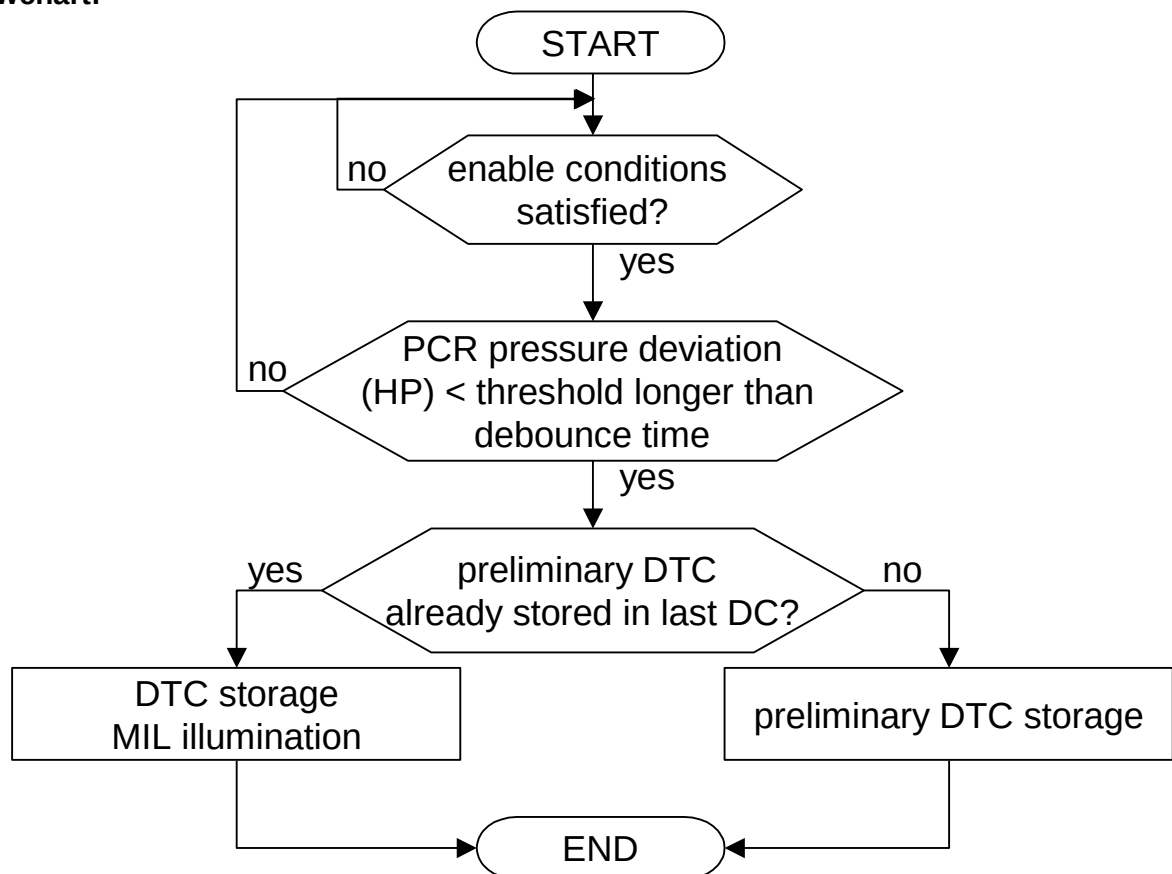
Flowchart:

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**15.28.5.2.2 Boost pressure governor deviation (minimum)
(P0234)****General description:**

If the PCR (high pressure (HP)) control deviation is below the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

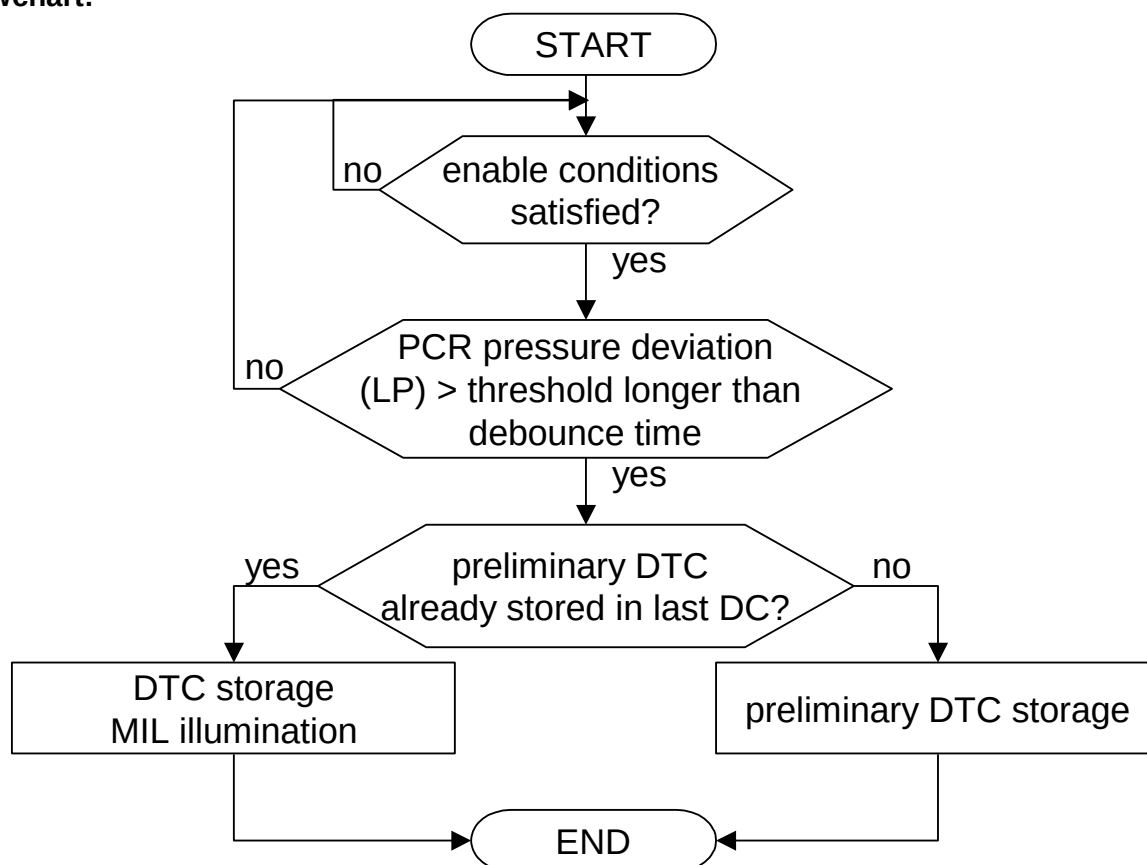
Flowchart:

**15.28.6 Manifold Absolute Pressure Regulation Low Stage****15.28.6.1 Functional check**

The functional check of the pressure control regulation (PCR) is based on the monitoring of the control deviation.

**15.28.6.1.1 Boost pressure governor deviation LP (maximum)
(P02CB)****General description:**

If the PCR (low pressure (LP)) control deviation is above the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:



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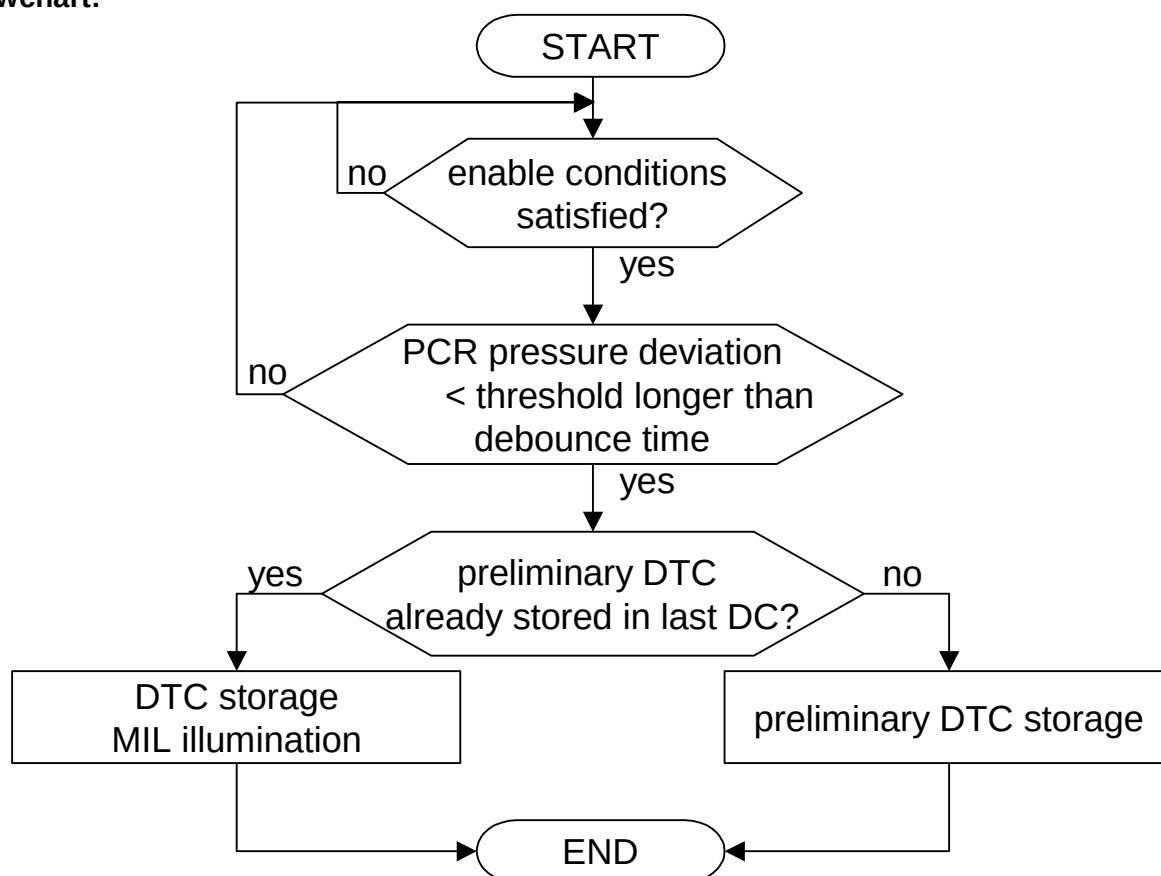
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15.28.6.1.2 Boost pressure governor deviation LP (minimum)
(P02CA)

General description:

If the PCR (low pressure (LP)) control deviation is below the calibrated threshold for longer than the calibrated time a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

Flowchart:



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15.29 Manifold Absolute Pressure Sensor

15.29.1 Circuit continuity (P0237, P0238)

General description:

If the raw voltage signal of the manifold absolute pressure sensor is above/below the applied thresholds or the interpreted raw voltage signal (hPA) is out of the applied range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

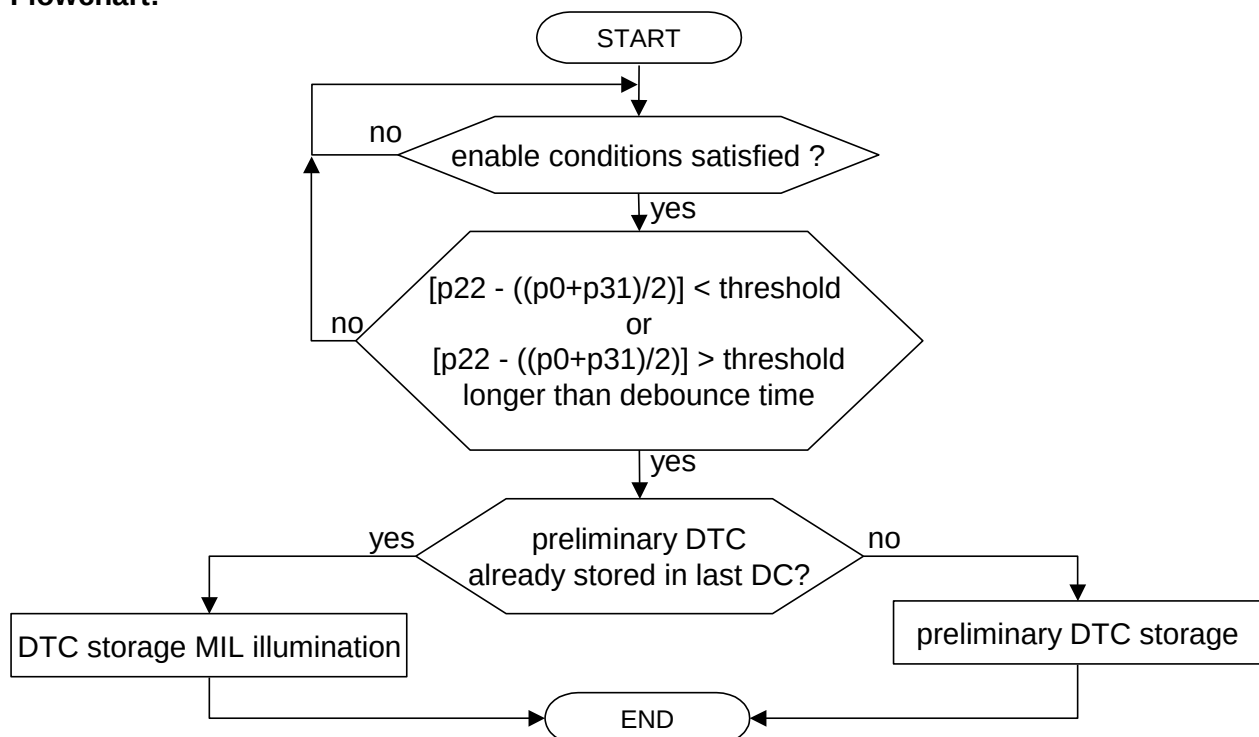
see 21.1.1Circuit continuity

15.29.2 Rationality check (P0069)

General description:

The plausibility check is executed with ignition on or in the afterrun. During the plausibility check of the manifold pressure sensor (p22) a deviation between the manifold pressure sensor and the other sensors (barometric pressure sensor (p0), exhaust manifold pressure sensor (p31)) is calculated. If the calculated deviation is above/below the thresholds for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This is a continuous diagnosis.

Flowchart:





15.30 Induction Air Temperature Sensor

15.30.1 Circuit continuity (P007C, P007D)

General description:

If the raw voltage signal of the induction air temperature sensor is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This is a continuous diagnosis.

Flowchart:

see 21.1.1 Circuit continuity

15.30.2 Rationality check (P007B)

General description:

A fault is detected via the cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated.

To enable the diagnosis the applicated engine off time has to be exceeded.

The induction air temperature sensor is compared to:

- ambient air temperature sensor
- exhaust temperature downstream EGR cooler
- exhaust temperature downstream EGR LP cooler(only X5 3.0sd)

Flowchart:

see 21.1.2 Cross-check of Temperature Sensors



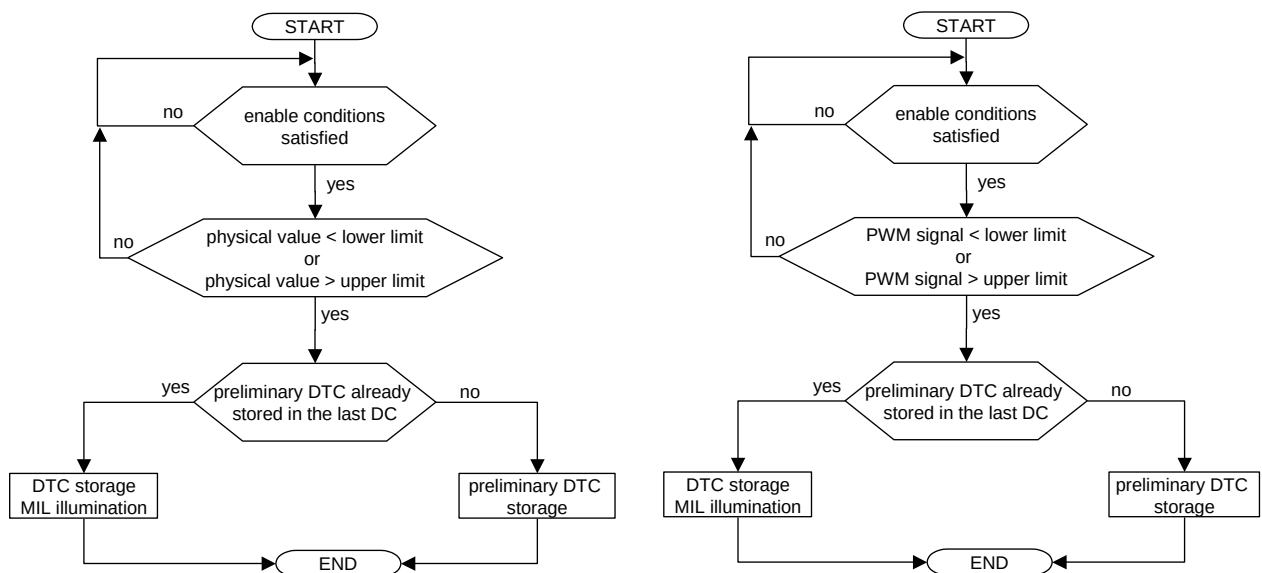
15.31 Mass Airflow Sensor

15.31.1 Circuit continuity (P0102, P0103)

General description:

If the PWM signal of the AFS sensor is above/below the applicated thresholds or the interpreted PWM signal (kg/h) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

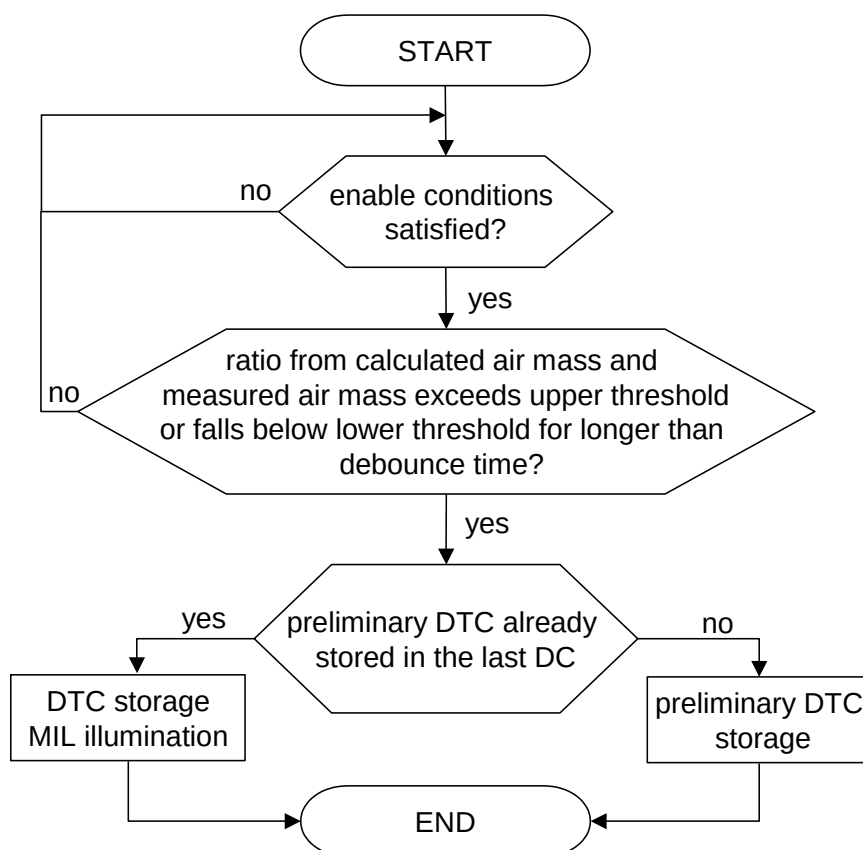


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15.31.2 Rationality check**15.31.2.1 Plausibility monitoring
(P0101)****General description:**

The measured air mass of the air flow sensor is monitored for plausibility by a calculated air mass value. Therefore a ratio of these two values is calculated. If that ratio exceeds a maximum threshold or falls below a minimum threshold a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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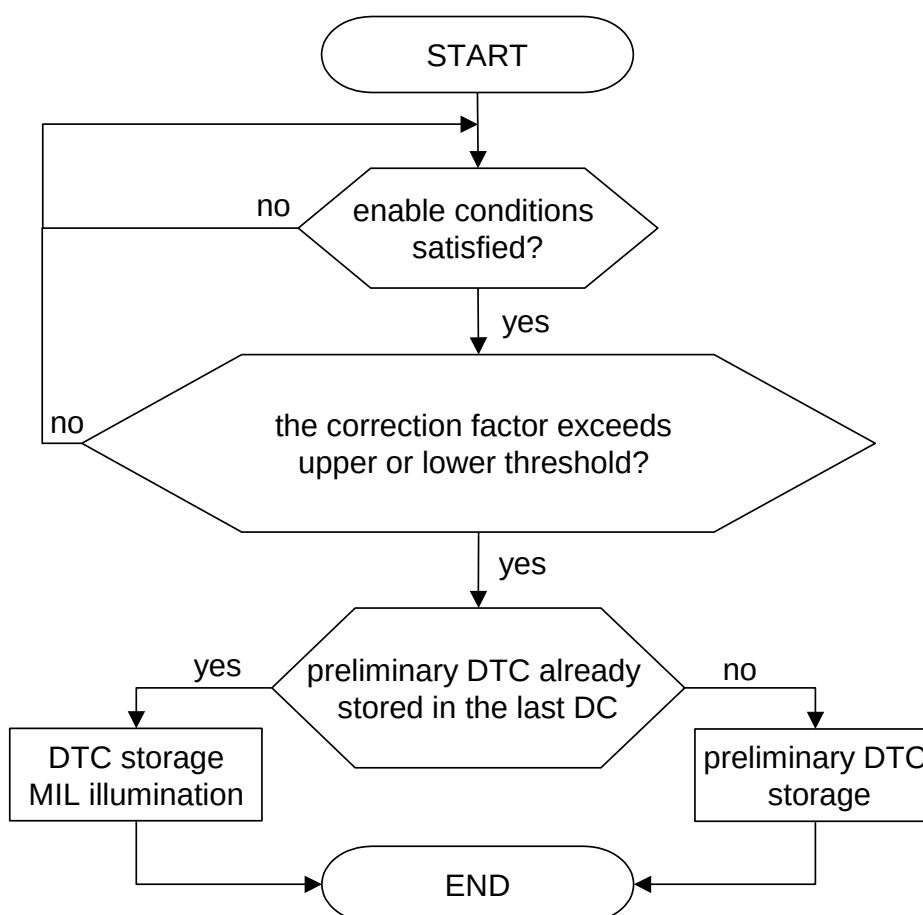
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**15.31.2.2 Signal adaption monitoring
(P0101)**

General description:

The correction factor for the AFS sensor is monitored. In specified engine operation points the AFS signal is compared to a calculated value. Based on that, a correction factor is calculated. If the calculated factor exceeds a maximum threshold or falls below a minimum threshold a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



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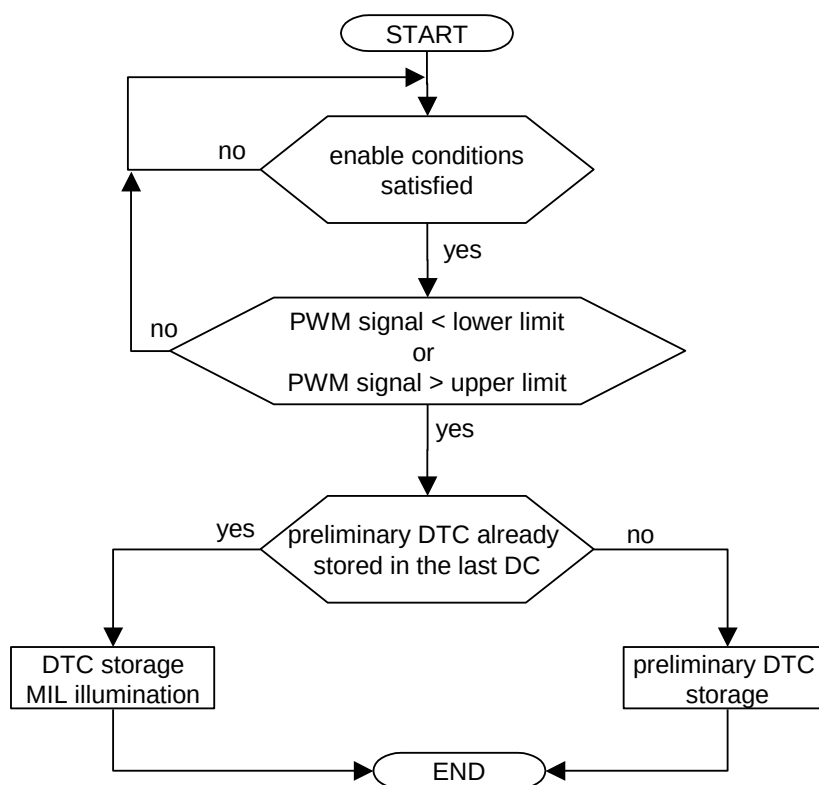
15.32 Mass Airflow Temperature Sensor

15.32.1 Circuit continuity (P0112, P0113)

General description:

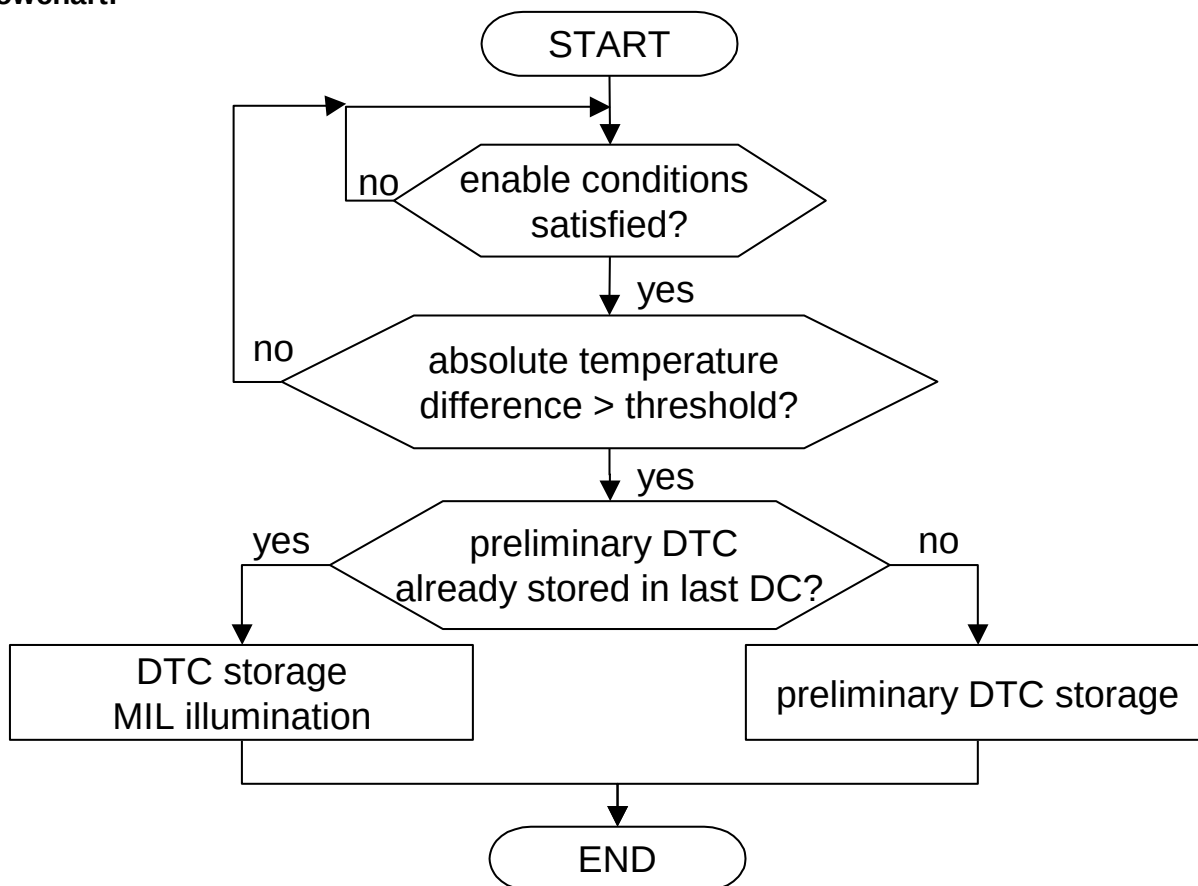
If the PWM signal of the mass airflow temperature sensor is above/below the applicated thresholds for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:



**15.32.2 Rationality check
(P009A)****General description:**

The diagnosis compares the mass airflow temperature value with the ambient air temperature value. If the absolute difference between the mass airflow temperature value and the ambient air temperature value is above the applied threshold a fault is detected and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles the MIL is illuminated. To enable the diagnosis a defined calculated airmass must have passed the mass airflow temperature sensor.

Flowchart:

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15.33 Exhaust Temperature Sensor Upstream DOC**15.33.1 Circuit continuity**
(P0545, P0546)**General description:**

If the raw voltage signal of the exhaust temperature sensor upstream DOC is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

see 21.1.1 Circuit continuity

15.33.2 Rationality check**15.33.2.1 Stuck in range high**
(P0425)**General description:**

There are two approaches to detect a stuck in range high temperature sensor.

In the first approach a fault is detected via cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles (CUC), the MIL is illuminated.

To enable the diagnosis, an applicated engine off time has to be exceeded.

The exhaust temperature sensor upstream DOC is compared to:

- exhaust temperature sensor upstream DPF
- exhaust temperature sensor upstream SCR

In the second approach a fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its positive threshold for longer than the applicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream DOC – modelled exhaust temperature upstream DOC

Due to the high temperatures and exothermic effects caused by the post injection during coldstart and regeneration the monitoring is deactivated during these conditions and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DOC and the exhaust temperature upstream DPF after regeneration the monitoring is delayed for a calibrated time.



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Flowchart:

see 21.1.2 Cross-check of Temperature Sensors

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**15.33.2.2 Stuck in range low
(P0425)****General description:**

A fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its negative threshold for longer than the applicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream DOC – modelled exhaust temperature upstream DOC

Due to the high temperatures and exothermic effects caused by the post injection during coldstart and regeneration the monitoring is deactivated during these conditions and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DOC and the exhaust temperature upstream DPF after regeneration the monitoring is delayed for a calibrated time.

Flowchart:

see 21.1.3 Rationality Check Low

**15.34 Particulate Matter Filter Differential Pressure Sensor
(P244A, P244B)****15.34.1 Circuit continuity****General description:**

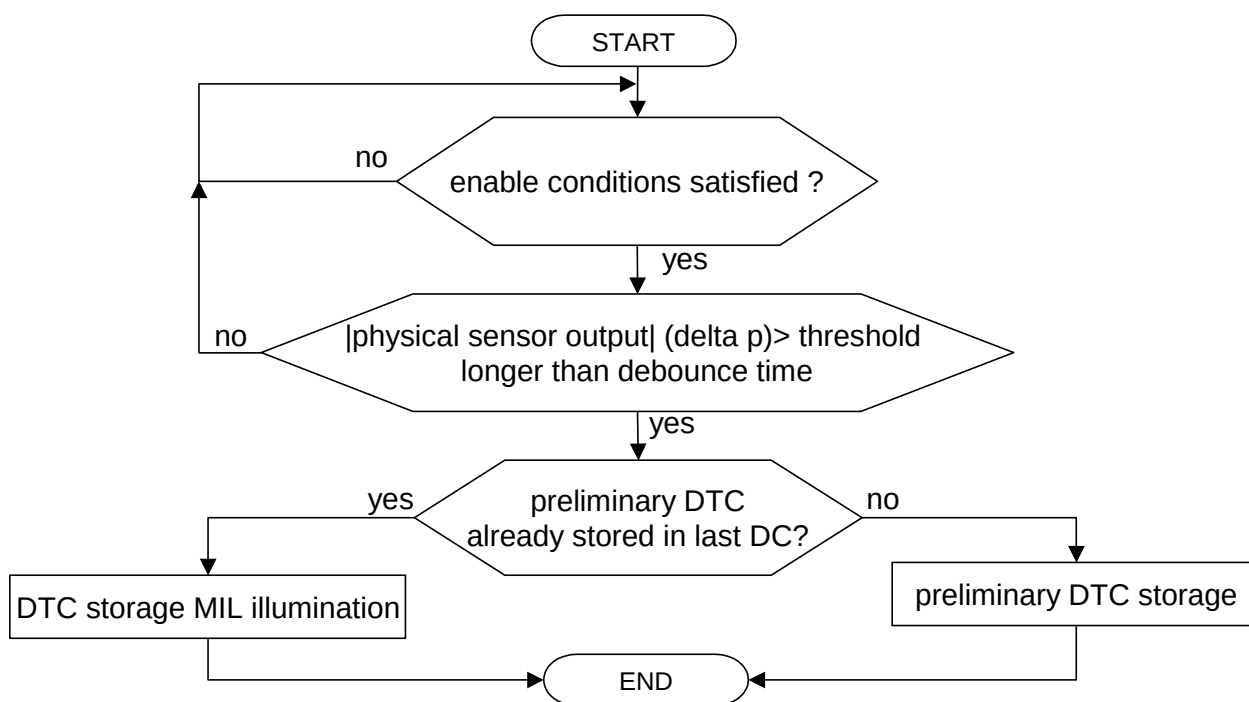
If the raw voltage signal of the differential pressure sensor is above/below the applicated thresholds or the interpreted raw voltage signal (hPa) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

see 21.1.1 Circuit continuity

**15.34.2 Rationality check (Offsettest of differential pressure sensor)
(P14A3)****General description:**

The diagnosis starts in the afterrun. During the afterrun the ECU compares the absolute raw differential pressure sensor value with a threshold. If the sum of the raw value is above the threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

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15.35 Exhaust Temperature Sensor Upstream DPF**15.35.1 Circuit continuity**
(P0545, P0546)**General description:**

If the raw voltage signal of the exhaust temperature sensor upstream DPF is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

see 21.1.1 Circuit continuity

15.35.2 Rationality check**15.35.2.1 Stuck in range high**
(P042A)**General description:**

There are two approaches to detect a stuck in range high temperature sensor.

In the first approach a fault is detected via cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles, the MIL is illuminated.

To enable the diagnosis, an applicated engine off time has to be exceeded.

The exhaust temperature sensor upstream DPF is compared to:

- exhaust temperature sensor upstream DOC
- exhaust temperature sensor upstream SCR

In the second approach a fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its positive threshold for longer than the applicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream DPF – modelled exhaust temperature upstream DPF

Due to the fact that the temperatures cannot be modelled adequately with the exothermic effects of the post injection during coldstart and regeneration the monitoring is inactive during regeneration and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DPF after regeneration the reactivation of the monitoring is delayed for a calibrated time.



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Flowchart:

see 21.1.2 Cross-check of Temperature Sensors

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**15.35.2.2 Stuck In Range Low
(P042A)**

General description:

A fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its negative threshold for longer than the appicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream DPF – modelled exhaust temperature upstream DPF

Due to the fact that the temperatures cannot be modelled adequately with the exothermic effects of the post injection during coldstart and regeneration the monitoring is inactive during regeneration and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DPF after regeneration the reactivation of the monitoring is delayed for a calibrated time.

Flowchart:

see 21.1.3 Rationality Check Low



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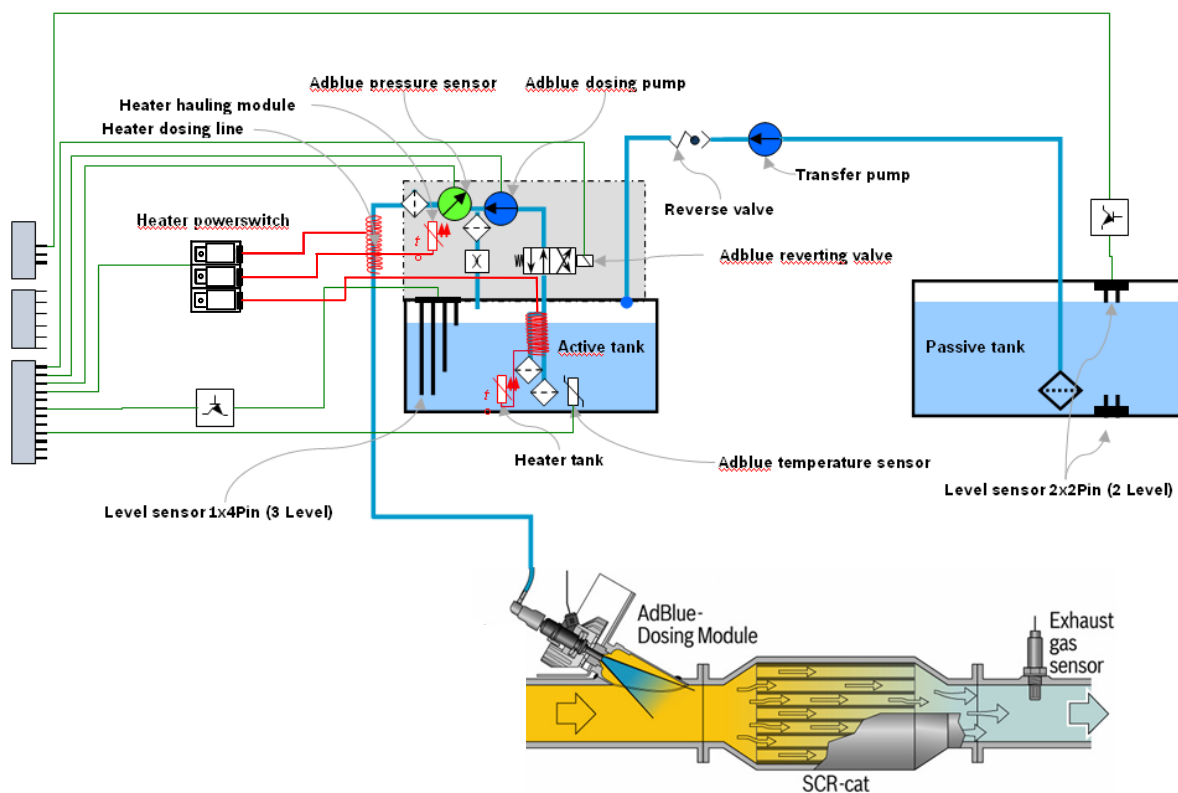
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15.36 Reductant Injection System



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15.36.1 Reductant Injection System Dosing Module

15.36.1.1 Circuit Continuity
(P2062, P2063, P2064)

General description:

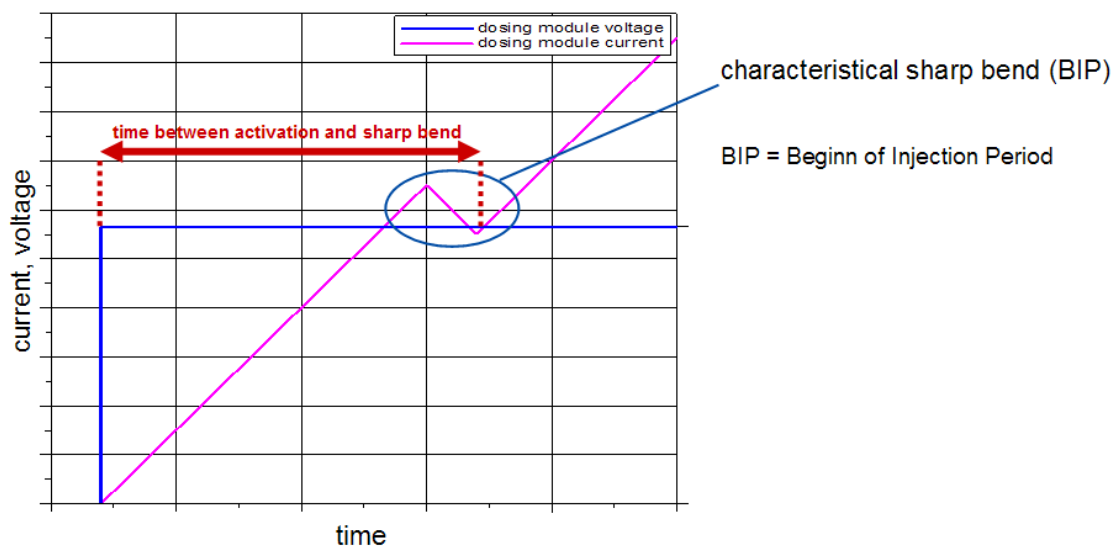
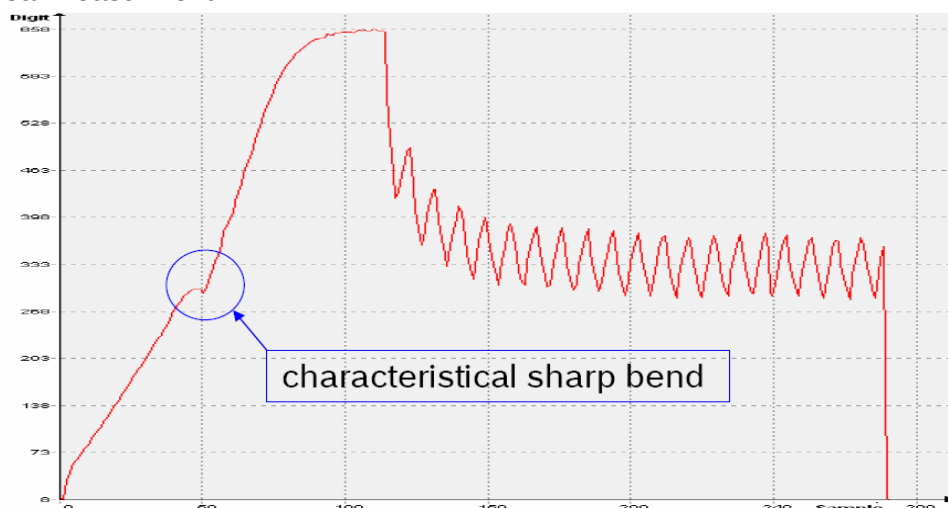
The dosing module power stage driver continuously and independently checks for short to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

see 21.1.1 Circuit continuity

**15.36.1.2 Rationality check**
(P208E)**General description:**

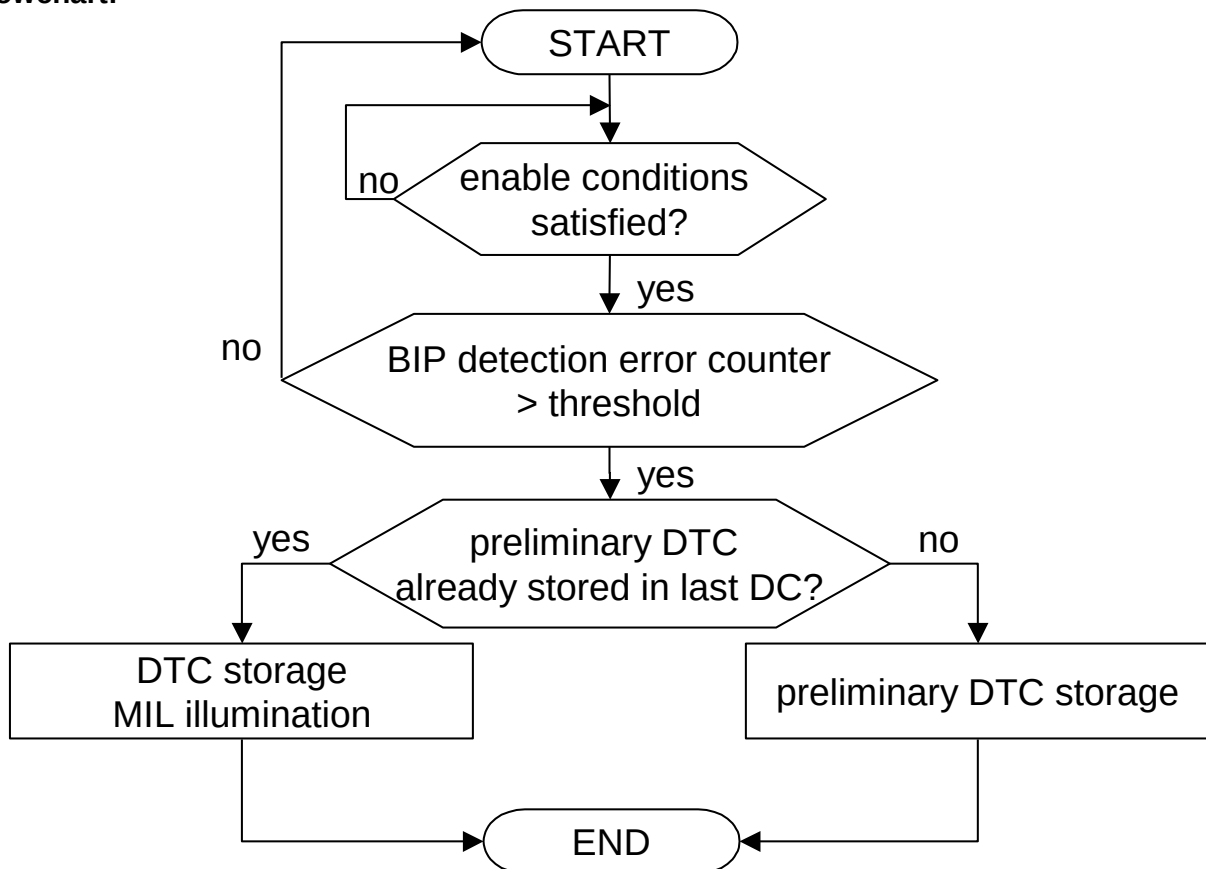
The monitor detects a non-opening dosing module via the current response of the module. During an opening process the time between the activation and the characteristically sharp bend (BIP) in the current profile is evaluated. The characteristically sharp bend results when the dosing module is fully opened and the needle in the module hit the limitation. If this bend is not identified or the current curve has a time delay due to the expected curve the component driver indicates a blocked dosing valve.

Schematic Overview:**Real Mesurment:**

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For monitoring the plausibility of the dosing module some enable conditions must be fulfilled. A fault is detected if the BIP (Begin of Injection Period) detection error counter is exceeded. The detection error counter counts if the dosing module did not open normally (without the characteristically sharp bend) in an opening processes. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

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15.36.2 Reductant Injection System Level Sensor Passive Tank**15.36.2.1 Tank Level Sensor Plausibility Monitoring for Passive Tank
(P21A9)****General description:**

The monitoring strategy is the same as for the active tank but the passive tank level sensor has only two single level sensors.

For further details see description and Flowchart of the active tank.

see 2.4.1 Tank Level Sensor Plausibility Monitoring for Active Tank

Following enable conditions have to be satisfied for this monitoring:

- engine is running
- ambient temperature above a specified threshold and

and

- engine off time above a specified threshold
- and urea tank temperature above a specified threshold
- and defrosting time of the pressure line above a specified threshold
- and ambient temperature above a specified threshold

or

- ambient temperature above a specified threshold
- and urea tank temperature above a specified threshold

**15.36.2.2 Tank Level Sensor Signal Monitoring for Passive Tank
(P21A8)****General description**

The monitoring strategie is the same as for the active tank but the passive tank level sensor has only two single level sensors.

For further details see description and Flowchart of the active tank.

see 2.4.2 Tank Level Sensor Signal Monitoring for Active Tank**Signal description**

There are two 2 pin level sensors mounted to the passive tank, which can show three levels, 0%, 55 % and 65 %, the monitoring strategy is compareable to the active tank monitoring strategy.
(see 2.4.2 Tank Level Sensor Signal Monitoring for Active Tank)

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The sens error P21A9 and the sens mon error P21A8 indicate the defect of an open circuit, shortcut to battery, shortcut to ground or shortcut between the sensors.

The level plaus error P21A9 indicates a plausibility fault of the sensor, here also the transformed raw sense signal is used.

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15.36.3 Reductant Injection System Pressure Line / Supply Module Heater**15.36.3.1 Circuit continuity****15.36.3.1.1 Power Stage**
(P20BD, P20BF, P20C0)**General description:**

The pressure line heater power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

see 21.1.1 Circuit continuity

15.36.3.1.2 Signal Range Check
(P20C0)**General description:**

If the raw voltage signal of the pressure line heater diagnostic line is above the applicated threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously. Only the maximum signal range check will detect because it is not usefull to detect the minimum value. If the heater is deactivated the diagnostic voltage will set to zero. In this case the system always should set the minimum signal range check error.



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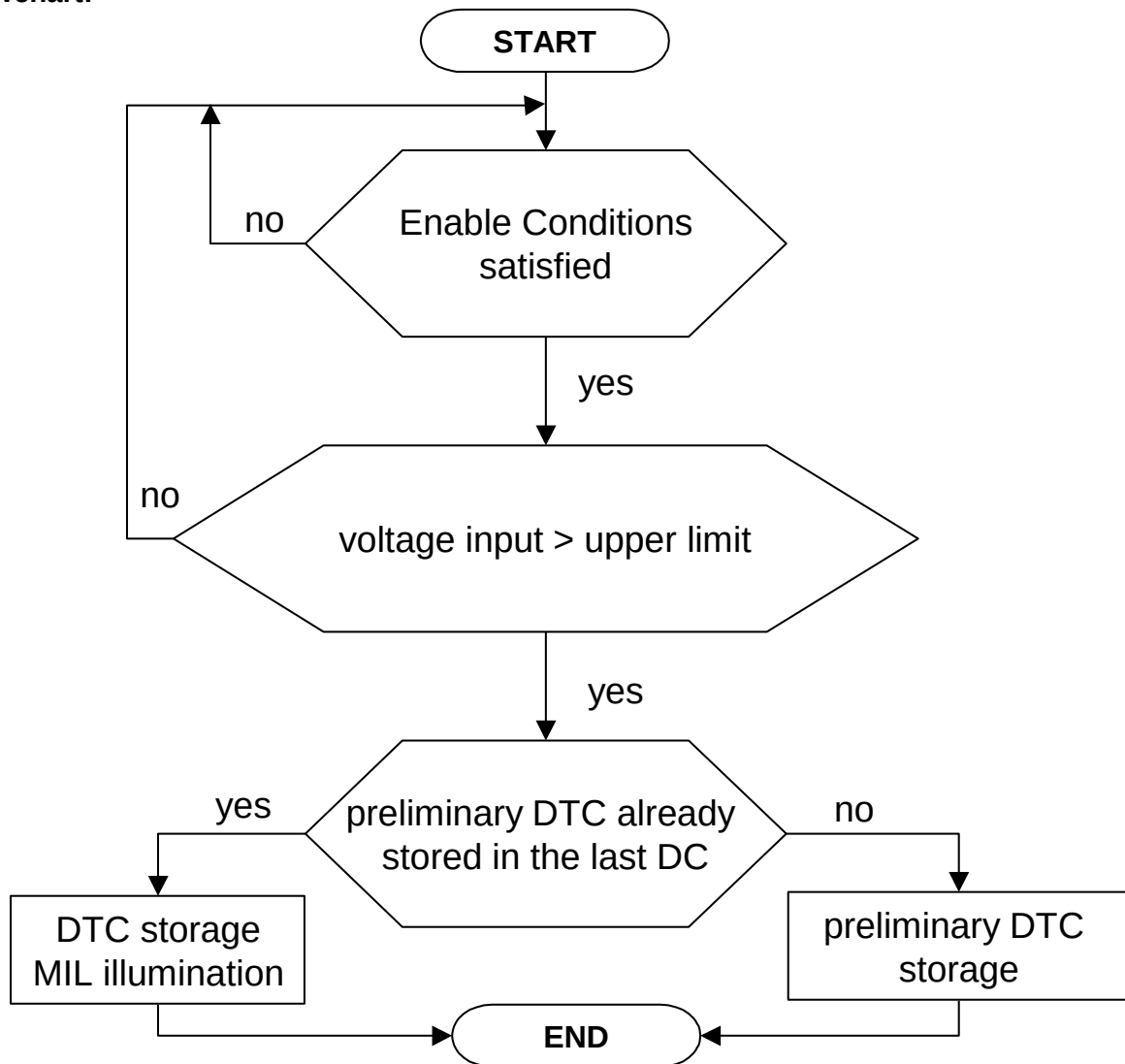
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Flowchart:



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15.36.3.2 Rationality Check**General description:**

For ensuring the functionality of the urea system in case of cold temperatures and a frozen urea tank it is necessary to heat the tank, the pressure line and the supply module to get a liquid medium for dosing. A diagnostic line exists for the monitoring of the pressure line and the supply module heater. The diagnostic line measures a diagnostic current. This current is monitored due to a maximum value in case of a deactivated heater and a minimum value in case of an activated heater. The measured current is divided with the measured voltage to get the conductance. Additionally it is possible to detect a short cut in a range of the tank PTC. After switching on the PTC heater a peak appears before the Power/conductivity is regulated to a constant value. In this range of the peak detection the system can detect a shortcut if the power/conductivity exceeds a specified threshold. If a fault is detected a preliminary DTC is stored. If a fault is detected in two consecutive driving cycles, the MIL is illuminated.

15.36.3.2.1 Current monitoring while pressure line heater is not active (P20B5)

The fault is detected if the heater is switched off and the diagnostic current is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Enable conditions: the pressure line heater must be switched off



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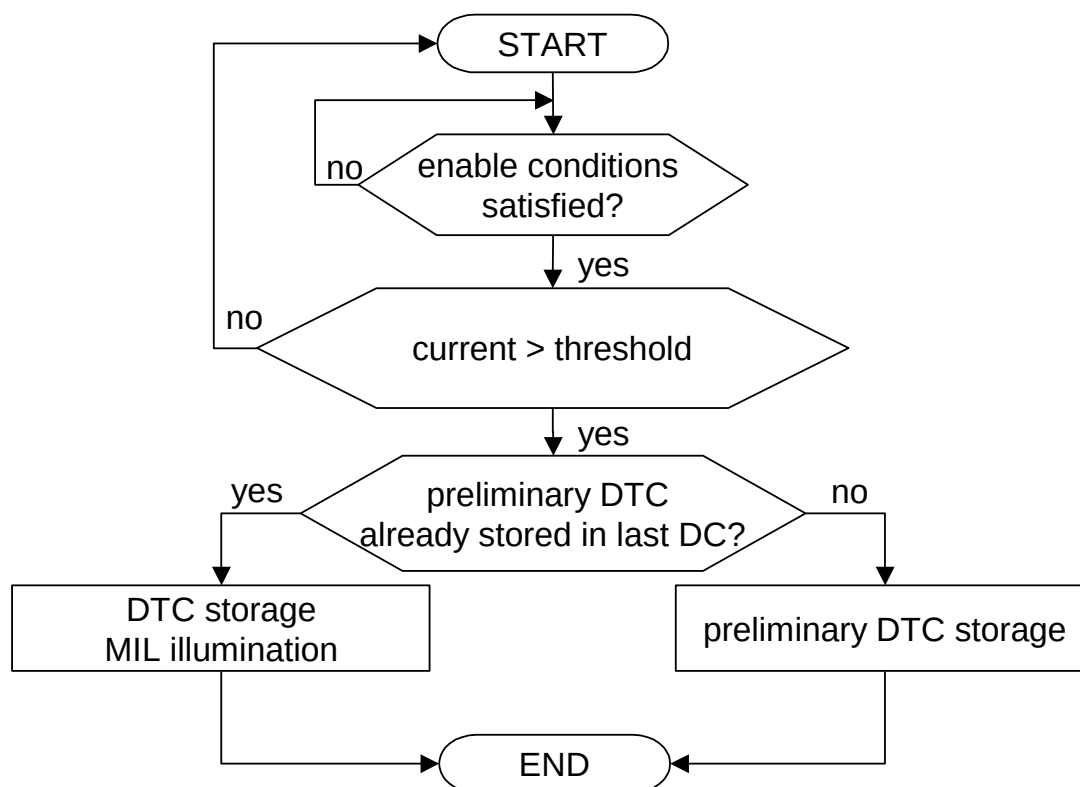
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Flowchart:

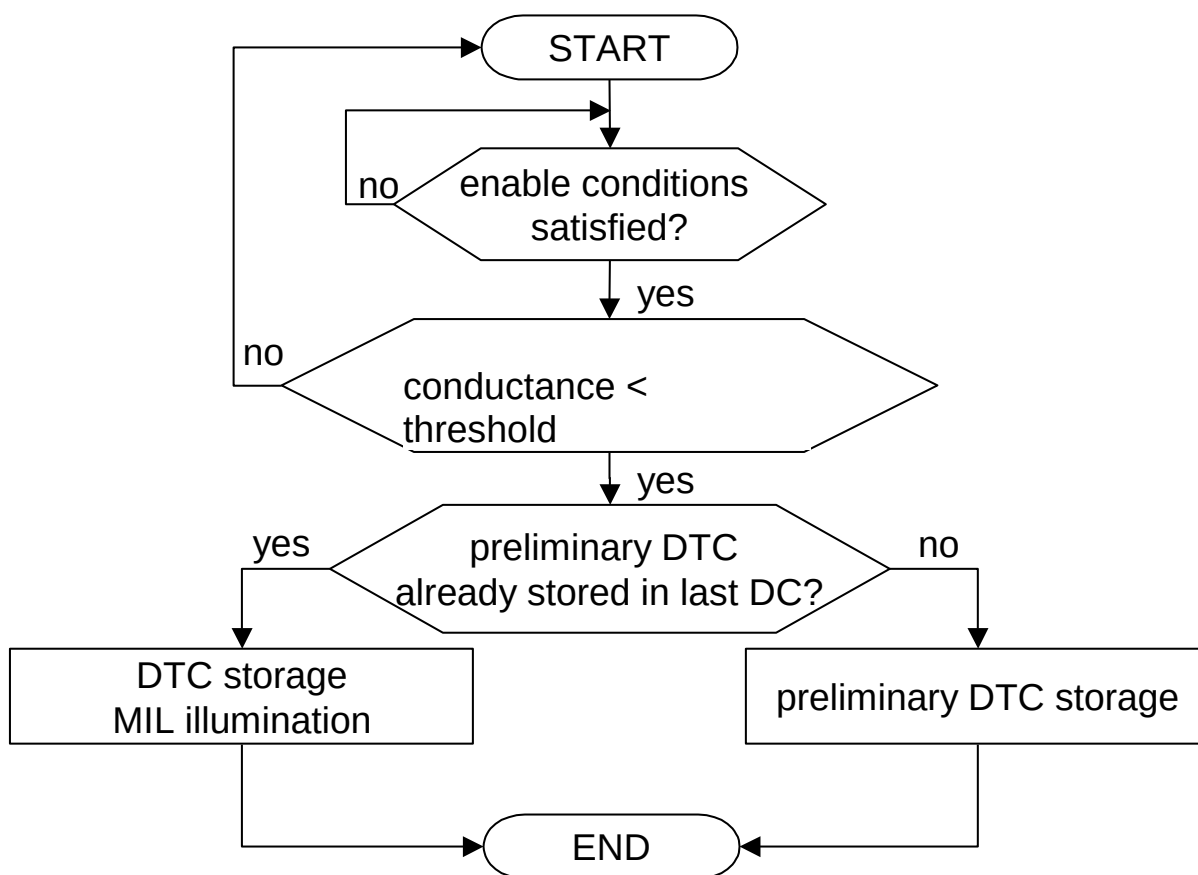


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**15.36.3.2.2 Conductivity monitoring while pressure line heater is active**
(P20B6)**General description:**

This fault is detected if the heater is switched on and the diagnostic conductivity is lower than a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Enable conditions: the pressure line heater must be active

Flowchart:



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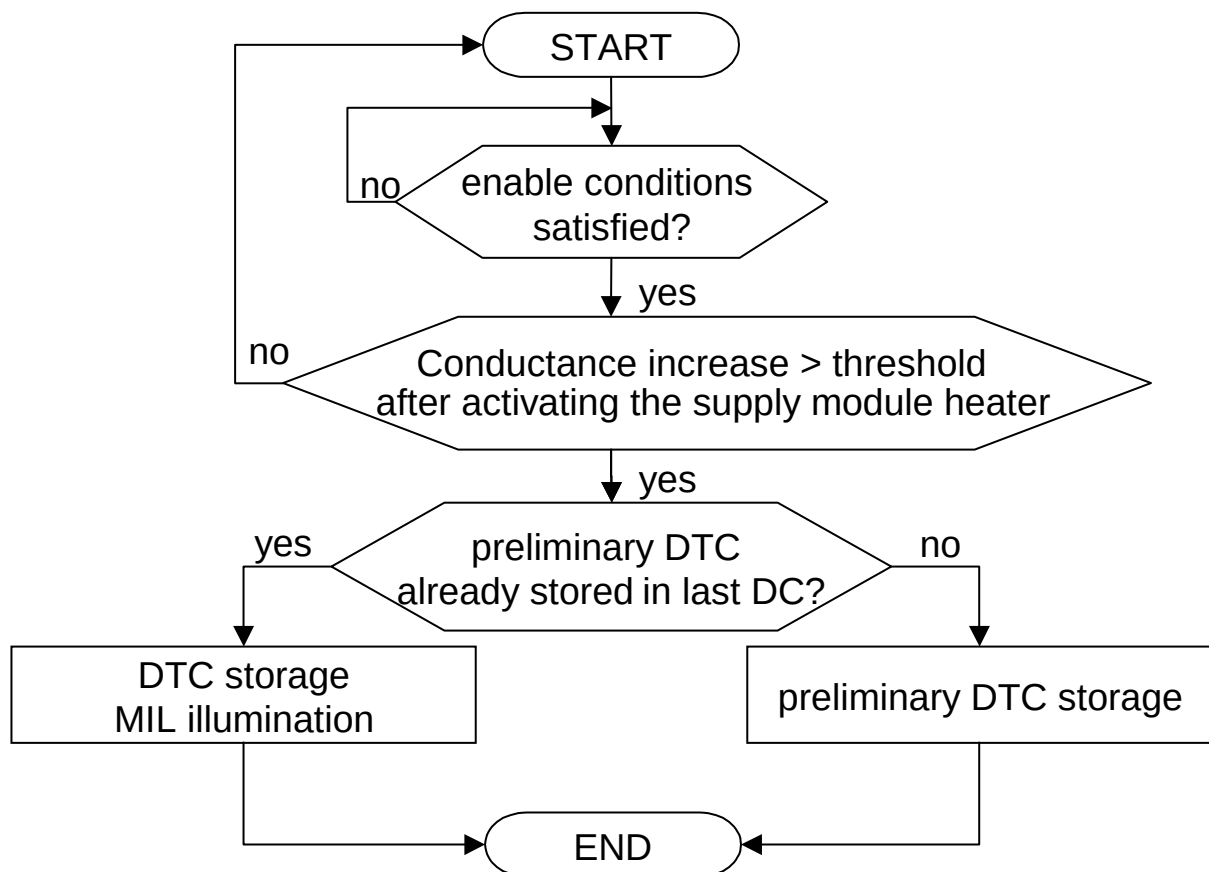
Issued:11/04/09

15.36.3.2.3 Monitoring of supply module heater regarding short cut interruption when supply module heater is activated.

(P20B8)

General description:

This diagnosis monitors the conductivity increase after activating the supply module heater. The fault is detected if the conductivity increase is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart



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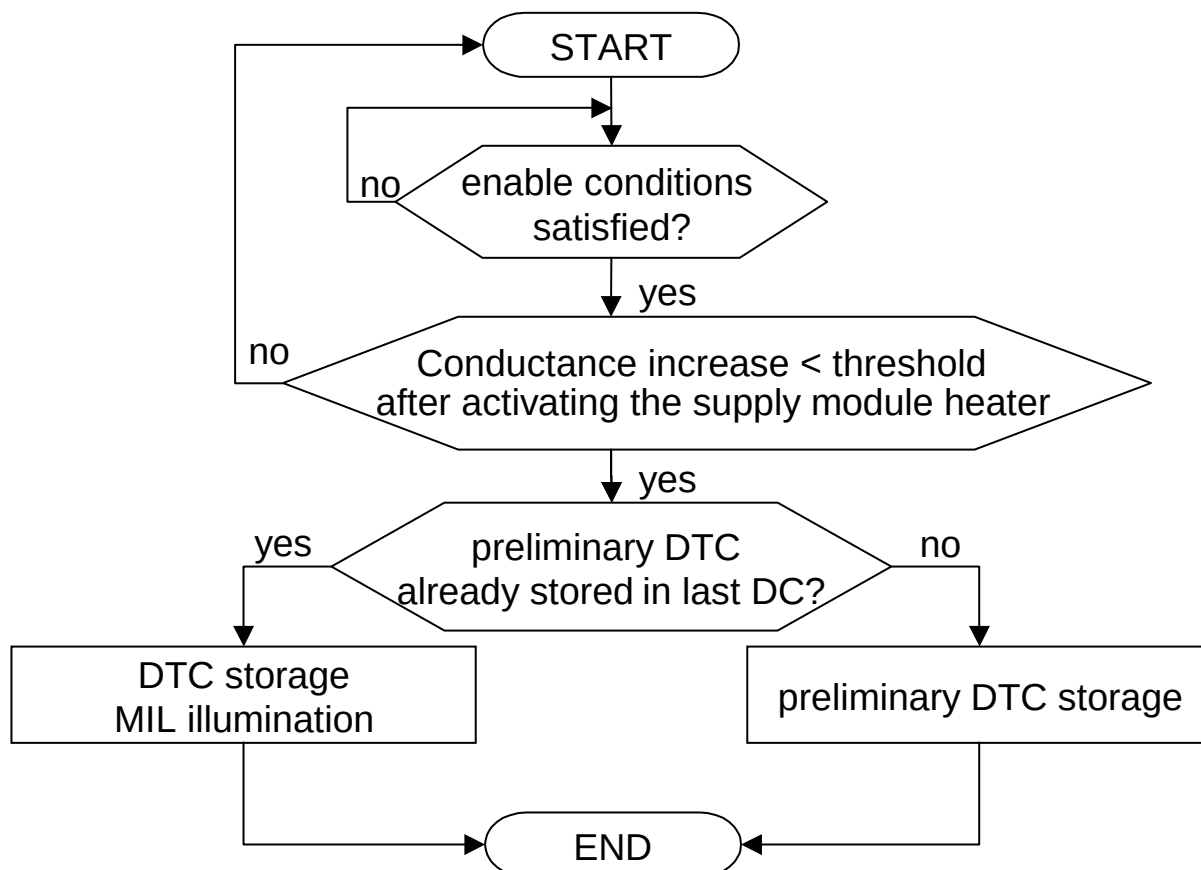
Issued:11/04/09

15.36.3.2.4 Monitoring of supply module heater regarding open circuit when supply module heater is activated.

(P20B5)

General description:

This diagnosis monitors the conductivity increase after activating the supply module heater. The fault is detected if there is no conductivity increase after activating the supply module heater. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart



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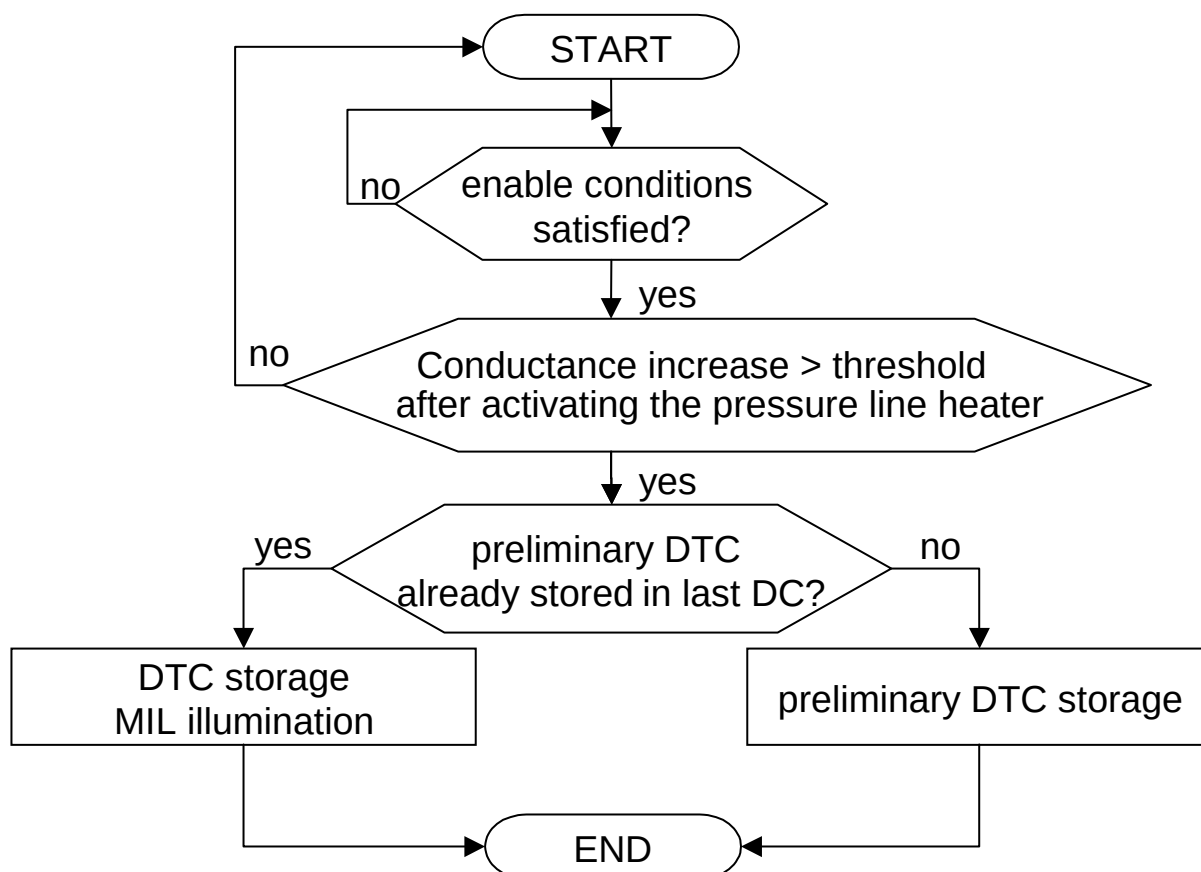
Issued:11/04/09

15.36.3.2.5 Monitoring of pressure line heater regarding short cut interruption when pressure line heater is activated.

(P10DB)

General description:

This diagnosis monitors the conductivity increase after activating the pressure line heater. The fault is detected if the conductivity increase is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart



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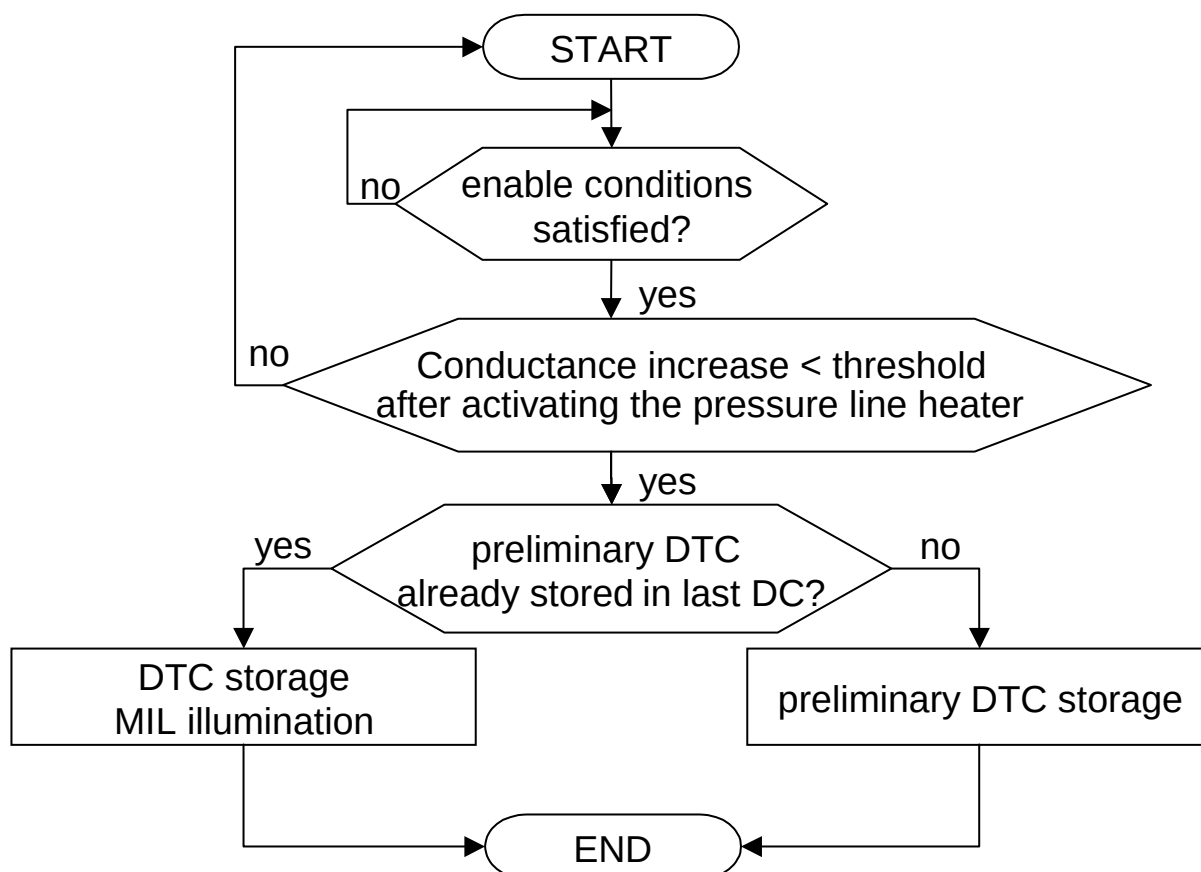
Issued:11/04/09

15.36.3.2.6 Monitoring of pressure line heater regarding open circuit when pressure line heater is activated.

(P10DA)

General description:

This diagnosis monitors the conductivity increase after activating the pressure line heater. The fault is detected if there is no conductivity increase after activating the pressure line heater. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart

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15.36.4 Reductant Injection System Pressure Pump**15.36.4.1 Circuit continuity****15.36.4.1.1 Power Stage**
(P208A, P208C, P208D)**General description:**

The pressure pump power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

see 21.1.1 Circuit continuity

15.36.4.1.2 Physical Range
(P204D)**General description:**

The fault path physical range has to check the pressure of the reductant injection system due to a maximum threshold in the whole operating states also for protection of the urea quality system. The fault is detected if the pressure exceeded a specified maximum threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.



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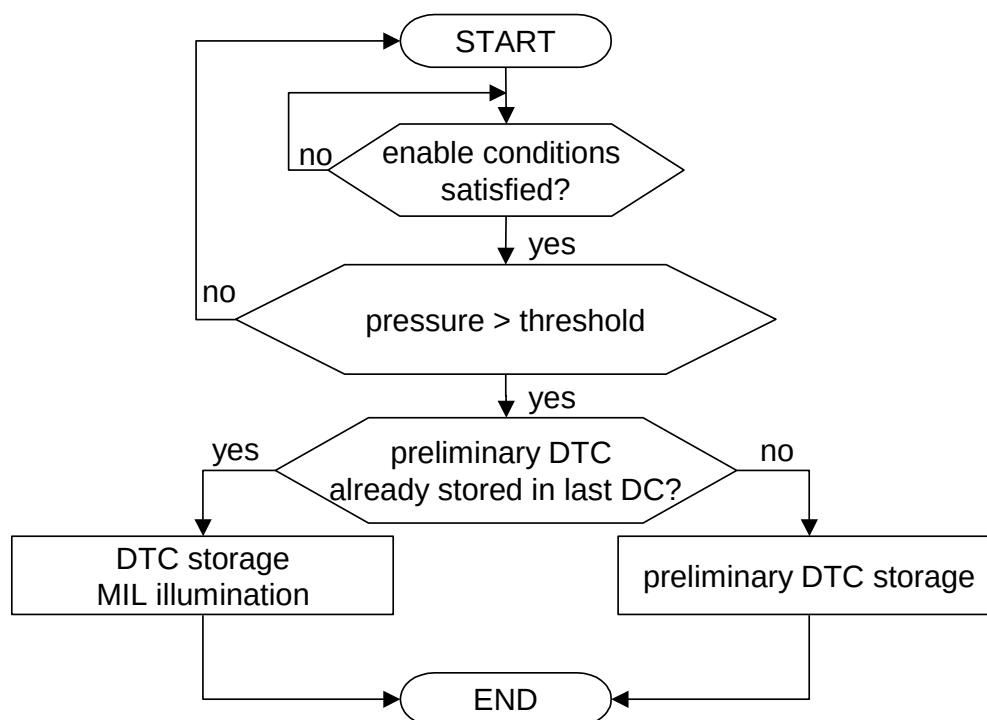
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Flowchart:



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15.36.5 Reductant Injection System Pressure Sensor**15.36.5.1 Circuit Continuity**

(P204C, P204D)

General description:

If the raw voltage signal of the pressure sensor is above/below the applied thresholds or the interpreted raw voltage signal (hPa) is out of the applied range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously

Flowchart:

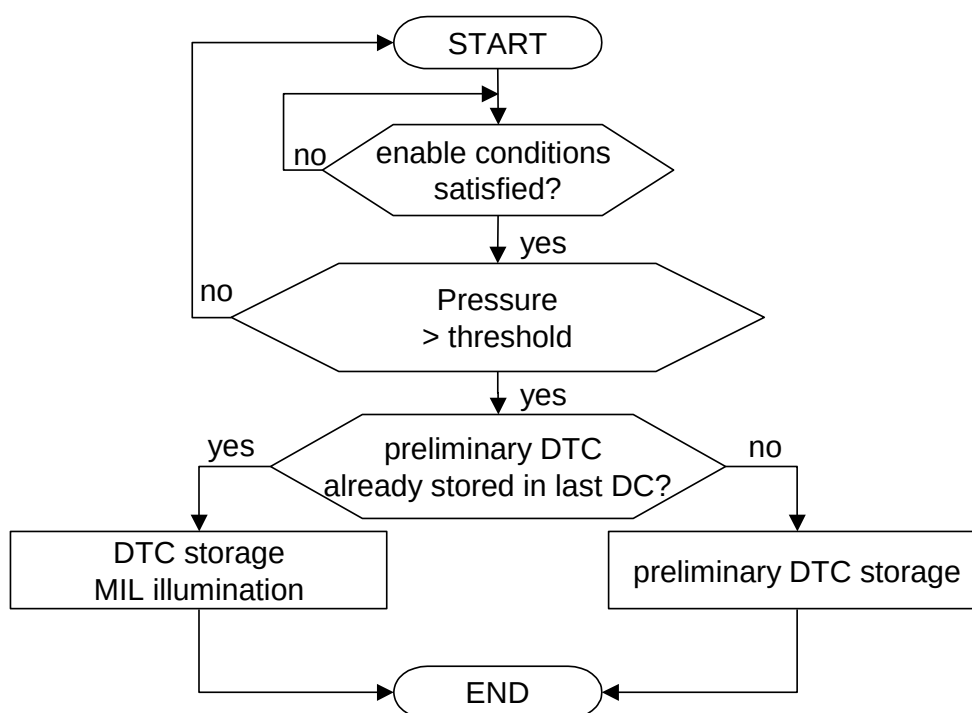
see 21.1.1 Circuit continuity

15.36.5.2 Rationality check

(P204B)

General description:

This diagnosis monitors a plausibility fault of the urea quality system pressure sensor. The enabled conditions are satisfied if the urea system is unloaded and unpressurized. The system is unpressurized if the catalyst temperature is below a specified threshold. A fault is detected if the pressure is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:**CONFIDENTIAL**

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15.36.6 Reductant Injection System Reverse Control Valve**15.36.6.1 Circuit Continuity**

(P20A0, P20A2, P20A3)

General description:

The reverse control valve power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

see 21.1.1 Circuit continuity

15.36.6.2 Rationality check

(P20A5)

General description:

In this function the plausibility of the reverse valve is checked. The monitoring checks in engine afterrun an urea pressure reduction. This is only possible, if the reverse control valve opens. Therefore a successful pressure build up during engine running is necessary. If the pressure build up is not successful P20E8 is set.

If the enable conditions are fulfilled the urea pressure value is stored. In case of a pressure reduction (due to the opened reverse control valve) the ECU will check the pressure difference between start value and current value. A fault is detected if the pressure difference is below a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.



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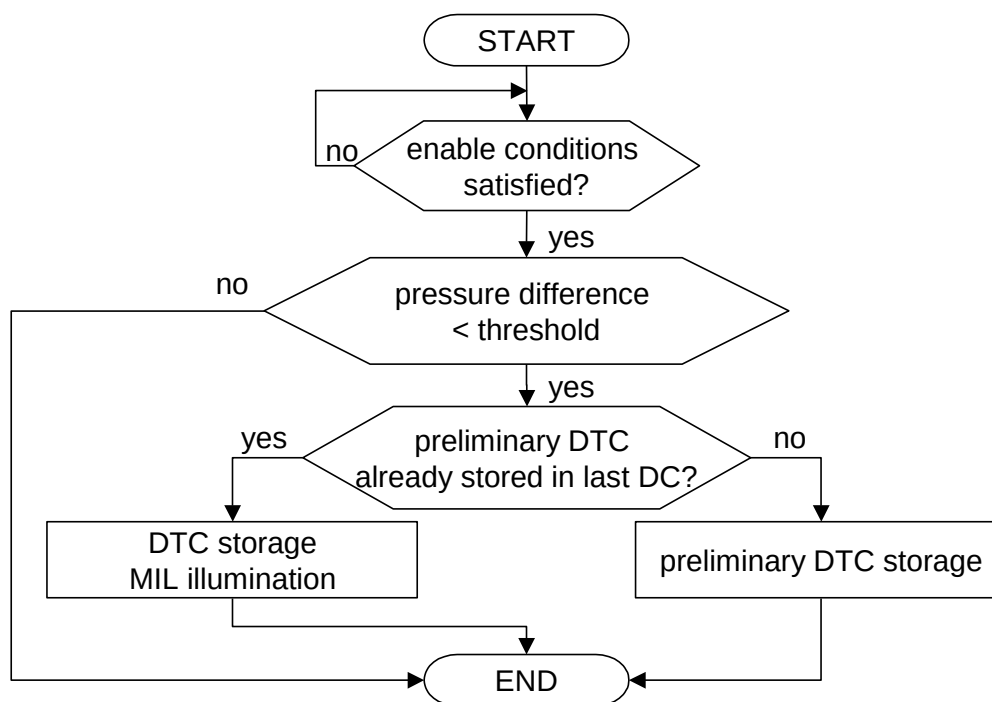
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Flowchart:



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15.36.7 Reductant Injection System Tank Heater**15.36.7.1 Circuit continuity****15.36.7.1.1 Power Stage**
(P202A, P202B, P202C)**General description:**

The tank heater power stage driver continuously and independently checks for short circuit to ground, short circuit to battery and open circuit faults. If a fault is detected a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated

Flowchart:

see 21.1.1 Circuit continuity

15.36.7.1.2 Signal Range Check
(P202A)**General description:**

If the raw voltage signal of the tank heater diagnostic line is above the applicated threshold for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously. Only the maximum signal range check will detect because it is not usefull to detect the minimum value. If the heater is deactivated the diagnostic voltage will set to zero. In this case the system always should set the minimum signal range check error.



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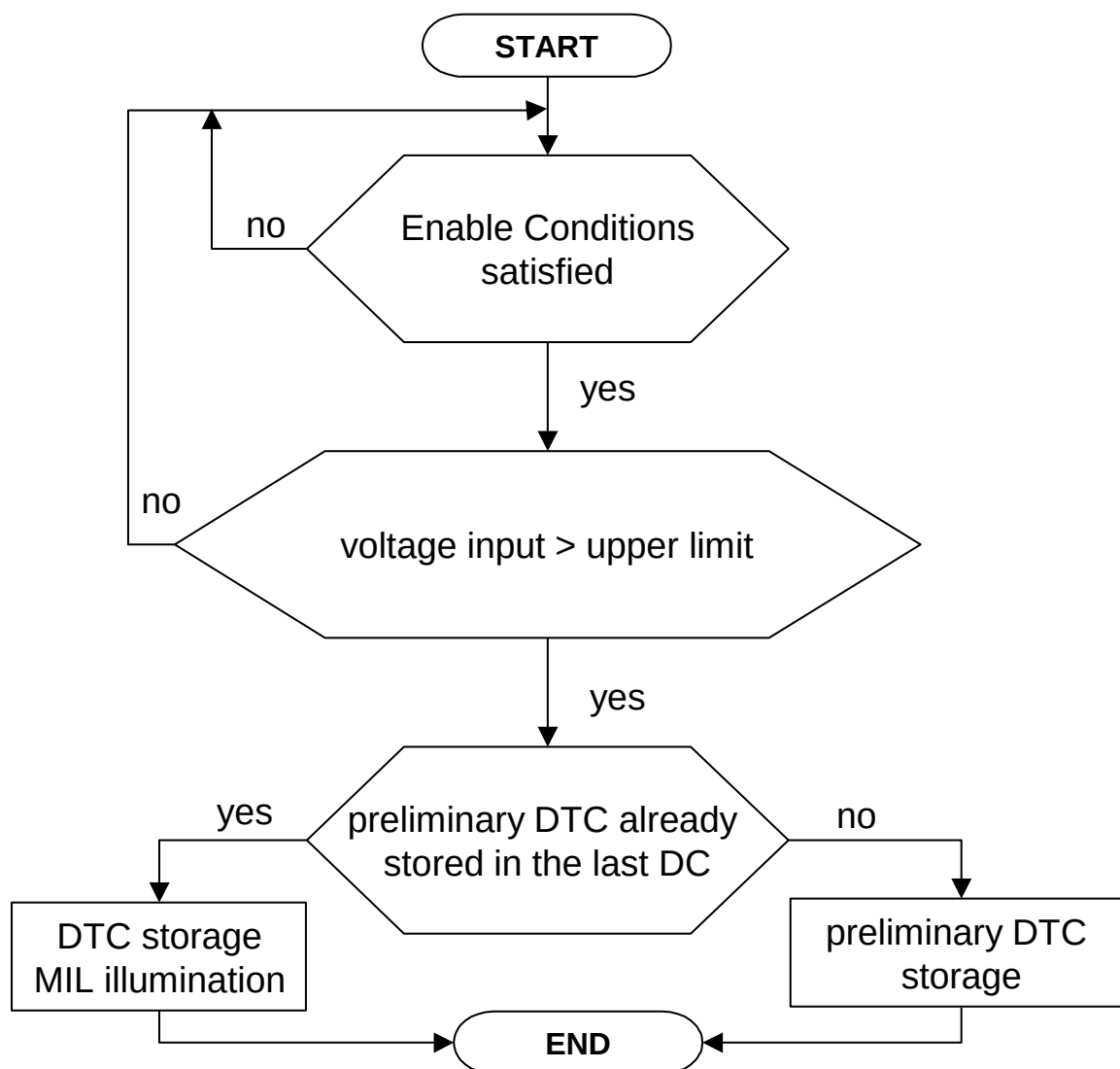
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Flowchart:



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**15.36.7.2 Rationality Check****General description:**

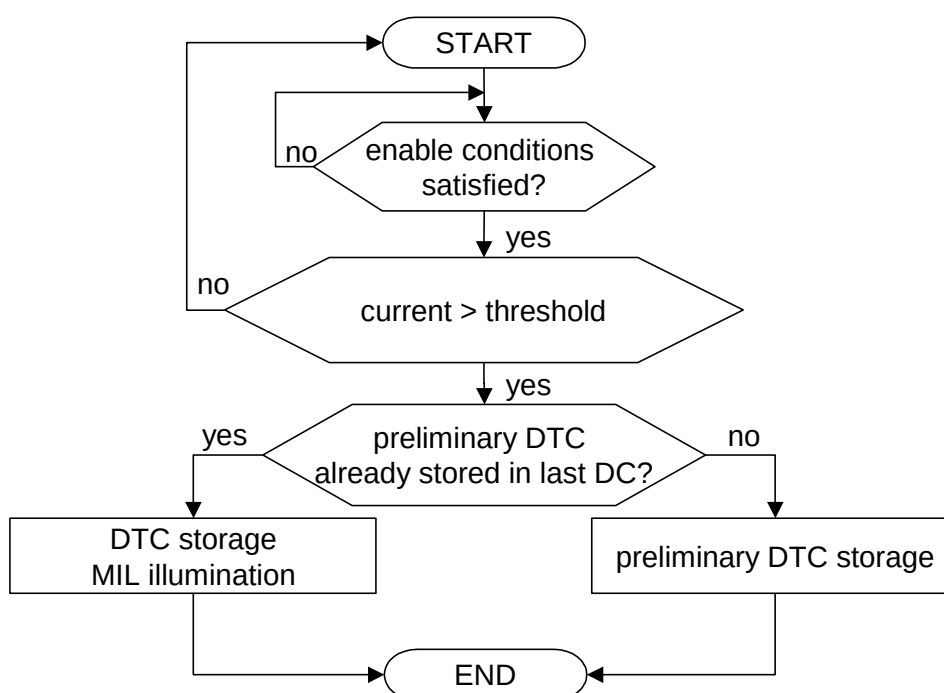
For ensuring the functionality of the urea system in case of cold temperatures and a frozen urea tank it is necessary to heat the tank to get a liquid medium for dosing. A diagnostic line exists for the monitoring of the tank heater. The diagnostic line measures a diagnostic current. This current is monitored due to a maximum value in case of a deactivated heater and a minimum value in case of an activated heater.

After switching on the heater the system regulates to a constant power/conductivity value without a peak. Therefore it is only possible to detect a short cut in case of an active tank heater but no open load. In this case it is possible to detect a shortcut if the power/conductivity exceeds a specified threshold.

**15.36.7.2.1 Current monitoring while urea tank heater is off
(P202A)**

This fault is detected if the heater is switched off and the diagnostic current is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

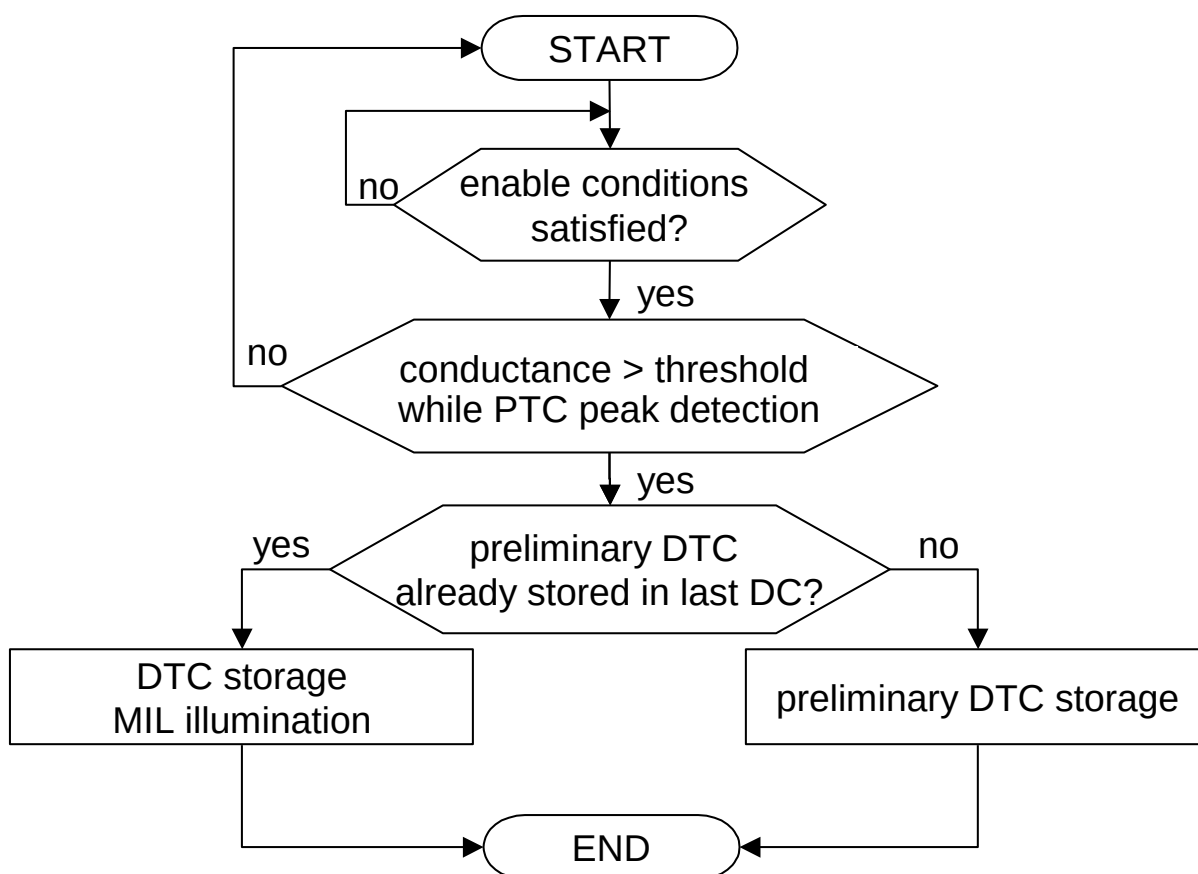
Enable conditions: the urea tank heater must be switched off

Flowchart:

**15.36.7.2.2**
(P202C)**Monitoring of urea tank heater short circuit while PTC peak detection****General description:**

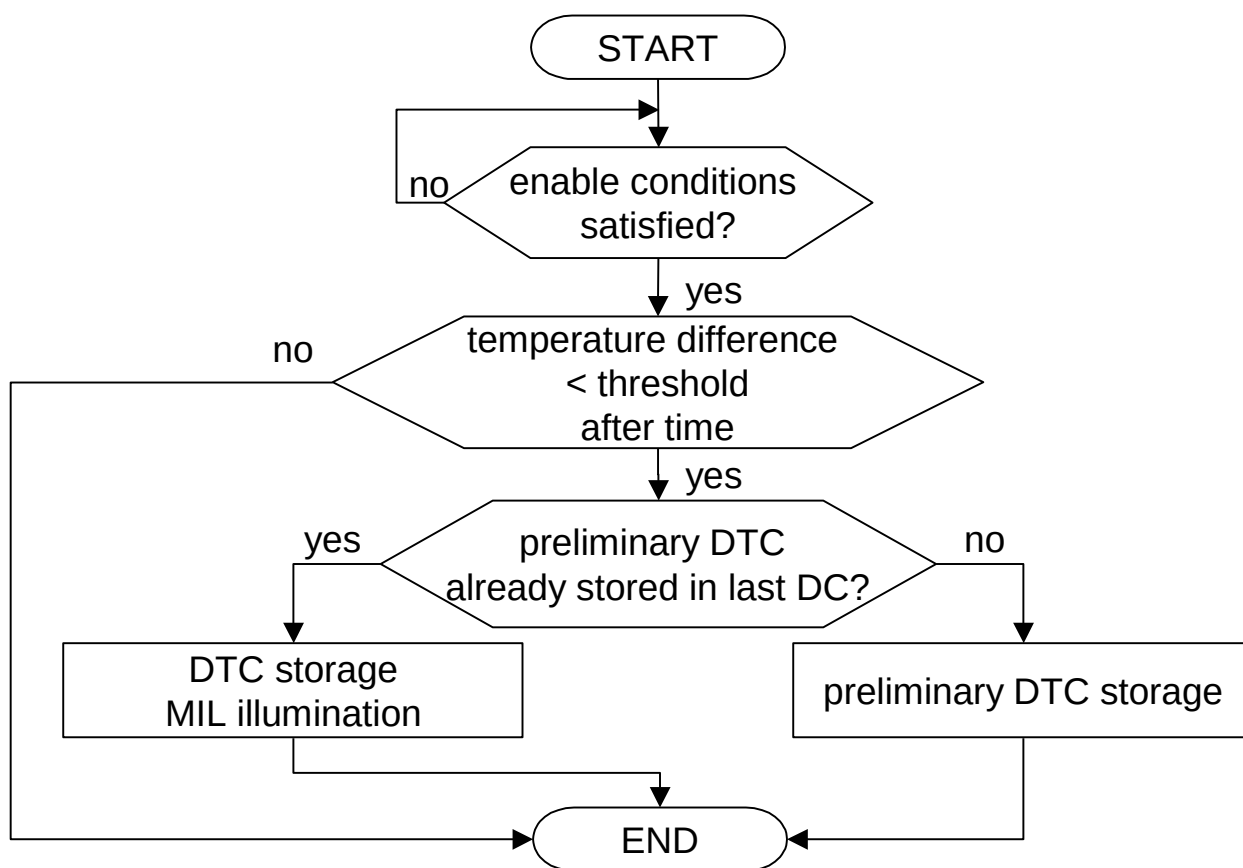
This fault monitors the conductivity in a range of the PTC peak detection. This fault is detected in the range of the PTC peak detection and if the conductivity is above a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Enable conditions: the tank heater must be active and the conductivity peak must be detected

Flowchart:

**15.36.7.2.3 Monitoring of urea tank heater plausibility
(P209F)****General description:**

The function monitors the increase of the urea tank temperature in case of heating. The monitoring will be executed if there is no fault due to the heater or temperature sensor. In case of heating a timer starts and the start value of the tank temperature is saved. A fault is detected when the timer is elapsed and the temperature difference between start value and current value is below a specified threshold. In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:

**15.36.8 Reductant Injection System Temperature Sensor****15.36.8.1 Circuit Continuity**

(P205C, P205D)

General description:

If the raw voltage signal of the urea tank temperature sensor is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:

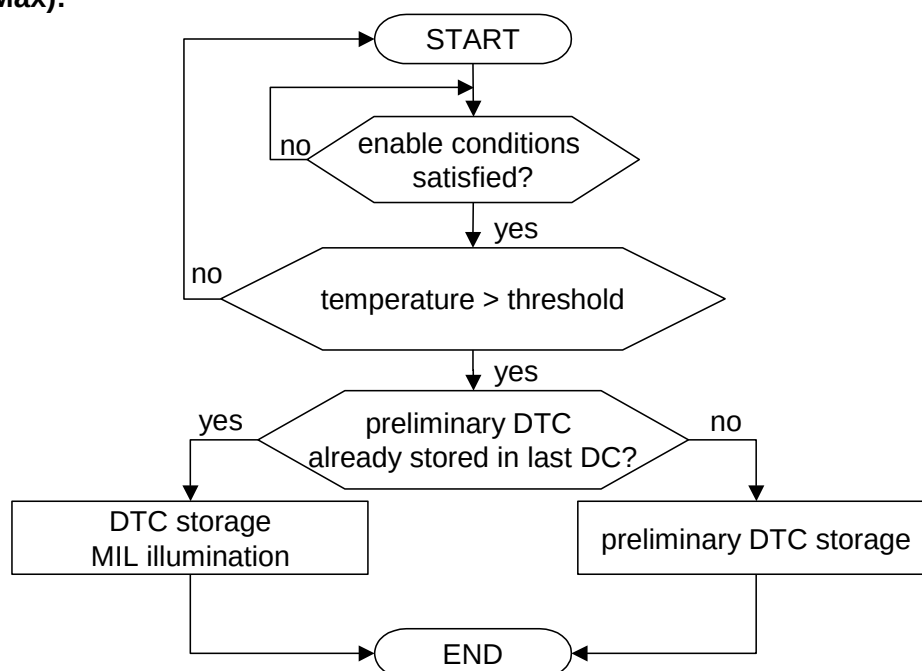
see 21.1.1 Circuit continuity

15.36.8.2 Rationality check**15.36.8.2.1 Plausibility of the temperature sensor (max)**

(P205B)

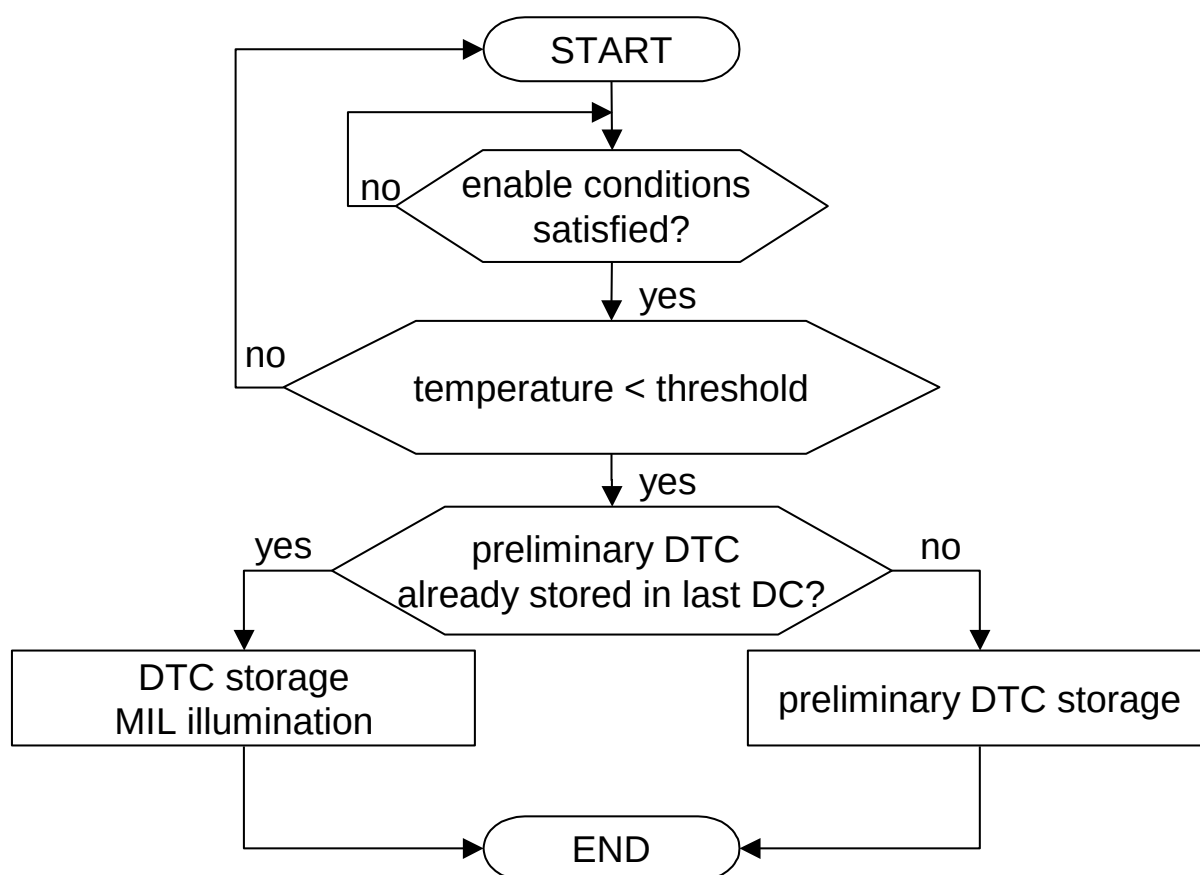
General description:

The diagnosis monitors a plausibility fault of the urea tank temperature sensor. The fault is monitored in case of cold start condition (The ambient air temperature, the engine coolant temperature and the SCR catalyst temperature have to be in a specified range). A fault is detected if the temperature deviation between urea tank temperature and ambient temperature is above a specified threshold (positive). In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart (Max):**CONFIDENTIAL**

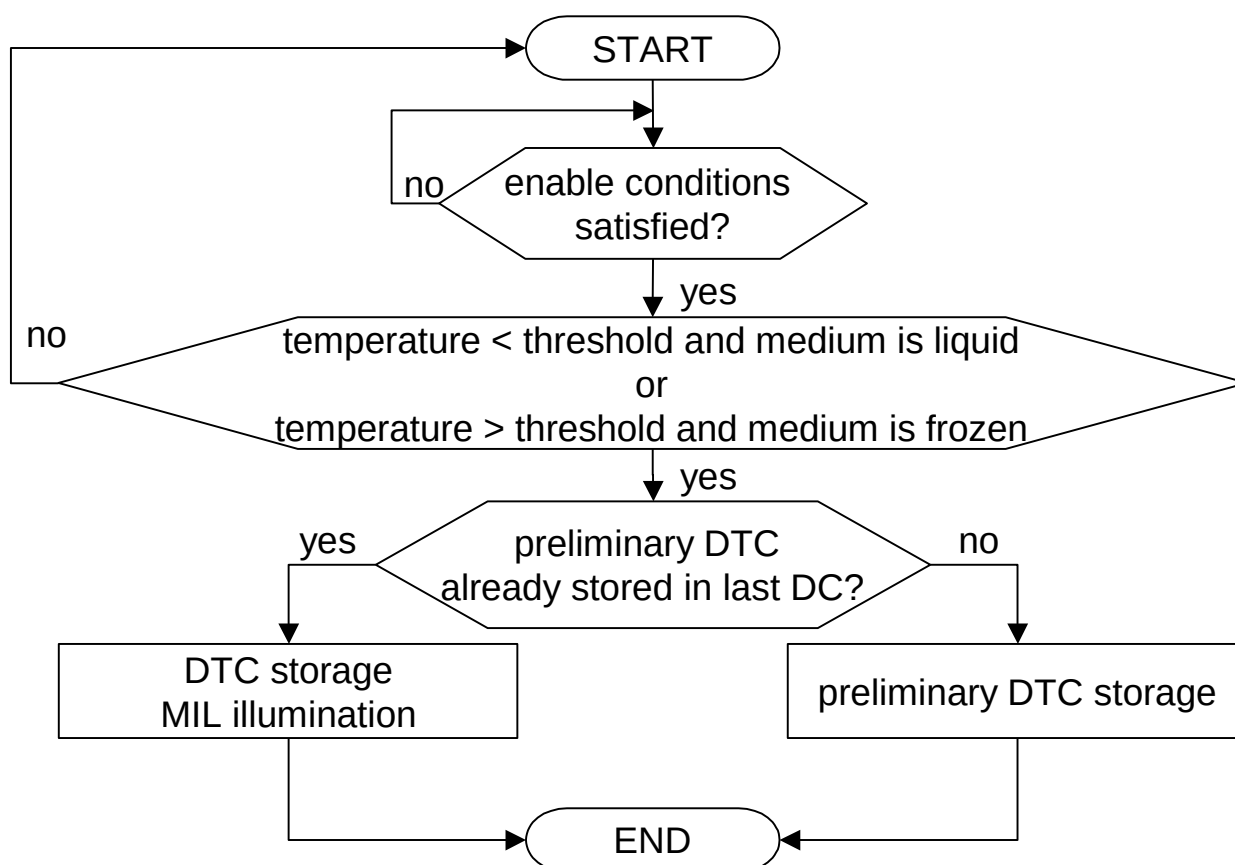
**15.36.8.2.2 Plausibility of the temperature sensor (min)**
(P205B)**General description:**

The diagnosis monitors a plausibility fault of the urea tank temperature sensor. The fault is monitored in case of cold condition, so the ambient air temperature, the engine coolant temperature and the SCR catalyst temperature have to be in a specified range. Also the system must be unpressurized, to secure the start plausibility conditions (cold urea injection system). A fault is detected if the temperature deviation between urea tank temperature and environmental temperature is below a specified minimum threshold (negative). In this case, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart (Min):

**15.36.8.2.3 Plausibility check of the Temperature tensor**
(P205B)**General description:**

This diagnosis monitors a plausibility fault of the urea tank temperature sensor. The fault is monitored in case of cold condition, so the ambient air temperature, the engine coolant temperature and the SCR catalyst temperature have to be in a specified range. Also the CAN bus must be active and the calculated level of the tank must be above a specified value. A fault is detected if the temperature is below a specified threshold and a valid tank level is detected (medium is liquid). A fault is also detected if the temperature is above a specified threshold and no valid tank level is detected (medium is frozen). In this cases, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

Flowchart:



15.37 Exhaust Temperature Sensor Upstream SCR

15.37.1 Circuit continuity

(P242D, P242C)

General description:

If the raw voltage signal of the exhaust temperature sensor upstream SCR is above/below the applicated thresholds or the interpreted raw voltage signal (°C) is out of the applicated range for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

see 21.1.1 Circuit continuity

15.37.2 Rationality check

15.37.2.1 Stuck in range high

(P242A)

General description:

There are two approaches to detect a stuck in range high temperature sensor.

In the first approach a fault is detected via cross checking of several temperature sensor values. During the cross checking phase every sensor value is compared to every other sensor value (within the function) and the difference to each sensor value is calculated. If the absolute difference of one sensor value to all other sensor values is above the applicated thresholds for each sensor value comparison a fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles, the MIL is illuminated.

To enable the diagnosis, an applicated engine off time has to be exceeded.

The exhaust temperature sensor upstream SCR is compared to:

- exhaust temperature sensor upstream DOC
- exhaust temperature sensor upstream DPF

In the second approach a fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its positive threshold for longer than the applicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream SCR – modelled exhaust temperature upstream SCR

Due to the fact that the temperatures cannot be modelled adequately with the exothermic effects of the post injection during coldstart and regeneration the monitoring is inactive during regeneration and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DPF after regeneration the reactivation of the monitoring is delayed for a calibrated time.

Flowchart:

21.1.2 Cross-check of Temperature Sensors

15.37.2.2 Stuck in range low

(P242A)

General description:



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A fault is detected via the comparison of two temperatures. If the difference between these temperatures exceeds its negative threshold for longer than the applicated time a fault is detected, and a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles (CUC) the MIL is illuminated.

The compared temperatures are the:

- exhaust temperature upstream SCR – modelled exhaust temperature upstream SCR

Due to the fact that the temperatures cannot be modelled adequately with the exothermic effects of the post injection during coldstart and regeneration the monitoring is inactive during regeneration and a calibrated time after engine start.

Regarding the cooling down of the exhaust temperature upstream DPF after regeneration the reactivation of the monitoring is delayed for a calibrated time.

Flowchart:

21.1.3 Rationality Check Low



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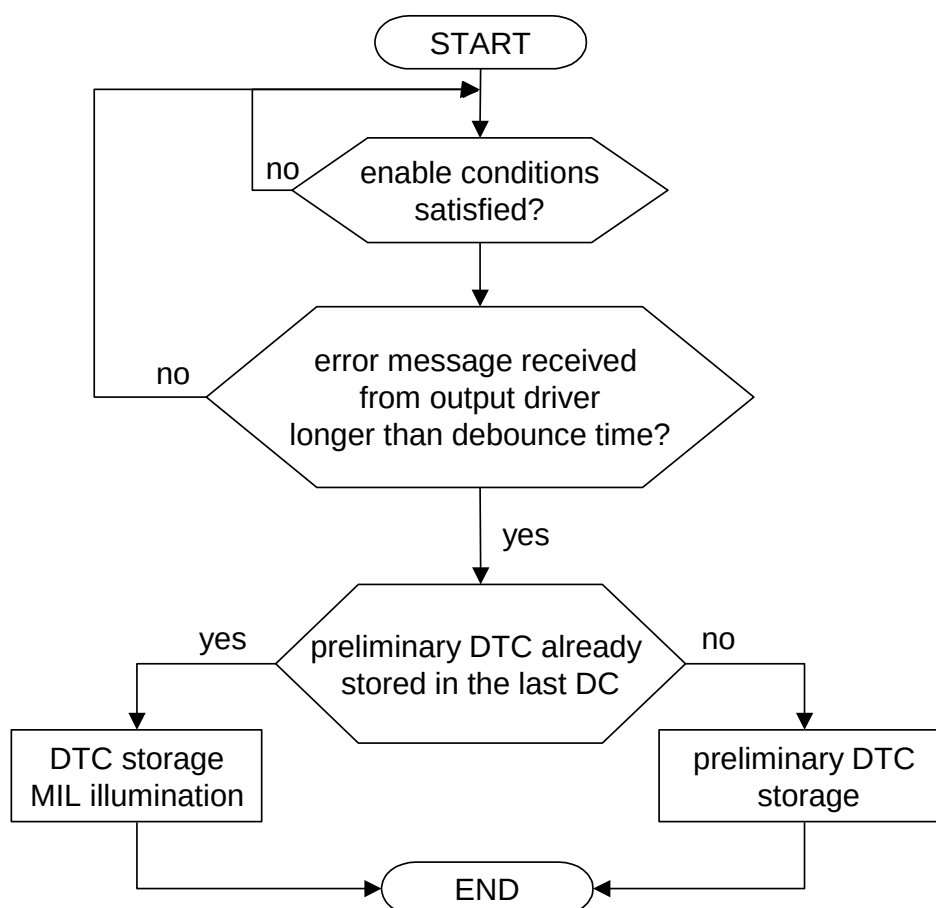
15.38 Sensor Supply Voltage

15.38.1 Circuit continuity
(P0641, P0651, P0697)

General description:

The sensor supply potential is monitored by a comparator in the hardware. The diagnosis is a power stage internal diagnosis. If the supply potential is below/above a threshold the comparator detects an error. If this error is present for longer than the allowed time, a preliminary DTC is stored. If this fault is detected in two consecutive driving cycles, the MIL is illuminated. This diagnosis is performed continuously.

Flowchart:



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15.39 Swirl Valve

15.39.1 Circuit continuity (P2008, P200A, P2009, P2010)

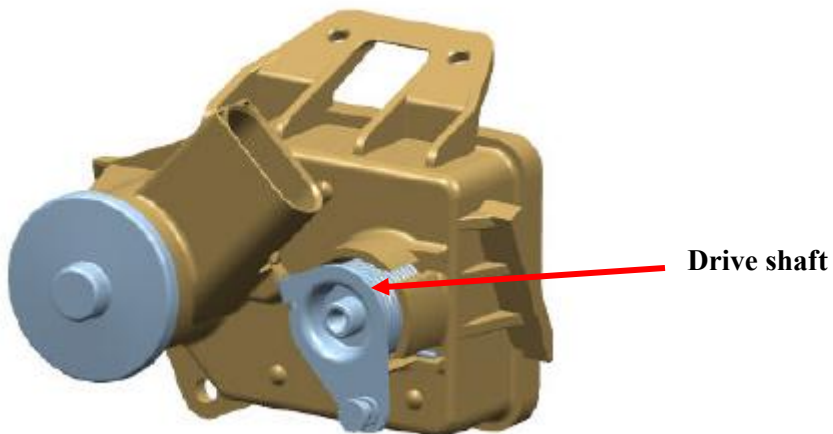
General description:

The swirl valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature from the power stage. It can also detect governor deviation, controller errors and communication errors of the swirl valve.

If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

The Swirl Valve actuator is a so called “smart device”. It is an actuator with an integrated position sensor. If the smart device detects a governor deviation (e.g. caused by a jammed flap) a failure is reported to the engine ECU applying a defined pulse duty factor.

The position sensor is positioned inside the device on the drive shaft:





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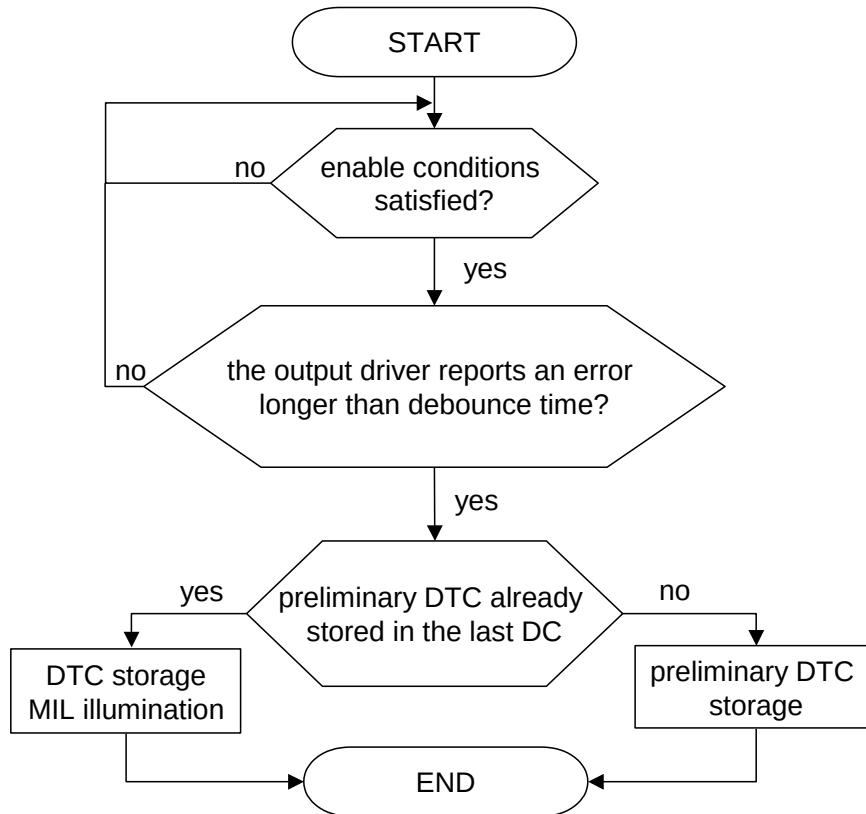
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Flowchart:



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15.40 Throttle Valve

15.40.1 Circuit continuity (P2620, P0638, P2621, P2622)

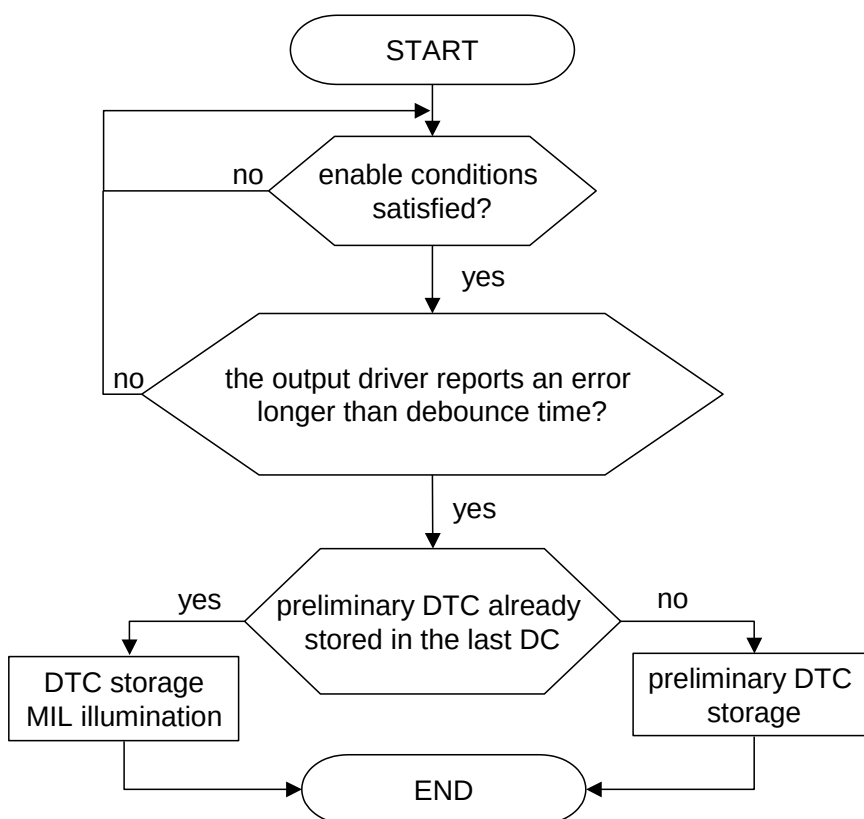
General description:

The throttle valve is controlled by an output driver with self-diagnosis capability. The driver can detect open circuit, short circuit to battery, short circuit to ground and overtemperature from the power stage. It can also detect governor deviation, controller errors and communication errors of the throttle valve.

If an error is present, the driver communicates this information to the processor, which recognizes a fault, and stores a preliminary DTC. If this fault is detected in two consecutive driving cycles, the MIL is illuminated.

The throttle valve actuator is a so called “smart device”. It is an actuator with an integrated position sensor. If the smart device detects a governor deviation (e.g. caused by a jammed throttle valve) a failure is reported to the engine ECU applying a defined pulse duty factor.

Flowchart:





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15.41 Vehicle Speed Sensor

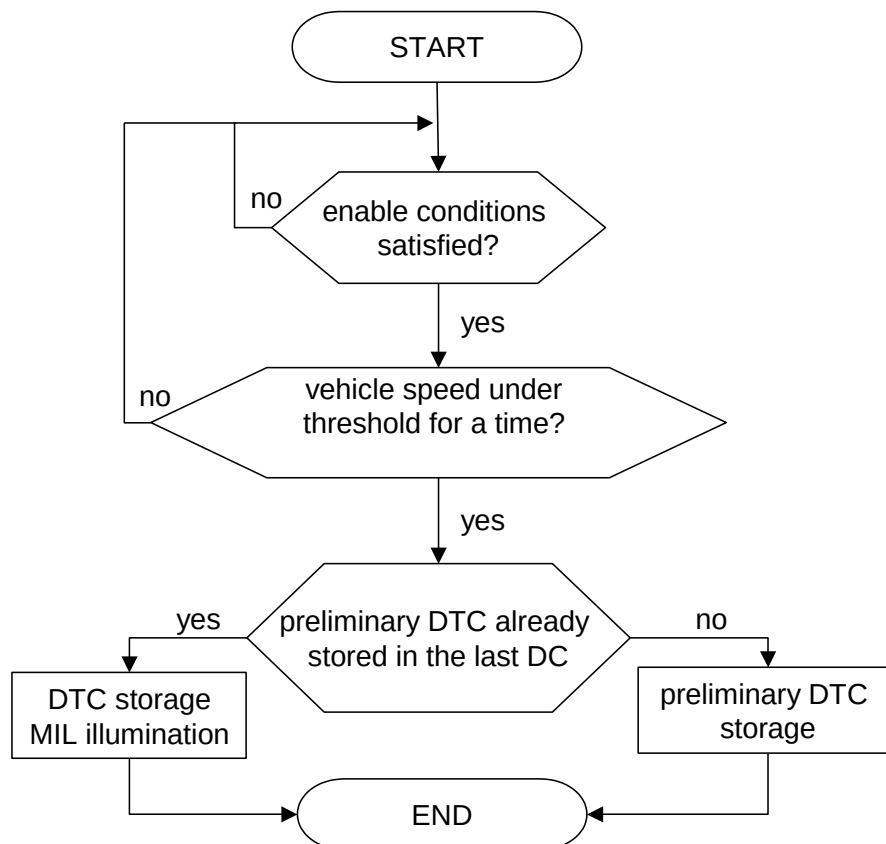
The information of the wheel speed sensors is provided by the Antilock Break System (ABS) control unit via CAN. The vehicle speed is determined with the average value of the designated wheel speed sensors.

15.41.1 Functional (P0501)

General description:

The vehicle speed is monitored for plausibility using engine speed und torque. If these two values are above a threshold and the vehicle speed is below a threshold longer than the debounce time, a fault is detected and a preliminary DTC is stored. If this malfunction is detected in two consecutive driving cycles the MIL is illuminated.

Flowcharts:





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15.41.2 Other

**15.41.2.1 Signal Fault
(P0500)**

General description:

If none of the CAN messages of the designated wheel speed sensors are plausible the vehicle speed signal fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles, the MIL is illuminated. This is a continuous diagnosis.

Flowchart:

see 21.1.4 CAN Signal Fault

**15.41.2.2 Timeout Fault
(P0500)**

General description:

If no CAN message of the designated wheel speed sensors is received the vehicle speed timeout fault is detected and a preliminary DTC is stored. If the fault is detected in two consecutive driving cycles, the MIL is illuminated. This is a continuous diagnosis.

Flowchart:

see 21.1.5 CAN Timeout Fault



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16 Pinning ECU

Pin	Description
101	injector 4, HS, cylinder 5
102	injector 1, HS, cylinder1
103	injector 4, LS, cylinder 5
104	injector 5, LS, cylinder 6
105	injector 5, HS, cylinder 6
106	injector 2, HS, cylinder 3
107	injector 3, LS, cylinder 2
108	injector 2, LS, cylinder 3
109	injector 6, HS, cylinder 4
110	injector 3, HS, cylinder 2
111	injector 6, LS, cylinder 4
112	injector 1, LS, cylinder 1
113	reductant injection system pressure pump, passive tank
114	reductant injection system pressure pump, active tank
116	reductant injection system tank heater, active tank
117	reductant injection system reverse control valve
118	reductant injection system pressure line heater
119	reductant injection system level sensor passive tank
120	reductant injection system level sensor active tank
202	mainrelay
203	battery ground
204	switched battery voltage (T87)
205	battery ground
206	switched battery voltage (T87)
207	battery ground
208	switched battery voltage (T87)
301	CAN (high) NOx sensor downstream
303	brake light switch
307	starter relay (automatic start)
309	CAN (low) NOx sensor downstream
310	engine fan control
311	brake light switch (test)
313	ignition switch (T15)
316	engine speed signal output
317	fuel filter heater (only X5)
318	accelerator pedal sensor 1, ground

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319	powertrain-CAN (low)
320	powertrain-CAN (high)
323	accelerator pedal sensor 2, ground
325	car access system (switch off)
327	crankcase ventilation heater
328	accelerator pedal sensor 1, signal
329	accelerator pedal sensor 2, signal
332	reductant injection system tank heater
333	engine fan supply voltage
335	battery sensor (BUS)
336	immobilizer
338	accelerator pedal sensor 1, supply
339	accelerator pedal sensor 2, supply
340	ECU-Box Fan (only 335d)
405	reductant injection system pressure sensor, signal
406	reductant injection system pressure sensor, sensor supply
407	reductant injection system pressure sensor, ground
408	exhaust temperature sensor upstream SCR, signal
409	exhaust temperature sensor upstream SCR, ground
410	reductant injection system temperature sensor, signal
411	reductant injection system temperature sensor, ground
414	glow control unit, ground
415	LIN-Bus / glow control
417	generator
418	oil quality level temperature sensor (only X5)
419	crankshaft position sensor, signal
420	crankshaft position sensor, sensor supply
421	crankshaft position sensor, ground
422	mass airflow sensor, signal
423	mass airflow temperature sensor
424	mass airflow sensor, ground
425	reductant injection system tank heater, diagnosis current for tank heater
426	reductant injection system tank heater, diagnosis current for pressureline heater
429	oil pressure sensor signal
433	lambda sensor heater
434	lambda sensor, pump current
435	lambda sensor, virtual ground
436	lambda sensor, compensation current
437	lambda sensor nernst voltage
441	throttle valve actuator

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444	exhaust temperature sensor downstream EGR cooler, signal
445	fuel temperature sensor, signal
446	fuel temperature sensor, ground
447	fuel pre supply pressure sensor, sensor supply
448	fuel pre supply pressure sensor, signal
449	induction air temperature sensor signal, ground
450	induction air temperature sensor signal, signal
451	low pressure EGR valve
452	EGR cooler bypass
453	exhaust temperature sensor downstream EGR cooler, ground
454	manifold absolute pressure sensor, sensor supply
455	manifold absolute pressure sensor, signal
456	manifold absolute pressure sensor, ground
457	high pressure EGR valve position sensor, signal
458	high pressure EGR valve position sensor, ground
459	high pressure EGR valve position sensor, sensor supply
501	rail pressure control valve
502	rail metering unit
503	high pressure EGR valve, h-bridge +
504	high pressure EGR valve, h-bridge -
506	low pressure EGR valve position sensor, sensor supply
507	low pressure EGR valve position sensor, ground
508	exhaust temperature sensor upstream DOC, signal
509	exhaust temperature sensor upstream DOC, ground
510	exhaust temperature upstream DPF, signal
511	exhaust temperature upstream DPF, ground
512	particulate matter filter differential pressure, ground
513	particulate matter filter differential pressure, signal
514	particulate matter filter differential pressure, sensor supply
515	oil quality level temperature sensor, ground (only X5)
518	reductant injection system dosing valve, HS
519	turbocharger bypass valve
520	reductant injection system dosing valve, LS
521	turbocharger low pressure wastegate regulating valve
524	engine suspension control 1
525	exhaust manifold pressure sensor, signal
526	exhaust temperature sensor downstream EGR LP cooler, ground
527	exhaust temperature sensor downstream EGR LP cooler, signal
529	engine coolant temperature sensor, ground
530	engine coolant temperature sensor, signal
531	CAN (low) NOx sensor upstream

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532	CAN (high) NOx sensor upstream
533	fuel rail pressure sensor, sensor supply
534	fuel rail pressure sensor, signal
535	fuel rail pressure sensor, ground
536	camshaft position sensor, signal
537	camshaft position sensor, ground
547	turbocharger high pressure wastegate regulating valve
548	swirl valve actuator
552	exhaust manifold pressure sensor, sensor supply
553	low pressure EGR valve position sensor, signal
554	exhaust manifold pressure sensor, ground

	= OBD relevant
EGR	= exhaust gas recirculation
DOC	= diesel oxidation catalyst (NMHC-catalyst)
DPF	= diesel particulate matter filter
SCR	= selective catalytic reduction
HS	= high side
LS	= low side

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17 Scan Tool communication (E8)

17.1 Standardization

Scantool-Communication has been implemented according to the following standards:

- SAE-J1979, Rev. May 2007, SAE-J2012
- ISO 15765

17.2 Service \$01: Current Powertrain Diagnostic Data

The following PIDs are supported in Service \$01:

Supported data streams	PID [hex]
number of stored confirmed fault codes	01
MIL status	01
Readiness status (f) 1 NMHC (f) 2 NOx Cat (f) 3 Misfire (f) 4 Fuel System (f) 5 EGS (f) 6 EGR (f) 7 Boost Control (f) 9 PM Filter (f) 15 CC	01
calculated load value	04
engine coolant temperature	05
engine speed	0C
vehicle speed	0D
intake air temperature	0F
air flow rate from mass air flow sensor	10
absolute throttle position	11
oxygen sensor location	13
OBD requirements to which the engine is certified	1C
time elapsed since engine start	1F
distance traveled while MIL activated	21
oxygen sensor output	24
commanded EGR valve duty cycle/position	2C
number of warm-up cycles since fault memory last cleared	30
distance traveled since fault memory last cleared	31
barometric pressure	33
monitor status since last engine shut-off for each monitor used for readiness status	41
engine control module system voltage	42

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relative throttle position	45
ambient air temperature	46
absolute pedal position	49
redundant absolute pedal position	4A
commanded throttle motor position	4C
fuel rate	5E
driver's demand engine torque	61
actual engine torque	62
commanded EGR valve duty cycle/position	69
actual EGR valve duty cycle/position	69
EGR error between actual and commanded	69
fuel rail pressure	6D
commanded fuel rail pressure	6D
boost pressure	70
commanded/target boost pressure	70
wastegate valve position	72
exhaust pressure sensor output	73
intercooler temperature	77
exhaust gas temperature sensor output (temperature upstream DOC --> EGT Bank 1, Sensor 1) (temperature upstream DPF --> EGT Bank 1, Sensor 2) (temperature upstream SCR --> EGT Bank 1, Sensor 3)	78
PM filter delta pressure	7A
total engine run time	7F
engine run time for AECD - deactivation of EGR due to excessive coolant temperature	81
NOx sensor output	83

Not supported datastreams (redundant information)	PID [hex]	see PID
manifold absolute pressure	0B	70
fuel pressure	23	6D
EGR error between actual and commanded	2D	69
turbocharger turbine outlet temperature	75	78
catalyst temperature (temperature upstream DOC --> PID 78 - EGT bank 1, Sensor 1) (temperature upstream SCR --> PID 78 - EGT bank 1, Sensor 3)	3C/3E	78
PM filter inlet temperature (PM inlet temperature --> PID 78 - EGT bank 1, Sensor 2)	7C	78
PM filter outlet temperature (PM outlet temperature --> PID 78 - EGT bank 1, Sensor 3)	7C	78



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17.3 Service \$02: Powertrain Freeze Frame Data

The following PIDs are supported in Service \$02:

Supported freeze frame data	PID [hex]
DTC that caused required freeze frame data storage	02
calculated load	04
engine coolant temperature	05
engine speed	0C
vehicle speed	0D
intake manifold temperature	0F
air flow rate from mass air flow sensor	10
absolute throttle position	11
time elapsed since engine start	1F
commanded EGR valve duty cycle/position	2C
barometric pressure	33
engine control module system voltage	42
relative throttle position	45
ambient air temperature	46
absolute pedal position	49
redundant absolute pedal position	4A
commanded throttle motor position	4C
fuel rate	5E
driver's demand engine torque	61
actual engine torque	62
commanded EGR valve duty cycle/position	69
actual EGR valve duty cycle/position	69
EGR error between actual and commanded	69
fuel rail pressure	6D
commanded fuel rail pressure	6D
boost pressure	70
commanded/target boost pressure	70
wastegate valve position	72
exhaust pressure sensor output	73
intercooler temperature	77
exhaust gas temperature sensor output (temperature upstream DOC --> EGT Bank 1, Sensor 1) (temperature upstream DPF --> EGT Bank 1, Sensor 2) (temperature upstream SCR --> EGT Bank 1, Sensor 3)	78
PM filter delta pressure	7A

Not supported freeze frame data (redundant information)	PID [hex]	see PID
manifold absolute pressure	0B	70
EGR error between actual and commanded	2D	69

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turbocharger turbine outlet temperature	75	78
fuel pressure	23	6D
PM filter inlet temperature (PM inlet temperature --> PID 78 - EGT bank 1, Sensor 2)	7C	78
PM filter outlet temperature (PM outlet temperature --> PID 78 - EGT bank 1, Sensor 3)	7C	78

17.4 Service \$06: On-Board Monitoring Test Results for Specific Monitored Systems

The following test results are supported in Service \$06:

OBD M ID	Test ID	Unit ID	Unit	Description Scan Tool	P- Code	Comment
01	E1	11	s	Exhaust Gas Sensor Monitor Bank 1 - Sensor 1	P0133	LSU Dynamic
01	E2	11	s	Exhaust Gas Sensor Monitor Bank 1 - Sensor 1	P0133	LSU Dynamic
01	E3	11	s	Exhaust Gas Sensor Monitor Bank 1 - Sensor 1	P0133	LSU Dynamic
01	E4	30	%	Exhaust Gas Sensor Monitor Bank 1 - Sensor 1	P2297	LSU Plaus Overrun
01	E5	30	%	Exhaust Gas Sensor Monitor Bank 1 - Sensor 1	P2A00	LSU Plaus Part Load
02	B1	11	s	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Dynamic
02	B2	11	s	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Dynamic
02	B3	81	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Offset Max
02	B4	81	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Offset Min
02	B5	86	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Plausibility Max
02	B6	86	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 2	P2201	NOx Us Plausibility Min
03	C1	30	%	Exhaust Gas Sensor Monitor Bank 1 - Sensor 3	P229F	NOx Ds Lam Lin Plaus
03	C2	81	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 3	P229F	NOx Ds Offset Max
03	C3	81	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 3	P229F	NOx Ds Offset Min
03	C4	81	raw	Exhaust Gas Sensor Monitor Bank 1 - Sensor 3	P229F	NOx Ds Stk Err
21	F2	05	raw	Catalyst Monitor Bank 1	P0420	NMHC Catalyst Efficiency
31	81	03	raw	EGR Monitor Bank 1	P0400	EGR Slow response neg.
31	82	03	raw	EGR Monitor Bank 1	P0400	EGR Slow response pos.
31	83	96	°C	EGR Monitor Bank 1	P2457	EGR HP Cooler
81	91	01	raw	Fuel System Monitor Bank 1	P02CD P02CC	Injection timing / quantity

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81	92	01	raw	Fuel System Monitor Bank 1	P02D5 P02D4	Injection timing / quantity
81	93	01	raw	Fuel System Monitor Bank 1	P02D1 P02D0	Injection timing / quantity
81	94	01	raw	Fuel System Monitor Bank 1	P02D7 P02D6	Injection timing / quantity
81	95	01	raw	Fuel System Monitor Bank 1	P02CF P02CE	Injection timing / quantity
81	96	01	raw	Fuel System Monitor Bank 1	P02D3 P02D2	Injection timing / quantity
85	F1	06	raw	Boost Pressure Control Monitor Bank 1	P026A	Charge Air Cooling Efficiency
98	D1	04	raw	NOx Catalyst Monitor Bank 1	P20EE	NOx Catalyst Efficiency
A2	0C	24	counts	Misfire Cylinder 1 Data	P0301	-
A2	0B	24	counts	Misfire Cylinder 1 Data		-
A3	0C	24	counts	Misfire Cylinder 2 Data	P0302	-
A3	0B	24	counts	Misfire Cylinder 2 Data		-
A4	0C	24	counts	Misfire Cylinder 3 Data	P0303	-
A4	0B	24	counts	Misfire Cylinder 3 Data		-
A5	0C	24	counts	Misfire Cylinder 4 Data	P0304	-
A5	0B	24	counts	Misfire Cylinder 4 Data		-
A6	0C	24	counts	Misfire Cylinder 5 Data	P0305	-
A6	0B	24	counts	Misfire Cylinder 5 Data		-
A7	0C	24	counts	Misfire Cylinder 6 Data	P0306	-
A7	0B	24	counts	Misfire Cylinder 6 Data		-
B2	A1	17	kPa	PM Filter Monitor Bank 1	P2002	PM Filter Efficiency
B2	A2	17	kPa	PM Filter Monitor Bank 1	P14A6	PM Filter Missing Substrat
B2	A3	36	g	PM Filter Monitor Bank 1	P2458	PM Filter Incomplete Regen.
B2	A4	2F	%	PM Filter Monitor Bank 1	P2459	PM Filter Regen. Frequency

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17.5 Service \$09: Vehicle information

The following InfoTypes are supported in Service \$09:

Description:

Infotype:

Vehicle Identification Number	\$02
Calibration Identifications (ECU and NOx sensors)	\$04
Calibration Verification Numbers (ECU and NOx sensors)	\$06
ECU Name	\$0A
In-use Performance Tracking	\$0B

The In-use Performance Tracking contains the following information:

Group	P-Code	Description	Denom
NMHC Catalyst Monitor	P0420	NMHC Catalyst Efficiency	500mi
NOx Catalyst Monitor	P20EE	NOx Catalyst Efficiency	-
NOx Adsorber Monitor	-	(No component in this category available)	-
PM Filter Monitor	P2002	PM Filter Efficiency	500mi
Exhaust Gas Sensor Monitor (EGS)	P0133	EGS Bank 1 Sensor 1: LSU Dynamic	-
	P2297	EGS Bank 1 Sensor 1: LSU Plaus Overrun Max	-
	P2297	EGS Bank 1 Sensor 1: LSU Plaus Overrun Min	-
	P2A00	EGS Bank 1 Sensor 1: LSU Part Load Max	-
	P2A00	EGS Bank 1 Sensor 1: LSU Part Load Min	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Offset Learn Max	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Offset Learn Min	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Dynamic	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Offset Max	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Offset Min	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Plausibility Max	-
	P2201	EGS Bank 1 Sensor 2: NOx Us Plausibility Min	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Lam Lin Min	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Lam Lin Max	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Lam Lin Plaus Max	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Lam Lin Plaus Min	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Offset Max	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Offset Min	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Offset Learn Max	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Offset Learn Min	-
	P229F	EGS Bank 1 Sensor 3: NOx Ds Stk Err	-
EGR/VVT Monitor	P0400	EGR Slow response neg.	-

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	P0400	EGR Slow response pos.	-
	P14D0	EGR LP Cooler (only X5)	-
	P2457	EGR HP Cooler	-
Boost Pressure Monitor	P026A	Charge Air Cooling Efficiency	-



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17.6 Similar Conditions

The following diagnosis supports similar conditions:

(f)(4) FUEL SYSTEM MONITORING
P0087
P0088

17.7 Permanent Trouble Codes

The OBDII-System supports 4 permanent trouble codes.

**18 In-use monitor performance ratio - kernel function**

The in-use monitor performance (IUMPR) kernel function represents the core of the software algorithms in the OBD II system implemented to individually track and report in-use monitor performance, in the standardized tracking and reporting format, for every monitor of the following components/systems:

- Catalyst Monitor Bank 1
- Catalyst Monitor Bank 2
- EGR/VVT Monitor

All monitors for which an in-use performance record is required do have an interface (a function identifier) through which they communicate with the IUMPR kernel function. It is this kernel function that does the actual tracking and preparation for reporting in the standardized format (see Figure below). The IUMPR kernel function additionally tracks and records the ignition cycle counter, the general denominator for every driving cycle and determines the monitor with the lowest numerical ratio within each group that has multiple monitors.

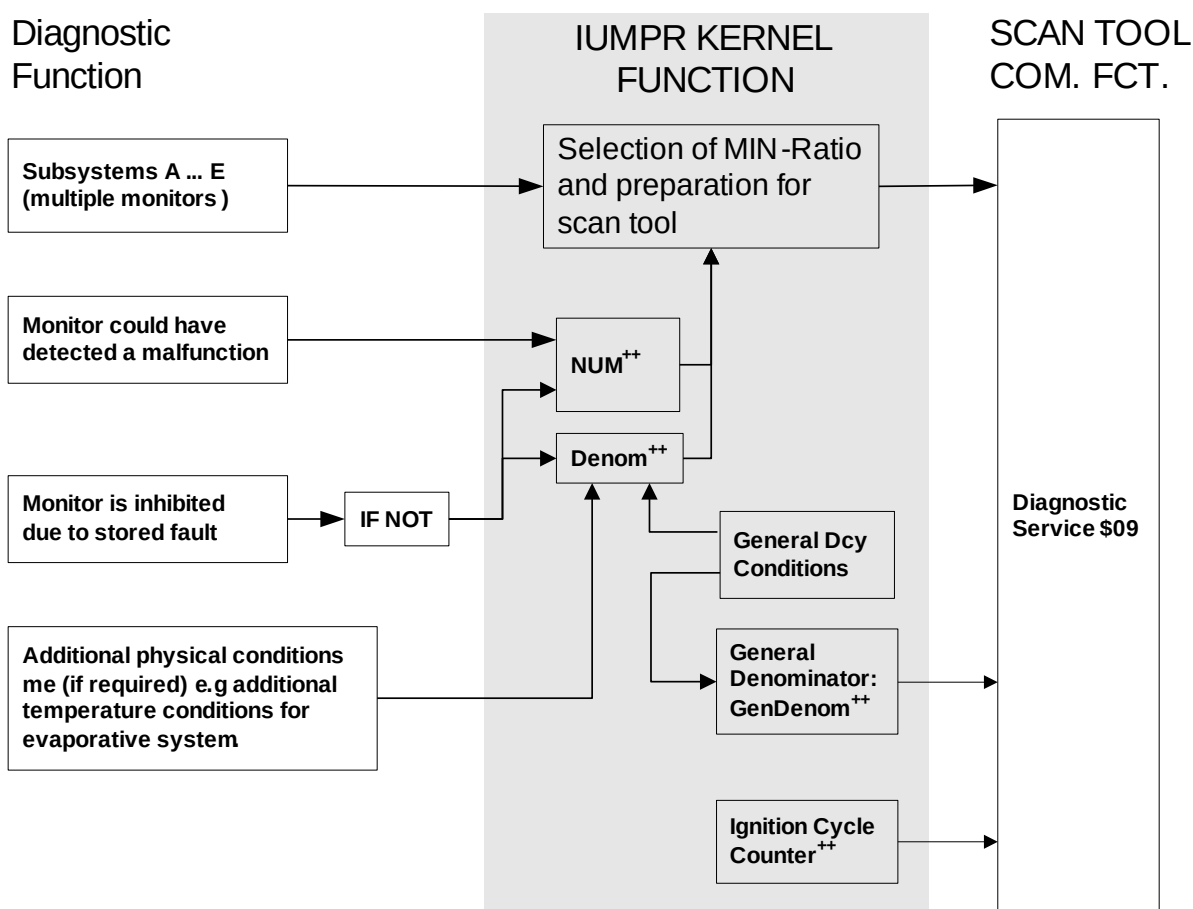


Figure: Schematic view of in-use monitor performance ratio implementation

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18.1 Ignition cycle counter

The ignition cycle counter, when incremented, is incremented by an integer of one and only once per driving cycle. If the ignition cycle counter reaches the maximum value of 65,535, it rolls over and increments to zero on the next ignition cycle to avoid overflow problems.

Conditions for incrementing the ignition cycle counter

- The ignition cycle counter is incremented within ten seconds if and only if the vehicle meets the engine start definition for at least two seconds plus or minus one second. The “Engine start” condition is determined via the engine speed sensor.
- Incrementing of the ignition cycle counter is disabled within ten seconds if a malfunction of the engine speed sensor has been detected and the corresponding pending fault code has been stored

18.2 General denominator

The general denominator, when incremented, is incremented by an integer of one and only once per driving cycle. If the general denominator reaches the maximum value of 65,535, it rolls over and increments to zero on the next driving cycle that meets the general denominator definition to avoid overflow problems.

Conditions for incrementing general denominator

The general denominator is incremented within ten seconds if and only if the following criteria are satisfied on a single driving cycle:

- cumulative time since engine start is above than or equal to 600 seconds while at an elevation of less than 8,000 feet above sea level and at an ambient temperature of greater than or equal to 20 degrees Fahrenheit
- cumulative vehicle operation at or above 25 miles per hour occurs for greater than or equal to 300 seconds while at an elevation of less than 8,000 feet above sea level and at an ambient temperature of greater than or equal to 20 degrees Fahrenheit
- continuous vehicle operation at idle (i.e., accelerator pedal released by driver and vehicle speed less than or equal to one mile per hour) for greater than or equal to 30 seconds while at an elevation of less than 8,000 feet above sea level and at an ambient temperature of greater than or equal to 20 degrees Fahrenheit
- No faults from those sensors used to determine ambient temperature, ambient pressure (elevation), vehicle speed and idle condition; in flowchart referred to as “global faults”.

Incrementing of the general denominator is disabled within ten seconds if a malfunction of one of the sensors mentioned above has been detected and the corresponding pending fault code has been stored.



18.3 IUMPR – Records

The kernel function maintains a record (a collection of elements from different types of arrays as depicted in figure below) for each monitor for which in-use performance ratio tracking is required. An update of a monitor's record is triggered or inhibited by the monitor itself. Each monitor's function identifier addresses its corresponding record via a pointer.

Each record holds the following information about the respective monitor:

- the function identifier (interface between monitor and IUMPR kernel function)
- the associated diagnostic fault path
- the numerator
- the denominator
- IUMPR status information from the diagnostic function
- the associated component/system group (necessary for selection of minimum ratio of multiple monitors of one of the subsystems).

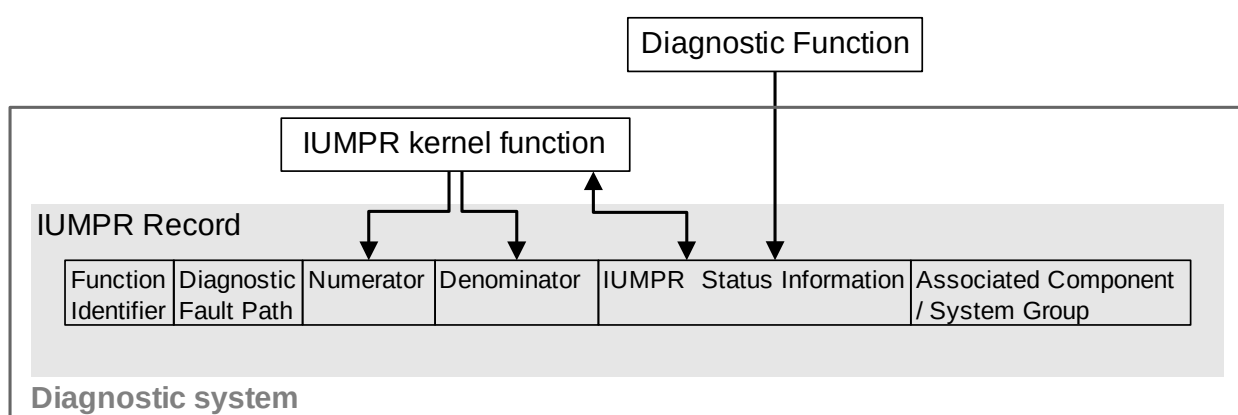


Figure: Illustration of IUMPR record

IUMPR Status Information
Fault is found / could have been found
Inhibition of denominator due to physical conditions if considered (e.g. for Evaporative System or Secondary Air System)
Inhibition of monitor due to stored fault
Numerator incremented in this driving cycle
Denominator incremented in this driving cycle

Figure: IUMPR status information of a monitor

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18.4 Incrementing the numerator and denominator

A cyclic check is performed to find out if all conditions necessary for incrementing the numerator and the denominator have been fulfilled.

Conditions for incrementing the numerator

- incrementing of the general denominator is not disabled by a fault
- no inhibition of the diagnostic function due to a fault
- diagnostic function has found or could have found a fault (includes all monitoring conditions)
- the numerator has not yet been incremented in this driving cycle.

Conditions for incrementing the denominator

- conditions for incrementing the general denominator have been fulfilled
- incrementing of the general denominator is not disabled by a fault
- no inhibition of the diagnostic function due to fault
- Additional physical conditions met (if required)
- denominator has not yet been incremented in this driving cycle.

If either the numerator or denominator for a specific component reaches or exceeds the maximum value of 65,535, both numbers are divided by two before either is incremented. This helps to avoid overflow problems.

18.5 Minimum ratio selection (multiple monitors)

The associated component/system group identifier in a record is a pointer to the group (subsystem A...E) a monitor belongs to. IUMPR ratios are continuously calculated for all monitors. The IUMPR kernel function continuously determines the monitor with the lowest ratio in each group and provides its numerator and denominator values to Service \$09 of the generic scan tool together with the ignition cycle counter and the general denominator.



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Flowchart

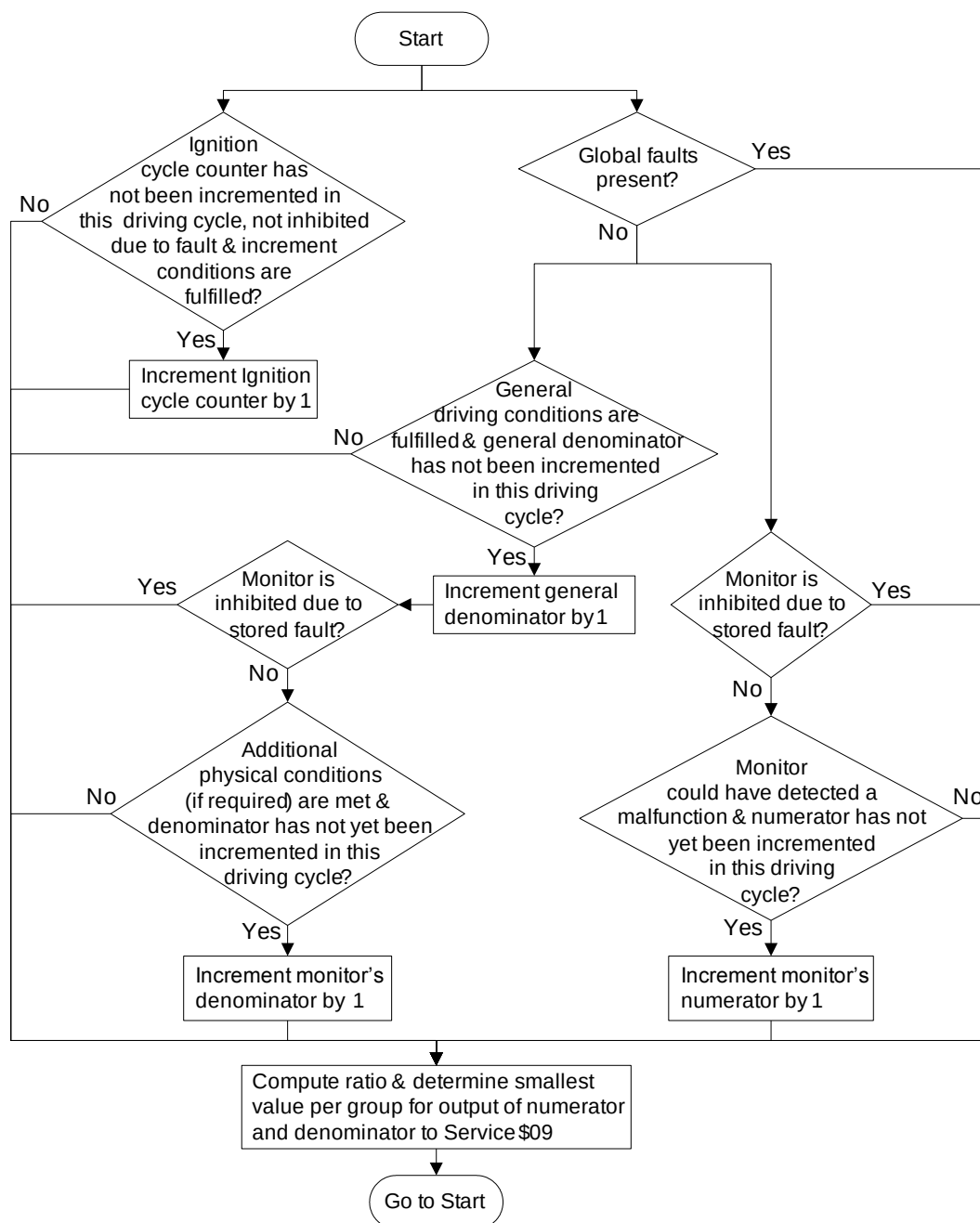


Figure: in use monitor performance ratio - kernel function - Flowchart

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19 Location of data link connector

19.1 Test Group 9BMXT04.8E70 (model X5 4.8i)



Position OBD II Connector

Figure 1: Position DLC

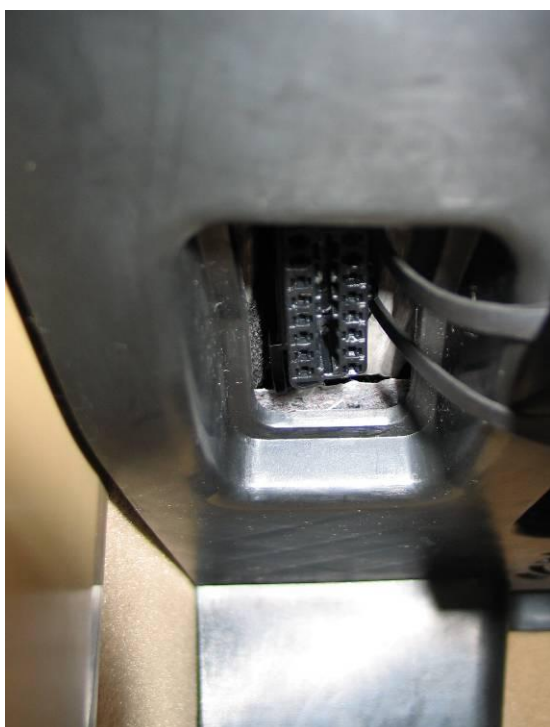


Figure 2: DLC and removed cover

The DLC is located at the lower left side of the instrument cluster. Actual still open, if there will be one cover with the letters "OBD" on it as shown removed or not any cover.

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19.2 Test Group (model 335td)



Figure: Position DLC for **3 series** and **1 series** models and closed cover

The DLC is located at the lower left A-pillar and under a cover. This cover has the letters OBD on it.

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20 Drawing and location of the Malfunction Indicator Lamp

The Diesel engine utilizes a standardized instrument panel (identical to gasoline vehicles) that turns the MIL on if the communication between engine and instrument panel is lost.

20.1 Test Group 9BMXT04.8E70 (model X5 4.8i)





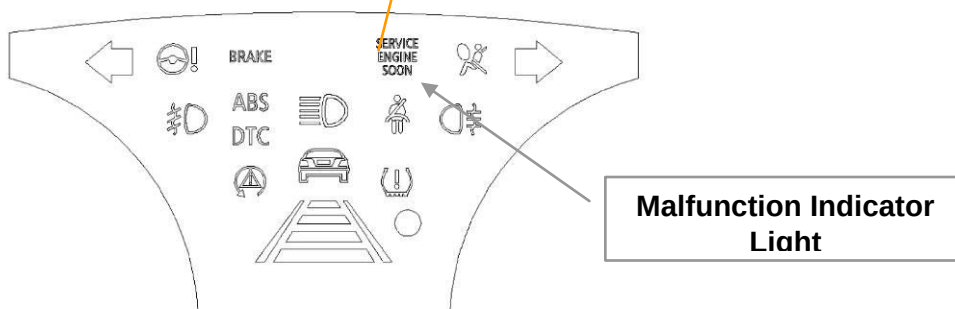
20.2 Test Group ... (model 335td)

20.2.1

Complete Instrument panel (European Version)



Detail (US Version)





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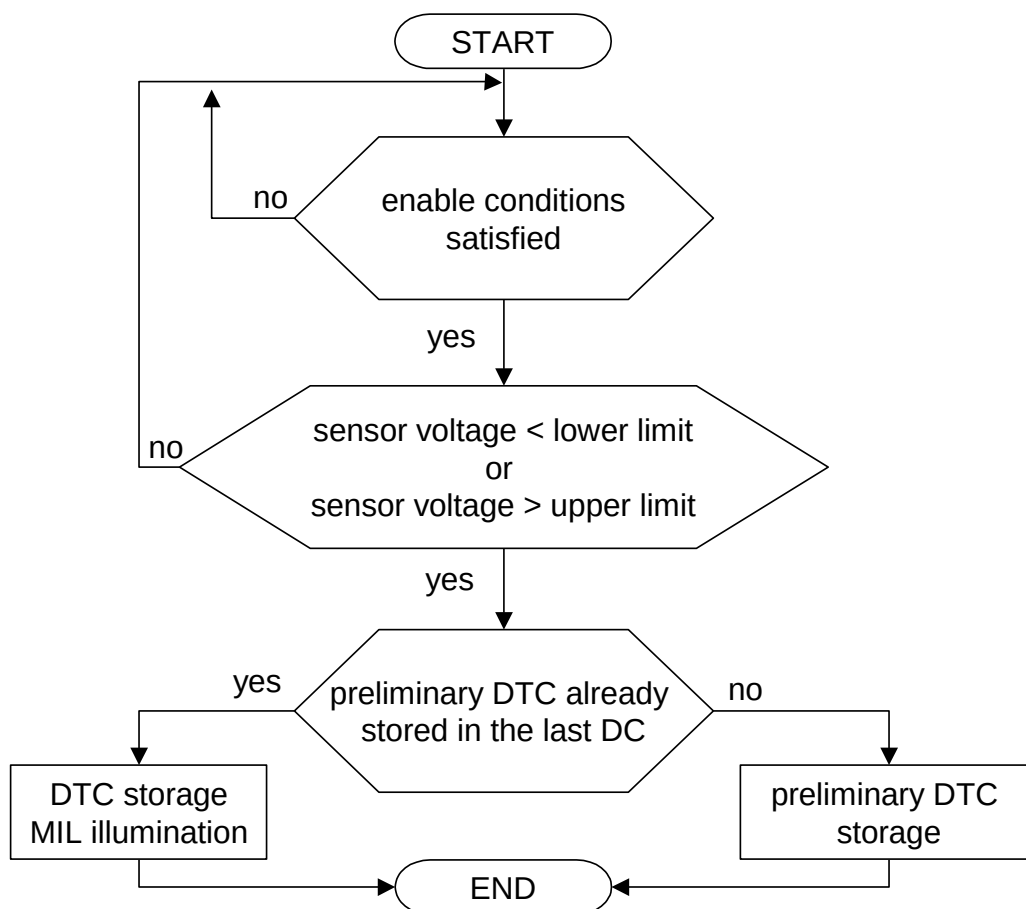
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21 Appendix

21.1 General Flowcharts

21.1.1 Circuit continuity

21.1.1.1 Sensor Voltage



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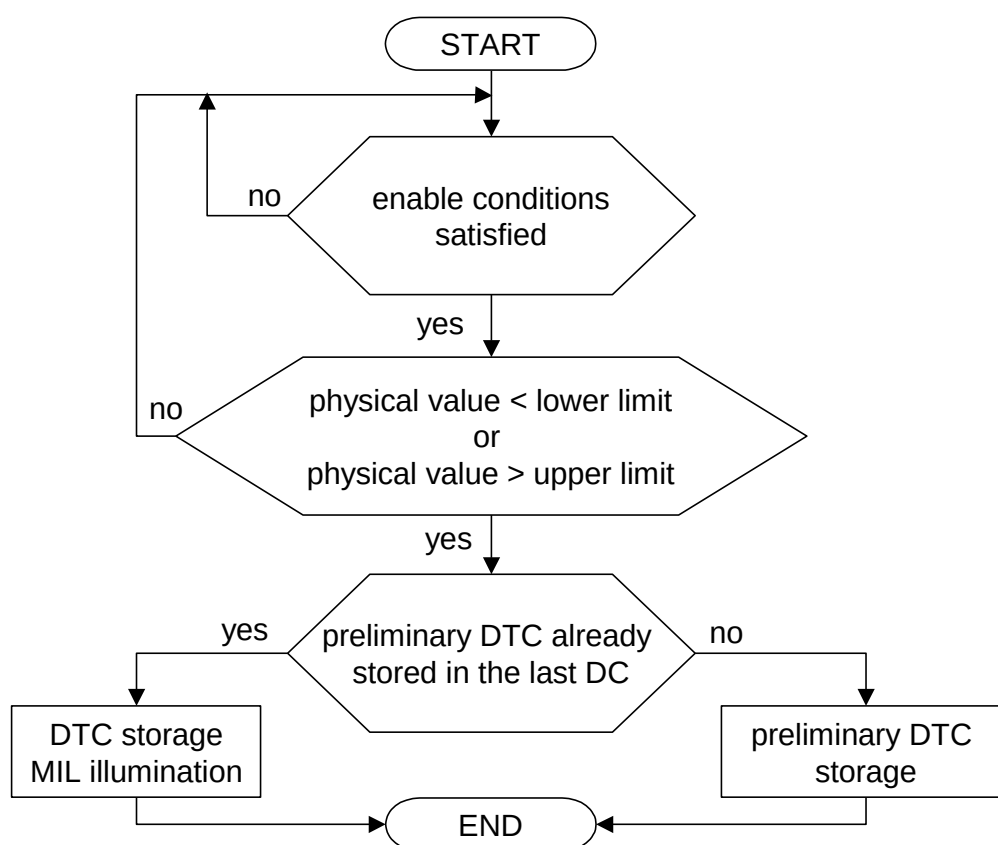
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21.1.1.2 Physical Value



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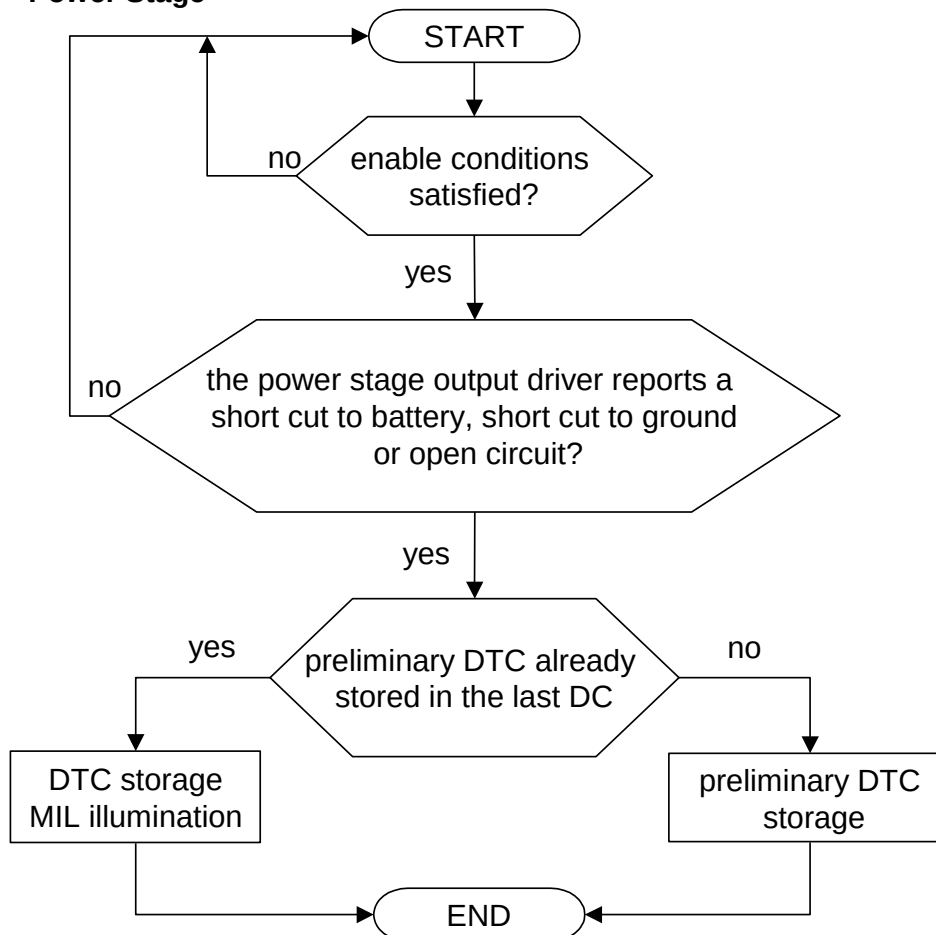
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21.1.1.3 Power Stage



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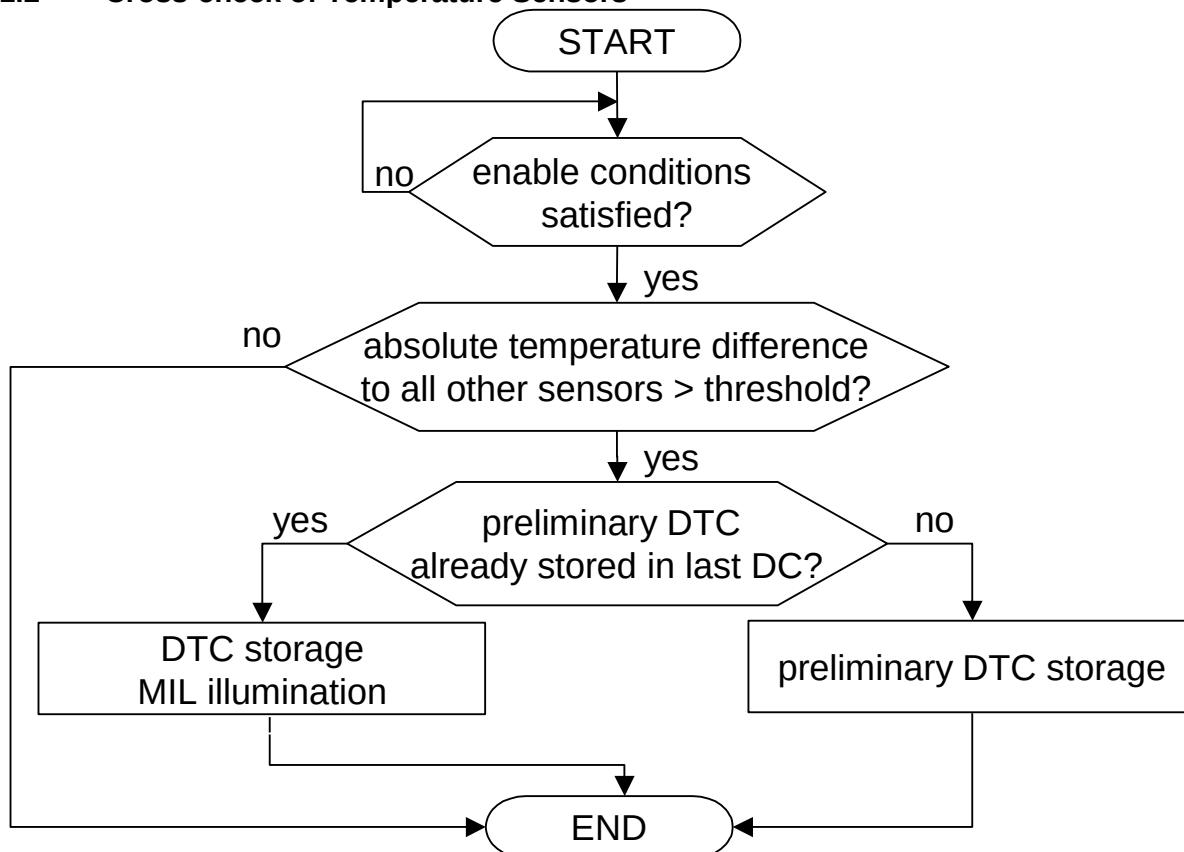
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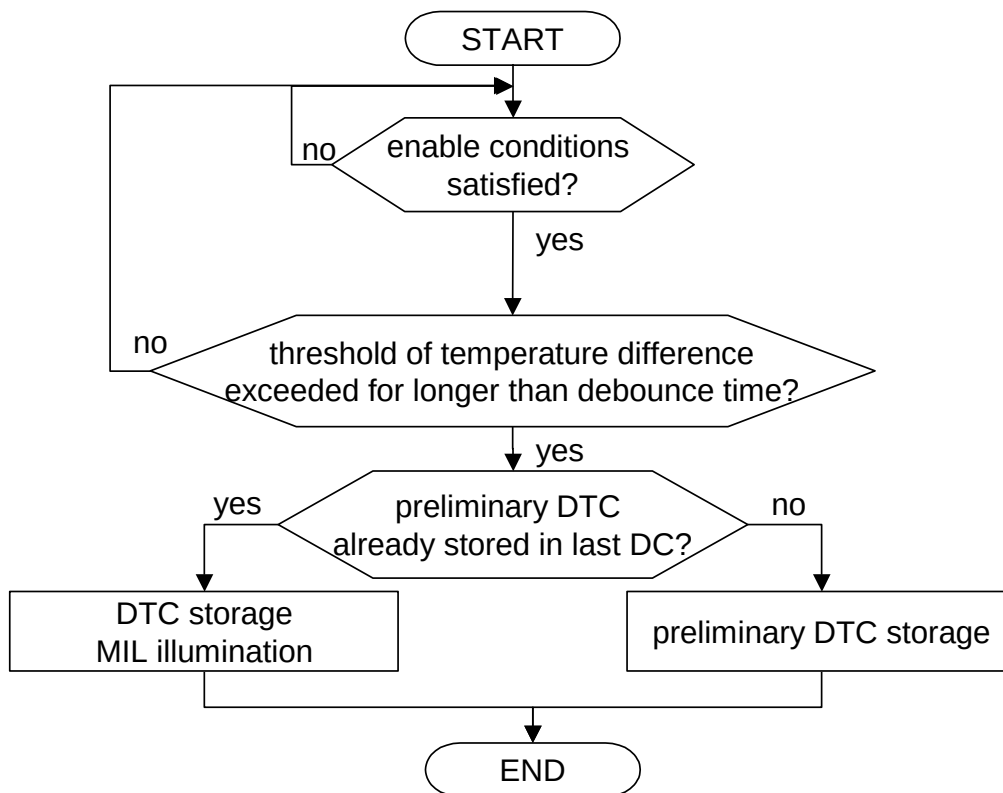
21.1.2 Cross-check of Temperature Sensors



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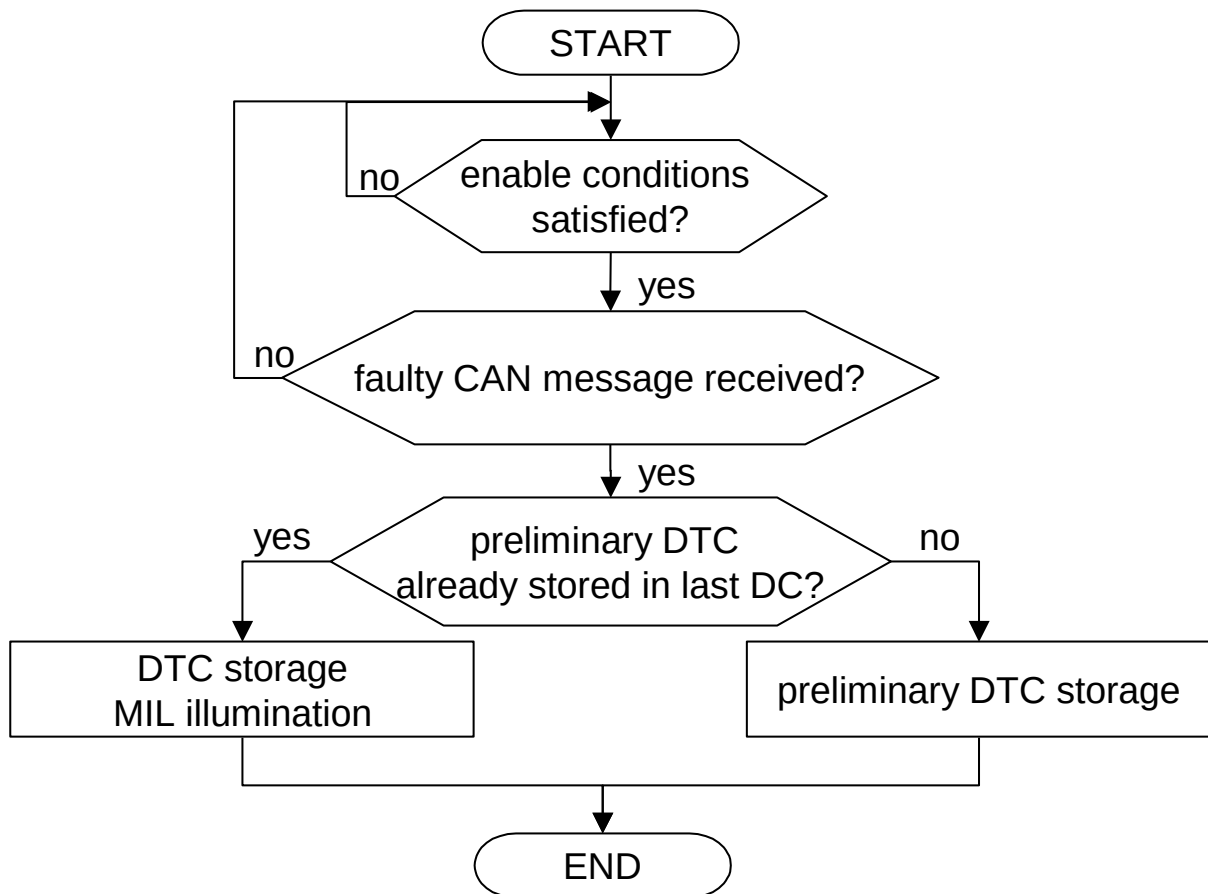


21.1.3 Rationality Check Low





21.1.4 CAN Signal Fault





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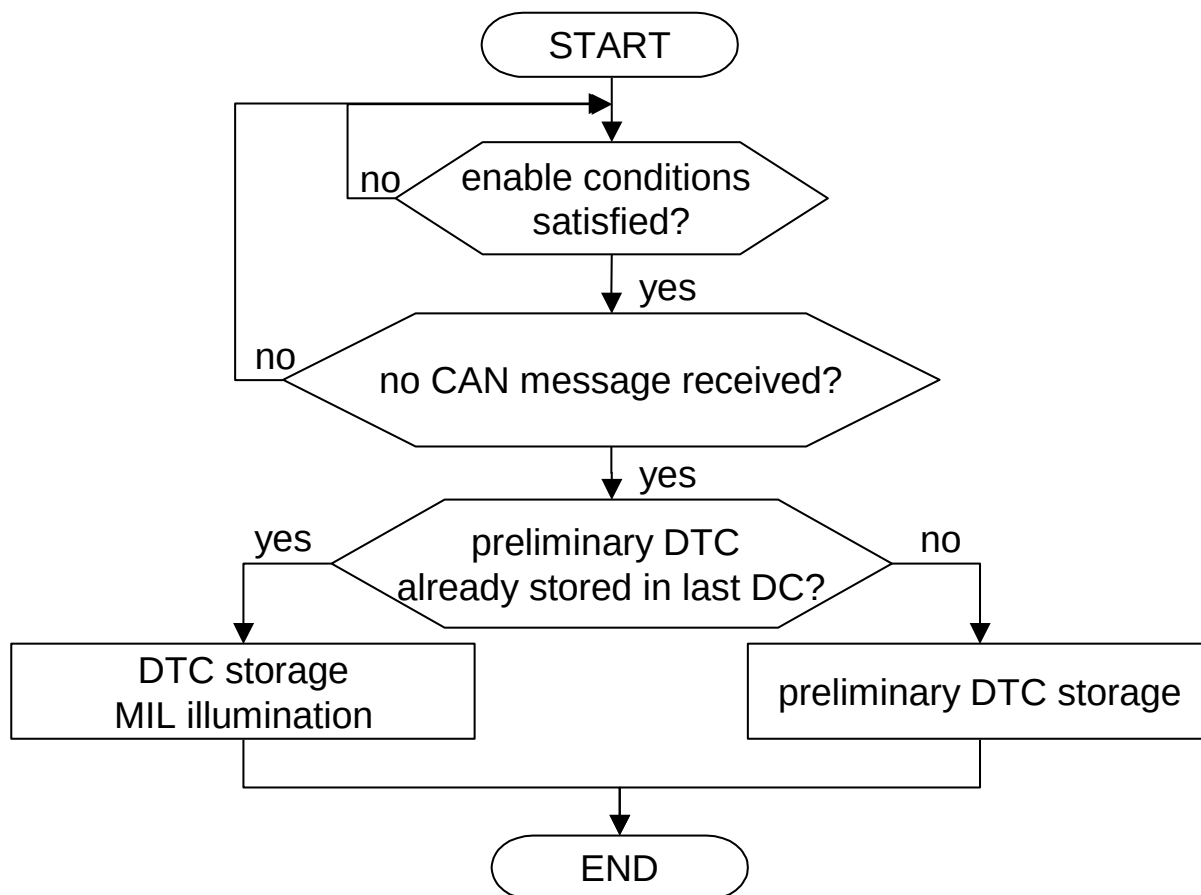
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21.1.5 CAN Timeout Fault



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